



2021 K-Innovation ODA Program with Lao PDR

National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022

2021 K-Innovation Partnership Program with Lao PDR

Title	National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022
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National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022

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Executive Summary

The role of Science, Technology, and Innovation (STI) is considered as a key tool to national economic and sustainable development under the 9th National Socio-Economic Development Plan. In order to advance the national innovation system in Lao PDR, the country has outlined five main goals to improve R&D, upgrade STI education, strengthen international and domestic STI networks, foster public research, and promote technology adaptation in its National Strategy for STI Development.

With the recognition of its weak STI institutions and a lack of STI resources to fulfill the national strategy, Lao PDR requested the STI policy training and consultation supports to the Science and Technology Policy Institute (STEPI) in 2017. After the timely short-term training services from STEPI to the Ministry of Science and Technology (MOST) in Lao PDR in 2018, the country's first-ever policy consultation project titled National Master Plan for STI Research & Development and Application in Lao PDR 2020-2022' was successfully initiated by the Science and Technology Policy Institute (STEPI) with funding from the Korean Ministry of Economy and Finance in 2020.

The aim of this joint project between STEPI and MOST was designed to support Lao government officials' task on developing the National Master Plan for STI Research & Development and Application and to promote a better understanding of STI systems and recent STI issues in both countries that will be the critical foundation for further bilateral cooperation in the future. The main components of this project are 1) mutual learning of Korean experience cases in STI-driven economic development and current STI-related challenges facing the Lao government, 2) policy consultations from STI policy experts in Korea and Lao PDR, and 3) capacity-building programs for policy planning.

It's been unfortunate that the project launched amid the COVID-19 outbreak in the spring of 2020, and so there haven't been any in-person meetings between Korean and Laotian

research teams for two years 2020-2021. During the first year of the project, the research team overviewed the national STI policy and education in Lao PDR and presented a brief case of Korea's STI development path with Korea's current status in the global research ecosystem.

In the spring of 2021, the second year of the project, the Laotian government has made a dramatic restructuring of its ministries. MOST, a counterpart of the K-Innovation Lao PDR Project, has been dissolved, and its main function for technology and innovation has been moved into the Department of Technology and Innovation (DTI) in the Ministry of Technology and Communications in Lao PDR. Due to this big change in the domestic governmental environment in Lao PDR, our project made a necessary reorganization of its work scope to fully reflect the most recent needs and demands from our partner organization and country.

In this second year, the research team has offered more in-depth case studies on several major STI topics such as research institutes, STI workforce, and STI-related sectoral development. Firstly, Mr. Myunghwan Oh delineates a political, economic, and societal history of Lao PDR with the most up-to-date environments of the country.

In the following STI-focused chapters, Professor Mahn Yong Moon delivers a historical analysis on the emergence and expansion of government-funded research institutes (GRIs) and explains how GRIs in the form of a science and technology complex paved the way forward for STI-oriented national economic development in Korea. Professor Jejeong Lee then examines the current structures and roles of major public STI research institutes in Lao PDR and points out the urgency of the digitalization of basic STI data produced by public organizations.

In the next issue of the STI workforce and role of higher education, Professor Eunkyong Lee delineates the decades-long efforts of the Korean government to foster highly qualified R&D personnel and research-oriented universities, and presents insightful implications for the policy of fostering scientific and technological workforce in Lao PDR. An overview of the higher education system and roles of universities for national development in Lao PDR is

also presented by Professor Ji Soon Chang.

As a sample case of the development of STI-applied sectors, Dr. Jeehyun Kim provides a rich historical analysis of R&D and workforce development in agriculture and biotechnology in Korea. Mr. Changho Shin also presents a brief investigation of the Laotian situation for the same sectors for benchmarking.

Despite the continuing global pandemic constraints hindering face-to-face interaction, the project has been carried out under the auspices and contribution of our research teams in both Korea and Lao PDR, and STI policy experts participating in the online high-level launching seminar, K-Innovation Dissemination Forum, and capacity-building program. STEPI would like to credit the tremendous support from Mr. Choulamany and DTI colleagues, the Embassy of Lao PDR in Korea, Korea-Lao PDR Friendship Association, and Korean Ministry of Economy and Finance. With the development of the National STI Master Plan draft awaited in this coming final year of our project, STEPI expects this report to be a quality analysis to guide our collaboration between Korean and Laotian experts to explore national innovation systems in both countries.

CONTENTS

Chapter 1. Project Overview	3
1. Introduction	3
2. Objectives	6
3. Project Framework.....	7
4. Project Team	9
5. Project Activities	11
6. Project Schedule.....	14
7. Project Outputs	16

Chapter 2. Macro-environments of Lao PDR.....	19
1. Lao PDR's National Development Strategies	19
2. Political History and Current Environment	21
3. Economic History and Current Environment.....	31
4. Education.....	40
5. International Relations (main foreign partners)	43
6. Lao PDR-Korea Collaborations.....	54
7. Current Issues	61
8. Conclusion	65

Chapter 3. Public STI Research Institutes in Korea	70
1. Growth of Science and Technology of Korea and Government-funded Research Institutes.....	70
2. Emergence of GRIs: Establishment of KIST	74
3. Expansion of GRIs.....	86
4. Construction and Growth of Science and Technology Complex	95
5. Characteristics and key implications.....	101

Chapter 4. Public STI Research Institutes in Lao PDR.....	108
1. Overview and Background.....	108
2. List and information of Public STI Research Institutes.....	112
3. Policy and Governance Environment.....	125
4. Role of research institutes on STI and Economic Development in Lao PDR.....	131
5. Conclusions and Implications	137

Chapter 5. STI Higher Education and Universities in Korea	142
1. Introduction	142
2. Overview of Science and Technology Manpower Policy in Korea	145
3. In the 1960s: Laying the Foundation for Science and Technology Manpower Policy	150
4. In the 1970s: Establishment of Two-Track Manpower Nurturing System based on Sophistication of Industrial Structure	156
5. In the 1980s: Reorganization of R&D System and Establishment of Advanced Scientific and Technological Manpower Development System	162

6. In the 1990s: Nurturing Research-Oriented Universities and Fostering Excellent R&D Personnel	169
7. Paradigm shift in the 2000s: from quantity to quality, from nurturing to utilization	175
8. Implications for the Policy of Fostering Scientific and Technological Manpower in Lao PDR	183

Chapter 6. STI Higher Education and Universities in Lao PDR188

1. The Role of the University for National Economic Development based on STI in Lao PDR	188
2. The Analysis of Status of Lao PDR.....	192
3. The Case of Korea.....	202
4. The Role of the University for National Development	212

Chapter 7. STI in Agriculture Sector during Korea's Structural Transformation Period.....216

1. Historical Trends in the Development of Agriculture Sector in Korea	216
2. Key Policies and Systems in Agriculture Sector in Korea	232
3. Major Universities in Agriculture Sector in Korea.....	250
4. Major Research Institutes in Agriculture Sector in Korea.....	258
5. Implications of STI in Agriculture Sector during Korea's Structural Transformation Period	269

Chapter 8. STI in Biotechnology Sector during Korea's High-Tech Development Period.....274

1. Korea's Biotechnology Sector	274
---------------------------------------	-----

2. Major Policies and Systems in Biotechnology Sector in Korea	286
3. Major Universities in Biotechnology Sector in Korea	296
4. Major Research Institutes in Biotechnology Sector in Korea.....	301
5. Implications of STI in Biotechnology Sector during Korea's High-tech Development Period	306
Chapter 9. STI in Agriculture and Biotechnology Sectors in Lao PDR.....	310
1. STI in Agriculture Sector in Lao PDR.....	310
2. STI in Biotechnology Sector in Lao PDR	352
3. Implications of STI in Agriculture and Biotechnology Sector in Lao PDR.....	360
Chapter 10. Capacity Development Process	364
1. Preparations for the Second-Year Project	364
2. High-Level Forum and Kick-off Meeting.....	371
3. Online K-Innovation Dissemination Workshop.....	376
4. STI Demand Survey	377
5. Basic Statistics Information.....	383
6. Result	388
Chapter 11. Conclusion	428
1. Conclusion	428
REFERENCES	430

List of Tables

[Table 1-1] Research Team	9
[Table 1-2] Local Research Team	10
[Table 1-3] Project Schedule	15
<Table 4-1> Lists of public STI research institutes in Lao PDR	112
<Table 5-1>Trend of employment of KAIS master program graduates, 1975-1980	160
<Table 6-1> Number of Degree of Staff in NUOL.....	196
<Table 6-2> Lecturers by Academic Degree and Faculties at SU (2018-2019)	197
<Table 6-3> Status on Staff in Champasak University	198
<Table 6-4. Population, GDP, Exports in Korea(1960-2020)	203
<Table 6-5> Status on External Assistance in Korea (1945-199).....	203
<Table 6-6> Basic STI Data in Korea (1963-2019)	209
<Table 7-1> Trends in GDP of agriculture/forestry/fisheries industry and manufacturing industry in Korea, 1970-2000	217
<Table 7-2> Trends in Korean farm households, 1970-2000	218
<Table 7-3> Trends in Korean farm population, 1970-2000	218
<Table 7-4> Trends in economically active population in Korean agriculture, 1970-2000	219
<Table 7-5> Trends in the area of farmland and composition in Korea, 1970-2000....	220
<Table 7-6> Trends in the area of cultivated land per farm household in Korea, 1970- 2000	221
<Table 7-7> Trends of the farm households by size of the cultivated land in Korea, 1980-2000	222
<Table 7-8> Trends in the number of major agricultural machines in Korea.....	223
<Table 7-9> Trends in the value and percentage of agricultural production by commodity in Korea.....	225
<Table 7-10> Trends in the composition of food crop cultivation area, 1970-2000	226
<Table 7-11> Trends in Korean agricultural trade, 1970-2000	227
<Table 7-12> Trends in rice yields with development of Tongil-type rice varieties.....	230
<Table 7-13> Major Achievements of Saemaul Undong, 1972-1982.....	236

<Table 7-14> Changes in the distribution/retail margin rate of fruits and vegetables before and after the opening of public wholesale market.....	239
<Table 7-15> Changes in the proportion of each shipment type in the wholesale market before and after the opening of public wholesale market	240
<Table 7-16> Comprehensive Measures and Burden Reduction Measures of Comprehensive Rural Measures	244
<Table 7-17> Basic Direction and Blueprint of Comprehensive Rural Development Measures	245
<Table 7-18> Idea of New Agriculture from Heo Shin-haeng, the then Minister of Agriculture, Forestry and Fisheries	247
<Table 7-19> The Key Contents of the Five Year Plan of New Agricultural Administration	247
<Table 7-20> newly established department in the College of Agriculture in Seoul National University between 1960s and 1970s.....	250
<Table 7-21> The composition of College of Agriculture at Seoul National University, 1962-1996.....	251
<Table 7-22> Number of instructors and students in the college of engineering and college of agriculture at Seoul National University 1963-1993	253
<Table 7-23> Core achievements of the Rural Development Administration according to changes in the internal and external environment and policies.	259
<Table 7-24> Changes in the purpose and concept of research and guidance and education projects of the Rural Development Administration.	261
<Table 7-25> Organizational change of Rural Development Administration, 1962-1994	263
<Table 8-1> Basic direction and action plans of the 1st Biotechnology Promotion Master Plan.....	292
<Table 8-2> National strategic areas and key R&D tasks of the 1st Biotechnology Promotion Master Plan.....	293
<Table 8-3> Status of professors in departments of biology field at SNU in the 1980s	297

<Table 8-4> Status of professors in Department of Biological Engineering at KAIS in the 1980s	299
<Table 8-5> Status of professors in Biology at KIT in the 1980s	300
<Table 8-6> Composition of KRIBB's Research Funds, 1985-2000.....	304
<Table 9-1> Current status of farmland and crop cultivation area in Lao PDR	312
<Table 9-2> Proportion of Agricultural Workers in Five Mekong Countries, 1998-2019	312
<Table 9-3> Agriculture, forestry, fisheries and manufacturing GDP of five Mekong countries in 2010 and 2019.....	313
<Table 9-4> Lao PDR' Gross Production Value (GPV) (based on nominal prices), 1998-2019	314
<Table 9-5> Top Agricultural Items by Gross Production Value in Lao PDR in 2018 (Based on Nominal Price).....	315
<Table 9-6> Top 10 export items in Lao PDR, 2013, 2016, 2019	323
<Table 9-7> Top 10 imported items in Lao PDR, 2013, 2016, 2019	324
< Table 9-8> 11 countries of the Lao Ministry of Agriculture and Forestry (MAF).....	327
<Table 9-9> Details of 8th NSEDP's Agricultural and Rural Policy.....	331
<Table 9-10> Contribution of Agriculture, Livestock and Forestry Sector to 9th NSEDP Goals	333
<Table9- 11> ADS 2025 Goals and achievements by target period	335
<Table 9-12> Key tasks for the agricultural sector in Lao PDR by 2023	336
<Table 9-13> Agricultural colleges and departments of National University of Lao PDR	338
<Table 9-14> National University of Lao PDR' Faculty of Agriculture Bachelor's and master's degree programs	339
<Table 9-15> Champasack University's Faculty of Agriculture and Forestry Bachelor's and master's degree programs	342
<Table 9-16> FRAME academic focus areas by partner institutions.....	343
<Table 9-17> NAFRI's organization composition	346
<Table 9-18> Distribution of NAFRI's doctoral degree personnel in headquarters and affiliated research centers	348
<Table 9-19> Distribution of NAFRI's doctoral degree personnel by age group	349

<Table 9-20> Distribution of Lao PDR' Agricultural Researchers by Institution and Degree as of 2017	350
<Table 9-21> Distribution of Lao PDR' Agricultural Researchers in Universities, 2010, 2013, 2017	351
<Table 9-22> Distribution of NAFRI's Agricultural Researchers by degree, 2010, 2013, 2017	351
<Table 9-23> Status of local medicines and medical devices manufacturers in Lao PDR	353
<Table 9-24> The distribution of registered medicines in Lao PDR by country as if 2017	354
<Table 9-25> The Size of Lao PDR' main medicine imports by country	355
<Table 9-26> The size of Lao PDR' main medical devices imports by country	356
<Table 9-27> Ongoing public and private hospital construction projects in Lao PDR..	359

List of Figures

[Figure 1-1] Conceptual Framework of the Project	8
[Figure 4-1] Organizational chart of MOST	110
[Figure 4-2] GDP per capita (PPP) in ASEAN countries.....	126
[Figure 4-3] GERD as a share of GDP (%) in ASEAN countries.....	128
[Figure 4-4] Internet Access per 100 inhabitants in ASEAN countries (2019)	130
[Figure 4-5] Distribution of tertiary graduates in Lao PDR (2018)	132
[Figure 4-6] Distribution of tertiary graduates in other ASEAN countries (2019)	133
[Figure 4-7] Volume of scientific publications in ASEAN countries	133
[Figure 4-8] GDP of Lao PDR by category (Unit: Billion Kip).....	134
[Figure 5-1] Supply vision of personnel for economic development	150
[Figure 5-2] Two track frame of S&T manpower training policy in Korea.....	158
[Figure 5-3] 1990s Two track policy for research university in Korea	172
[Figure 5-4] Two track frame of S&T manpower training policy in Korea.....	179
[Figure 7-1] Trend of GDP by Sector (in %) of Agriculture, Mining and Manufacturing Industry, Service Industry in Korea	217
[Figure 7-2] Trend in the Proportion of Leased Land out of Total Agricultural Land in Korea.....	221
[Figure 7-3] Trends in Production of Rice (Milled Grain) in Korea	224
[Figure 7-4] Trends in Composition of Agricultural Products in Korea.....	224
[Figure 7-5] Trends in Agricultural Total Factor Productivity (TFP) and Labor Productivity in Korea	227
[Figure 7-6] Trends in Food and Rice Self-Sufficiency Rates in Korea	228
[Figure 8-1] The size of domestic bio-health industry market in Korea in 2019	275
[Figure 8-2] Status of pharmaceutical, biotech, and medical device industries in Korea in 2019	276
[Figure 8-3] Statistics on bioindustry production scale in Korea, 1992-2000	277
[Figure 8-4] Trend of bioindustry trade in Korea, 1992-2000	278

[Figure 8-5] Investment trends in genetic engineering field by industry in Korea, 1982-1988	279
[Figure 8-6] Trends of Korea's bioindustry investment, 1994-2000	280
[Figure 8-7] Trends in research manpower in the genetic engineering field in Korea's industry, 1982-1988	281
[Figure 8-8] Trends of bioindustry workforce in Korea, 1994-2000	282
[Figure 8-9] Trends of research manpower in Korea's bioindustry, 1992-2000	283
[Figure 8-10] Trends of production manpower in Korea's industry, 1994-2000	284
[Figure 8-11] Status of genetic engineering research funding by the Ministry of Culture & Education, 1985-1989	289
[Figure 8-12] Government's investment by stage of the 1st Biotechnology Promotion Master Plan	294
[Figure 8-13] Human Resources Trend of Korea Research Institute of Bioscience and Biotechnology (KRIBB), 1985-2000	303
[Figure 8-14] The progress of KRIBB's total budget and research funds, 1985-2000	304
[Figure 8-15] Trends of KRIBB's domestic and international journal publications performance, 1985-2000	305
[Figure 9-1] Lao PDR' agricultural topography	311
[Figure 9-2] Trend of Rice Production and Production Area in Lao PDR, 1998-2019	317
[Figure 9-3] Trends in vegetable production and production area in Lao PDR, 1998-2019	318
[Figure 9-4] Trend of Banana Production and Production Area in Lao PDR, 1998-2019	319
[Figure 9-5] Trends in coffee production and production area in Lao PDR, 1998-2019	320
[Figure 9-6] Trends in Tobacco Production and Production Area in Lao PDR, 1998-2019	321
[Figure 9-7] Agricultural products trade in Lao PDR, 1998-2019	322
[Figure 9-8] ForHeal project's academic focus areas and framework	341
[Figure 9-9] The location of NAFRI's headquarters and affiliated research centers (research institute)	347
[Figure 10-1] Kick-off Meeting	373



CHAPTER 1

Transformation PeriodProject Overview

in Lao PDR 2020-2022



Chapter 1. Project Overview

1. Introduction

The Government of Lao PDR plans to transition to emerging country status from a least-developed (LDC) one and focuses on the significance of science, technology, and innovation as a tool for economic growth and its national development plan. The National Master Plan for Technology & Innovation Development & Application in Lao PDR (NMTIDL) and National Master Plan for Policy on Science and Technology Research (NMPSTR) are both going to be top-priority projects, which will incorporate the relevant national development plan of science and technology between 2019-2025.

The set goals are to formulate the relevant master plan enhancing and supporting the development of the 8th National Socio-Economic Development Plan, and contribute to the preparation of the 9th National Socio-Economic Development Plan, which will be followed in the coming years, as well as the country's development strategy and vision for 2030 which has identified the National Science and Technology Development Program as a driving force for economic growth to be sustainable, eco-friendly, and consistent with the SMEs' development strategies associated with SMEs' transformation from the existing manufacturing base to the new modern economics of science, technology, and innovation.

For the second master plan to reach the set goal, however, this project's result really requires support from various sectors such as governments, research institutions, banking, private and public sectors, in addition to international cooperation, particularly the support from the Republic of Korea's STEPI in the role of technical consulting along with providing some budget subsidies and other assistance.

To achieve the implementation of a 5-year National Socio-Economic Development Plan, period VIII, the national strategy on science and technology development, and National Master Plan for Technology & Innovation Development and Application is necessary and important to encourage and promote the implementation of sartorial strategies through researching and selecting advanced and applicable technologies. To determine the project on technological development and application for top-priority sectors, R&D for innovative products, and research on science and technology constitute a major task for promoting the sustained growth of national socio-economic development, so that the state party - government takes up the research and development of science and technology as the main driving force of the commercial productivity production process which has affected the changing economic structure of sustainable growth, thereby aiming to achieve the overall development goal of transitioning to emerging middle-income country status from a least-developed one in 2020. Green growth and sustainability in 2030 is stated in its 'Vision for 2030' and 10-year socio-economic development strategy (2016-2025).

In order to ensure the achievement of this goal, the National Master Plan for Technology & Innovation Development and Application in Lao PDR and a national policy on science and technology research must be developed with the aim of focusing development on a science and technology research master plan, survey & study development indicator of relevant sectors, human resources, science and technology research innovation networking, and infrastructure development, all of which are facilitated by research facilities, while building a research center for science and technology and knowledge transfers leading to increased economic trade, research competitiveness and more innovative industrial development.

The Department of Technology and Innovation (DTI) acts as a secretariat to the Ministry of Technology and Communications (MTC) to set up policy, strategy, and manage and promote technology transfers and innovation in Lao PDR and the Cabinet of Academy of Science and Technology (CAST) acts as a secretariat to the Ministry of Science and Technology and National Science Council (NSC) in order to manage the research on science and technology, and build connections to science sectors and related organizations within the central and local government to implement and certify S&T research projects and set up a national R&D

master plan and technology application for sectional of Lao PDR.

The Science and Technology Policy Institute (STEPI) is a state-sponsored research institute under the auspice of the Prime Minister's Office, and a leading global think tank that conducts research and analysis on issues pertaining to science, technology, and innovation. STEPI works closely with dedicated ministries, regional governments, and relevant institutes to support policy design. It also has extensive experience in providing STI policy consulting for developing countries, taking on the role of conducting STI development cooperation activities with partner countries.

2. Objectives

The project is titled "National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022." The program duration is from January 1, 2021 to December 31, 2021 (12 months).

1) The research objective of the K-Innovation Partnership Program with Lao PDR (2020-2022) is to set up the first National Master Plan for STI R&D and Application in Lao PDR and Master plan on Application of Technology and Innovation for Sectorial which both underlie the National Social-Economic Development Growth. Moreover, the project aims to strengthen the capacity of key institutions of the Ministry of Technology and Communications in the mid-and long-term strategic planning of STI in Lao PDR.

2) The detailed objectives are as follows:

- Review the current status of STI policy and infrastructure in Lao PDR
- Strengthen the capacity building of policymakers, developers and scientists, young researcher, identify the priority areas of the research and development of the National STI Research Program.
- Share Korea's experience in developing master plan for national STI
- Conduct a capacity-building workshops
- Provide technical advice on the draft report of an STI masterplan
- Provide a roadmap to establish the National R&D Institute

3. Project Framework

The Project's main policy consultation highlights will be framed as follows:

- Sharing the current status of STI development and policy issues of Korea and Lao PDR
 - Select strategic industrial technologies;
 - Set the role of science and engineering colleges and national research institutes for training experts
 - Establish a national R&D plan for internalization of industrial technology
 - Introduce benchmarking case of Korea.
- Support for establishing National STI research and development masterplan of Lao PDR
 - Develop draft Masterplan by Lao PDR Research Team
 - Hold a policy planning and methodology workshop to share ideas and reviews the draft master plan

The Project's expected output will be as follows:

- Find directions for national STI masterplan in Lao PDR and enhancing implementation capability
- Capacity-building on policy planning and analyzing capabilities of STI policy makers and practitioners in Lao PDR
- Advancing national economic development leveraging STI effectively

[Figure 1-1] Conceptual Framework of the Project



Source: author

4. Project Team

4.1 Research Team

The research team is composed of three full-time STEPI researchers and three experts from other institutes. The experts are invited to share their knowledge on Korean development experiences. The STEPI research team consists of three researchers, including a STEPI project manager (PM) Dr. Inkyoung Sun and project coordinator Jiyong Ahn. The research team is responsible for researching and reviewing Lao STI. Other Korean experts will review and provide consulting on STI policy strategies, sharing Korean experience cases to be benchmarked for the Lao STI master plan.

[Table 1-1] Research Team

Name	Organization	Position
Inkyoung Sun	STEPI, Associate Research Fellow	Project Manager
Jiyong An	STEPI, Researcher	Project Coordinator
Je-jeung Lee	Former Professor / National University University of Lao PDR	Advisory on Science and Engineering University Research in Lao PDR
Mahn Yong Moon	Professor / University of Jeonnam	cases of training Korean science and technology experts
Eunkyong Lee	Professor / University of Jeonnam	Cases on establishment and development of public research institutes in Korea
Myunghwan Oh	Head / Korea-Lao PDR Friendship Association	Background Information and mentoring for cooperation
Youngjoong Jang	Former ICT Advisor to KOICA	Advisory on Lao PDR ICT Policy
Hyungjoon Lee	Former Advisor to KOICA	Advisory on STI Governance
Jeehyun Kim	Head/ B.T. Insight	Bio and medical R&D
Jisoon Chang	Industry-Academic Cooperation Team , Sangmyung University	STI Demand Survey

4.2 Local Research Team

The Ministry of Science and Technology (MOST) of Lao PDR was the main counterpart from January 2020 to February 2021, and then MOST was dissolved into five separate ministries. The main counterpart for this project from March 2021 is the Ministry of Technology and Communications (MTC) which is responsible for developing the national STI master plan in Lao PDR. The local research team will work in collaboration with the STEPI research team.

[Table 1-2] Local Research Team

Name	Office	Position
Soumana CHOUNLAMANY	Department of Technology and Innovation, MTC	Director General
Sengchanh PHAXAYASENG	Department of Technology and Innovation, MTC	Deputy Director General
Viengvilay THONGMANYLA	Department of Technology and Innovation, MTC	Deputy director
Pisa KEOMANILAY	Department of Technology and Innovation, MTC	Project Coordinator/ Technical Officer
Chongnengxiong NENGXAY	Department of Technology and Innovation, MTC	Technical Officer
Phongsavanh KHATTIYAVONG	Department of Technology and Innovation, MTC	Technical Officer

5. Project Activities

5.1 Activity 1: Preparation of the Project

From January to March 2021, the STEPI team made contact with the Lao PDR Ministry of Science and Technology (MOST) which later was dissolved into five separate ministries, while the Ministry of Technology and Communications (MTC) became the new counterpart ministry for the project in March 2022. Given that it's the second year to initiate the ODA partnership program with the Lao PDR, STEPI and MTC had a working group meeting to introduce the new research team and share the research plan based on the Project Concept Papers written in 2019, as well as the project in its first year through emails.

The uncertainty caused by COVID-19 would affect the project plan's format and schedule for the year as well as its way of implementation, and the research team from both countries needed to wait and see on the feasibility of visiting Lao PDR and Korea. In the meantime, STEPI drafted a contingency plan for the projects to be shared with its counterpart. The research teams from both sides decided to postpone the kick-off meeting to August/September, and wait until the Lao government could adapt to the internal reshuffling of the ministries.

5.2 Activity 2: Outlining the Workslope

The STEPI and DTI under MTC outlined the scope of work of the project in the COVID-19 situation, the contingency plan with a changed schedule and plan under the new counterpart ministries, the Department of Technology and Innovation in MTC. Moreover, STEPI invited the Lao PDR embassy in the Republic of Korea and Korean embassy in Lao PDR to engage in active cooperation and collaboration under the program. Instead of traveling to both countries, STEPI and MTC decided to actively host and hold a series of activities online and hybrid format such as kick-off meeting, STI capacity-building workshop, dissemination workshop etc.

The research teams in STEPI and MOST started literature review and research on current status of STI, STI policy and education, its governance and legislative infrastructure etc to understand the current situation and status of Lao PDR' STI policy status. STEPI reached out to hire local consultants from Lao PDR to research on baseline data and information on STI and related information in Lao PDR.

5.3 Activity 3: High-level Forum and Kick-off Meeting

Instead of traveling to both countries, STEPI and MTC decided to hold an online hybrid kick-off meeting where participants from each sides host the event online while inviting some participants for face-to-face meetings as well. The online hybrid high-level forum and kick-off meeting was held on September 1, 2021 through an online conferencing platform. The Lao PDR embassy in the Republic of Korea attended the meeting offline with the participation of a number of Lao international students in Korea. The online kick-off meeting is followed by introductory, welcome, and congratulatory remarks of each respective side. The presentations on the project introduction, current status of Lao STI and keynote presentations on Korea's STI development and economic growth. About 100 participants attended the meeting both online and offline. The team took part in Q&As regarding the presentations afterward.

5.4 Activity 4: STI Demand Survey

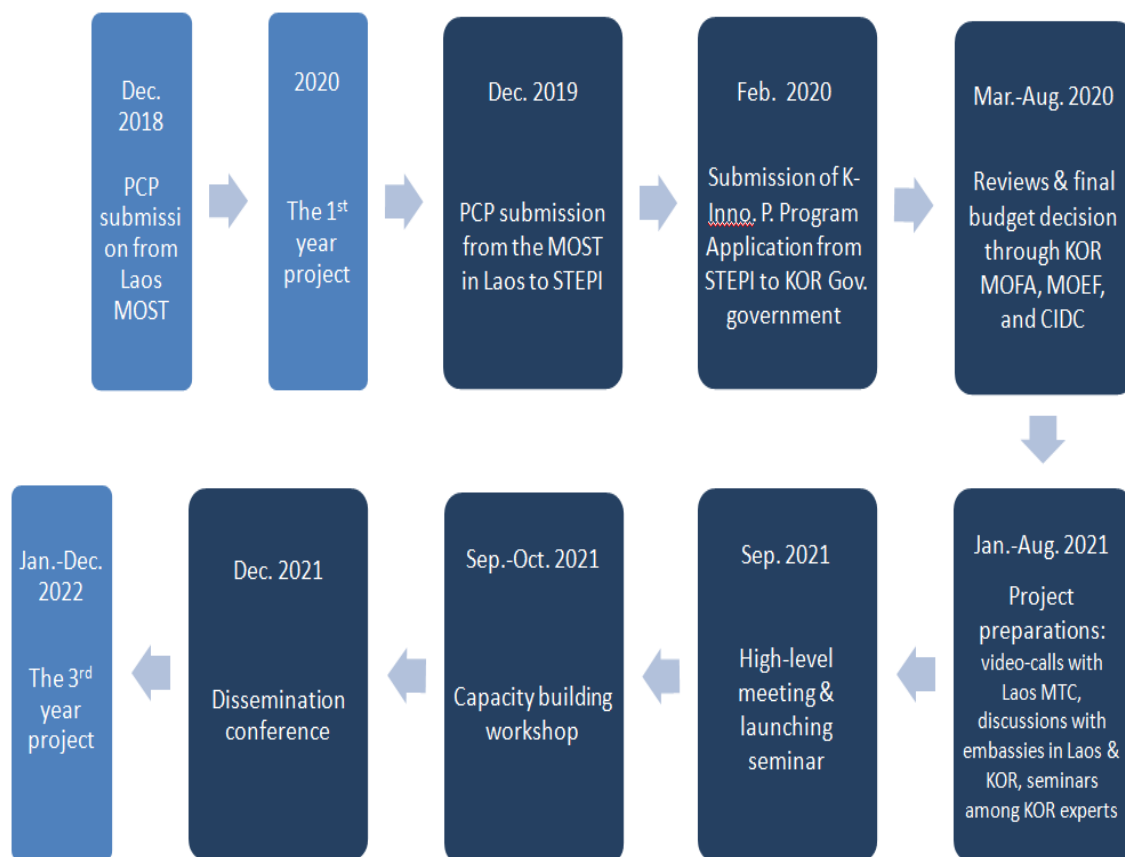
The STI demand survey was conducted in order to find the key challenges in Science and Technology in Lao PDR. Forty-six responses were collected from government officials and experts and Lao master's degree students in Korea. The results showed and identified some key challenges in S&T, the necessary capacity of science and technology, insufficient human resources, living standard disparity between urban and rural segments, main challenges/obstacles for STI development, and some STI development experience in Korea, etc.

5.5 Activity 5: Online K-Innovation Dissemination Workshop

An online expert workshop was held on December 15, 2021. During the workshop, STEPI and Korean experts made presentations on methodology to develop national STI master plan strategies and share experiences in the highlighted topics, including national STI strategic planning. At the same time, experts and Lao government officials will also share their own current STI status and STI policy & demands in Lao PDR in order to enhance an in-depth understanding of Lao PDR' current status and experiences. Afterward, STEPI and MTC will form a consensus on the expected results and outcome. STEPI will present the results and achievement of the joint project with the Lao PDR, accompanying a follow-up plan for 2022.

6. Project Schedule

The 2021 Project Backgrounds and Process is as below:



[Table 1-3] Project Schedule

Activities	Jan. '21	Feb. '21	Mar. '21	Apr. '21	May '21	Jun. '21	Jul. '21	Aug. '21	Sep. '21	Oct. '21	Nov. '21	Dec. '21
Project Identification												
Literature Review and research												
Kick-off Meeting and Workshop												
Interim Report												
Comments on Progress												
Dissemination Workshop												
Final Report												

7. Project Outputs

STEPI and MTC will draw up the following deliverables:

- **Deliverable 1:** Project proposal
- **Deliverable 2:** Program agenda for kick-off meeting
- **Deliverable 3:** Presentations and training materials for online workshop (STEPI)
- **Deliverable 4:** Draft of report on Lao PDR's STI data and information
- **Deliverable 5:** Korean and English version final report (2021)

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CHAPTER 2

Macro-Environments of Lao PDR





Chapter 2. Macro-environments of Lao PDR

1. Lao PDR's National Development Strategies

1.1 Overview

The Lao People's Democratic Republic (Lao PDR) is a small country located at the center of Indochina, surrounded by another country to its east, west, south and north. It has many charms yet to be discovered, including a fascinating culture, landscape, and attractions waiting to be unveiled. Once, New York Times reporter Harrison Salisbury called it one of the most mysterious nations in the world.

From the reign of the Kingdom of Lane Xang, Lao PDR had been divided into 3 kingdoms; Kingdom of Luang Prabang, Kingdom of Vientiane and Kingdom of Champasak until the late 19th century, and then it became subject to colonial rule by France in the late 19th century up until the modern-day Lao PDR was established after a series of political turmoils and upheavals.

In 1975, after South Vietnam lost in the Vietnam War, the other countries in Indochina of Vietnam, Lao PDR, and Cambodia became communist in succession. That year, the Lao PDR was founded on December 2 by the Lao People's Revolutionary Party.

Having Vientiane at the center, the southern part of Lao PDR was most affected by the Khmer and the northern part was affected by China. There are 3 major ethnic groups in the nation, including the Lao Loum (tallying more than 60% of the population) mostly living in the plains, the Lao Theung living in the mid-to-low areas and the Lao Soung living in the highlands and mountainous regions, all of whom are further divided into 49 minority groups.

Bordering Vietnam in the east, Thailand in the west, Cambodia in the south, China in the northeast and Myanmar in the northwest, Lao PDR is a landlocked country with no access

to the sea with its border extending longer than 5,000km, surrounded by 5 countries. Out of the entire Mekong River, running 4,200km in total, 1,800 km is running through the border of the nation offering abundant water resources. As late as the mid-1990s, Lao PDR was known as a locked country but is now transforming into a regional hub or battery country of Indochina. Since joining ASEAN 1997, as one of the 10 ASEAN member countries, Lao PDR has been strengthening its presence, seeking continuous growth on many fronts after opening up its economy in 1990 and strengthening the membership in ASEAN.

Lao PDR is a landlocked country with a landmass of 16,000 km sq.m which is larger than South Korea (similar in size to Gangwon Province in South Korea). It is geographically similar to the Korean Peninsula, having almost 75% of landmass in mountainous areas. The infrastructure of the nation is still underdeveloped with unpaved roads and transportation conditions. There is no coastline and its industries are still underdeveloped due to restrictions on domestic consumption and exports from a small population relative to the territory size.

After its communization in 1975, Lao PDR opened its markets around 1987 and has been witnessing an impressive 6% or higher economic growth rate yearly since the early 1990s. With investment and support from China, a high speed railway is set to be completed around the end of 2021 (connecting China-Vientiane, Lao PDR), and the nation is transforming from a locked country to a regional hub of Indochina.

Most Lao people are known for being easy-going and soft-spoken but are full of pride. Over the past 26 years of relations with the Lao PDR, I've never seen any inebriated local wobbling in the street and seen them publicly fighting outside. They wouldn't say 'no' directly upfront. That is why it is known as a nation of hidden charms yet to be discovered. However, with progressive economic development and wealth accumulation, it seems that some of these charming virtues are gradually fading in the urban sprawl.

With economic development, wealth accumulation, lure of capital flows, and speculative investment with an influx of Chinese influence, the market order has been somewhat disturbed, and this led to challenges in seeking stable economic growth. At the same time, the gap and tension between urban and rural segments and the haves and have-nots are constantly growing.

2. Political History and Current Environment

2.1 The Monarchy Era

Lao people originated from the Thai ethnicity, residing in Yunnan Province in China, and around the 7th century, ethnic Thais founded the ‘Nanzhao Kingdom’, which was growing into a powerful nation. It kept expanding toward the southern region but around the 13th century, as Mongolia conquered China and Vietnam, the Nanzhao Kingdom was also destroyed by Mongolian invaders and the ethnic Thais became scattered across present-day Thailand, Lao PDR, Myanmar, Vietnam, and India.

A century later, a national hero, Fa Ngum appeared and conquered Vientiane, Xiangkhouang, and Khorat Plateau in the northeast of Thailand in succession, and finally founded the Kingdom of Lane Xang.

- Lane means million and Xang (象 in Chinese character) means elephant.

In this way, the first unified state of Lao PDR was thus founded. Fa Ngum, the first king of the kingdom designated Luang Prabang as its capital and embraced Theravada Buddhism. Since then, the religious system of Buddhism was converged with the political system, becoming the state religion of Lao PDR.

Two centuries later, the capital was moved to the present-day capital of Vientiane, and another century later, during the reign of Sourigna Vongsa for more than 57 years since rising to the throne in 1637, Lao PDR enjoyed its own golden age. Back then, the kingdom of Lang Xang became the largest kingdom in Indochina in terms of territory. Another two centuries later, it was destroyed by the Kingdom of Siam of Myanmar in 1829 and around 1893, the Lao monarchy ended after enduring colonization by France.

2.2 Establishment of a Modern Republic

2.2.1 Early stage

Even after the end of the World War II, unlike Korea, Lao PDR was not able to establish an independent nation due to the constant intervention of France. Thus, the Lao people organized the socialist liberation army Pathet Lao (the Lao People's Revolutionary Party- Pathet means a state) and after a series of armed revolts against France, in October 1954, the nation took its first step as a constitutional monarchy.

However, even after its independence, a civil war continued in a confrontation between the left, middle, and right-wing parties. Amid the turmoil, Lao PDR was drawn into the Vietnam War regardless of its intent to do so. The U.S. started bombing raids across Lao PDR in 1964 to cut the supply routes of the Viet Cong and dropped more than 2 million tons of bombs until 1973, taking so many precious lives. Until the early 1990s the nation suffered injuries and deaths from unexploded ordnance (UXO). To this day, the U.S. still provides fund to remove UXO for Lao PDR.

2.2.2 Birth of a Republic

In August, 1975, the Lao PDR Patriotic Front- Liberation Army led by Pathet Lao prevailed against the government and achieved a bloodless revolution. That year in December, the monarchy was abolished and the Lao People's Democratic Republic was established.

Under the leadership of Pathet Lao's leader Kaysone Phomvihane, who served as Secretary General of the Lao People's Revolutionary Party (LPRP) and Prime Minister, a socialist government was organized and on December 2, 1975, it was designated as the National Independence Day.

In the 4th National Congress of LPRP 1986, a new economic mechanism focusing on economic reform and market-based economy was introduced to facilitate the privatization of state-run companies, bank system, & taxation reform and the enactment of foreign direct investment promotion Act. In 1991, a new constitution was established and reform &

opening-up policy was adopted. After that, the first election of the members of the National Assembly was held in 1992 to establish the framework of a modern, law-governed country.

2.2.3 Building Framework as a Modern Country with Open Structure

Since Lao PDR put an end to its confrontation with the West, adopting a reform and opening-up policy, the Korean embassy in Lao PDR opened in Vientiane in 1996 and with Secretary of State Hillary Clinton's visit to Lao PDR in 2012, as the first U.S. secretary of state to visit the country, formal relations between Lao PDR and the U.S. was fully normalized.

After that, Lao PDR took the opportunity of the 9th ASEM (Asia-Europe Meeting) held under the slogan of 'Friends for Peace, Partners for Prosperity' on November 5~6, 2012 to inform the world that formal relations with European countries were also fully restored, with the Korean Prime Minister also in attendance at the meeting.

In a press conference right after ASEM attended by 52 heads of state of Asia and Europe, the Lao PDR Prime Minister Thongsing Thammavong expressed his content, affirming that ASEM this time was meaningful as it offered an opportunity to strengthen the presence of Lao PDR on the global stage and added that an unprecedentedly large number of political leaders visited Lao PDR from around the world and the meeting set the stage for many global leaders to see the passion and enthusiasm of the Lao people for growth and prosperity.

That was the largest event in Lao PDR in the history of the nation. Next year in 2013, Lao PDR joined the WTO as an official member to strengthen its presence in the global community. In April 2016, President Bounnhang Vorachith was elected and in September, the country hosted the 28th ASEAN Summit and EAS (East Asia Summit) as a chair country of ASEAN. The then-U.S. President Barack Obama attended the meeting for an even more meaningful occasion.

2.2.4 National Congress of the Lao People's Revolutionary Party

NO.	Year	Comment
1 st	1955	- Became independent from France 1954 - - The first National Congress of LPRP 1955 - - The 2nd National Congress was held in 20 years 1975 due to civil wars and upheaval - - Since then, holding the congress in every 5 years and organized a new cabinet - - The incumbent cabinet was organized in the 11th congress 2021 (October)
2 nd	1975	
3 rd	1981	
4 th	1986	
5 th	1991	
6 th	1996	
7 th	2001	
8 th	2006	
9 th	2011	
10 th	2016	
11 th	2021	

Source : Korea-Lao PDR Friendship Association (KLFA)

2.3 Current Political System

2.3.1 Political System

1) People's Democratic Republic ;

In the National People's Congress in December 1975, the Lao monarchy officially ended and the Lao People's Democratic Republic (LPDR) emerged to govern.

In Chapter 1 of the constitution, it states that the rights of the multi-ethnic people to be the masters of the country are equally endowed to be exercised and guaranteed through the functioning of the political system with the Lao People's Revolutionary Party and all powers originate and reside in the people and in the interests of the multi-ethnic people across all strata of society with the laborers, farmers and intellectuals as key constituents.

Under the constitution, with the LPRP's single party system, the national president serves

as the head of state and the prime minister as the head of administration and president of the National Assembly, all serving as politburo members to be the highest decision-making body.

2) Administration ;

- ① Central government; President is elected by the vote of the National Assembly (getting more than 2/3 votes of members present) and serves for 5 years. Incumbent President, Thongloun SISOULITH was elected in the 11th National Congress of LPRP January, 2021. Prime Minister is appointed by the president with an approval of the National Assembly and leads and supervises the administration overall.
Incumbent Prime Minister, Phankham VIPHAVANH was elected 2021 together with the president.
- ② Local Governments ; It has 16 provinces and 1 special city and governors and the mayor of special city of Vientiane are appointed by the president with recommendation of Prime Minister.

3) Legislative Body ;

The National Assembly has the single chamber system and the number of lawmakers total 132 with a 5-year tenure. As of October, 2021, Xaysomphone Phomvihan is the chair of the National Assembly and is a politburo member with the 4th highest position in the party. All lawmakers are LPRP members and also members of the Lao-Front for National Development under the party. In the 8th general election in 2016, 2 lawmakers from state-run companies and 6 business men from private companies were elected, creating a new trend in the political arena of Lao PDR. The regular session of the National Assembly is held twice a year and the National Assembly has the authority to approve or modify the constitution, enact or abolish laws, approve major national policies, elect the president, and approve the nominated prime minister.

The parliamentary election is generally by direct and popular voting and those aged 18 or older are given the voting right and those aged 21 or older are eligible to run for election.

The status of the National Assembly is somewhat different from that of other democratic countries and under the constitution, there are the following restrictions. The National Assembly and all the other national agencies shall be established and governed according to the principles of democratic centralism.

4) Judicial and Legal System ;

① Judicial system ;

With the Supreme Court as the highest court, there are people's court in each province and county and head and judges of the highest court are appointed by the standing committee of the National Assembly.

Trials are held as an open trial except for extraordinary cases and the legal advisory council defends the case for the defendant as a public defender.

As an independent body, there are the Supreme Public Prosecutors and local prosecutors office in each province and county.

② Constitution and other laws ;

Even after its independence from France, due to protracted civil wars and political uncertainty, the written law was established not until the late 1980 and the official gazette started to be published as late as 1993. Since the 2,000's written law system has been adopted.

The constitution was established and announced August 1991 and was first amendment was made April, 2003.

The basic principle of Lao PDR constitution is to follow the principles of the 5th National Congress March 1991 to maintain the single party system and on economic front, pursue market and open economy. Also, it stipulates the principles of respect for human rights, freedom of residential mobility, freedom of speech and public order, right to education and employment.

The constitution was amended 2nd time 2015, in the 7th National Assembly to stipulate the restrictions on the 2-term presidency twice in a row.

Presidents of Lao PDR and Tenure			
No.	Name	Tenure	Others
1 st	Souphanouvong	1975-1989	1 st president after the collapse of monarchy
2 nd	Kaysone PHONVIHANE	1991-1992	Designating December 2, 1975 as an official Independence Day
3 rd	Nouhak PHOUNSAVANH	1992-1998	Relation with Korea resumed in full fledge from the tenure of the 3rd president.
4 th	Khamtay SIPHANDONE	1998-2006	
5 th	Choummaly SAYASONE	2006-2015	Visiting Korea November 2013
6 th	Bounhyang VORACHIT	2016-2020	
7 th	Thongloun SISULITH	2021-present	Visiting Korea several times when he served as the Foreign Minister and Prime Minister

Source : Korea-Lao PDR Friendship Association (KLFA)

5) Political party and organizations ;

① LPRP (Lao People's Revolutionary Party) ;

Originated from the Indochinese Communist Party of Vietnam established 1930, LPRP was established in March 1955.

As of 2021, president Thongloun SISOULITH lead the party as general secretary of the party and it has about 300,000 members in the party.

The basic guideline of the party is to build Lao PDR in peace, independence, unity and prosperity with establishment of people's democratic system and eventually completes establishment of a socialist country.

Other than the central committee, there are local committee in each province and sub-administrative entity and the National Congress is held in every 5 years while the central committee meeting is held every two years to discuss and decide the guidelines of the party and basic policy direction of the government.

Affiliated organizations of the party include Lao Front for National Development, Lao Federation of Trade Unions, Lao People's Revolutionary Youth Union, Lao People's Revolutionary Women's Union, Lao Peace and Solidarity Committee, Lao Buddhist Association and friendship associations with each country. .

② LFND ; Lao Front for National Development

- It is an organization for the front of national unification and replacing the Front organization in 1979, it was established to pursue peace, independence, completion of a socialist state.
- Major code of conduct
 - With the force of national unity
 - strengthening the front of the national unification with labor federation
 - promoting socialist democracy
 - contributing to establishment and development of national defence, economy and culture
 - contributing to implementation of diplomatic policies of the party and the nation.
- Incumbent chair of the National Assembly,

Xaysomphone PHOMVIHANE served as the chair of the front before March 2021 before Dr Sinlavong KHOUTPHAYTHOUNE (former mayor of Vientiane) was appointed.

6) Recent political circumstances in the nation ;

LPRP adopted the 7th 5-year National Socioeconomic Development Plan (NSEDPlan) at the 9th

National Congress March 2011.

Under the influence of Buddhism, people of Lao PDR tend to have positive view, be self-satisfied and try not to be greedy. Most of people live in poverty but the sense of comparative deprivation is not so strong and this seems to be one of the reasons behind social stability in the nation.

As secretary of the party, Bounnhang Vorachith who has close ties with Vietnam was elected as president 2016 and as many of the key political leaders studied in Vietnam, the Lao PDR government maintained friendly relations with Vietnam but for stable economic growth, the nation needs the capital of China.

In the 9th National Assembly organized as a result of the general election March, 2021, President Bounnhang stepped down and former prime minister , Thongloun Sisoulith was elected as general secretary of the party and president of the country while former vice president, Phankham Viphavanh was elected as prime minister.

2.3.2 Military

1) Lao People's Armed Forces

Since established as 'Lao People's Liberation Army' January 1949, it was renamed 'Pathet Lao Fighters', 'The Army of Lao Front for National Development' and again in 1982, Lao People's Armed Forces and it is led by the Minister of Defence, following the order of LPRP.

Under the system of universal conscription stipulated by the constitution, the country is running draft system (for the men 18~28 years old, for 24 months) and voluntary service system. Just as other socialist countries, many of Politburo Members have military backgrounds, including the president also serving as secretary of the party, thus the military has significant influence over the politics.

2) Organization of the Ministry of Defence

① Organizational Chart of the Ministry of Defence ;

- Ministry of Defence ----- ▲The General Chief of Staff (Deputy Minister)
 - ▲Director of General Politics Bureau (Deputy Minister)
 - ▲Director of General Logistics (Deputy Minister)
 - ▲Chief Secretary (Undersecretary)

② Military Power Status ;

In Lao PDR, the boundary between military organizations and police organizations are blurred and specific figures on military forces and budget are national confidential information. As paramilitary organization, it has more than 100,000 self-defence forces.
- They are voluntarily organized at community level for the purpose of regional defence.

Source; <Sabaidee Lao PDR> 2008, Park Jae Hyun <Culture and Economy of Lao PDR> 2012, Jang Tae Gu

<Overview on Lao PDR> 2019, Ministry of Foreign Affairs

<Korea-ASEAN Statistics> 2020, Korea-ASEAN Center <Others> Korea-Lao PDR Friendship Association/KLFA

3. Economic History and Current Environment

3.1 Economic Circumstances ;

Lao PDR is one of the least-developed countries, and setting its goal to reach emerging country status from a least-developed one, it is implementing its 5-Year National Socioeconomic Development Plan (NSEDPlan) for 2020.

From 2011 to 2013, the Lao PDR economy witnessed annual economic growth rates of 8.0 percent or higher on average and in 2014 and 2015, despite the global economic slowdown, its economy grew by 7.8 and 7.6 percent, respectively, backed by consistent foreign direct investment into hydroelectric power generation and growth in the service sector.

The biggest contributor to drive economic growth is large-scale foreign direct investment, leveraging natural resources of Lao PDR, and ODA from donor countries. To secure stable growth, it is urgent to strengthen capabilities in each sector, improve transparency, and execute reform measures to achieve financial soundness and build pro-investment environments in particular.

As of 2019, the major trade partners of Lao PDR were mostly neighboring countries, including Thailand and China and major investors were China, Thailand, Vietnam, Malaysia and Korea in order of significance.

- 10 major trade partners of Lao PDR 2019 were as follows.
 - Export: Thailand, China, Vietnam, Japan, India, Germany, Hong Kong, Switzerland, U.S., Italy
 - Import: Thailand, China, Vietnam, Japan, Singapore, U.S., Korea, Australia, U.K., India

3.2 Recent Economic Trend

3.2.1 Overview

With growth in major mines, inflow of foreign investment and USD3.8 billion hydroelectric power generation project (Xayaburi dam), from 2011~2013, Lao PDR achieved a remarkable 8% or even higher economic growth in annual average. In 2014, it witnessed a mild decline with 7.8% economic growth and in 2015, witnessed a 7.6% robust growth, supported by continued growth in service and tourism sector and electricity production increase with operation of the dam.

Due to increase in import of capital goods for project implementation and consumer goods due to low competitiveness of local products, despite increase in tourism income and assistance from overseas, the country recorded current account deficit in 2018.

As LPRP is implementing policies like adoption of market economy and opening up of the market as top priority and is strengthening stable political base, no major changes are expected in the regime and in the overall direction. Lao PDR has excessive external debt for its economic scale and as it is highly dependent on foreign assistance, concessional loans account for approximately 40% of the debt.

In 2013, due to strong Bhat of its major import partner, Thailand and inflow of foreign investment, the country recorded 6.4% inflation but with oil price hike and decline in import prices, the price level declined down to 4% in 2014 and still remained as low as 1.7% 2015. As the Central Bank of Lao PDR implemented the managed floating exchange rate and foreign investment kept flowing in, the exchange rate of LAK remained stable in 2016.

Due to fiscal deficit of the government, 59 mega projects for infrastructure building including road, airport, bridge under the 7th NSEDP (2011~2015) were delayed and the followings were the large-scale investment in infrastructure and energy, still going on in 2016.

- China-Lao PDR High-speed Railway
(Funded by China, project budget USD 6.9 billion, kick-off 2016 end and scheduled to complete 2021 end)
- East-West Corridor Railway (funded by the capital of Malaysia, seeking ways to secure USD 5 billion fund for the project)
- Vientiane- China Boten Highway (Funded by China, project budget USD 2.5 billion, running 460Km in total, construction for the 1st phase 109Km started H12018)
- Xayaburi Hydroelectric Power Plant (1,285MW capacity, funded by Thailand, project budget USD3.8 billion, started 2012 and originally planned to be completed 2019 but still in progress)
 - Lao PDR government is planning to build 183 power plants to have 20,691 MW capacity by 2030 and generate 9,470GWH electricity every year.
After the accident in the project led by a Korean company, it is expected to go slow in dam construction for the time being, thus rather than competing with large-scale dam construction led by China and Japan, power transmission and distribution network seems to be a better choice.
- HongSa thermal power plant (1,878MW capacity, funded by Thailand, project budget 3.7 billion, started 2011 and completed 2016)

If the large projects listed above are to be implemented as planned, Lao PDR will be able to overcome the limitations of a landlocked country and transform into a hub country of Indochina and better country for ASEAN, to achieve remarkable economic growth.

3.2.2 Status by Sector

1) Economic Growth

Supported by robust mining industry and growth in mining production, hydroelectric power generation, strong construction industry and active tourism, Lao PDR has been witnessing 7% or higher economic growth every year since 2005.

Thanks to its low dependence on external trade, Lao PDR was able to maintain robust growth despite global economic slowdown.

With increasing global metal price and construction of infrastructure and facilities for Southeast Asian Games 2009, the country recorded 7.5% and 8.5% economic growth respectively 2009 and 2010. In 2011, gold production declined, copper price turned to slide down and Lao PDR was struck by natural disasters including typhoon. Despite these unfavorable conditions, the economy grew 8%, driven by increase in copper production and hydroelectric power generation growth.

In 2012, with rising mining production and growing hydroelectric power generation and boost in construction in the run up to ASEAN Universiade, the country witnessed a robust 8.3% economic growth.

In 2016, with hosting of ASEAN Summit and construction of social infrastructure in full scale, the economy grew about 7% and in 2017, the growth rate slowed down a bit, to 6.9% and down to 6.5% in 2018 as the country was hit by a huge flood. Energy and mining are driving economic growth of Lao PDR, increasing additional 50% during the 8th NSEDP (2016~2020), compared to the period of the 7th NSEDP (2011~2015). Tourism is one of the major industries of the country, along with mining and hydroelectric power generation and tourism boom is the main driver of economic growth. However, the COVID19 pandemic plaguing the entire world since early 2020 gave a devastating blow and the industry suffered a lot even until 2021.

2) Inflation and Exchange Rate

In 2014, supported by global oil price and decline in import prices, price index declined 4.1% but as the structural issue with heavy dependence on import from neighbouring countries, was the key factor to determine inflation up and down of Lao PDR, in 2015, inflation rate was 1.7% as import prices from Thailand and other neighbouring counties declined and the rete hovered around 2% in 2018 thanks to strong price stabilization policies. The value of LAK against USD remained strong since 2006 and after 2014, it still remained stable.

However, since 2020, the country has been suffering from the impact of COVID 19 and global oil price hike. Especially, as Lao PDR has weak manufacturing base and is highly dependent on import, depreciation of the currency directly led to price hike. Exchange rate of LAK against USD was 8,100 in 2014 but it increased up to 9,300 January 2021 and 9,550 July same year, putting more and more pressure on the country.

3) Balance of International Payments

With its weak manufacturing base, export of electricity continuously increased 2016~2018 but due to slowdown in mining export, increase in capital and consumer goods import, reduction in agricultural export due to climate change and falling price of copper, the country has been in chronic deficit in goods balance.

4) Tourism

Along with water hydro power generation, tourism has been one of the main drivers of the economy and how to vitalize tourism is the key factor to economic growth of the country. As of 2019, GDP in travel/tourism amounted to USD 1.8 billion, accounting for 9.1% of total GDP. This is the highest percentage among 10 ASEAN countries.

The number of people working in the industry is around 300,000, accounting for 9.6% of total employment.

No. of visitors 2019 was 4.8 million.

▲2,513,000 in 2010 ▲ 4,684,000 in 2015 ▲3,869,000 in 2017 ▲ 4,186,000 in 2018

Korean tourists visiting Lao PDR 2019 was about 230,000.

Indicator	Unit	2015	2016	2017	2018	2019
Population	1,000	674	685	696	706	716
GDP	USD Bil.	14.4	15.9	17.1	18.1	19.1
Per capita GDP	USD	2,131	2,323	2,453	2,568	2,661
Export	USD Mil.	3,653	4,245	4,873	5,484	5,653
Export to Korea	USD Mil.	6	2	10	4	5
Total import	USD Mil.	5,675	5,372	5,667	6,164	6,783
Import from Korea	USD Mil.	48	80	68	67	71
E/X	Kip	8,128	8,124	8,245	8,401	8,679
Foreign direct investment	USD 100 Mil.	0.39	0,15	0.1		

3.3 Major Economic Indicators

Source : 2020 Korea-ASEAN Statistics

◇ Top 10 Trade Partners of Lao PDR (2019, USD Mil.)

Rank	Export		Import	
	Country	Value	Country	Value
1	Thailand	2,406	Thailand	2,916
2	China	1,672	China	1,681
3	Vietnam	1,055	Vietnam	451
4	Japan	94	Japan	118
5	India	91	Singapore	88
6	Germany	82	U.S.	78
7	Hong Kong	59	Korea	71
8	Switzerland	58	Australia	52
9	U.S.	30	U.K.	36
10	Italy	25	India	34

Source : 2020 Korea-ASEAN Statistics

3.4 Economic Policy

3.4.1 Objectives of Economic Development

Lifting itself from the status of the least developed country by 2020 and entering the group of middle income countries

Based on strong political and social stability, pursuing balanced development between cities and rural areas

Seeking balanced development of agriculture, forestry, engineering and service industries

- Attracting capital and aid from Western countries, based on improvement in relations with the countries
- Actively attracting foreign investment
- Sustainable development toward market economy
- Promoting domestic industrial advancement

3.4.2 The 8th National Socio-Economic Development Plan (2016~2020)

■ Macro Economy

- Emerging from the status of the least developed country, achieving at least 7.5% economic growth annual average
- Per capita GDP USD2,661 in 2019, keeping inflation below 5%
- 15% annual growth in export and 6% annual growth in import
- Fiscal revenue: 19 ~ 20% of GDP, fiscal deficit: aimed to be in the range of 5% of GDP

■ Social Environments

- Achieving below 10% poverty rate by 2020
- Achieving 95% literacy rate among those under 15 or younger

- Infant mortality rate below 30/1,000, maternal mortality rate below 160/100,000
- Keeping low weight rate among children under the age of 5 below 20%, undergrowth rate below 32%
- Tap water supply rate 90%, social security and health insurance 80%

3.4.3 2025 Strategy (2016 ~ 2025 10-year Socioeconomic development strategy)

- Achieving at least 7.5% economic growth rate in annual average
- Increasing per capital GDP twice higher than the level of 2015, pulling down poverty rate below 5%
- Efforts to settle down legal system, gradual advancement into a low-governed country with legal system in place

3.4.4 2030 Vision (2016 ~ 2030 15-year National Vision)

- Entering the group of middle income countries by 2030
- Increasing per capital GDP, 4 times higher than the level of 2015 by 2030
- Establishing legal systems in every sector by 2030, becoming a country with legal systems in place

3.4.5 Economic Outlook after 2021

- With the devastating impact of COVID 19, market consumption and export demand slowed down and tourism and service industries are still suffering, and this is expected to have negative impact on trade balance and FX market.
- As investment into China is declining amid the trade war between US and China, if some of the foreign investment companies move from China over to Southeast Asia, Lao PDR may be benefited from the trend.

- Since it started investment into Lao PDR in 1993, Korea ranked 4th 2015, among foreign investors for Lao PDR. But since then, investment into massive hydroelectric power plant construction has been declining and overall investment into Lao PDR has been sliding down in 2019-2020, due to impact from COVID 19 and stagnant domestic economy.

Source; <Sabaidee Lao PDR> 2008, Park Jae Hyun <Culture and Economy of Lao PDR> 2012, Jang Tae Gu

<Lao PDR Overview> 2019, Ministry of Foreign Affairs <Korea-ASEAN Statistics> 2020, Korea-ASEAN Center <Others> Korea-Lao PDR Friendship Association (KLFA)

4. Education

4.1 General

School System ; 5 years at primary school, 6 years at middle school (high school course in Korea)

The constitution stipulates that the primary school education is mandatory and as of 2018, school attendance rate of the nation was around 99%.

Middle school attendance rate ; 78% for the first 3-year course, 46% for the last 3-year course

University entrance rate ; 19%, as of 2019, the number of university students was around 40,000.

Considering the country is at the stage of just emerging from the status of the least developed country, the figure is quite high.

Literacy rate among adults was almost 84% in 2018.

4.2 Educational Institutions

	No. of schools	No. of students	No. of teachers
Primary	8,854	770,659	36,127
Middle	1,792	659,850	38,500

■ Primary/middle schools – as of 2019

Source : Ministry of Education and Sports, LPDR

■ Universities – As of 2019

	No. of universities	No. of students	No. of teachers
Universities	5	39,497	2,170

Source : Ministry of Education and Sports, LPDR

- According to the university reform plan of Asian Development Bank (ADB) 1996, 3 colleges in Vientiane (Vientiane Educational College, College of Engineering, College of Medical) and 9 higher education institutes (in the field of electric, electronic, architecture, transportation, logistics, irrigation, forestry, agriculture and others) were integrated to establish National University of Lao PDR.
- The Korean government also participated with EDCF loans to open the 2nd national university, Souphanouvong University Campus (construction kicked off December 2005, completed February 2008) and has continued academic, research and human exchanges with the university, and the Korean language course was opened.

■ **There are 39 vocational schools and 111 technical colleges in Lao PDR.**

4.3 Media

4.3.1 TV, Radio

3 TV broadcasting agencies (National LNTV, Lao Star TV, TV Lao)

1 radio broadcasting agency (National LNR)

LNTV branch for each province

Using satellite antenna, people can watch programs of neighbouring countries and global channels, including the Australian broadcasting channel, CNN, YTN and Arirang TV.

✱ LNTV is running 2 channels and is provided with global news and documentary programs from advanced countries.

Korea also provided 2 TV dramas and 1 documentary program every year 1999~2003 and the programs were aired, earning great popularity for an extended period of time as 6 TV dramas and 5 documentary programs were provided in 2007 and 4 TV dramas in 2008 in free of charge and with support of voice over.

In cooperation of Korea Communications Commission, broadcasting equipment (USD 300,000) was provided for broadcasting production and airing in Lao PDR 2008~2010 to facilitate development in TV and radio broadcasting and continuous support has been made

to promote cultural exchange of the two nations.

4.3.2 Newspaper

- All the media agencies are government related or have the nature of institutional journal of LPRP.
- Major newspapers include PASAXON, KPL, Vientiane Mai and Vientiane Times.
 - PASAXON ; Founded 1950, institutional journal of LPRP (PASAXON means the ‘people’), most of the articles are about promoting the policies of the party
 - KPL ; Founded 1968, published by the national news agency (Lao News Agency), published in English and French version.
 - Vientiane Mai ; Founded 1960, managed by the Ministry of Information, Culture and Tourism, the most popular and general newspaper, covering the areas of politics, economy, culture and others
 - Vientiane Times ; Founded 1994, the biggest English daily in Lao PDR, published by the Ministry of Information, Culture and Tourism

Source; <Sabaidee Lao PDR> 2008, Park Jae Hyun <Culture and Economy of Lao PDR> 2012, Jang Tae Gu

<Overview on Lao PDR> 2019, Ministry of Foreign Affairs

<Korea-ASEAN Statistics> 2020, Korea-ASEAN Center <Others> Korea-Lao PDR Friendship Association/KLFA

5. International Relations (main foreign partners)

5.1 Diplomatic Policy Stance and Key Initiatives

Being surrounded by 5 countries of China in the northeast, Vietnam in the east, Cambodia in the south, Thailand in the west and Myanmar in the northwest, Lao has been deploying a well-balanced diplomatic policy with the principles of ‘peace, independence, friendship, cooperation and non-intervention.’

Since it became a communist country 1975 with active support of Vietnam, Lao PDR has been maintaining close relations with socialist countries and after the demise of former Soviet Unions, it has been maintaining a friendly and balanced relations with its neighbors, Vietnam, China and Thailand for economic development and national security of the nation. Meanwhile for international issues, it is responding jointly with ASEAN countries since it joined the group 1997.

- Also to attract foreign investment and implement economic development policies with support of ODA (Official Development Assistance- assistance given by the governments of richer countries to promote the economic development of developing countries), Lao PDR has been maintaining friendly relations with major donor countries including Korea.

As part of its efforts to overcome its restrictions as a landlocked country, Lao PDR hosted GMS (Greater Mekong Subregion) summit to implement regional diplomatic policy to build ‘East-West Corridor’ and ‘South-North Corridor’ and to grow into a logistic hub of the region, leveraging its location at the center of logistics path in the region. In 2010, the country hosted the 1st meeting of the states parties on CCM (Convention on Cluster Munitions) and the 9th ASEM Summit, making efforts to expand its roles and presence on the global stage.

5.1.1 Relations with Neighboring Countries and Major Countries

A) Vietnam ;

(Vietnam and Lao PDR are blood alliances and in terms of politics and friendship, the two countries are the closest neighbors for each other.)

- LPRP has been maintaining close relationships with the communist party of Vietnam as it provided extensive support from the independence movement in the 60's to the communist revolution 1975 and the joint statement of the two countries 1976, stipulated that the relations between the two was 'special ties.' They also signed an agreement of friendship and cooperation 1977 to strengthen the bilateral relations.
 - Lao PDR sided with Vietnam in the conflict between China and Vietnam 1979.

For Lao PDR joining ASEAN later than others, Vietnam is a neighboring socialist country and a role model to benchmark for successful economic development policy and market opening. While expanding exchange between higher-ranking officials of the party and the government with Vietnam, Lao PDR is also actively working on large-scale project development.

- 2017 was the year of 'Lao PDR-Vietnam Friendship' and Vietnam supported USD 100 million to help kick off the construction of new building for the National Assembly of Lao PDR.
- 2018 they signed an agreement for joint development of Vung Ang Port in Ha Tinh Province of Vietnam to allow access to marine trade for Lao PDR and facilitate an agreement on investment into power distribution network for electricity export of Lao PDR.
- In 2019, Nguyen Phu Trong, General Secretary of Vietnam visited Lao PDR for the first time since he took office to discuss and agree on 9 cooperation projects, including the Vientiane-Vung Ang railway construction.
 - Railway construction project ; 555Km in total with 452Km running across Lao PDR and 103Km running across Vietnam

B) Thailand ;

(Looking at the modern and contemporary history, the relations of Lao PDR and Thailand are not so close but Lao PDR is highly dependent on Thailand in terms of economy.)

It has close ties with Thailand historically and economically and especially, it is closely connected to and highly dependent on Thailand on economic front as most of the imported goods into Lao PDR are coming through Thailand and most of the daily goods are also imported from Thailand.

- Thailand is one of the largest trade partners and investors of Lao PDR, along with Vietnam and most of the hydroelectric power generated in Lao PDR is exported to Thailand.
- Having the same origination with the language used in northeast part of Thailand, Lao PDR language has many similar words and is under the cultural influence of Thailand.

Since the communists took power in Lao PDR (early 1980's), tension was rising in the relations with Thailand with a series of small conflicts and fights, across the Mekong but as Lao PDR adopted the opening-up economic policy, the relationships with Thailand turned round and since then the two countries hosted a joint committee to discuss expansion of the Mekong national boundary, economic aid and others. The relations keep improving and evolving, indicated by the fact that the prime ministers of Thailand mostly choose Lao PDR as the first country to visit after inauguration.

- In 2009, the government of Thailand repatriated all 4,000 Hmong people living in Thailand to Lao PDR following the agreement with Lao PDR government, despite criticism from the global society including UNHCR.
 - They were the people who helped U.S. during the Vietnam War and after U.S. left 1973, most of them also fled the nation to overseas and some of them moved to Thailand.
- The government of Thailand has provided loans since 2012 for expansion of Pakse International Airport, supply of water, road construction and others and the

construction of the bridge connecting Bokeo Province, Lao PDR and Chiang Rai of Thailand was completed 2013.

- Prime Minister of Thailand officially visited Lao PDR 2014
- Thong Lun, Prime Minister of Lao PDR officially visited Thailand 2015.

In 2016, Prime Minister Thong Lun visited Thailand again to sign an agreement of power export to expand the amount of power trade from 7,000MW to 9,000MW by 2025. In 2019, a construction agreement was signed to build 'the 5th Bridge of Friendship' connecting Bungkan (150km in the east of Vientiane) and Bolikhamsai Province of Lao PDR with the budget of USD130 million. When completed, the bridge is expected to offer the main route from Thailand to Da Nang, the port city of Vietnam, through Lao PDR, allowing Bolikhamsai to become a land transportation hub.

Meanwhile, if the Vientiane-China high speed railway is completed along the border (to be completed by the end of 2021) Lao PDR will be able to overcome its limitation as a landlocked country to grow into a logistics hub country in Indochina.

C) China ;

(Since early 2020, China has been making extensive investment and providing support with grant and loans and this comes with side effect in the development of market economy in Lao PDR).

As Lao PDR sided with its partner, Vietnam in 1979 China-Vietnam conflict, the relations with China was tightened for a while but as the conflict ended, the bilateral relations have been normalized, improved, and expanded under the slogan of ' extensive cooperation with socialist countries including China.'

Based on the rapid economic development in the nation and close relations with high-ranking officials in Lao PDR leadership, China has expanded trade and exchange in various fields, including manufacturing, mining and wholesale & retail and is strengthening its geopolitical influence.

- As China's trade with Lao PDR is rapidly growing, in 2019, China ranked 2nd with trade amount of USD3.4 billion, after No.1 trade partner Thailand with USD5.3 billion and ahead of No. 3 trade partner Vietnam with USD1.5 billion.
- China is also emerging as the largest investor with its investment into Lao PDR also fast growing from a mere USD75 million in 2001 to USD5 billion for the period from 2000 to 2014.

Jiang Zemin, General Secretary of the Chinese Communist Party visited Lao PDR November 2000, for the first time as a Chinese national leader and since then, state leaders of the two countries frequently visited the other and more recently, President and Prime Minister of Lao PDR visited China almost every year, for various purposes including participating in multilateral conferences, visiting trade fairs or others.

China has been expanding ODA for Lao PDR in various fields, including USD 80million aid in total for construction of the main stadium for Southeast Asian Games held in Lao PDR 2009 and provided USD8.6 million loan for expansion of Luang Prabang International Airport.

With the high speed railway construction project starting 2015 to connect Boten in Yunnan Province to Vientiane, Lao PDR is part of the Trans-Asia Railway project and under the project, 414km-long railway is scheduled to complete by 2021 end.

- Total project budget is USD6billion (70% covered by China and 30% covered by Lao PDR) and as of 2021, despite the impact of COVID 19, it is proceeded as planned. As of September 2020, 91% of the planned railway was completed. Completion of Lao PDR-China logistic path has greater meaning than simply connecting the two countries.

2017, the incumbent General Secretary is Xi Jinping paid a state visit to Lao PDR in 11 years since the visit of former General Secretary, Hu Jintao 2006 and signed 17 MOUs to strengthen and expand the bilateral relations with Lao PDR.

- To strengthen cooperation with riparian countries along the Mekong, China launched Lancang-Mekong Cooperation (LMC)' 2015 and promised to provide preferential loans

and establish ‘Mekong Cooperation Fund’ for cooperative development to seek political security, sustainable economic development and social & cultural exchange with Lao PDR, Myanmar, Thailand and Vietnam and Lao PDR served as a co-chair country along with China for Lancang-Mekong Cooperation from 2018 to 2020.

In 2018, Minister of Foreign Affairs of China Wang Yi visited Lao PDR to attend the 4th Lancang-Mekong Cooperation Foreign Ministers’ Meeting and discussed the foundation of a railway operations training college in Lao PDR.

2019 marked the 10th anniversary of comprehensive strategic partnership between Lao PDR and China and in April of the year, President Bounnhang Vorachith of Lao PDR visited China to attend the 2nd Belt and Road Forum for International Cooperation summit and signed an agreement of ‘Action Plan for China and Lao PDR to Build a Community of Common Destiny’ for the coming 5 years.

D) U.S. ;

The relations between Lao PDR and U.S. has never been severed since the opening of U.S. consulate in Lao PDR 1950 but the two countries have been at odds due to the issue of military support for Lao PDR monarchy during the Vietnam War and especially due to the issue of unexploded ordnance (UXO) from the bombardment over the territory of Lao PDR and the relocation of Hmong people over to U.S. after communization of Lao PDR.

However, as the ambassadorial relations were recovered 1992, efforts to strengthen the ties have been made in diverse ways, including U.S. government’s support for Lao PDR mostly for MIA (missing in action) and the drug issue and now the relations are fully normalized.

U.S. granted Normal Trade Relations (NTR) to Lao PDR December 2004 and significant lowering the export duty on Lao PDR goods and the bilateral trade agreement took effect February 2005, strengthening the trade relations even further between the two countries.

The relations between the two countries rapidly advanced as the Lao PDR government allowed to have embassy attache in U.S. embassy in Lao PDR 2008 and in 2019 U.S. government excluded Lao PDR companies from the list of capital movement restrictions for companies in communist countries.

- Especially, as the foreign minister of Lao PDR visited U.S. for the first time July 2010 and in September of the year, the president of Lao PDR visited U.S. to attend UN General Assembly and U.S.-ASEAN Summit and that was the first time since the establishment of the people republic.
- After upgrading the comprehensive bilateral meeting between the two up to the level of assistant secretary/vice minister meeting March 2010, high-level exchange and visits also expanded as U.S. Assistant Secretary of State for Refugees visited Lao PDR, June of the year.

State Secretary Hillary Clinton's visit to Lao PDR July 2012 was meaningful as it took 57 years since the visit of former U.S. Secretary of State last time and at the same time, it was expected to be an important event to further advance the bilateral relations. During the visit, an agreement was made on UXO support and expanded efforts for remains recovery mission for those missing in action during the Indochina War.

E) Japan :

Japan temporarily occupied Lao PDR during World War II and since the peace treaty August 1951, it opened a diplomatic office in Lao PDR 1955 and has been operating it since then. Japan is one of the major donor countries, accounting for about 30% of ODA for Lao PDR and is providing massive amount of ODA investment to keep China in check in its expansion towards South and maintain and expand its influence in Southeast Asia.

Especially, along with the 'bilateral relations', it is seeking to expand multilateral relations in diverse angles, launching 'Japan-ASEAN Cooperation' and 'Japan-Mekong Cooperation.'

- As the 1st Japan-Mekong Ministerial Meeting was held in Japan 2008, the meeting for working-level preparation for the meeting was held in Lao PDR.

- After the 2nd Japan-Mekong Foreign Ministerial Meeting October 2009, the first Japan-Mekong Summit was held in November.
- High-level exchanges were active between the two countries including President Choummaly Sayasone of Lao PDR's visit to Japan March 2010, Prime Minister Bouasone Bouphavanh's visit in May, Thongsing Thammavong's visit as the chairperson of the National Assembly and later as Prime Minister March 2012 and Lao PDR-Japan Summit November.

F) Russia ;

Lao PDR designated the relations with former Soviet Unions as 'Special Friendship' just as its relations with Vietnam and has been strengthening cooperation in politics, economy and military. At the incumbent government and political sector of Lao PDR, there is a group of people who studies in former Soviet Unions in the past, indicating the close relations with the entity.

After the demise of Soviet Unions, support for Lao PDR also almost stopped but the exchange among the parties and government still remained to some extent.

G) Korea ;

(After resumption of the relations with Lao PDR October 25, 1995, the two countries have been making remarkable progress on political, economic and diplomatic front.)

During the visit to Lao PDR September 9, 2016, President Park Geun Hye had a meeting with President Bounnhang Vorachith of Lao PDR to assess the bilateral relations after resumption of the diplomatic ties 1995 and discuss ways to further promote the areas of common interest in trade, investment, energy and development cooperation.

Korean companies entry into the market

- Since the first investment 1992, the total investment until 2018 remained at USD1.1 billion.

- Korea Exchange (financial) CJ Logistics (logistics) KOLAO(automobile, financial)

SK E&C(hydroelectric power generation) Korea Western Power CO. (Hydroelectric power generation) Booyoung Co. (Construction, financial)

HHI Co. (construction) Hotel COSMO(hospitality, restaurant)

Gelachon (ascon production) BNK Capital (lease financial) Welcome Savings Bank (financial) DGB(lease financial) KB(lease financial)

144 corporations/branches/regional offices

Grants

- From 1987 to 2018, more than USD235 million was provided through KOICA projects in the field of rural development, healthcare and education.

Loans

- In terms of loans, since the first loan for ‘Luang Prabang Souphanouvong University foundation project’, approximately USD566 million has been approved for 15 projects in total until January 2018.

The list of the project since 2011

Description	Year of Approval	Amount Approved (USD mil.)
① Tax Information System Building	2011	29
② Savannakhet Comprehensive Rural Development	2014	46
③ Xe Pian-Xe Namnoy Hydroplant	2015	81
④ Champasak Mekong River Integrated Management	2015	53
⑤ Vientiane Mekong River Integrated Management (2 nd round)	2016	57
⑥ Public Security Office Modern Hospital Construction	2016	65
⑦ National Medical University Hospital Construction	2017	103
⑧ Fire Extinction System Building at Cultural Heritage Sites	2019	30

Source: Korea-Lao PDR Friendship Association. Reference (Lao PDR Overview 2019, Korea-ASEAN Statistics 2020)

KOICA ODA projects since 2019

Projects	Year	Budget (USD 10,000)
Attapeu Climate Change Resilience and Self-sufficiency Competence Building	2019-2023	1,150
Waste Water and Waste Disposal Capability Improvement for Urban Environments Enhancement	2019-2024	650
2nd Round of UXO Removal Support	2019-2022	550
Service Advancement for Maternity Health Access Improvement	2019-2024	750
Violence against Women Prevention/Response System and Policy Capabilities Improvement	2020-2024	700
Efficient Water Management System Building for NamNgum River	2020-2025	1,100
Rural Development Projects for 3 Southern Provinces	2020-2024	900
2020 COVID 19 Comprehensive Emergency Support	2020	USD 526,696
2020 COVID 19 Walk-through Test Station Support	2020	USD 175,087

Source: Korea-Lao PDR Friendship Association. Reference (Lao PDR Overview 2019, Korea-ASEAN Statistics 2020)

Cultural/Human Exchange

- No. of Korean visitors to Lao PDR was 13,000 in 2007 but in 2019 alone more than 200,000 Koreans visited the country.
 - Other than general tourists, No. of people visiting Lao PDR in groups for volunteer activities from schools, hospitals or volunteer organizations is also increasing.
- In the early stage, most of Lao PDR people visiting Korea were the students invited on the government scholarship or those in political sector or the government but in 2018, more than 12,000 people visited Korea from Lao PDR. As the economy of Lao PDR further develops and Korea-Lao PDR relations deepen on political and economic front, visit from Lao PDR is expected to grow continuously.

■ Korea and Lao PDR amid COVID19 (2020)

- Supplying COVID19 prevention goods (KOICA) to Children's Hospital of Lao PDR (April)
- COVID19 response thrill and goods supply (USD600,000) (May)
- COVID19 goods supply for Lao-Korea Skills Development Institute (KOICA) (May)
- (USD 37,000) goods supply including medical goods in northern region (KOFIH) (June)
- COVID 19 prevention goods for refugees affected and minority ethnic groups (KOICA) (July)
- In Lao PDR, the number of foreign visitors dropped 82% 2020 compared to 2019 (In Korea, the number declined 88%.)
- In cooperation of Korea-Lao PDR Friendship Association and Korea Love Sharing Community /KOLOS, supplying medical goods and devices (USD850,000) in 2020 and 2021

Employment Permit System

Since Korea-Lao PDR MOU 2016, workers came to Korea from February 2018 and has been working in the nation (as of October 2021, 287 workers), but it is still in the infant stage and they do not meet the quota as the number of Lao PDR workers passing the Korean language competence test is not so many. It is expected to grow gradually with the efforts of Korean language education and others.

6. Lao PDR-Korea Collaborations

6.1 Backgrounds of the Two Countries

Korea forged diplomatic ties with Lao PDR June 1974 to open representatives office but severed the relations 1975 as countries in Indochina including Vietnam, Lao PDR and Cambodia were all turned to Communist countries after South Vietnam lost in the war.

In 20 years, Korea resumed the relations with Lao PDR October 25, 1995 and at that time, then Vice Minister of Foreign Affairs, Phongsavath BOUPHA visited Korea to sign an agreement to reopen the relations.



Source : Korea-Lao PDR Friendship Association; Minister to President's Office, Phongsavath BOUPHA's visit to Korea to commemorate the 20th Anniversary of Diplomatic Ties November 2015

Looking at the modern history of Lao PDR, the French colonial rule started 1893, and it was invaded by Japan 1945 and finally 1954, it achieved full independence from France.

With the ideology conflicts between LPRP and Lao PDR internal unities 1954 and conflict with the monarchy, Lao PDR went through a turbulent and dynamic time in political arena and since its independence from France until establishment of Lao PDR, Lao PDR was inclined more to the liberal front.

After South Vietnam went down and Vietnam, Lao PDR and Cambodia in Indochina becoming a communist country, LPRP having close relations with Vietnam, founded Lao PDR (Lao People's Democratic Republic) December 2, 1975.

Souphanouvong was the first president of the republic, followed by the 2nd president Kaysone PHONVIHANE.

President of Lao PDR							
No.	Term of Office	Name	Photo	No.	Term of Office	Name	Photo
1 st president	1975-1989	Souphanouvong		5th	2006-2015	Choummaly SAYASONE	
2 nd	1991-1992	Kaysone PHONVIHANE		6th	2016-2020	Bounhyang VORACHIT	
3 rd	1992-1998	Nouhak PHOUNSAVANH		7th	2021-	Thongloun SISULITH	
4 th	1998-2006	Khamtay SIPHANDONE		Others			

Source: Korea-Lao PDR Friendship Association/ Data in archive

6.2 Bilateral Relations after Forging Diplomatic Ties

After resuming diplomatic relations October 25, 1995, Korea opened an embassy 1996 in Vientiane and since the first Ambassador to Lao PDR Mr. Choi Geun Bae, now the 11th Ambassador to Lao PDR, Lim Mu Hong is serving at the embassy as of October 2021. Lao PDR opened an embassy to Korea 2001 in Seoul and since the first ambassador, Mr. Thongsavath PRASEUTH, currently, the 5th ambassador Mr. Thieng BOUPHA is serving in Korea as of October 2021.

6.3 Bilateral Cooperation

Korean companies entered the Lao PDR market in the early 90's even before the resumption of the ties, serving as a bridge between the two countries. Especially, Daewoo entering the market in the early 1990 completed the construction of Houay-Ho hydropower plant in Attapeu Province in southern Lao PDR (150MW capacity at the time of completion) but it was sold to other country in the aftermath of financial crisis in Korea. With the surprising result of a Korean company Daewoo, successfully building a hydropower plant, overcoming all the hardship and difficulties in the middle of the jungle, Lao PDR began to have positive image and built trust in Korea since then.



Houay Ho Hydropower Plant built by Daewoo 1993-1999 in Attapeu, Lao PDR /Source : Houay-ho-power-plant

Lao PDR established diplomatic ties with North Korea, almost at the same time as its relations with South Korea. While the relations with South Korea was discontinued 1975, Lao PDR and North Korea have been maintaining political relations, so at the time of resuming relations with South Korea 1995, Lao PDR stood closer to North Korea than the South in diplomatic terms.

However, as a growing No. of Korean companies entered the market, the presence of South Korea was renewed in Lao PDR and these factors contributed to resuming the relation 1995. In 26 years since resumption of the relations, the two countries have been making remarkable progress in political, cultural, economic and diplomatic relations.

There have been active exchanges between the two countries including Korean presidents' visit to Lao PDR including President Rho Mu Hyun 2004, President Park Geun Hye 2016 and President Moon Jae In 2019 and Lao PDR leaders' visit to Korea including President Choummaly Sayasone 2013, Prime Minister, Bouasone Bouphavanh, Prime Minister, Thongsing Thammavong and Prime Minister Thongloun Sisoulith (incumbent president) and despite relatively short period of diplomatic ties, the two counties are working together in mutual partnership on the global stage.

Korea has been providing grants for Lao PDR through KOICA and loans to build infrastructure in Lao PDR through EDCF and the government and private institutions also make diverse efforts to strengthen presence of Korea in Lao PDR and presence of Lao PDR in Korea, through various ODA projects and tighten the relations as comprehensive partners.

As civilian air routes opened 2011, running direct flight to Lao PDR 5 times per day before the COVID 19 pandemic, about 230,000 Koreans visited Lao PDR for travel. With the growth of national income and improvement in living standards, the number of people visiting Korea from Lao PDR also increased. However, since the outbreak of the COVID 19 pandemic early 2020, tourism between the two countries almost came to a halt.



Source: Korea-Lao PDR Friendship Association/ December 2011

Especially with strong interest of the Korean government and the private sector in educational sector in Lao PDR, they invited students of Lao PDR over to Korea since 2002 to support them to study in graduate and doctoral courses on scholarship in Korea. Currently (October 1, 2021) 110 Lao PDR students are studying in Korea on scholarship programs and this is expected to make huge contribution to driving mutual advancement of the two countries in the future by promoting understanding on Korea and enhancing Korea-friendly sentiment, with educational support.

Also Korea-Lao PDR Friendship Association has continued non-profit diplomatic activities over the past 17 years in private sector, including the project to improve education and livelihood in remote areas in Lao PDR, medical devices and medicine support program, the project to support the disabled association, supply of emergency rescue goods, support programs for students of Lao PDR studying in Korea, voluntary activities, and others. At the same time, with management of human network, the association has been building grounds to expand bilateral friendship and exchanges.



Source: Korea-Lao PDR Friendship Association/
11th Korea Culture Experience Day for Lao PDR Students Studying in Korea May 2019



Source: Korea-Lao PDR Friendship Association/ 11th Conference in Vientiane of the two Associations December 2019

Rather than expecting cooperation on an equal footing with Lao PDR, it seems better to maintain the partnership between Korea and Lao PDR at global stage and take time to support Lao PDR to help the country have the opportunity for self-sufficiency and further development, thereby seeking cooperation and partnership on equal footing in the future when Lao PDR becomes economically fully independent and advanced.

For the outcomes from Korea-Lao PDR cooperation now, it is more important to pursue intangible outcomes rather than tangible outcomes and to achieve both tangible and intangible outcomes in the future, steady cooperation both from the government and private sector is required.

7. Current Issues

Located at the center of Indochina, historically Lao PDR was under the influence of China in the North and Khmer in the South, having Vientiane in the middle. It was under the colonial rule of France from the end of the 19th century to mid 20th century. It has close political ties with Vietnam as both countries were occupied by France and after the victory against U.S. in the Vietnam War, Lao PDR and Vietnam have been maintaining the relationship of blood alliance.

Vietnam played the key role in foundation of Lao PDR December 1995 and out of the 5 surrounding countries, Lao PDR has dispute over the territory with Thailand in the early 1980's and is now having dispute with Cambodia over the border.

Against this historical backgrounds, Lao PDR is making diverse efforts to emerge from the status of the least developed country, including Three-build Directive starting 2012 in full scale. This is a project to transfer the authority of the project from the province to district and to village to promote the fight against poverty, educational environment improvement and literacy rate increase by 2020, led by National Committee for Rural Development and Poverty Eradication (NRDPE) but the country has yet to fully emerge from the status of the least developed country.

Infrastructure in Lao PDR is still underdeveloped and it is a landlocked country, with almost 75% of its territory in mountain areas. It puts priority on peace and unity among so many minority groups and rather than applying high-hand, forceful policies, it prefers appeasement policies in politics.

As it joined ASEAN 1997, Lao PDR, once called a Locked Country is now going through massive changes and is growing into a Hub Country, located at the center of Indochina.

Lao PDR has been focusing on infrastructure building and economic development since the mid 1980's with grants and loans from ADB and World Bank. Since it is still an

underdeveloped country, it is dependent on the aid and loans from foreign countries for national infrastructure building to overcome shortage of capital.

As a member of ASEAN, Lao PDR is striving to strengthen its role and awareness at global stage and the conflict with U.S. after the Vietnam War began to ease since the early 1990's and the relations was normalized with the State Secretary Hillary Clinton's visit to Lao PDR 2012. Nowadays, the country is free from any economic sanctions and has no restrictions in export and import with Europe and any other country in the world.

Lao PDR has been receiving extensive economic aid and support for infrastructure building from China since 2010. As the high speed railway connecting China and Vientiane is completed as scheduled by 2021 end, huge changes are expected in Lao PDR where there has been no railway yet and the construction of highway connecting Vientiane and Vangvieng is also bringing about many changes.

It is sure that the aggressive investment and support from the Chinese government helped the economic development of Lao PDR but it did not come without a cost. This led to the confusion in the existing market order and disruption in the Lao PDR economic system and market and especially, the real estate price downtown in Vientiane skyrocketed. With these changes, some politicians and businessmen in Lao PDR began to worry about the future of the country.

Korea has been providing grants and loans through various channels since 1995, including EDCF loans, grants through KOICA and investment through private channels for the Mekong Bank Protection in Vientiane, riverside protection in Champasak Province, farming land arrangement in Savannakhet, the 2nd national university, Souphanouvong University construction in Luang Prabang and construction of roads and hospitals and others, to strengthen and further advance bilateral partnership.

With the construction of hydro power plants using water resources since the early 1990, power export has been the main source of revenue of Lao PDR and as Korean companies, Daewoo entered the market in the early 1990 and built Houay-Ho hydropower plant with

150MW capacity. Since the entry of Daewoo into the market in the early 1990, currently SK and Korean Western Power are investing in Lao PDR, and Xe Pian Xe Namnoy hydro power plant will be completed in consortium of Lao PDR and Thailand and soon start power generation.

Korea played the key role in building hydro power plants in many locations and helping power export of Lao PDR since the early stage.

In the aftermath of COVID 19 outbreak, Lao PDR also witnessed a fall in GDP 2020 from 2019 (USD2,654) and even further decline is expected in 2021. In the meantime, Lao PDR economy is expected to slow down, ending the steady growth before the pandemic, and a turbulent road ahead is expected due to restrictions on trade with foreign countries, decline in No. of foreign visitors and uncertainty in domestic market.

Major issues in internal Lao PDR

- Emerging from the status of LDCs → Overcoming poverty
- Rural development → Development in the main industry, agriculture
- Infrastructure building → Economic development with establishment of transportation network and Mekong River development
- Educational environment improvement and literacy improvement → Nurturing talents with education
- Economic independence
- Addressing dispute arising from the gap between haves and have-nots

Global Issues

- Conflict over border with a neighboring country (Cambodia)
- Border management surrounded by 5 neighboring countries
- Infrastructure expansion with foreign aid
- Boosting awareness of Lao PDR at global stage



2021 K-Innovation Partnership Program with Lao PDR

- Attracting foreign capital and private investment, with special economic zone
- Activation of tourism industry in Lao PDR

8. Conclusion

For constructive relations between Korea and Lao PDR, it is important to understand historical and cultural differences and each other's traditions, as well as strengthen cooperative relations as global partners to seek win-win prosperity and development.

The Lao PDR has been maintaining close political relations with North Korea. But since resuming the diplomatic ties in 1995 with the Lao PDR, South Korea has made diverse efforts, including ODA policies, private investment and exchange and much progress has been made in the state of relations between the two countries. There was a time when South Korea ranked 4th in foreign direct investments to Lao PDR (around 2015), and these activities have contributed a lot to boosting the national image of South Korea in Lao PDR.

When Lao PDR became a communist country in 1975, diplomatic relations with South Korea was severed while Lao PDR still maintained close political relations with North Korea. In the 20 years since the discontinuation, Korea resumed its diplomatic ties with the Lao PDR 1995 and since then, there have been active exchanges, including grants through KOICA, EDCF loans, a series of ODA projects and investment from the private sector, while to date, Lao PDR has been showing much more interest in relations with South Korea than the North.

Since becoming independent from France in 1954 until the founding of Lao PDR on December 2 1975, Lao PDR had gone through a turbulent time of political upheaval (clashes between the right and the left), restoration of monarchy and economic difficulties. During the Vietnam War, Lao PDR sided with Vietnam to fight against the U.S. and since the fall of South Vietnam, the two countries have been maintaining close political relations as a brotherly alliance.

To promote cross-border ties and expand exchange with the Lao PDR and pursue win-win benefits as partners, it is important to understand the historic backgrounds of the Indochina countries from the viewpoint of diplomatic relations of Lao PDR with Vietnam, Thailand and Cambodia, respectively (as mentioned in the session of issues in Lao PDR).

At the 11th National Congress of the Lao People's Revolutionary Party (LPRP) in January 2021, Thongluon Sisulith was elected as the 7th Lao PDR president while Phankhaun was elected as the prime minister and Xaysomphone Phomvihane as the president of the National Assembly to establish a new government cabinet of the Lao PDR. This year marks the 26th anniversary of resuming diplomatic ties between Korea and Lao PDR. During this period, the two countries have made much progress in cordial relationships based on mutual reciprocity through many state visits among many political leaders, including presidents, to each other's country.

Notably, President Thongloun Sisoulith is known to be pro-Korea and have a positive stance toward Korea. The two countries have made much progress in bilateral relations in the areas of politics, diplomacy and the economy. Based on the experiences of the past 26 years, more matured and advanced relations are expected between the two.

Lao PDR, once dubbed a locked country, has made steady economic growth, active exchange & cooperation with foreign countries and steady & consistent policy execution since it joined ASEAN in 1997. Now as a member of ASEAN, it is growing to become a regional hub or battery country, taking advantage of its location at the center of Indochina.

Over the past 26 years, Korea has made diverse efforts to expand investment, market entry, bilateral exchange and cordial relations through grants, loans, ODA projects and investment through private channels. Since opening direct flight to Lao PDR on December 11, 2011, there are 5 direct flights operating between Korea and Lao PDR before the outbreak of the pandemic, and in 2019 alone, almost 230,000 Korean tourists visited Lao PDR with a growing number of people visiting Korea from Lao PDR as well.

It has been 26 years since diplomatic ties were resumed, but despite a relatively short period of time, the two countries have made major strides in wide-ranging political, diplomatic, cultural, and economic progress. To further advance and develop these relations, it seems that government-level support for Lao PDR, continuous exchange & investment and long-

term plans are required, and by the same token, to ensure that diplomatic relations evolve from government-to-government relations in the past into government-private, private-government or private-private partnerships, it is essential to allow public diplomatic policies implemented at the government level to reap meaningful results leveraging well-coordinated public-private partnerships.



CHAPTER 3

Public STI Research Institutes in Korea

Chapter 3. Public STI Research Institutes in Korea

1.

Growth of Science and Technology of Korea and Government-funded Research Institutes

After the independence from colonial rule 1945, the science and technology sector in Korea was in a dire situation. During the 35-year Japanese occupation (1910-45) only a handful of researchers received doctoral degrees in the field of natural science and engineering. Among those 400 students who majored in natural science and engineering, only half of them lived in Korea at the time of liberation, and they even had to be divided as the nation was divided into the South and North. The science and technology sector, let alone Korean society was hit hard by the Korean War (1950-53). In the wake of the war, many of the scientists and engineers in South Korea went to North Korea or were abducted for the sake of ideology or science and technology researches. Even in this situation, scientists and engineers in Korea emphasized the importance of science and technology development to rebuild the nation and urged the government to take action accordingly. In other words, it could be seen as requests for strengthening science and technology education, establishing full-fledged research centers, and organizing an administrative body to oversee relevant policies. However, back then, science and technology were not a top priority of public policies. The government founded national universities with the college of engineering and science in each region to cultivate talents in this sector, but it failed to set up research centers or state agencies to oversee science and technology matters.

The efforts were gradually made in earnest to promote science and technology, starting in the mid-1960s. The Korea Institute of Science and Technology (KIST) boasting advanced research facilities and competent talents was established in 1966, followed up next year by establishing the Ministry of Science and Technology (MOST) to oversee science and

technology policies. In just half a century since then, the science and technology sector of Korea took an unprecedented leap forward to be on a par with many other developed nations in the world in terms of quantity, as shown by hard figures such as amount of R&D investment and number of researchers, papers, & patents, etc. In the World Competitiveness Rank of the International Institute for Management Development (IMD) released in 2021, Korea ranked 23rd overall while taking the 2nd and 17th highest place in science infrastructure and technology infrastructure, respectively.¹ The impressive rapid growth of Korea's national science and technology was led by many drivers, notably the government's effective growth policies and efforts of Korean scientist and engineers who called for pursuing and promoting science and technology even before the government adopted related policies and actively engaged in setting policy, as well as scientific cooperation with advanced countries on sharing know-how and resources to help realize their vision.

One of the recent studies on the drivers of compressed growth of national science and technology, (Kim Geun Bae, 2016) suggested that the advancement was backed by 'Institution & Practice', emphasizing the importance efficient institution adopted by the government and commitment to follow through on results with the system.² For developing countries playing catch-up in science and technology development, it is more important than anything else to put the institutions in place, and so Korea was able to take a big leap forward by establishing an effective system specific to the Korean domestic situation and realizing economies of scale in science and technology through focused R&D investment and mobilization of research talents. In fact, to nurture new industries and science & technology, Korea has built a framework including establishment of research institutes and pro-science and technology acts and adopted effective strategies to level up in a short period of time with intensive and focused support. Also, with the approach of creating many similar institutions in tandem, based on the system proven to be effective, it was possible to drive quantitative growth in science and technology in general. It was supported by the establishment of the leading GRIs (government-funded research

¹ <https://worldcompetitiveness.imd.org/countryprofile/KR/wcy>

² Kim Geun Bae, *The Structure of Scientific and Technological Revolution in Korea*, Deulneuk, 2016

institutes), which are the main research systems of Korea, the science and technology research complex, science and technology-specialized universities, and national R&D programs.³ Notably, GRIs served as a key mechanism to lead a compressed science and technology growth of Korea and build the platform to execute and realize relevant science and technology policies. With the establishment of science and technology infrastructure and cultivation of talents, GRIs became the important launch pad and served as the most effective groundworks and tools in the process of pursuing rapid development in science and technology. That being said, this chapter shall review the emergence and growth of public R&D systems, focusing on GRIs.

Some of the national research institutes of Korea started during the days of Japanese colonial rule, but due to the disconnection and shortage of human resources and materials during the Korean War, these institutes were in a dire situation both in research environments and resources. In this situation, scientists and engineers urged the government to establish ‘full-fledged research institutes’ to build research environments and conduct researches required at a national level. At the U.N. General Assembly 1954, President Eisenhower delivered his ‘Atoms for Peace’ speech, and in his speech, he proposed that the U.S. would share nuclear materials and technologies with countries in need. That being said, Korea immediately moved to reach an agreement with the U.S. and began nuclear projects in 1956. After that, the Office of Atomic Energy (OAE) was established in 1959 as a state-run agency and Atomic Energy Research Institute (AERI) was founded under the body.⁴ At AERI, TRIGA Mark-II, the first nuclear reactor in Korea was adopted. Also, considering the nature of the nuclear sector as a comprehensive science connected to so many other fields, the ‘nuclear conference’ was held every year to invite scientists and engineers from various fields to make academic discussions together. It was rare to see such an academic conference at that time, and scientists and engineers at the conference announced ‘proposals’ to the government and called for enacting pro-science

³ Moon Manyong, *The Evolution of Science and Technology Research Systems in South Korea*, Deulneuk, 2017

⁴ John DiMoia, “Atoms for Sale?: Cold War Institution-Building and the South Korean Atomic Energy Project, 1945–1965,” *Technology and Culture* 51-3 (2010), 589-618.

and technology policies for advancement.

Despite the high expectation on AERI, it had its limitations since it was a national research institute. All researchers at the institute were civil service workers, and they had a pre-determined headcount with compartmentalized duties. As they were applied with the Act on Government Workers for their salary, etc., it was not possible to attract dedicated and competent researchers. Also, as the institute was run by the Budget and Accounts Act, it had no liberty at all in accounting for the operation of institutes, making it almost impossible to start new and creative research activities. One of the cases would be the request to purchase refrigerators for research activities rejected several times, as refrigerator was on the list of luxury goods designated by the military government in the early 1960s. Scientists and engineers thought it was needed to establish different type of research institutes, and so the government understood and moved to build a different type of institute, namely non-national institutes. However, the plan did not go far due to the lack of budget and differences in opinions of researchers. The situation turned around as the U.S. made the offer to help Korea establish a new research institute during the Korea-U.S. Summit in 1965. As a result, KIST was established, marking the first time the new type of public research system called GRIs would emerge.

In Section 2, we would like to look into the backgrounds of the establishment of KIST, the first GRI in Korea and the starting point to build a modern national science and technology system. We shall review what issues were raised in the process of establishment and how they were resolved. In Section 3, the process of GRIs' expansion by establishing research institutes specialized in each sector. We also reviewed how the roles and relevant systems have been evolving for GRIs since the 1980s up to now. Section 4 shall explain how the R&D cluster of Korea was established from the Seoul R&D Complex, Daedeok Science Town to the recently established R&D specialized zone (Innopolis). Lastly, Section 5 shall review the implications drawn from the history of the public research system of Korea.

2. Emergence of GRIs: Establishment of KIST

In the Korea-U.S. Summit in May 1965, President Lyndon B. Johnson made an unexpected proposal that the U.S. would support the establishment of a new research institute in Korea. Since he was well aware of how the nation was struggling to overcome the limitation of national institutes and found new type of research institute to support the technological development of the industrial sector, President Park Chung Hee wasted no time to agree with this proposal. The proposal was brought up by the U.S. side at the end of the summit and this idea was formulated by Dr. Donald Hornig, the presidential science advisor of the U.S., following an order from President Johnson, who was mulling over a gift for President Park.⁵

Looking at the context of the U.S. proposal to support the establishment of a new research institute in Korea, it was aimed to facilitate the process of normalizing the relationship between Korea and Japan at that time. After Korea's independence from colonial rule in 1945, diplomatic ties between Korea and Japan had been severed, and the U.S. hoped to establish a triangular alliance of Korea, the U.S. and Japan to fight against Communist nations. Toward this end, the U.S. needed to normalize Korea-Japan relations. At that time, the negotiation to resume relations between Korea and Japan was at the final stage, but Koreans did not welcome the idea. On top of the lingering animosity of historical colonial occupation, there was a concern about whether U.S. support for Korea would end after normalizing Korea-Japan relations, so that Korea could become economically dependent on Japan. Thus, the U.S. promised to continue support even after normalizing diplomatic relations, and as a token of the promise, the U.S. agreed to provide a USD 150million loan and support establishment of a new research institutes during the summit. Foremost, a new research institute building was a useful way to support President Park's government with no political partisanship. Next month after the Korea-U.S. Summit, Korea and Japan agreed

⁵ For the process of establishing KIST, see Manyong Moon, "Postcolonial Desire and the Tripartite Alliance in East Asia: The Hybrid Origins of a Modern Scientific and Technological System in S. Korea," in Hiromi Mizuno, Aaron S. Moore, and John DiMoia eds., *Engineering Asia: Technology, Colonial Development, and the Cold War Order* (Bloomsbury Academic, 2018), 165-188.

to and signed a treaty to normalize diplomatic relations between the two.

The second factor in the context of the U.S. proposal was the U.S. intention to build a case of reverse brain drain. At the time, criticism was running high across the world against the U.S. being the largest beneficiary of an international brain drain. They criticized that the U.S. passively watched and did nothing to stall an exodus of human capital, while reaping the most benefits. Thus, by establishing a research institute in Korea where brain drain had been a serious issue and luring away Korean scientists working in the U.S. as core researchers, the U.S. intended to use it as a model case to address this issue. In fact, after establishing KIST, the U.S. cited the phrase 'reverse brain-drain center' to present this real-world case to the international community.

Lastly, the proposal from the U.S. was a reciprocal gesture to Korea for its decision to send military personnel in the Vietnam War. One of the key decisions in the 1965 summit was for Korea to dispatch military personnel to Vietnam. There was no official document between the two nations, saying that the U.S. decided to support establishing a research institute in exchange for dispatching military personnel, but it was widely understood and acknowledged as the case among the public. Thus, in the process of KIST establishment and operations at the early stage, KIST researchers believed that failure was not an option since the institute was built and paid for with the blood and tears of young Korean servicemen in Vietnam. There have been many arguments over the legitimacy of the decision of Korea to join the Vietnam War but it was an undeniable truth that USD 1 billion provided in return for the sacrifices of the young Koreans became the seed money to drive the economic growth of the nation. In the end, Korea and the U.S. both had its own goal to realize through establishing a new research institute, KIST.

The proposal of the U.S. was made in this context and was very timely. As there were strong needs for creating a new research institute and the basic plans for achieving this, financial support and know-how shared by the U.S. facilitated putting the plan into practice since the Korean government had conceived it up without hands-on experience. In just 15 days after the joint statement of Korea and the U.S., the Economic Planning Board (EPB) of Korea delivered a comprehensive science and technology development plan to the U.S., including

a revised plan for establishment of a new research institute. Korea was able to respond quickly, presenting a detailed vision and plan as it already had been preparing to execute the plan to build new research institute for several years. The plan indicated that the new research institute would build technological competency and conduct research activities to directly contribute to the industrial development of the nation.

In July 1967, two months after the summit, Dr. Hornig visited Korea again for a feasibility study on a new research institute. Dr. Hornig and his fellows in the investigation group stayed for a week to meet with the ROK president and people from science & technology and industrial sectors, and also visit companies and universities to review the status of science and technology fields of Korea. After returning to the U.S., Dr. Hornig submitted a report to President Johnson, saying that a new research institute was needed for Korea.⁶ Following the suggestion in this report, the Battelle Memorial Institute (BMI) was selected as an affiliated institute to assist the establishment and operations of a new institute in Korea. After appointed as an affiliated institute to help KIST establishment, BMI organized a group of experts and dispatched them to Korea in September 1965. They conducted an investigation for its establishment, and after coordination with the Korean government, submitted a report in December.⁷ The report offered basic guidelines for KIST establishment, including the basic principles suggested by Korea during the visit of Dr. Hornig and his team.

Finally, in February 1966, an official announcement on the founding and establishment of KIST was disseminated. KIST was established as a result of a joint Korea-U.S. project and organized with government support, but the institute was established as a private foundation. President Park also was on the list of KIST founders as an individual. Hongneung in Seoul, managed by the Experiment Forest Station under the Ministry of Agriculture and Fisheries was selected as the location for KIST. However, the ministry did not want to give up the site in Seoul with its extensive history just for a new research institute. After political

6 Donald F. Hornig, "Report to the President: Regarding the Feasibility of Establishing in Korea with U.S. Cooperation an Institute for Industrial Technology and Applied Science" (1965).

7 E. E. Slowter, J. L. Gray, W. J. Harris and D. D. Evans, "Report on the Establishment and Organization of a Korean Institute of Industrial Technology and Applied Science" (Battelle Memorial Institute, 1965).

tussles with the ministry's opposition over the site selection, President Park Chung Hee intervened and Hongneung was decided as the site for the institute. The ROK president said the ministry should yield since it was an important national project and the minister had to abide by an executive order.

As in this case, there were differences in opinions among the ministries and governmental departments in the process, but the ROK president as a founder of the institute, supported the project as an active mediator. He visited the construction site often and offered his verbal encouragement for those in the field and even the officers of army corps of engineers were dispatched to accelerate the construction process. There was an anonymous letter saying that the compensation for the KIST researchers were excessive, but he again sided with the institute, saying that it was necessary to attract competent researchers. It is business as usual to see conflicting interests between new and existing institutes or organizations, as is normally the case when new institutes receive support even to the level of appearing as preferential treatment. In this situation, the strong support from the head of state was required to convince that the new institute would become an exemplary model to be followed for succeeding ones. In 1963, Stevan Dedijer published an article titled 'Underdeveloped Science in Underdeveloped Countries' arguing the more underdeveloped a country is, the more important the head of state's interest and support for scientific advancement is.⁸ President Park's support and interest in KIST helped boost the social status of KIST researchers and give them special roles and responsibilities. This showed that the intervening and support from the head of state is essential in building a new public research system.

As the first KIST president, Choi Hyung Sup, a metallurgist who had served as the president of AERI, was appointed. Some argued that an administrative officer, instead of a scientist would a better option for the first president of the institute but the government believed that appointing a person with a science and technology background would help boost the social awareness on them and contribute to facilitating communication with the science and technology sector. Dr. Choi was a leader with extensive experiences in academia, industrial,

⁸ Stevan Dedijer, "Underdeveloped Science in Underdeveloped Countries," *Minerva* 2-1 (1963), 61-81.

government, and research sectors, and living up to expectation, he successfully led the establishment of KIST. At the early stage after KIST establishment, some in the science and technology community held a pessimistic outlook, but Dr. Choi readily communicated and presented his vision and philosophy, while proactively asking for cooperation. He pointed out the problems the science and technology sector of Korea had back then and persuaded people that new institutes like KIST were required to renew research activities and environments. This showed that appointing an accomplished science and technology leader as the first president of the institute would contribute to facilitating interaction with fellow scientists and engineers and be more advantageous in earning their support and dedication to achieving goals.⁹

In its form and nature, KIST was a non-profit independent organization, and so this became a general rule to be applied to all GRIs that have been established after KIST. Although it was an institute established and organized with government budgetary support, it was established as a private foundation to eliminate bureaucratic inefficiency in personnel management & accounting and guarantee autonomy & independency in operation,. In officially naming this organization, the term GRI (government-funded research institute) was initially coined.

To legally guarantee stable support from the government for the private institute by law, the Act on the Support of KIST was enacted. The first version of the Act instated by the government was designed to guarantee autonomy in operating the institute to the utmost, but in the process of passing the bill, a provision was added by lawmakers, which stipulated that the annual business plan of the institute shall be approved by the government and it shall be subject to an audit every year. They argued that strict management and supervision by the government was required as it was operated with large government support. However, with this supervision, it was inevitable that the institute would naturally suffer

⁹ Choi Hyung Sup served as the minister of science and technology for 7 and a half years after his term as the first KIST president. After stepping down from the ministry, he published 3 books on science and technology development strategy in developing countries and all 3 books were published in English versions: Choi Hyung Sup, *Bases for Science and Technology Promotion in Developing Countries* (1983); *Technology Development in Developing Countries* (1986); *Springboard Measures for Becoming Highly Industrialized Society* (1989).

from bureaucratic operation, not unlike other public research institutes. President Choi Hyung Sup met with the politicians and persuaded them for a revision of the Act, saying that it would be a huge obstacle to independently operating the institute. Notably, he sought support on pushing for a revision of the Act from the U.S. embassy in Seoul and USOM (U.S. Operations Mission). In the end, J. Bernstein, the director of USOM sent a letter to the Presidential Office of the ROK, stating that there was concern over the Act passed. The Korean government could not ignore the opinion of the U.S. as it was the source of the funding for the research institute establishment. After a series of arguments, the revised Act was enacted in 3 months, eliminating the provision of a business plan approval and reducing the requirement level for audits. It was rare to see a newly enacted Act to be revised in just 3 months even before being implemented. This showed the extraordinary attention and interest of the Korean government in KIST and by the same token, how important the role of the U.S. aid organization was as a buffer to protect the institute.

Since KIST was established through a joint project involving Korea, the U.S., the U.S. aid organization, the head of USOM (later USAID/ Korea), and the president of BMI, the affiliated research institute also participated in the board of directors (BOD) of KIST. They helped and sided with KIST when there was a conflict between KIST and the Korean government. The foreign board members stood by the institute whenever the central government meddled in the institute's affairs with an unreasonable request, allowing it to retain its independence without being compromised by conflicting interests with the government. The foreign directors played an essential role to protect the status of the institute and strengthen the connection between a donor and recipient countries.

KIST adopted the contract-based research system as a key basis of its operation. Under this system, the institute signed a contract with a requestor of a research project providing the research budget and submitted the results to the sponsor. At that time, Koreans were not familiar with the system. Some in the science and technology community criticized the contract-based system as being too profit-seeking and not working for the institute, only pursuing financial interests in the name of the project budget. However, the Korean government and President Choi Hyung Sup accepted the system, arguing that it could offer

an opportunity to reshape the overall research environment and practices in Korea. This was aimed to have researchers be actively engaged in practical research activities and the companies in industrial sector to adopt the results of the researches commissioned in the production field, and as a result, the belief that R&D was an essential investment for business was reinforced. Also, there was another justification that the burden on the government could be reduced if KIST became financially independent based on the research contracts.

With the assistance of the affiliated institute BMI, KIST recruited dedicated and competent researchers from home and abroad. In a meeting with the final candidates, President Choi said, “If you seek a Nobel Prize, do not join us. If you come to Korea, you can’t just study the fields you want. Instead, you have to run projects needed for Korean society. We will guarantee a stable and comfortable livelihood here and as a research director, I promise to give you as much independence as possible.” This was because KIST needed to fulfill the various requests of the industrial sectors in Korea as the institute adopted a contract-based system, which meant that individual researchers may not focus only on the fields of their personal interest. In fact, when selecting key researchers, KIST did not insist on doctorates and respected more the experiences for a certain period in industrial sectors for the requisite qualifications. It was fairly common to see cases where new research institutes of developing countries recruited researchers having doctorates of prestigious universities overseas but their research outcomes did not contribute much to the science & technology or industrial sectors of the nation because their researches did not apply well with the domestic situation. This has been called the ‘internal brain drain’ or ‘marginalization of science and technology in underdeveloped countries’. Thus, KIST’s clear goal of researches for the industrial sector was highly important since it clearly indicated the nature of research projects the researchers had to do when they returned to Korea.

For its recruited researchers, the institute provided great benefits. Senior researchers received 2-3 times higher salaries than professors at national universities. In return, all researchers at KIST were banned from doing side jobs. This was intended to ensure that they stay focused on the research activities. For those returning from overseas, air tickets,

relocation stipends and an apartment at KIST were provided. For senior-level researchers, a sabbatical year after every several years was allotted. The conditions and treatment for KIST researchers was rare to find anywhere in Korea at that time. It is hard to say definitively what constitutes a fair compensation package, since the criteria is different by country, but it is important for a new institute to provide an exemplary model capable of competing with incumbent research institutes or other researchers. That being said, despite of the risk of being criticized as preferential treatment, generous salaries and perks should be provided for researchers to ensure they are fully committed to pursuing the institute's research activities.

At that time, media outlets also paid keen attention to the researchers recruited by KIST, making it headline news in the society columns. The general public expressed their gratitude to them for giving up their stable careers and comfortable lives overseas by repatriating for the development and national modernization. There was one story of a scientist recruited by KIST in the earlier days where a taxi driver realized that he was a KIST researcher and did not receive the taxi fare to show appreciation for his work. Episodes like this showed how much the public respected and held high hopes from these researchers. The fact that KIST researchers received great perks and support from society, including the ROK president and the general public, made them feel proud and a weight of responsible altogether. The researchers worked hard and were so committed to living up to their expectations. This may come out of a pure sense of patriotism, but it would more fitting to say that they wanted to seize the opportunity to unleash their capabilities to their fullest as researchers. Providing a place where they could show their potential for those who needed their experiences and knowledge the most is something to be emphasized when establishing a new research institute or recruiting researchers in developing countries.

As KIST adopted a contract-based system, the contract amount is a clear figure to indicate how many research activities KIST has performed. Categorizing the research projects performed by KIST until the 1980s since its founding, by the source of project budget, it is grouped into either state-commissioned, industry-commissioned, or state endowment projects. From KIST's founding to around 1972, the state-commissioned projects were the

main driver, and between 1973-1978, industry-commissioned projects were the main focus, while since 1979, state endowment projects took the largest portion. This showed that during the relatively short 15-year period, there was a significant change in KIST research activities.

Dividing the contract amount signed by KIST since its founding to 1980, by the source of budget, state-commissioned projects accounted for only 14 percent. Still, state-commissioned projects were attributed to important research activities during the early stage after establishment when there were not many contracts with the industry sector. The state-commissioned projects covered diverse contents from product development for the entrusting institutes, as well as plant design and construction to research for policy development. One of the key contributions KIST made in its earlier days was the researches resulting in the founding of POSCO. KIST researchers performed an analysis on the technological and economic feasibility for POSCO as a business and formulated a report, suggesting that total annual steelmaking production capacity should be increased to 1.03 million tons, as well as investigating the status of facilities and equipment. Based on their research, foreign investment, including the funds from Korea's detailed property claims against Japan was introduced, and so the founding of POSCO was able to be completed successfully.¹⁰ It went so successfully that even the deputy prime minister of the Economic Planning Board publicly said that with the success of POSCO, investment into KIST had already reaped returns. Also, KIST played a key role in developing and executing related public policies as a thinktank in science & technology and industrial sectors. In fact, at the earlier stage of KIST's founding, those type of researches took a larger portion than development projects. KIST policy studies built key foundations for the growth of electronics, automobile, ship building and heavy machinery industries later on.¹¹

Since KIST was built to contribute to the economic growth of the nation with industrial

¹⁰ Song Sungsoo, "An Analysis on the Role of KIST for the Early Development of Iron and Steel Industry in Korea: Focusing on the Planning and Supporting of Pohang Steelworks," *Journal of Technology Innovation* 27-5 (2019), 75-98 (in Korean).

¹¹ Kwak Kiho, "Long-term Plan for Promotion of Automobile Industry and Development of Own Models in Korea: A Historical Review," *Journal of Technology Innovation* 27-5 (2019), 39-74 (in Korean); Kim Jong-Hyun and Lee Yong-Gil, "Role of the Korean Institute of Science and Technology in the Early Development Process on Korean Shipbuilding Industry," *Journal of Technology Innovation* 27-5 (2019), 99-122 (in Korean).

technology studies, the main focus had to be on the projects commissioned by the industrial sector. However, as Korean companies did not have much interest in R&D back then, KIST made diverse efforts to promote new research projects. In the beginning, most of the tasks were about providing technological guidelines to address the issues at the production site. Factories adopting production facilities and technologies from overseas experienced a series of minor or bigger issues from breakdowns, shortages of material supplies or a lack of technological competency, and KIST helped the companies experiencing difficulties by providing technology or business management guidance. There were many simple items like folding straws or desktop calculators among the new technologies KIST developed at the request of companies. This reminded us of the comment of President Choi Hyung Sup that their research should be aimed at market commercialization, not a Nobel Prize. However, responding to various requests from the industry put much pressure on the researchers.

When developing countries adopt overseas technology and succeed in localization, it is foreseeable that the company having the original technology would try to prevent commercialization of a local technology through dumping or a technology partnership with other companies. KIST also experienced such instances, and after discussion with the government, it built a system to protect domestically developed technologies. With a revision of the Technology Development Promotion Act, a certain period of protection was applied for domestically developed technologies. Thanks to the provision, PET film production technology that was developed by KIST under the commission of Sunkyung Chemical (present-day SK) at the end of 1977 was able to complete successful market commercialization with a 4-year patent protection guaranteed. As shown in this instance, besides the value of products and technologies developed by KIST R&D, it was also meaningful that KIST itself served as a catalyst to build various policies and systems to facilitate and support R&D.¹²

With the founding of KIST, the Korean government developed and executed a plan to promote the Korean science and technology sector in general. For starters, the government

¹² Moon Manyong, "KIST Model: Its Formation and Changes", *Journal of Technology Innovation* 27-5 (2019), 123-154 (in Korean).

argued that a state-run science and technology agency was required to support and manage the research institute(s) and established the Ministry of Science and Technology (MOST) in April 1967 as a ministerial-level administrative body. MOST supervised and oversaw the administration of science and technology in Korea and served as the key actor to instate and execute science and technology policies.¹³ Since establishing MOST, it became possible to recruit more systematically Korean scientist working overseas and develop various policies to support the research activities of scientists and establish an organization to introduce science and technology to the public at large, implementing various projects for science and technology advancement. At the end of 1967, a newspaper article cited the term ‘science and technology boom’, evaluating the performance of the sector that year. The science and technology boom indicated that a modern science and technology system had been built in Korea.¹⁴

In general, a science and technology system consists of S&T administrative agencies, research institutes and overall supporting system to assist scientific activities. In Korea, KIST was established in 1966 as a modern research institute, and soon, the Korean Federation of Science and Technology Societies (KOFST) was founded as an organization to lead and represent Korea’s scientists and engineers. Next year, the Science and Technology Promotion Act was enacted and the Ministry of Science and Technology (MOST) was established as an administrative body to oversee the sector.

MOST established the Korean Association for Supporting Science and Technology (present-day Korea Foundation for the Advancement of Science and Creativity) to support the livelihood of senior scientists and engineers by promoting public awareness of the science field, and it started an initiative to lure back Korean scientists and engineers working abroad back to Korea. During 1966-67, various systems and institutes to support the science and

¹³ In 2004, the minister of science and technology was promoted to be Deputy Prime Minister. Being merged with the Ministry of Education & Human Resources Development 2008, it became the Ministry of Education, Science and Technology and 2013, and later as education and science & technology were separated, it was renamed to the Ministry of Science, ICT and Future Planning to oversee science and technology policies. Since being reorganized as the Ministry of Science and ICT 2017, the name remains unchanged still.

¹⁴ Moon Manyong, “‘Science and Technology Boom’ in the 1960s: The Formation of Modern Science and Technology System in Korea,” *The Korean Journal for the History of Science* 29-1 (2007), 67-96 (in Korean).

technology sector were adopted. The structure of the modern and contemporary science and technology system built at that time has lasted to this day, and KIST was the very launch pad for all of them. The government intended to grow KIST as an exemplary institute to realize its own catchphrase ‘modernization of the homeland’, and KIST with the best facilities and its dedicated competent researchers recruited from overseas became a must-visit venue for foreign officials on state visits.

KIST, founded as a result of the U.S. proposal, set a new exemplary model for the GRI-oriented research system, and with research activities to support the industrial sector, it contributed to disseminating the value of science and technology R&D across local society. With successfully recruiting competent scientists and engineers back from overseas, it offered a solution to address the issue of Korea’s brain drain. Moreover, KIST did not stay complacent as an individual research institute, but rather it served as a catalyst to advancing the establishment of a modern science and technology framework at home. This was possible as diverse factors attributed to the various sectors working in collaboration, of the government’s commitment and investment, and the ROK president actively pushing to drive science and technology advancement in general, instead of getting complacent with the support received from the U.S. for a new research institute. Other drivers were the efforts of scientists and engineers to emphasize the necessity of a new institute and lead pro-science and technology policies by the government with the role of the press to actively support and boost awareness of the movement.

When a new institute was established and received preferential treatment, it was inevitable that it would be held in check by incumbent organizations or departments. In this situation, the ROK president served as a support and mediator for KIST and its members, including the first KIST president who did his utmost to live up to expectations and reciprocate the public support for KIST. In the process, KIST foreign directors and the U.S. aid organization sided with KIST to rebuff the Korean government’s unreasonable meddling or requests, and KIST members took full advantage of these supports to strengthen the independence and autonomy of the institute.

3. Expansion of GRIs

KIST was the first GRI in Korea and afterward, based on KIST as an exemplary model, research institutes specialized in each field were then established. The Korea Scientific & Technological Information Center (KORSTIC), starting as a foundation right next to KIST with its opening ceremony held on the same day in 1969 with KIST, became the second GRI in Korea.¹⁵ Later on, the Agency for Defense Development (ADD) to study the defense industry, the Korea Development Institute (KDI) to study national economic development plans and policies, and Korea Advance Institute of Science (KAIS), a specialized graduate school to nurture the top talents in science and engineering, were established near KIST.¹⁶ These institutes were organized in the form of foundations by law just as KIST was, and were provided with the government's support upon enacting a special Act. Accordingly, the structure of GRIs, first starting with KIST, had become an exemplary research institute model in Korea to be applied to educational and social science institutes rather than remaining as a single isolated case, and Hongneung became the first science and technology research complex (Seoul Research Park) in Korea.

As a plan to build Daedeok Science Town was implemented as a 2nd research complex in 1973, a series of GRIs were established in various fields, and they all had the same legal status as KIST. The creation of specialized GRIs was implemented in three ways in general. Firstly, as in the case of the Korea Atomic Energy Research Institute or Korea Research Institute of Geoscience and Mineral Resources, incumbent national research institutes were expanded and reorganized to establish a new institute. In this case, as the status of existing researchers were changed from public workers to private researchers, proper incentives and vision of the institutes enough to compensate the change were essential to persuade

¹⁵ KORSTIC was the former institute of KISTI (Korea Institute of Science and Technology Information).

¹⁶ KAIS was the former institute of KAIST (Korea Advanced Institute of Science and Technology). As the Ministry of Education strongly opposed to establishment of a new graduate school, the president ordered the Ministry of Science and Technology to oversee and lead the establishment of KAIS. Due to push back from the Ministry of Education, KAIS could not have the term 'university or graduate school' in its name despite being founded as a graduate school.

them. The second case was the Korea Ocean Research and Development Institute and Korea Research Institute of Ships established as an affiliated institute of KIST, and after a period of growth, they were separated into an independent institute, or as in the case of Korea Electronics Technology Institute, KIST's organization and personnel were split off to establish a separate institute. In fact, due to KIST not having a long history, KIST's BOD expressed concerns over the idea of establishing affiliates and splitting off the human resources of KIST, but they had no choice but to accept it as it was aimed at expanding the research base at a national level. Also, there was a similar case of the Korea Research Institute of Chemical Technology or Korea Institute of Machinery & Materials, in which they were established independently with no direct link with KIST. That being said, even in the last cases, as they adopted similar operation principles, researchers from KIST moved to join the management and share their experiences at KIST. While the number of research institutes increased rapidly, instead of establishing a separate pro-science and technology act for each research institute at Seoul Research Park and KIST. The Support of Specialized Research Institute Act was established based on the KIST Promotion Act through an addendum of the Act encompassing the target institute to define the government's support for new research institutes. In the process, the GRIs model was disseminated to become Korea's leading research system. This was a representative case of Korean-style science and technology growth strategy to build and replicate a success model in a short time period for quantitative expansion.

The various experiences of KIST including the way KIST selected researchers, treatment for researchers and researcher contract-based system provided highly effective guidelines for GRIs established later on. Also, as many researchers moved to other institutes, companies or universities, after accumulating experiences at KIST, they became the key leaders in Korea's science and technology sector. In the end, KIST served as a brain hub to nurture competent researchers and dispatch them to other institutes of Korea. This might seem like a brain drain for KIST, but for local society in general, it instilled a positive impact nationwide as the experiences and know-how of KIST were shared to many other institutes. In other words, KIST served as beacon tower to attract scientists and engineers from overseas and overcome the issue of brain drain in the nation.

Generally, the government plays a key role in the process of science and technology development for late-comer developing countries in catch-up, while in Korea, it was led by multiple non-profit research institutes studying industrial technologies, commissioned by the government, to drive research in the private sector. In fact, back then, establishment of GRIs was the No.1 priority in science and technology policies in Korea, and it was not an exaggeration to say that the history of science and technology policies in the 1970s can be referred to as ‘the history of GRIs establishment’.¹⁷ In the situation where industries and universities did not have strong capabilities and commitment to R&D, GRIs were expected to serve as an engine for national R&D.

Requiring a significant budget allocation to build a new institute, the idea of building a new institute, was not welcomed unconditionally by the government. Therefore, scientists and engineers had to draft a plan to explain in detail why a specialized research center for the field was needed, what role it would play, and how the institute would grow to persuade the government departments concerned. Once officials in the departments agreed with the necessity, then they had to visit the budget bureau to persuade them to provide financial resources for establishing the institute. Generally, scientists and engineers did not like to go out and meet with the government officials and discuss with them. However, when they determined that it was necessary, they would actively participate and persuade the importance of the institute or pro-science and technology policies. As they were able to be rewarded with a sense of pride that they could make a direct impact on the industries and national science & technology policies, as opposed to doing researches assigned to them at some well-established institute in an advanced country, many Korean scientists decided to give up the stable career overseas and repatriated to Korea.

The number of GRIs in the science and technology sector established back-to-back in the 1970s tallied 19 in total in the late 1980s, including one affiliated institute. However, establishing so many institutions in a short time period caused both minor and major problems. Above all, it was not easy to secure operating funds for the institutes. As the

¹⁷ Song Sungsoo, “A Study on the Characteristics of Science and Technology Policy in Korea,” *Journal of Science & Technology Studies* 2-1 (2002), 63-83 (in Korean).

government already had to spend huge amounts of money for establishment of the institutes, it was too much of a burden covering the entire research budget. Most of the institutions were built with the government subsidies and support loans, and several institutes including the Korea Research Institute of Chemical Technology were promised to get support funding from relevant industries. However, with the 2nd oil crisis starting from late 1978 and the ensuing economic slowdown, companies became hesitant to make R&D investments and the new institutes endured considerable financial difficulties.

As many departments competed with each other to found new research institutes in relevant fields, it triggered the issue that the supervisory body was divided and overall coordination and management were not enough to ensure uninterrupted establishment and operations later on. This made it more challenging to define clear roles among institutions and caused controversies as to their overlapping roles, raising the issue of integration, even before the establishment was completed. Also, as the competition to recruit researchers was intensifying, while some of the institutes were built in research or industrial complexes in remote regions with unfavorable living surroundings and conditions, so that it became more difficult to attract experienced scientists and engineers from overseas. Despite such challenges, they appointed a number of directors for each new institute and most of them had to be occupied with institution management, and so this restricted the capabilities of the competent researchers who had been recruited with much time and effort from abroad even in a situation where high-level researchers were not enough in the nation, resulting in the researchers feeling out of place.

This showed that incorporating more GRIs into each sector is a necessary policy, but in the process, a well-organized plan is required. That being said, it is important to define clear roles between new and existing institutes. As the institutes spinning off from KIST and newly established institutions are all independent entities, having the same legal status as KIST, it did cause redundant roles or unnecessary competition among them. This means that there was no system to comprehensively coordinate and manage the roles and operations of the entire institutions. There was an attempt to have comprehensive coordination in the late 1990s in establishing GRIs integrated with the BOD, but it was not easy to coordinate and

align these institutes with a history of independence operation. That being said, it is necessary to build a system or control center to efficiently oversee and manage the process from the establishment of a new institute. Also, a detailed master plan on operations after establishment is essential. At that time, as government agencies were busy competing to build new institutions, they failed to build detailed plans on human resources and research budget allocation for successfully operating these institutes. It is true there were unforeseen externally driven changes including the oil crisis and the sudden demise of President Park Chung Hee 1979, and it was also true that the counter-measures to nimbly respond to these changes were not sufficient.

After the death of President Park, a founder and strong supporter of KIST and GRIs, the new government pushed ahead with the integration and shutdown of GRIs to reorganize them into nine institutions 1980. The government wanted to improve operational efficiencies with a reshuffle, and for this purpose, supervision of GRIs was strengthened. Also, the management of these institutes was co-delegated with MOST in overseeing the science and engineering institutes and the Ministry of Education in overseeing educational institutes. The means of funding support also changed from lump-sum amount to itemizing basic operational expenses for disbursement, including labor costs based on the headcount of researchers. It was similar to operations of a state-run institute rather than an independent one. Instead, the government tried to have GRIs at the center as a key organization to implement new national R&D programs.

MOST began specialized R&D programs from 1982, as it shifted from the approach of establishing and nurturing individual institutes in the 1970s to that of the government providing direct funding for specific projects. In other words, expanding the mechanism for the government to set the research budget allocation and individual research institutes developed a plan to earn and allocate the funding support according to the Support of Specific Research Institutes Act, it was designed to expand the coverage of funding to the industry sector and academia to develop core industrial technologies. With this, the aim was to expand and grow the overall national research system, beyond GRIs, to include universities and companies. At the early stage of specialized R&D programs, with an aim to

improve productivity and overcome export limitations through industrial advancement, it was divided into national R&D projects with full-scale subsidies of the government and projects with co-funding support between government and companies. The state-run national R&D projects were mostly running by GRIs and most of them were mid- to long-term large-scale projects of a public nature and fully funded by the government, including those for public welfare or future-oriented advanced technologies as they were too large in scale to be run by the private sector. Meanwhile, for the public-private joint projects, funding support was shared by the government and private sector to develop core industrial technologies that could not be developed entirely by the private sector. Under the supervision of GRIs, private companies participated to focus on quality and productivity improvement in the industrial field and new product development.

Throughout the 1980s, national R&D programs became diversified as other state-run ministries, including the Ministry of Trade and Industry (MITI) and Ministry of Energy and Resources, realized the value of state-run national R&D programs with implementing specialized R&D programs. For national R&D programs, it was natural that GRIs served as the overseeing body as the interests of private companies were intertwined. Being in unstable conditions after the integration and shutdown in the 1980s, GRIs were able to redefine their roles with each specialized R&D program. For example, the Electronics and Telecommunications Research Institute (ETRI), growing its scale in the integration and shutdown process, guided the nation to the ranks of an IT powerhouse through developing electronic switching system, computer, semiconductor, and telecommunication technologies.

Although the number of GRIs declined in the integration and shutdown in the 1980s, new GRIs were continuously founded. MOST established new institutes in the genetic engineering field or aerospace, and industrial ministries also established institutes for relevant technologies, including the Korea Food Research Institute or Korea Institute of Civil Engineering and Building Technology. Notably, as with the goal of adopting field-oriented production technology development and strengthening the global competitiveness of industries, MITI established the Korea Institute of Industrial Technology (KITECH) in 1989,

and there was competition between MOST and MITI for R&D planning and management of the institutes.¹⁸ In the end, the roles were co-delegated with MOST in overseeing large-scale basic science and MITI in focusing on field-oriented industrial technology agenda.

In the mid 1980s, as a new type of GRIs emerged, the function and mission of GRIs began to be further expanded from economic development through industrial technology development. For starters, a plan to found basic science institutes was adopted and implemented, including the Korea Basic Science Institute and Korea Astronomy and Space Science Institute. Also, each state-run ministry established policy institutes to oversee research planning, management, and evaluation. With the Economic Planning Board creating KDI, MITI established the Korea Institute for Industrial Economics & Trade and the Ministry of Communication established the Korea Information Society Development Institute as a for the ministerial thinktank, respectively. MOST founded a technological development evaluation center at KAIST in 1984 and expanded it into KAIST Center for Science-Technology Policy. The center was in charge of policy study & research planning and assessment for technology sectors and now is reorganized into the Science and Technology Policy Institute (STEPI).

However, as Korean companies' R&D investment significantly increased and internal R&D capabilities strengthened, the issue of renewing the roles and mission of GRIs to be differentiated from the private sector then emerged. There was criticism that GRIs did not present research productivities, living up to the scale of investment they received since they mostly pursued quantitative growth until then. With this, the government conducted the 2nd round of restructuring for GRIs in 1991, but did not achieve the expected goal of redefining new roles and improving the efficiency of the institutes since it focused on reducing the number of institutes just as it did in the first round of restructuring. This showed it is not easy to restructure and reorganize incumbent institutes with their own independent history and legacy during integration and shutdown.

Thus, the government adopted a project-based system (PBS) for financial support to GRIs to

18 Joel R. Campbell, *The Technology Policy of the Korean State Since 1961* (The Edwin Mellen Press, 2009), 110-111.

reflect the overall project cost in a research budget. PBS is a system to align research projects to budget flow and subsidies by each project, setting the base price based on the total cost allocated into direct and indirect cost, including labor costs for each research project. The original goal of adopting PBS was to improve the efficiency and competitiveness of GRIs by adopting ‘competition’, but researchers in the field argued that the competition to earn projects to justify labor costs negatively affected the quality of researches, and until now, PBS has stirred up the main controversy among the policies on GRIs. Well-aware of the issues with PBS, the government has been reducing the size of PBS gradually while expanding block funding to be executed independently by the institutes.

In the wake of the financial crisis of the late 1990s hitting the nation hard, Korean society implemented corporate restructuring, while science and technology policies also sought a new turnaround to pursue a new goal of advancing a national innovation system based on a stable knowledge-based economy. Later on, instead of separate BODs for each GRI, an integrated board was established to ensure comprehensive coordination and cooperation among GRIs. This policy was drafted based on the understanding that the GRI policy had been allowing for competitive expansion while it pursued quantitative growth as a main priority, and individual institutions established as a result of the competition experienced unnecessary competition with the lack of integrated coordination, during which they each grew into their own independent entities. Boards for GRIs in the science and technology field had been divided into the Basic Technology Research Council, Public Technology Research Council and Industrial Technology Research Council, but they were integrated into the National Research Council of Science & Technology 2014 overseeing 25 research institutes altogether.

Currently, GRIs covers a large spectrum from certain institutes conducting traditional industrial technology researches including machine, chemical, and electronic communication, institutes researching areas of a public nature, including nuclear fusion, power, and standardization, etc. to those of a basic research nature, including the Korea Institute for Advanced Study, Korea Basic Science Institute or Korea Astronomy, and Space Science Institute. Within an individual institute, they conduct various functions including

support for the industrial sector to long-term basic research. In some cases, a researcher works on multiple projects with a different nature. This is because the legacy of R&D history in diverse fields and sectors covered by GRIs still carries on.

The controversies over the role of GRIs is still ongoing, but the basic goal of creating new knowledge at a national level to be effectively disseminated has not changed. However, moving away from its original goal to seek individual technological development for national industrialization and responding to the changing environment, GRIs is now advancing toward a new goal of serving as a science and technology hub to meet Korea's future demand.¹⁹ For a national innovation system, a platform of knowledge where diverse innovation players share and exchange knowledge and information is necessary, and an entity with the status and capabilities to coordinate and align diverse players should play a role to serve as a national innovation hub. That being said, GRIs are best-suited to take the role of national knowledge hub. With its role as a platform to share and exchange knowledge and information, GRIs should be able to understand the nature, needs, and shortage of each field, and the areas led by GRIs should also be defined to coordinate and align the functionality and roles of other R&D players. This means GRIs should be able to play diverse roles including basic research, applied research, support for the industries, and infrastructure management, and rather than sticking to the designated roles and responsibilities, GRIs should be flexible enough to adopt new assignments, depending on the field and circumstances. Since it emerged first in the 1960s, GRIs have been readjusting and realigning their new roles to fulfill the mission of developing facets of Korean society, and continues to make efforts in earnest. As Korea's leading public research system, allocated 40% of national R&D funds, GRIs need to assess their roles and should never stop striving to meet the needs of society in general.

¹⁹ OECD, *Public Research Institutions: Mapping Sector Trends* (OECD Publishing, 2011), 61.

4.

Construction and Growth of Science and Technology Complex

The idea was raised in 1965 to put research institutes and universities together in one location to share research facilities and exchange human resources and information to improve research efficiency and conduct comprehensive researches on relevant agendas. To Dr. Hornig, visiting Korea for a feasibility study on research institute establishment discussed at the Korea-U.S. Summit, the Korean government presented a science and technology development plan to build a new institute near Gongneung-dong where the College of Engineering of Seoul National University and Atomic Energy Research Institute (AERI) were located to develop the site into a 'science & technology town'. The plan was not realized as Hongneung was selected for the site for KIST, but because the Korea Institute of Science and Technology Information Center (KORSTIC), Agency for Defense Development (ADD), Korea Advanced Institute of Science (KAIS) and Korea Development Institute (KDI) were located near KIST, Hongneung naturally became the first R&D complex in Korea. The institutes at the Hongneung site known as 'Seoul Research Park', covered diverse areas of science & technology, economics, and national defense, while claiming the prestige of being the first research complex in Korea where the best and brightest of the nation gathered together to serve as a national state-run thinktank.

As it was not aimed to build a complex from the beginning, the Seoul R&D Complex had restrictions in space expansion. Thus, MOST planned to build a new research and academic town in non-metropolitan areas. After Dr. Choi Hyung Sup, the first KIST president was appointed as the 2nd minister of Science and Technology in June 1971, he began to work in earnest on constructing a complex. During his tenure of 7 and a half years, Minister Choi built a systemized procedure on science and technology policy and developed an R&D framework and construction of the complex was one of the key policies he pursued. After several months to review and understand the works of MOST after appointment as the minister, Minister Choi set the three policy goals for MOST, namely building and strengthening a science and technology base, strategic development of industrial

technologies, and creating scientific and technological climate. During his entire tenure serving as the minister, he consistently emphasized these three goals.²⁰ Since new research institutes in each field of industrial technology would be established in the new research and academic town, construction of the town was the key agenda for the science and technology groundwork and strategic development of industrial technologies.

In 1973, the government decided to build a research and academic town under a national project. Daedeok in Chungnam Province (currently Youseong-gu, Daejeon) strategically located at the center of the country, was selected as the site for the complex, and it was decided that MOST would lead the planning and construction of the complex in cooperation with the Ministry of Construction and Ministry of Internal Affairs. In fact, considering the economic conditions of Korea at that time, construction of a research complex was not an urgent priority. Ministries other than MOST were not eager to proceed with constructing a complex, but Minister Choi directly reported to President Park and received permission before persuading relevant agencies and the National Assembly. With the concentration of population in metropolitan area being a social issue by that time, it was also emphasized that construction of a research complex in a non-metropolitan area would better allocate and dispatch science and technology talents to other regions. According to the Daedeok Science Town Basic plan finalized in late 1973, it was originally planned to establish five new institutes, namely the Korea Research Institute of Ships, Korea Ocean Research & Development Institute, Korea Institute of Machinery & Materials, Electronics and Telecommunications Research Institute, and Korea Research Institute of Chemical Technology with the aim of actively supporting the six key strategic industries in the heavy industry development plan announced by the government. Also, with relocating 12 national institutes, the plan was to establish or relocate one science & engineering university, build common facilities, establish industrial research institutes, and build self-sufficient living environments in the town to accommodate about 50,000 residents.

Although it was reported that the institutes in the Daedeok Complex were founded to

20 Moon Manyong and Kang Mi Hwa, "The Institutions Builders of Science and Technology during the Park Chung Hee Era: Choi Hyung Sup and O Won-Chol," *The Korean Journal for the History of Science* 35-1 (2013), 225-244 (in Korean).

support heavy industry development, in early 1973, R&D or research institute construction was not considered from the beginning at the time of the government announcement of a heavy industry development plan. In fact, when the plan for the Daedeok Complex was discussed, relevant agencies, including MITI, were opposed to the idea of building new institutes when the demand for KIST was still low. However, scientists and engineers at KIST persuaded the state-run ministries with a report on the necessity for specialized research institutes and a establishment & operation plan. Notably, as the head of MOST in charge of science and technology administration, Minister Choi firmly argued that establishment of a research complex and GRIs in each field was essential to building the science and technology groundwork and seeking the strategic development of industrial advancement and directly persuaded President Park to proceed with the plan.

As the original name of 'Daedeok Research & University City' implies, the plan was to build a town centered on science and technology research and education with research institutes and universities at the center, while a production base was not taken into consideration. After the project was launched, in the aftermath of the first oil shock, it became difficult to execute the investment planned into construction of the complex due to the economic slowdown. In April, 1976, the 'Daedeok Research & University City Plan' was changed into the 'Daedeok Research Complex Construction Plan' with some overall modifications. While the plan to build a town was suspended, the original plan of 'research and university cities' was reduced to a 'research complex' and plans for the construction of several research institutes were revised after the decision to build institutes within the allocated budget in stages.

Construction of Daedeok Science Town started in 1974, but due to modification in basic plans and delays in construction, relocation of institutions began as late as 1978. Starting with relocation of the Korea Research Institute of Standards and Science in 1978, many GRIs moved over and many corporate affiliated institutes also settled in Daedeok, starting from the Ssangyong Research Institute in 1979. For higher education institutions, Chungnam National University moved over in 1978 and the Korea Institute of Technology (KIT) was established in 1984. KIT was later integrated with KAIST relocated from Seoul in 1990 to be

reorganized into the KAIST undergraduate school.

However, with the completion of Daedeok Science Town being delayed from the original plan and living environments not established with the suspension of downtown urban development, those entering the complex had to experience much inconveniences. Thus, various measures were taken to promote construction and facilitation of the complex. In a policy study in 1984, it was suggested that building a small-scale new business town, focusing on advanced technologies near Daedeok Complex to nurture and support the technology-based new start-ups, creating business with research results to support the research activities of institutes and the academia and contribute to advancing the industrial structure as well. This means the nature of the complex, beyond the production base, needed to be changed. In fact, there was criticism that Daedeok Science Town was essentially an isolated island, away from other sectors of society with only universities and research institutes in place. As the National Science Museum was relocated from Seoul to Daedeok in 1990, it built a contact point between Daedeok Science Town and the public at large. The museum was expected to narrow the distance with the public and disseminate science culture, providing an exhibition on research achievements of the complex.

Throughout the 1980s, the complex's construction plan continued to be partially modified to expand research and education facilities, and notably, in 1989, a development plan was adopted to increase accommodation capacity of the complex which was selected as the 1993 EXPO site. As a result, the number of institutes in the complex increased from 50 to 60, and the accommodation capacity increased from 50,000 to 70,000 people while the development period was expanded for an additional 3 years from the original period from 1981 thru 1990 until 1993. The opening ceremony of Daedeok Science Town was held on November 27, 1992, putting an end to the long journey starting from 1974. At that time, 33 institutions stayed or finished their relocation to the complex, including three government institutes, 15 GRIs, eight private research institutes and three higher educational organizations.

With the steadily increasing number of institutes entering the complex after completion, it is now called the 'mecca' of Korea's national science and technology. In 2000, the

government renamed it to Daedeok Valley, developing it into a complex boasting the convergence of industry, academy, and research. Modeled and named after Silicon Valley, Daedeok Valley strove to be a complex for education, research, and venture start-ups. Daedeok Science Town was situated at the center of Daedeok Valley, and the site was expanded with formation of Daedeok Techno Valley. In 2004, commemorating the 30th anniversary of construction of Daedeok Science Town, a plan was adopted to develop it into the Daedeok R&D Special Zone. With announcement of the ‘Special Act on Nurturing the Daedeok R&D Special Zone’ in 2005, the government planned to develop this R&D special zone into a cluster of advanced science and industries by developing a comprehensive plan to cultivate the R&D special zone, establishing an R&D special zone committee (chaired by the deputy prime minister of Science and Technology), allowing establishment of lab-based companies, providing preferential treatment for companies with advanced technologies or foreign investment, executing R&D-specific projects, and establishing a support HQ for the Daedeok R&D Special Zone. The R&D special zone was designated as a district created for developing new technology with R&D, expanding dissemination of R&D results, and facilitating business creation. Besides the Daedeok Research Complex (27.8k m²), there was an expansion of 67.8k m² in total area, including Daedeok Techno Valley, the Daejeon Industrial Complex 3 & 4, northern green belt, and part of the Agency for Defense Development. As of 2017, there were 26 GRIs, seven educational organizations, three national research institutes, nine other research institutes, ten government agencies and 14 public institutions at the special R&D zone, turning it into the largest national science and industrial complex. Since the construction completion of Daedeok Science Town, the site has been continuously expanding and the overall nature of the complex and supporting measures also have been evolving.

With the consensus that Daedeok R&D Special Zone has been well-established, the government appointed an additional four sites in Gwangju, Daegu, Busan, and Jeonbuk to roll out the Daedeok model based on its experience and increase R&D and industrialization capacity with advanced technologies nationwide. Most recently, the government designated a small but strong specialized zone, rather than an R&D special zone to designate and nurture them into R&D zones on a smaller scale with high density, featuring the major

hub institutes with core technologies at the center, including universities, research institutes, or public corporations.

Ultimately, Daedeok Science Town has become a role model for an R&D cluster of Korea as KIST became a model for national GRIs. There were people expressing skeptical views in the process, but with MOST at the center, scientists and engineers persuaded the necessity of a research complex and research institutes and foremost, the minister of MOST earned the permission of the ROK president to accelerate the procedures. The original goal was more skewed to an ideal target of having solely research institutes and educational organizations, but in the process of actual construction and operation, the goal was readjusted to allow production facilities. By the same token, the government has been providing continued support by renewing relevant systems and expanding overall space to ensure the Daedeok Complex to serve as the leading science and industry complex in Korea. As a result, Daedeok Science Town was able to fulfill its mandated roles and set purpose as a provider of science and technology knowledge at a national level for creation and dissemination. Notably, in the national strategic industries like information industry and bio engineering, as well as in the area of large-scale comprehensive technology, such as nuclear power and aerospace, the complex made a major contribution to building a national base for creating and disseminating core technologies.

5. Characteristics and key implications

In the compressed growth of science and technology in Korea, the public research system represented by GRIs has played a key role. Starting from KIST, GRIs worked hard to live up to its goal of contributing to economic growth with science and technology researches, and played a leading role to facilitate quantitative growth in national science and technology competency. It also generated pioneering research achievements including an electronic switching system TDX, telecommunication CDMA commercialization, Korea-style nuclear powerplant development technologies, etc. Starting with the goal of building a research and education town, Daedeok Science Town was constructed as a research complex led by GRIs, then transformed, and is still evolving as an innovation cluster connected with diverse industrial functions, encompassing R&D, production, and commercialization. Based on its history, we would like to review the insights from the history of Korea's public research system by the category of KIST's founding, expansion of GRIs, and formation of a national research complex.

Now, we would like to look into the insights of KIST's founding in terms of groundwork and research resources building. Firstly, it is important to note that KIST became the starting point to build science and technology research groundwork, policy and support system in a series and in the process, it served as a catalyst. With establishment of KIST, a chain reaction was created in science and technology policies to introduce many systems and policies. KIST's founding initiated with an unexpected proposal by the U.S. president, and the ensuing KIST project was aimed to build a research institution but the Korean government took the opportunity to pursue comprehensive science and technology development and build relevant systems and organizations. That being said, the government's strong commitment, including the president, was most important. Although the support from the U.S. stopped at the establishment of KIST, the Korean government determined that advancement of science and technology was required for economic growth and the national goal of 'national modernization' and proceeded with to build legal grounds, various institutes including the government agencies and a research town to realize this goal. Instead of sitting complacent with the support from the advanced nation, the government showed strong commitment

and made investments to build an overall system to drive science and technology advancement.

Secondly, recruiting competent talents and providing satisfactory benefits were the second factor. The key to determine the success of research institutes are research activities and the 'researchers' who are going to lead the activities. Recruiting the talents suitable for set objectives and agenda is the utmost important factor in deciding the future of the institute. KIST strove hard to recruit researchers in specific fields of high demand in Korean society, as opposed to looking for higher level degree holders. Given that 'internal brain drains' is a common occurrence where developing nations recruit researchers mostly for their high degrees and end up with a situation where their research activities do not match with the country-specific circumstances and needs, the choice of KIST was very effective. Also, it was equally important to provide great benefits for the recruited researchers, even to the point that it looked like preferential treatment. As a new research institute becomes a model for other institutes to follow and repeat, it is necessary to boost positive awareness of science and technology as well as scientists and engineers by guaranteeing enough benefits for the researchers. By the same token, it is necessary to define key roles for them to unleash their potential to the fullest and get recognition on the value of their activities. It is likely to see conflicts and differing opinions in the process, so the role of the head of the state as a mediator is also important. The preferential treatment would not only help the researchers feel proud but also imposes a weight of responsibility on them to keep working harder to keep achieving.

Next, we would like to review the characteristics and insights drawn in the process of expanding GRIs. Firstly, a comprehensive system was built to coordinate the roles of various institutes aligned with mid- and long-term policies and social needs. It is natural that research institutes are expanding as the society evolves. However, GRIs growing exponentially in the 1970s were built as an independent entity without being associated with the original model, KIST, or other institutions. This led to unnecessary competition for limited resources and human capital, and caused redundancies in research activities. Later on, restructuring occurred in the name of convergence and shutdown, and only in the late

1990s it was possible to have comprehensive coordination on operations, and researches of these institutes organized as an integrated board. That being said, while it is important to guarantee autonomous operation of each institute, it is also essential to build a system to coordinate their roles in a bigger picture.

The second factor is changes and adaptation in research activities. Ultimately, ‘research activities’ are the final grounds to evaluate the achievements. Even if all the others are well-established, we cannot consider an institute as successful without major research outcomes. The key focus of KIST, the first GRIs in Korea has been changing since right after its founding. KIST has been working as a multiplayer conducting various research activities for the first 15 years, serving as a ‘thinktank’ to help the government’s policy building and execution, while meeting diverse technological development needs of companies and establishing mid- to long-term national initiatives to meet national needs. The specialized institutes established in the 1970s also conducted research activities to meet the needs of the business sector, and after the 1980s, they turned to drive a larger-scale research projects as the research capabilities of companies and universities improved in the private sector. Also, as the capabilities of companies and universities were strengthened, GRIs are making continued efforts to diversify their roles, seeking differentiation from private sector institutes. In the 1970s, the research departments of KIST mostly covered industrial technologies, including electronics, machine, food, chemical engineering, and computing, but now they are transformed, expanding into brain science, next-generation semiconductor, biomedical convergence, and national backbone technology research. Responding to the dynamic development of the society, GRIs also need to keep evolving and adapting. In terms of public policy, it is also essential to stay flexible enough to recognize diversely evolving roles, reflecting social changes rather than setting just one objective or agenda. It is true that changing roles would put a significant burden on the members of institutes, but that is the responsibility of GRIs in charge of leading national R&D activities.

Lastly, the insights we could learn from the history of the national research complex would be that long-term investment and sufficient support is essential for establishing such a complex. It took 20 years to complete Daedeok Science Town, dubbed the ‘mecca’ of

national science and technology. It became a long-term project as the construction plan had to be modified, responding to the changes in political and economic environments. But essentially, significant investment of funds needs to be made to build an infrastructure for research complex and invite institutes and organizations to join the complex. Thus, it is essential to have a mid- to long-term investment plan and continued support.

Secondly, contact points with various fields of society needs to be built to prevent the research complex from being an isolated island onto itself. This means that it is important to have a marketing unit to make sure that research results lead to production and new products are effectively delivered to society, as well as the unit or channel to transfer the research results to society. This is essential to ensure the sustainability of the complex, which in this respect, Daedeok Science Town is in fact relatively weak.



CHAPTER 4

Public STI Research Institutes in Lao PDR





Chapter 4. Public STI Research Institutes in Lao PDR

1. Overview and Background

The Lao PDR has recognized that STI capacity building is essential in establishing country's socio-economic development. It has been a policy basis for the establishment of Lao PDR-based public STI research institutes and the Lao government has tried to improve the country's STI research capacity as it specified STI initiatives in the national development plan.

According to the '8th National Socio-Economic Development Plan (NSED), 2016–2020' prepared in cooperation with UNDP, the Lao government set the pursuit of local innovation and utilization of science and technology as one of the priority outputs. The plan emphasized the development of science and technology as a key driving force for rapid and stable economic growth, and thus strengthening STI for industrialization and modernization is a crucial path to promote the national economy.

In order to provide a legal framework for promoting STI, six pieces of legislation and two decrees have been developed and amended as the legal basis for science and technology, including the Law on Science and Technology, Law on Biotechnology Safety, Law on Electronic Transactions, Amended Law on Measurement, Amended Law on Intellectual Property Rights, Law on Standards, Decree on the Development of Science and Technology, and Decree on Enforcement of the Law on Standards. This is a clear indication of policy support for STI capacity building (MPI, 2016).

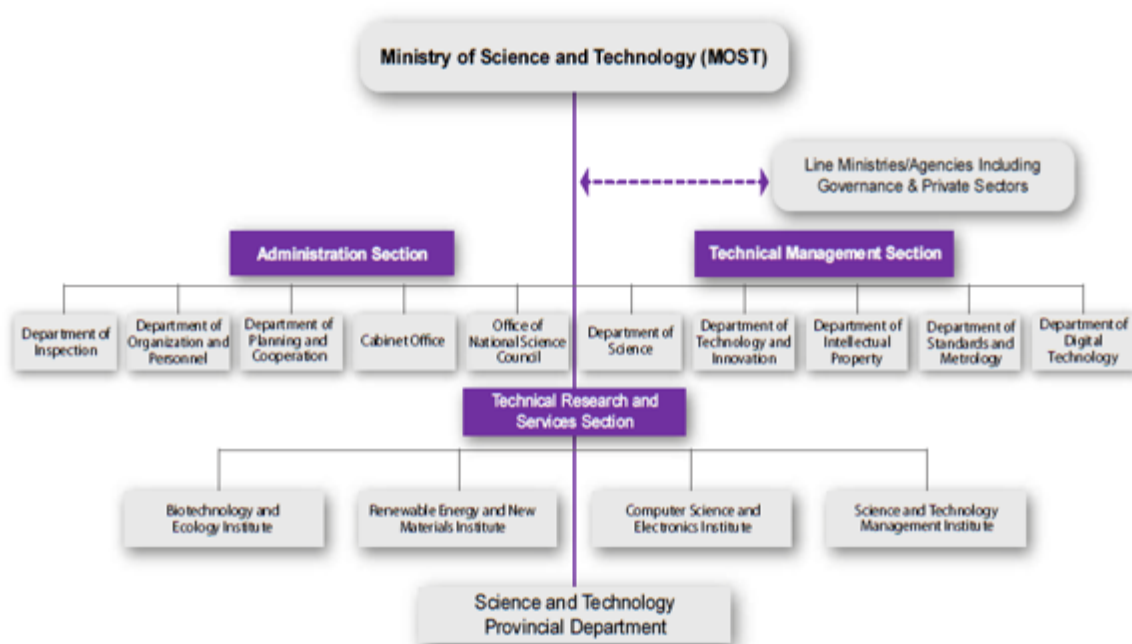
The National Science Council (NSC) was established in 2002 to promote the development of scientific research activities in Lao PDR. The Council is under the Prime Minister's Office and responsible for academic and professional organizations in science and technology.

Therefore, all public science, engineering and technology research institutes come under the NSC's umbrella. The main activity of NSC is to manage the Science Development Fund for which the government budgeted annually 1 percent of total public investment. NSC also provides advice on research activities and certifies the research results carried out nationwide.

In terms of organizational aspect implementing the national plan, the Ministry of Science and Technology (MOST) plays a central role in coordinating various STI-related public research institutes. Most of the Lao PDR's public research institutes are under the relative ministries, each of whom has roles and functions pertinent to its own characteristics. Through the Sectoral Science Council which is a technical organization convening representatives of STI-related ministries, agencies, and authorities, MOST can assess and evaluate the achievement of supported research and analyze the developing applications of technology with a view to ensuring consistency with the policy objectives, strategic goals, national social-economic development plan, and sectoral development plans.

While the research institutes on a central government level seem to be well-structured based on their roles and specialty, the Provincial Department for Science and Technology (PDST) in each province is the sole government institution which is responsible for locally directed STI research activities and policy. PDST acts as a secretary to provincial governor and manages STI issues such as intellectual property, standardization and metrology in the province. Figure 1 shows the organizational structure of MOST, including PDST.

[Figure 4-1] Organizational chart of MOST



Source: Mapping research and development in Lao PDR (UNESCO, 2017)

In March 2021, there was an important reorganization in the Lao PDR's government structure. The Lao government announced that it would dissolve the Ministry of Science and Technology. According to its announcement, the dissolution has been implemented for consolidation purposes and to ensure each ministry can efficiently deal with science and technology matters pertaining to its core purpose. The departments under the ministry will be transferred, along with all civil service workers, to five departments under corresponding ministries, including the Ministry of Education and Sports, Ministry of Post, Telecom and Communications, Ministry of Industry and Commerce, Ministry of Agriculture and Forestry, and Ministry of Energy and Mines.²¹

The dissolution of MOST would lead to organizational restructuring for research institutes as well, which were previously under the ministry. More importantly, the STI policy decision-

²¹ Laotian Times, March 1, 2021. <https://Laotimes.com/2021/03/01/government-of-Lao-PDR-dissolves-ministry-of-science-and-technology/>

making process would be rearranged since the new ministry shall assume the role previously done by MOST. For example, NSC is now under the Ministry of Education and Sports according to Mr. Soumana Chulamany, Director General of the Department of Technology and Innovation in the Ministry of Technology and Communications, which has been renamed after the transfer of department from MOST. From the dissolution, we can expect more decisions on organizational restructuring in STI institutes with a further announcement made from the Lao government.

Based on the above background for Lao PDR-based public STI policies, we examine the following subjects to assess the research capacity of Lao PDR-based public STI research institutes.

- List and information of public STI research institutes
- STI policy and governance environment
- Role of STI on economic development of Lao PDR
- Conclusions and Implications

2.

List and information of Public STI Research Institutes

Following the government's interest in developing STI for country's social and economic development, the government of Lao PDR has founded various research institutes which would enhance the R&D capacity in STI for the country.

Table 4-1 shows the lists of public STI research institutes in Lao PDR with affiliated ministry. In terms of institutional and legal position, they are all public institutes under the ministry of central government and located in Vientiane Capital. Many research institutes carry the role of training as well as research in their fields.

<Table 4-1> Lists of public STI research institutes in Lao PDR

Name of Institute	Affiliated Ministry	Organization Website
Biotechnology and Ecology Institute	Ministry of Agriculture and Forestry	N/A
Computer Science and Electronic Institute	Ministry of Technology and Communication	http://tcei.most.gov.la
Council for Science and Technology	Ministry of Agriculture and Forestry	http://cst-maf.la
Institute of Information and Communication Technology	Ministry of Technology and Communication	N/A
Institute of Renewable Energy Promotion	Ministry of Energy and Mines	N/A
Institute of Science and Technology Management	Ministry of Education and Sports	N/A
Lao Tropical Medicine and Public Health Institute	Ministry of Health	www.nioph.gov.la
National Agriculture and Forestry Research Institute	Ministry of Agriculture and Forestry	www.nafri.org.la
National Economic Research Institute	Ministry of Planning and Investment	www.neri.gov.la

Name of Institute	Affiliated Ministry	Organization Website
National Institute for Economic Research	N/A	N/A
Natural Resources and Environment Research Institute	Ministry of Natural Resources and Environment	http://nrei.monre.gov.la
Pasteur Institute of Lao PDR	Ministry of Health	www.pasteur.la
Public Works and Transport Research Institute	Ministry of Public Work and Transport	N/A
Renewable Energy and New Materials Institute	Ministry of Energy and Mines	N/A
Research Institute for Educational Sciences	Ministry of Education and Sports	www.moe.gov.la/ries
Research Institute for Energy and Mines	Ministry of Energy and Mines	N/A

Much of the information on institutes are extracted from a UNESCO report²². However, some information was already outdated and hence revised with new information. The information on each institute are as follows:

■ Biotechnology and Ecology Institute

The Biotechnology and Ecology Institute (BEI) was formed in 2011. Its former name was the Science and Technology Research Institute (STRI), and it was under National Authority for Science and Technology (NAST). When NAST was transformed into MOST, STRI became BEI in 2011. Since the recent dissolution of MOST, the institute has been reshuffled into the Ministry of Agriculture and Forestry.

BEI has played its role in research, development, application, and service in biotechnology and ecology. BEI consists of six divisions and one center, namely the Division of General

²² Except for Institute of Information and Communication Technology, Institute of Science and Technology Management, National Institute for Economic Research, Research Institute for Energy and Mines, information on institutes was extracted from Mapping research and development in Lao PDR (UNESCO, 2018).

Affairs, Division of Biotechnology, Division of Genetic Resources, Division of Ecology, Division of Chemical Technology, Division of Technical Service, and the Parksan Ecology Center.

BEI has established collections of orchid flowers consisting of more than 140 species and has more than 165 other species in its glasshouse. BEI was set up as an herbarium and natural history museum, and operates to disseminate and raise awareness of science and ecological systems. BEI carried out several research projects such as studies on species of *Delbergia*, Bryophytes, Zingiberaceae, Begoniaceae, Areceae, and Ebenaceae in different provinces. BEI also conducted some research activities to apply biotechnology to identify plant and animal diseases.

■ Computer Science and Electronic Institute

The Computer Science and Electronic Institute (CSEI) was established in 2011 by transforming the Institute of Information Technology which had been created in 2008 under the NAST. The total number of staffs as of 2018 were 34 persons.

CSEI acts as the secretariat to MOST in research, development software, hardware, network, automation, transfer and providing technical services related to computer science and electronics. CSEI consists of five divisions and one center: The Administration Division, the Software R&D Division, the Computer and Electronics Division, the Network Division, the Automation Division and the Transfer and Service Center.

CSEI sets up and maintains the network system of MOST, develops training materials and training courses on IT and its applications. CSEI conducts range of research and development projects such as development of Lao Fonts, E-book and E-dictionary. It also provides training manuals for Microsoft Office written in Lao language to provincial S&T Departments and participated in the Network-based ASEAN Languages Translation Public Service Project.

After the dissolution of MOST, CSEI has moved to MOTC along with the Department of Technology and Innovation.

■ Council for Science and Technology

The Council for Sciences and Technology (CST) was established in March 2003. Initially the secretariat team had been functioning as a division under the permanent secretary and was chaired by the Minister of Agriculture and Forestry (MAF). In 2013, the secretariat unit became equivalent to a department, managed independently and chaired by the Minister. The total number of staffs as of 2018 were 25 persons.

CST has the main role to formulate the policies and regulations relating to promote local brand of agriculture products and to develop the agriculture technology and other. CST is different from the NSC which is responsible for academic and professional organizations in science and technology of the whole country, but the similarity of their names can create confusion.

The core mandate of CST is to facilitate science and technology research and innovation in the MAF. CST also contributes to the achievement of the MAF research outcomes.

■ Institute of Information and Communication Technology

The Institute of Information and Communication Technology (IICT) was founded with support from UNDP as a training institute of telecommunication in 1970 with a name as Centre Nationale des Formations des Postes et des Telecommunications (CNFPT). In 2011, the name was changed to Institute of Posts and Telecommunications (IPT) and then it was changed again as IICT in 2017.

Initially, its main role was for developing human resource in the area of posts and ICT. Later, research for posts and ICT infrastructure development was added to its mandate. As for training program, it provides 3-year diploma program and short training program in ICT area. Through cooperation agreement, many lecturers have training experience from overseas ICT related institution in Korea, Japan and India, etc. so that they can transfer their knowledge acquired from overseas training to students.

As for R&D program, IICT takes part in quality assurance of ICT, development of ministry's

information system, and formulation of regulatory framework and establishment of management mechanism for ICT development.

■ Institute of Renewable Energy Promotion

The Institute of Renewable Energy Promotion (IREP) has role to promote, manage, monitor, inspect, develop and provide technical service on renewable energy, energy efficiency, rural electrification, alternative energy and manage the energy promotion fund in the country. The total number of staffs as of 2018 were 34 persons.

IREP consists of six divisions and one office as follows: The Executive Planning Division, the Division of Wind, Solar and Small Hydro Power Promotion, the Division of Energy Efficiency and Conservation Promotion, the Division of Alternative Energy “Hydrogen, Methanization and Nuclear” promotion, the Bio-Energy Promotion Division, the Rural Electrification Promotion Division, and the Energy Management Fund Office.

Its main activities are to (a) develop policy, strategy, planning and action plan on renewable energy, alternative energy, energy efficiency, rural electrification and energy promotion fund; (b) promote application of renewable energy to contribute to poverty eradication and sectoral development; (c) develop necessary legislation, regulation and mechanism for macro management and promotion of renewable energy and (d) carry out other duties as mentioned in the decree.

■ Institute of Science and Technology Management

The Institute of Science and Technology Management (ISTM) was established by the Prime Ministerial decree no. 319 in 2017. It has the responsibility to research, develop curriculum, teaching, train, and provide high quality and effective information and technology services to improve staff capacity in civil service for science and technology. ISTM consists of four divisions and one center: The Administration Division, the Academic Division, the Training Division, the Cooperation Division, and the Information and Library Center.

Previously, ISTM was supported by Vietnamese government with initial developing infrastructure such as buildings and facilities, a number of training courses and some trainers, and a number of equipment. In 2019, Korea International Cooperation Agency (KOICA) provided the advisor for technology management to help develop science and technology management policies, curriculum design, training and research about training systems and video conference.

Since the dissolution of MOST, ISTM has moved to become a part of Ministry of Education and Sports along with the Office of the Council of Science. The Ministry decides to change the name of ISTM to the Scientific Research Institute as well as its role. The decision is under review for the final approval.

■ Lao Tropical Medicine and Public Health Institute

The roles and functions of the Lao Tropical and Public Health Institute (Lao TPHI) were approved by Minister of Health (MOH) agreement no. 2979 in 2017. Previously, the National Institute of Public Health (NIOPH) had assumed the role which acted as secretariat for the Council of Medical Sciences of MOH before Lao TPHI was created. The total number of staffs as of 2018 were 51 persons.

Its mandates include: (a) serving as a central focal point for promoting, supporting, consulting, advising, coordinating and implementing health research; (b) developing regulation and standard measures to protect benefits for human subjects participating in research; (c) providing evidence based scientific research to decision makers for developing, improving health policy, strategies, action plans, and health work plans of the MOH in order to enhancing quality of health services for better health of the population; (d) developing and improving health research data bases as a center of excellence, providing the evidence base and research outputs that can be synthesizes so as to offer policy recommendations; (e) providing in-country short-term training on public health management for health professionals at all levels, according to each five year socio-economic plan of the country; (f) providing regional long-term training on public health and tropical medicine.

Through its roles and functions, the goals of Lao TPHI are: (1) to improve the people health status through a better understanding of public health disease priorities in the country; (2) to create national leadership in research by providing short and long term training for health institutions in Lao PDR at all levels as needed; (3) to serve the research findings to MOH for policy making in order to improve the health system in the country; (4) to collaborate with internal and external institutions for strengthening research capacity.

■ National Agriculture and Forestry Research Institute

Established in 1999, National Agriculture and Forestry Research Institute (NAFRI) is a research institute in the field of agriculture and forestry. As of 2018, NAFRI employed 332 staffs; 25 PhDs (and 8 doctoral students), 84 Masters' and 152 BScs.

Its main mission is to promote private agriculture by providing good-quality and various kinds of crop seeds. In addition, it mandates to contribute to Agricultural Development Strategy 2025, by carry out research for development on agro-biodiversity, sustainable use and conservation, productivity improvement, climate change adaptation and mitigation, and the supportive policy environment.

The annual budget of NAFRI is about 2–3 million US dollars. Assistance from the Japan International Cooperation Agency (JICA) and other international co-operation agencies accounts for 50% of the budget. All other funds come from the public sector. Research funds from the MOST have been increasing over time. On the other hand, only a small part of funds from the MAF can be spent on research. It should be noted that provinces in Lao PDR have relatively abundant agricultural research related budgets. Each province has a research institute, which receives funds from the central government.

Major activities of NAFRI include developing modernized agriculture technologies for improvement of agricultural production systems which support national food security and nutrition, and for commercialized production. Also, it manages the use of agricultural resources at a sustainable level.

The main research themes are the following: (a) promotion of bio-diversity to create a gene bank, while deepening the understanding of native plants and wild animals; (b) improvement of agricultural productivity, including a program for each rice-plant and livestock species and other programs which focus on maize, cassava, and coffee; (c) response to climate change is given priority, such as research on protecting crops from floods; and (d) information provision to farmers.

NAFRI has two divisions and 14 central and regional research centers across the country.

I. Divisions:

1. Administration and Personnel Division (APD)
2. Planning and Cooperation Division (PCD)

II. Commodity based Research Centers:

1. Agriculture Research Center (ARC)
2. Horticulture Research Center (HRC)
3. Livestock Research Center (LRC)
4. Forestry Science Research Center (FSRC)
5. Maize and Cash-crop Research Center (MCRC)
6. Research Center to Climate Change Resilience in Agriculture (RCRA)
7. Living Aquatic Resources Research Center (LARReC)

III. Non-commodity-based Research Centers:

1. Agriculture and Forestry Policy Research Center (AFPRC)
2. Center for Agriculture and Forestry Information and Communication (CAFIC)

IV. Regional Research Centers and Agencies:

1. Upland Agriculture Research Center (UARC)
2. Coffee Research and Multiplication Center (CRMC)
3. Thasano Agriculture Research Center (TARC)
4. Nong Daeng Agriculture Research Center (NARC)
5. Luangnamtha Agriculture Research Center (LTARC)

The detail information on regional research centers are not available.

■ National Economic Research Institute

The National Economic Research Institute (NERI) was established in 1997 as a strategic research institute under the supervision of the Committee for Planning and Investment (CPI) which became the Ministry of Planning and Investment (MPI) later. As a think-tank of the MPI, NERI's main functions are to formulate the long-term national and regional socioeconomic development strategy.

NERI provides research-based policy recommendations and advocacy at the macro level in globalization context. They follow a country-specific tailor-made advocacy methodology so as to make the best use of potentiality of the country. NERI has a vast network of co-operation and partnership with various ministries, institutes and universities in the country, as well as international organizations like UNDP, World Bank, SIDA, and ADB etc. NERI has implemented many projects supported by international donors and made substantial progress in socioeconomic development in Lao PDR. NERI has engaged on many micro-economic projects, treating specific sectors, agricultural and rural development, economic integration, SME development, financing and on other issues.

■ National Institute for Economic Research

National Institute for Economic Research (NIER) is a research organization under the Government of Lao PDR which is located in Xaythany District, Vientiane Capital, with a status equivalent to other institutes that are under the direct supervision of the Party Central Committee and the Government of Lao PDR. It conducts researches on Lao and international economies; conducts researches on how to improve mechanisms and policies related to economic management in Lao PDR in each period; provides information on researches conducted on economy for which, thorough scientific analysis has been carried out, will contribute to the provision of orientation for policies, strategic plans, master plans and the five-year National Socio-Economic Development Plan in each period; and provides economic advisory services and organizes training to enhance the economic knowledge of

public officials and researchers.

■ Natural Resources and Environment Research Institute

The Natural Resources and Environment Research Institute (NREI) play a role as secretariat to Ministry of Natural Resources and Environment (MONRE) in studying, research on natural resources and analysis environment quality and provide technical services in the area of natural resources and environment. The total number of staffs as of 2018 were 47 persons.

Its mandate includes implementing the policy, strategy, regulation, action plan on natural resources and environment in order to promote research, development, and the utilization of prototype on natural resources and environment which best suitable for Lao context.

The NREI also provides several training courses and materials, and publishes and distributes publications for natural resources and environment.

■ Pasteur Institute of Lao PDR

The Pasteur Institute of Lao PDR (Institut Pasteur du Lao PDR – IPL) is a Lao National Institution created by Prime Ministerial decree no. 402 in 2007. As its name indicates, IPL has been given a permission from the Pasteur Institute in France (PIF) to use the name of Pasteur. The cooperation between the Pasteur Institute and the Lao government started in 2004 when the Lao MOH requested the French government for counter-measures and capacity-building against SAAS and H5N1. Later, following the request from the Lao government for a long-term support, cooperation started with the PIF consequently lending its name. The total number of staffs as of 2018 were 70 persons.

The IPL has a mandate from MOH to fulfill activities of public services: (a) Research and diagnostic on emerging infectious diseases and vector borne diseases; (b) training, Education and Capacity building and (c) technical assistance to National Centre for Laboratory and Epidemiology (NCLE) for investigation of epidemics.

The IPL's main activities include engaging freely in collaborative research and investigations

with other Lao and international research and public health organizations. Its budget is independent from the Lao public finance system. The IPL is able to receive outside funding (donations, grants, bequeaths, etc.) and to generate its own resources through its own discoveries to insure its sustainability. The IPL operations started in December 2011 with 4 laboratories: 1) arbovirus & emerging viral diseases, 2) vaccine preventable and infectious diseases, 3) medical entomology and 4) parasitology.

■ Public Works and Transport Research Institute

The Integrated Institute for Survey, Design and Construction (IISDC) was established in 1982 under the Urban Planning Division of the former Ministry of Construction. In 1985, the Institute for Urban Technique was established, separated from the IISDC. In 1994, it was renamed to the Institute for Urban Planning. In 1999, the institute was upgraded to Urban Research Institute (URI) under the former Ministry of Communication, Transport, Post and Construction (MCTPC). In 2008, it changed its name to the Public Works and Transport Institute (PTI) under the Ministry of Public Works and Transport (MPWT). In 2017, the PTI was upgraded to Public Works and Transport Research Institute (PTRI). The total number of staffs as of 2018 were 47 persons.

The main responsibility of PTRI is to assist the ministry in the planning, analyzing and studying, as well as carrying out research in the areas of infrastructure (road-bridge), transport systems, housing and urban planning, water supply and sanitation, as well as the environment and disaster prevention. Furthermore, PTRI also has roles to evaluate, promote, provide consultation services, disseminate scientific advances, techniques and technology as well as research outcomes in the public works and transport sector. In addition, PTRI is also responsible for other tasks as to be directed and assigned by the MPWT.

Some major activities the PTRI conducted with government funds are: (a) development of regulation on environmental impact assessment of road project in Lao PDR; (b) study on setup of the unit cost for urban planning; (c) study on urban parking design manual; (c) study on an appropriate of urban road asset design; (d) study on the risk of landslide along road No. 13 north; (e) study on building construction safety in Vientiane Capital; (f) study on fire

risks in buildings in Vientiane Capital and (g) study on roundabout design manual.

■ Renewable Energy and New Materials Institute

The Renewable Energy and New Material Institute (REMI) was established by the Ministerial decree no. 0841/MOST in 2011. In the period 2007–2010, the Technology Research Institute (TRI) and the Science Research Institute (SRI) were transformed into the Science and Technology Research Institute (STRI) that was operating within the National Authority for Science and Technology (NAST). Some functions were separated, and REMI was formed when NAST became MOST.

REMI has played its role as secretariat to MOST in research, development, transfer, promotion, applications and services related to renewable energy and new materials. The main mandate of REMI is conducting applied and adaptive research on renewable energy and new materials.

REMI consists of five divisions such as the General Affairs Division, the Bio-energy Division, the Alternative Energy Division, the Mechanical Engineering Division and the New Materials Division. Several research and demonstration projects were conducted by REMI in the last five years including on solar photovoltaic, cooking stoves, gasification technologies and agriculture wastes as an energy source. In addition, REMI also organized several training courses on renewable energy technology and viable applications.

List of projects conducted by REMI: (a) development of a bio-diesel production process; (b) demonstration of a solar home system and solar pumping; (c) research on agricultural waste for solid fuel, and (d) research and development of other appropriate technologies.

After the dissolution of MOST, REMI becomes a part of the Ministry of Energy and Mines (MEM).

■ Research Institute for Educational Sciences

Research Institute for Educational Sciences (RIES) is the successor institute to the Department for Textbook Compilation and (Pedagogical) Research founded in 1975, with no substantive change in function.

Its major responsibilities are: (a) Study and develop curriculum for formal school; (b) draft and improve legislation for use and determination of curriculum and (c) train on application of developed curriculum. Therefore, researches conducted are for most part, curriculum and textbook related. Its professional staff cover all the subjects of general education as well as evaluation and research. The institute also has a long track record of designing and delivering teacher education programs.

■ Research Institute of Energy and Mines

No public information is available except for the name and the Ministry which it belongs. Further investigation is needed for the information of the institute.

3. Policy and Governance Environment

As mentioned in overview and background section, Lao government have recognized the importance of STI development and set the promotion of local innovation and utilization of science and technology as one of the priority outputs in the 8th NSEDP. This is a clear policy indication for developing STI capacity so that STI development can contribute to the social and economic growth of the country. Therefore, strengthening the capacity of public STI research institutes should be an important step Lao government need to take action to realize the national development goal of STI development.

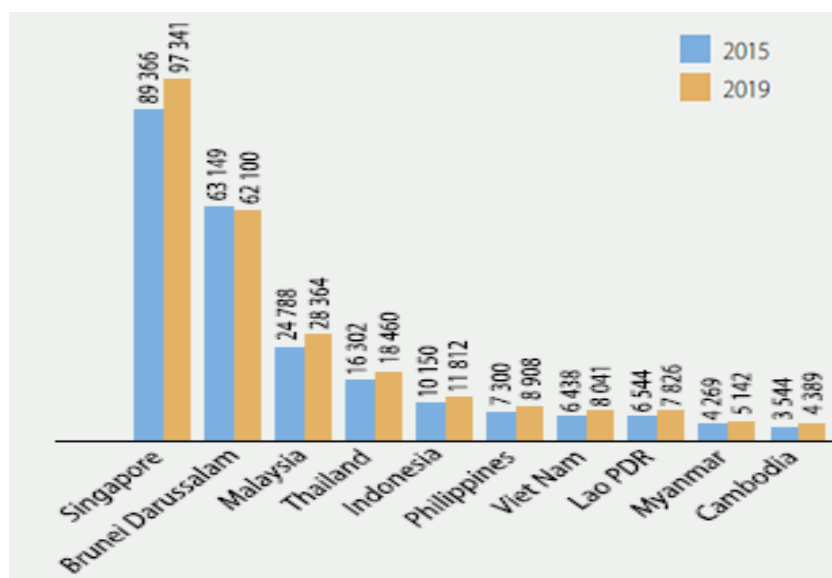
However, in reaching the planned vision and goals of STI development, certain limitations exist. In improving STI capacity of Lao PDR, the country faces the issues of 1) lack of public funding, 2) lack of public data (information) and 3) Lack of infrastructure in R&D.

■ Lack of public funding

Lao PDR is categorized as a least developed country and its GDP per capita lags behind the most neighboring countries in the region. Naturally, Lao government's funding allocation for STI capacity development is limited and the amount is relatively small.

[Figure 4-2] GDP per capita (PPP) in ASEAN countries

(Unit; US dollar)



Source: UNESCO Science Report 2021

According to the 8th NSEDP, in order to promote STI development, Lao government set the target of increasing the investment in R&D (gross expenditure on R&D, GERD) from 1 to 2 percent of public investment by 2020. We do not know whether the target is achieved since there is no report or data available. The last available data for GERD in Lao PDR dates back to 2002 with about US\$ 2.7 million (or 0.04% of GDP)²³.

Due to the lack of resources including public funding, international organizations such as UN and World Bank have played a major role in establishing and supporting the public research institutes. Many research institutes such as IICT were founded due to the direct support from those international organizations. Also, research institutes from developed countries have also given direct support and contribution to Lao partner institutes. IPL is a good example of this case. Support and contribution in form of ODA including financial aid from international organizations still continue to be one of major resources for public research institutes in Lao PDR and therefore, the roles of international organization in STI

²³ Extracted from Facts on Science, Technology and Innovation
Southeast Asia and European Union, [https://sea-eu.archiv.zsi.at/facts/sea/Lao PDR.html](https://sea-eu.archiv.zsi.at/facts/sea/Lao%20PDR.html)

development considers significant.

Lao government should facilitate the support and cooperation of international organization for local research institutes so that limited resource provided by Lao government can be compensated (UNESCO, 2021). Also, opportunities such as joint research provided by international research institutes can give research experience, hence improve research capacity for local research institutes.

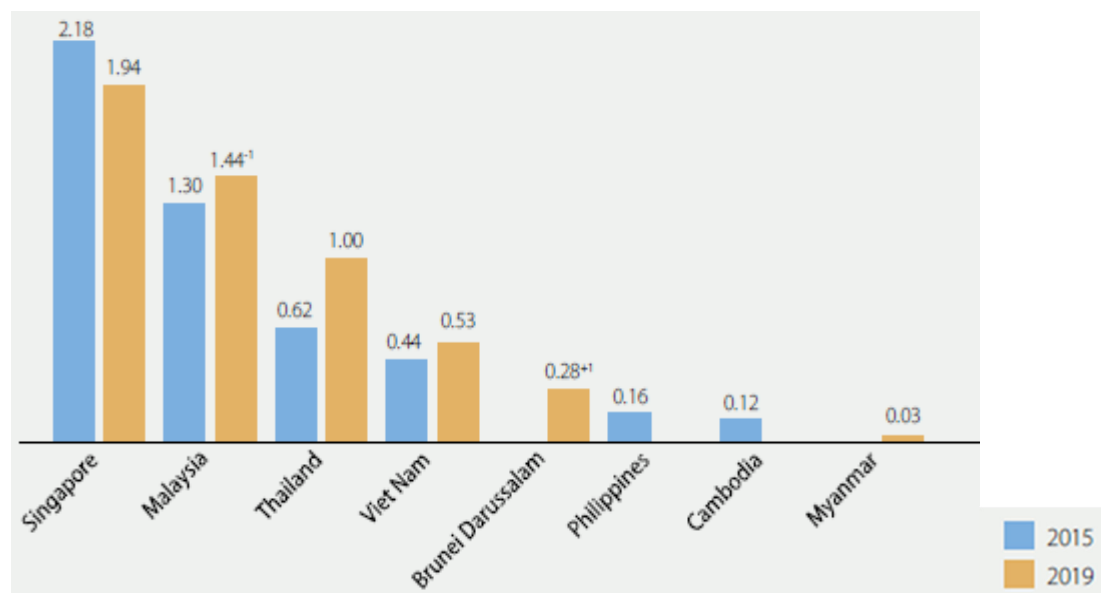
■ Lack of public data (information)

In order to conduct the research and evaluate the current situation on STI capacity, the availability of reliable data is quintessential requirement. Building those data are time consuming and can be costly but without them, conducting research with good output is very difficult to achieve. Therefore, data (or information) is an imperative source for research. However, one of the major challenges in developing STI capacity for public research institutes in Lao PDR is the lack of public data available.

At the 8th NSEDP, it is stated that establishing a national science and technology data center by 2020 is another target for achieving STI development. But it seems that the target has not met since there is no news or announcement on the national science and technology data center. Clearly, Lao government has recognized the importance of science and technology data so that it aims to build the data center but it is not realized yet.

Let's take the example of showing the lack of data: The missed data or information of the country in the official report published by UNESCO. UNESCO publishes the Science Report in every 5 years to monitor the evolution of the support system for science, technology and innovation worldwide over time. Key emerging trends are identified and placed within their socio-economic and political context.

[Figure 4-3] GERD as a share of GDP (%) in ASEAN countries



Source: UNESCO Science Report 2021

Figure 4-3 is the GERD data in the UNESCO Science Report which is lately published. But, the data for Lao PDR is not shown here even though the other ASEAN countries data are presented. Including GERD, other STI indicators such as researcher's specialty and field of activity are not available.

As for the public research institutes listed in Table 4-1, more than half of institutes do not have their websites which can show the organization information and their research activities available to the public whether in English or Lao language. Considering the internet accessibility and online public information are the norm in these days, lack of information or data for public research institutes should be considered a major issue that ministries overseeing those institutes should pay a close attention to improve the situation. This is the evidence that Lao PDR is lack of data in STI indicators and it presents the challenge in the capacity development because no basis or current status on STI is not clearly known.

While high-level plans such as NSEDP is necessary to set and guide the STI development goals, they tends to remain as political vision (UNESCO, 2021). In order to make the plan to execute effectively, Lao government should put an emphasis on detail action on the data

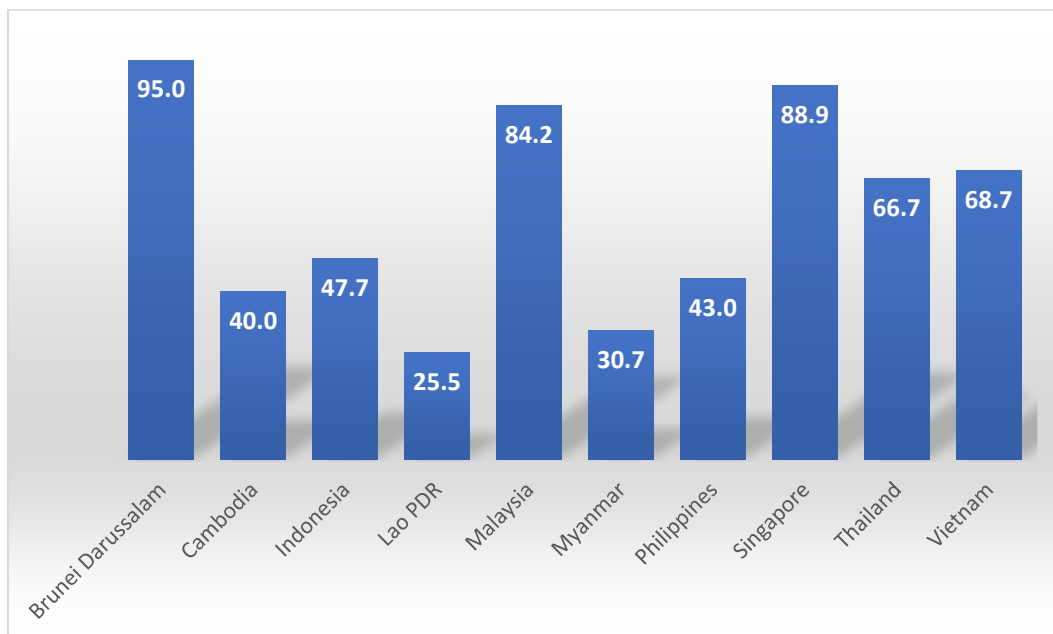
and information management as a priority policy and monitor the process so that the implementing organization can feel responsibility for the output. International organizations and research institutes could help and support in preparing the data and information management process through their experience and resources. This will improve the research environment for the public STI institutes of Lao PDR.

■ Lack of infrastructure in R&D

The lack of infrastructure in R&D is another major hurdle that Lao PDR is facing in developing STI capacity. The Lao government has acknowledged the necessity of R&D infrastructure in STI development hence tried to prepare for it. According to the 8th NSEDP, as a goal to promote local STI, the Lao government has endorsed establishing a science and technology zone and its construction by 2018. In addition to that, it support to establish a laboratory to develop technology for forecasting and early warning of climate change by 2020 considering that the climate change becomes a major issue in Lao PDR. However, the above goals have yet realized.

As for the ICT infrastructure which is a vital infrastructure-network necessary for not only STI but socio-economic development in a country, it can represent the barometer of infrastructure development level.

[Figure 4-4] Internet Access per 100 inhabitants in ASEAN countries (2019)



Source: UNESCO Science Report 2021

Figure 4-4 shows the internet accessibility among ASEAN member states. Lao PDR ranks last. In the era of industry 4.0 where ICT is at the core of its stage, establishing ICT infrastructure is an imperative to improve the national competitiveness. This is an example of showing the current situation on ICT infrastructure in Lao PDR and a great deal of work is required in developing and improving the national infrastructure which is essential for STI as well as economic development.

4.

Role of research institutes on STI and Economic Development in Lao PDR

In any country, research institutes have played a central role in STI and economic development. That is the reason any country tries to encourage and support establishing research institutes for STI development. The Lao government also set the plan to improve the research capacity of institutes. According to the 8th NSEDP, it aims to improve and upgrade research institutes under the MOST so at least one research institute is comparable with international standards by 2020. However, despite the Lao government's intention to improve research institutes for developing STI and economic development, the capacity improvement for research institutes are constrained due to the following factors: Lack of human resource capacity and rely on certain industries for economic development.

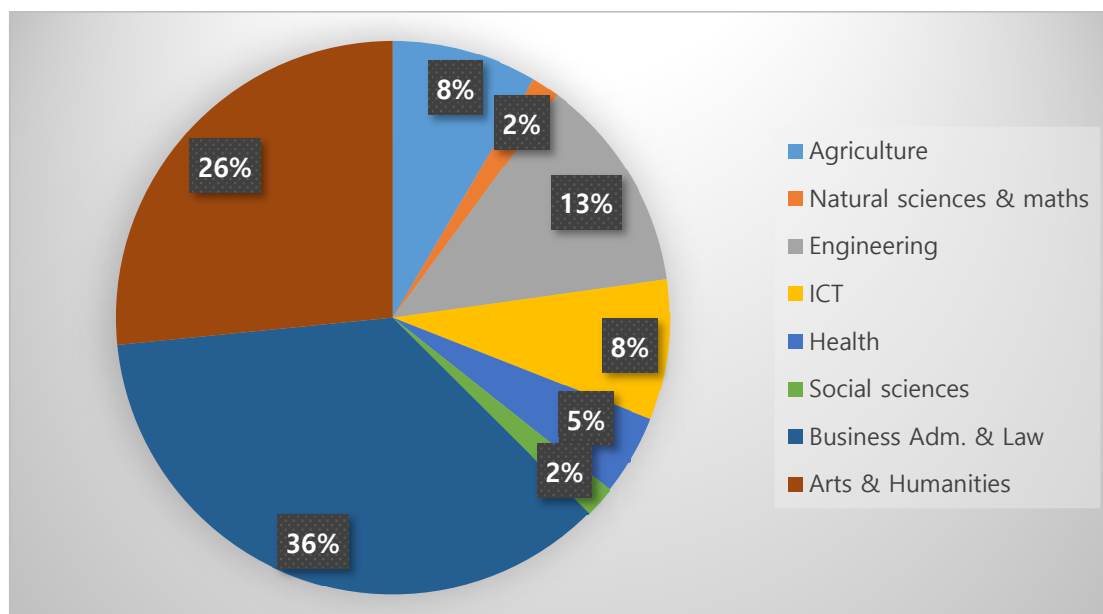
■ Lack of human resources and research capacity

One of the major challenges in STI capacity improvement in Lao PDR is the lack of human resources in terms of quality and quantity. Without adequate human capital, research institutes are not able to enhance their research capacity which can contribute to the social and economic development. In order to be qualified to be a researcher, one has to have higher education including technical junior college.

According to Lao statistics bureau, the total number of students enrolled in universities and junior colleges (2019) were 101,342 which was only 1.4% of total population²⁴. In addition to that, the majority of Lao students choose to study in the fields that are not related to STI (MPI, 2020).

²⁴ In case of Republic of Korea, the total number of students enrolled were 3,276,327 which was 6.3% of total population in 2020.

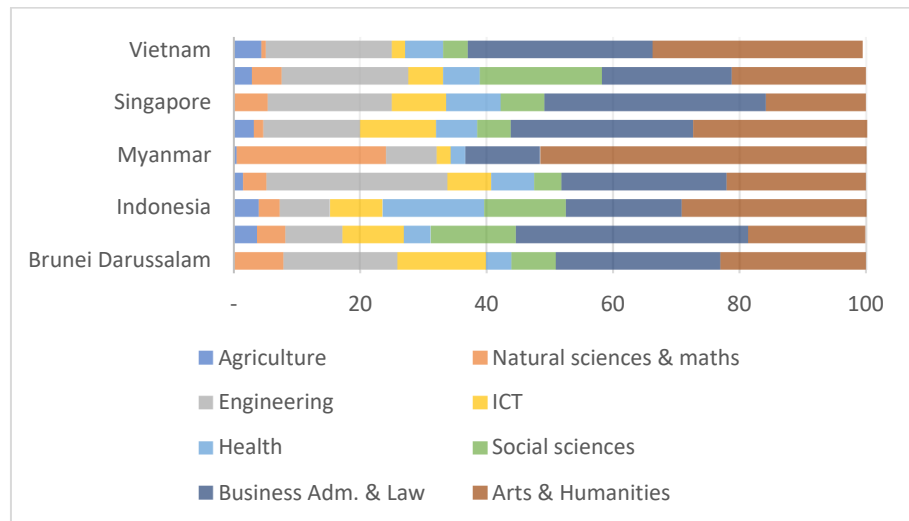
[Figure 4-5] Distribution of tertiary graduates in Lao PDR (2018)



Source: UNESCO Science Report 2021

Figure 4-5 shows the distribution of tertiary graduates in Lao PDR. Among various fields, agriculture, natural science, engineering, ICT and health can be considered as STI related areas. Less than one-third of total student study graduated in STI related fields. Among those graduated, statistics for how many did continue studying graduate level program such as Master's or PhDs are not available. Compared with other ASEAN countries where more students study in STI related fields, Lao PDR lags behind producing more human resources capable of working in the field of STI.

[Figure 4-6] Distribution of tertiary graduates in other ASEAN countries (2019)



Source: UNESCO Science Report 2021

According to Figure 4-6, Most of ASEAN countries have more tertiary graduates in STI field than Lao PDR. Especially, Malaysia, Singapore, Thailand and Vietnam all produce more engineering graduates than Lao PDR. The lack of inflow for human resource in STI can lead to the shortage of possible research activities and capacity for STI. As a result, the research outcome and performance of Lao PDR has not been very productive hence reveals the lack of STI research capacity.

[Figure 4-7] Volume of scientific publications in ASEAN countries

	2011	2015	2019
Brunei Darussalam	118	329	451
Cambodia	177	301	439
Indonesia	2,875	6,080	37,513
Lao PDR	139	209	300
Malaysia	16,489	22,405	30,172
Myanmar	155	190	639
Philippines	1,280	2,013	4,104
Singapore	13,880	17,099	19,437
Thailand	9,342	11,298	17,172
Vietnam	2,149	4,051	10,924

Source: UNESCO Science Report 2021

Figure 4-7 shows the volume of scientific publications in ASEAN countries over 10-year period and Lao PDR has been the least performing country during that time. Scientific publications are the means for measuring the degree of research activities and the fact that Lao PDR produces very small number of research papers indicates its research activities are very limited. Considering that scientific papers are usually written by academic scholars as well as researchers from public institutes, the number of publications from the researchers would be even more reduced.

■ Rely on certain industries in economic development

The Lao government has put an emphasis on developing more human resources in the hydropower, renewable energy, ICTs, agriculture and health sectors because those sectors are the major industry in Lao PDR and hence need more labor force (UNESCO, 2018).

[Figure 4-8] GDP of Lao PDR by category (Unit: Billion Kip)

Activity	2019**	2020
Agricultural	26,107	28,505
Agricultural cropping	17,859	19,196
Livestock & livestock products	3,188	3,747
Forestry & logging	1,608	1,762
Fishing	3,452	3,799
Industry	51,288	57,419
Mining & Quarrying	9,251	7,897
Manufacture of Food Products	2,436	2,742
Manufacture of Beverages & Tobacco	2,101	2,197
Manufacture of Textiles, Clothing, Footwear & Leather Goods	1,863	1,944
Other Manufacturing	6,033	7,584
Electricity	16,204	19,741
Water Supply; Sewerage, Waste Management & Remediation Activities	420	444
Construction	12,980	14,870

Services	67,526	68,294
Wholesale & Retail Trade; Repair of Motor Vehicles & Motorcycles	19,589	20,405
Transportation & Storage	2,190	2,173
Accommodation & Food Service Activities	4,370	2,020
Information & Communications	2,610	2,716
Financial & Insurance Activities	5,162	5,136
Real Estate Activities	9,286	9,689
Professional, scientific & technical activities	2,177	2,324
Public Administration & Defence; Compulsory Social Security	14,664	15,983
Education	3,737	3,855
Human Health & Social Work Activities	1,035	1,109
Other Services	2,707	2,883
Totals (At Basic Prices)	144,921	154,217
Taxes on Products and Import Duties, Net	17,736	18,395
Gross Domestic Product, (GDP), current price	162,657	172,612

Source: Department of Economic Statistics, Lao Statistics Bureau, MPI

Figure 4-8 shows the GDP of Lao PDR. The electricity made from hydropower is the biggest industrial product and the construction sector follows. Agricultural crop and mining are also the major part of domestic products. According to the 8th NSEDP, the Lao government has promoted national and international networking to support public institutes, universities

and the manufacturing sector in order to foster the collaboration between each other. While the sectors which the Lao government has a great deal of interest in developing further for the purpose of national economic development draw more attention and support, the other sectors which are not considered as major prospects of growth for national development can be neglected in the government policy and support.

Even for the sectors which the Lao government has expressed its commitment for capacity development, the effect of the government policy can be questionable. For example, international cooperation with European companies has increased with regard to the transfer of agricultural product's processing technology to Lao PDR. However, Laotian companies have difficulties in meeting the requirement of ISO certification (ERIA, 2019).

In order to increase the effectiveness of cooperation, for any given projects, clear goals and incentives are needed to support research activities of public institutes so that the researchers can have motivation for the outcome of the projects.

5.

Conclusions and Implications

The Laos government has recognized the importance of STI in national development and hence established the policy to promote the STI in the national development plan (MPI, 2016). The public research institutes are founded to support the government policy for the STI development and the legal framework has been provided for having organizational basis under the ministries of the central government with budget allocation.

Despite the government's interest and plan, the research institutes' role and contribution in STI development has generally been limited due to the lack of resources, in both human and financial, and the absence of detail implementation process for the plan. In order to realize the actual progress in STI, the following suggestions need to be considered for discussion among policy makers in the Lao government.

■ **Need for more public information about the research capacity including public institutes**

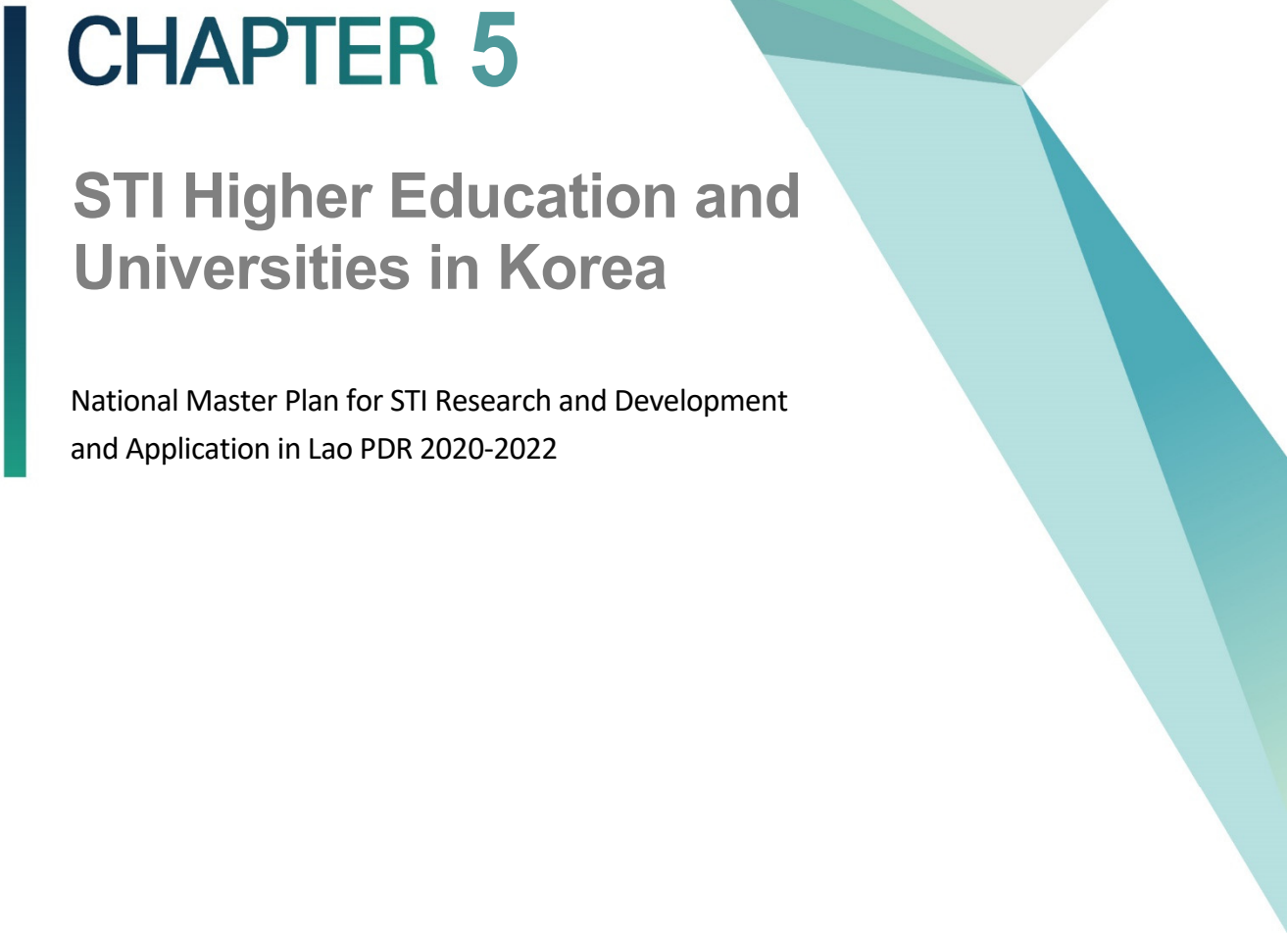
The availability of public information for general public in Lao PDR is still in its basic stage. Whether for the purpose of research or for personal curiosity, the public information about government organizations and projects are difficult to get. And the information about public research institutes is no exception. Much of institutes do not disclose their information, which many would consider, should be open to the public. Therefore, the information such as organizational structure, personnel and budgetary data are not known unless you have a contact within the organizations. It is not clear, though, whether the lack of information for public institutes is caused by the lack of resources or implementation policy. More attention and commitment should be given to make the public information readily available for the general use by the public.

■ **Collaboration and co-work with international organizations including research institutes**

The issue for the lack of resources in both human and financial capital won't be resolved in short term and takes time and efforts to improve. The local research institutes should be involved in the collaboration with international organizations in order to learn the research experience of international research institutes and to improve their research capacity. Various international organizations are actively working in Lao PDR to support the capacity development of the Lao government and organizations. The research institutes of STI should take advantage of these opportunities by actively conducting the joint research or getting training support for their research staffs.

■ **Need incentive system for STI research institutes staffs**

In order for staff members of institutes to get involved in research activities or capacity building, a certain and specific incentive need to be provided so that the researchers can have motivation to perform their jobs with clear goals in mind. The proposal of incentives should be informed along with clear job description and method of measuring for research output. This will be able to make staffs of institutes to understand the evaluation of their performance clearly and to let them to accept the incentive results without much of complain. The incentive system will not only improve the efficiency of organization in terms of research productivity but support the capacity building of staff members by accomplishing more research work successfully.

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CHAPTER 5

STI Higher Education and Universities in Korea

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022



Chapter 5.

STI Higher Education and Universities in Korea

1. Introduction

Korea achieved industrialization within such a short period of time that its growth has been referred to as ‘compressed growth’. Since industrialization began in earnest in the 1960s, Korea has gone through the technology stage of imitating and catching up to the level of advanced countries and moved toward the stage of post-catch-up where it develops homegrown cutting-edge science and technology on its own. The industrialization, economic growth, and social development are historical outcomes of the interaction of many domestic and foreign factors. Among them, factors such as political leadership, early establishment of a national science and technology system, innovative companies, risk-taking investments, civil society's value placed on education, advancement of science & technology, and excellent human resources are especially important factors.

A science and technology workforce policy is one of these factors and the result of interaction with other factors altogether. Notably, the science and technology workforce required for each stage of industrial development was properly nurtured, and they were able to work in the field of science and technology in Korea. The quantitative and qualitative supply and demand of a science and technology workforce was balanced up until the 1990s. The background behind such an achievement was that the government was one of the major economic actors since it led the industrialization process and with the education system under the control of the government. In Korea, the government took the initiative in setting the medium and long-term industry direction, planning the investment volume and setting growth targets. These targets were not the guidelines for the private sector, but the target that should be achieved by the government mandatorily. Therefore, it was possible to make a relatively accurate forecast on the demand for a science and technology

workforce by area and level of skills, while policy measures were mobilized to achieve the goal. It was also possible for the government to nurture a science and technology workforce who can preemptively meet the demand from the field through its national education system.²⁵ In addition to this, the factors, which allowed people to participate in education according to the government's manpower planning, incorporated local strong value placed on education, social belief that science and technology is the basis for national growth and special circumstances of national division.

A science and technology workforce policy has been created according to the needs of the stage of industrial and economic development. As embodied by the phrase 'from catch-up to post catch-up', the development process of science and technology innovation in Korea can be divided into two major stages. The first one is the stage where learning imported technologies and localization was important and the other one is the stage where homegrown technological innovation is important. Such a science and technology workforce policy sets the goals for each stage and utilizes various strategies and policy tools to achieve them. Based on this Korea's science and technology workforce policy was very effective for technology catch-up phase. It is desirable to achieve a quantitative and qualitative balance in the fostering and utilization of a science and technology workforce. As mentioned, the government led industrialization and economic policies and promoted education policies for manpower training aligned with the goals set by the government during the technology catch-up phase. On the other hand, it is relatively difficult for the government to predict the demand for a science and technology workforce because factors such as an increase in the role of companies, global competition, and accumulated knowledge base actively work during the post-catch-up phase. Therefore, the government faced limitations to establish a human resources policy based on demand forecasts, and it has been exploring ways to find measures to deal with this issue²⁶

²⁵ Meansun Noh, Byoung-Ho Son, Jongmin Lee, "A Comparative Study on the R&D Alternative Military Services of Korea and Taiwan", *Innovation studies* 10(1), 2015.6, 73-98. (in Korean)

²⁶ There are many literatures that summarizes the science and technology manpower policies in Korea in a chronological order, and representative literatures include Hong Seong-joo, Song Wi-jin, 『Science Technology Policy in Modern Korea』 (2017, Deulnyeok), Chapter 4; HRST Joint Research Center, 『Human Resources in Science and Technology』 (2007, Korea Research Institute for Vocational Education & Training), Chapter 11. This chapter is based on the content in the two literatures.

This chapter examines the development process of Korea's science and technology workforce policy since the 1960s and its characteristics in each stage of development. Since the 1960s, Korea's economic policy was based on the 'Five-Year Economic Development Plan' and has gone through major noteworthy changes every 10 years. As both the science and technology policy and science and technology workforce policy are considered a subcategory under economic policy, this chapter analyzes the characteristics of such policies at a 10-year interval. Based on the level of technology, the period between the 1960s to 1980s can be regarded as the catching-up stage of technology. The period of 1990s can be referred as the transition stage toward the post-catch-up phase, while the period of 2000s and afterward can be seen as the post-catch-up phase. This chapter particularly focuses on the characteristics of the science and technology workforce policy planned and promoted by the government as well as the factors that affected a change in the innovation activities of participating companies in which the youth and young professionals responded to and participated in the human resources fostering policy at that time. Investments based on the government's plan and establishment of an education system are only prerequisite conditions for nurturing a science and technology workforce. The reason is that effective implementation of the science and technology workforce policy is mainly influenced by social expectations and prospects for science and technology for the youth and young professionals, corporate employment policies & technology investment, and the dynamics of the professional labor force.

Firstly, the background information on understanding Korea's science and technology workforce policy, the recognition of Korean society on the role of education and science & technology, the education system, process of developing government's administrative organizations, science and technology policy and manpower development policy governance are introduced briefly. Major science and technology workforce policies between the 1960s to 2000s at a 10-year interval are introduced, analyzed and evaluated. This study defines what problems the science and technology workforce policies for each period had to solve, what policy measures were taken to resolve such problems, and what the impacts were. Finally, the implications for other developing countries are presented based on the characteristics of Korea's science and technology workforce policy.

2.

Overview of Science and Technology Manpower Policy in Korea

2.1 Social perception and expectation on science and technology

In Korean traditional society, education has been a ladder for upgrading one's status in the society. The Joseon Dynasty, which lasted for over 500 years, established a centralized administrative system in its early days of its founding and established a system of selecting administrative officials through state civil exams. However, science and technology were not included in the examinations or education of government officials. However, the country realized the importance of science and technology education when it opened its ports. Politicians, enlightenment activists, and social leaders emphasized modern education, with an emphasis on science and technology, as a means of fighting foreign forces requiring the Joseon Dynasty to open up its port at the end of the 19th century and increasing the strength to achieve independence during Japanese colonial rule in the first half of the 20th century. Unlike the West, where science and engineering were separated, Korea frequently used the term science and technology in tandem in the early 20th century. Science and technology became an umbrella term for natural science, engineering, technology, industry, mechanical equipment, and technical products. To date, the term science and technology is used to express a comprehensive meaning, but it also expresses the narrow meaning of referring just natural science (or science) and engineering.

The government, which was promoting industrialization made an effort to create the culture where social expectation of science and technology is preserved and the public understands the importance of science and technology education. Slogans such as 'excellent human capital in Korea with scarce natural resources is the key to economic survival' and 'national prestige of Korea through science and technology' are designed to stress the importance of science and technology education for individuals and the nation. Korean people responded to such a public policy with a strong value placed on education. There was a time when poor rural families were labeled 'cow bone tower' rather than ivory tower,

referring to the poor rural families' common livelihood of selling cows as the most important ticket to a college education for their children.

Foremost, the government took a number of drastic measures to raise social expectations of science and technology and secure an excellent scientific and technological workforce with the start of industrialization. One of the prime examples is the project to recruit top-notch expatriate scientists and engineers. The government implemented a policy of recruiting and repatriating Korean scientists and engineers actively working abroad after completing their studies overseas. In order to repatriate them to Korea hampered by a poor research environment unlike advanced countries, the Korean government provided high salaries and large-scale research funds to those scientists and engineers. The scientists and engineers who returned to Korea were hailed as patriots as they sacrificed their self-interests to contribute to national development and became a symbol of the nation's focus on science and technology. In addition, as a divided country that has a conscription system, Korea introduced a military exemption system that replaces mandatory service with engaging in R&D activities or industrial field service. This system encouraged young students to choose a research career by reducing the burden of a 33-month military service at that time. In addition, the recognition that R&D activities were as important as military activities for the country took root socially. The special exemption system is a representative type of support for world-class science and technology workforce and has been in place to date through expanding its scale and participating organizations.

2.2 Education System

The basis of nurturing a science and technology workforce is the public education system, with an emphasis on proceeding onto higher education. Elementary and secondary education (K thru 12) in Korea falls under state-funded public education. It is called a 6-3-3 system with six years of elementary school, three years of middle school, and three years of high school. The constitution established in 1948 made six years of elementary education compulsory and compulsory education was expanded to secondary school in 1984. As of 2021, high school education is not compulsory. However, since 2000, 95 percent or over of secondary school graduates have entered high school. High schools are divided into

academic and vocational high schools. Academic high schools provide education for students to enter universities or colleges, and vocational high schools run programs for vocational training. As the demand for technicians decreases along with industrial changes, the number of vocational high school graduates who enter college has been on the rise since the 2000s. Although some of the middle and high schools are privately run and established by the private sector, they are highly dependent on state finances and abide by government guidelines for student selection and curriculum. Thus, there is no substantial difference between public schools and private schools in terms of operations and curriculum. The secondary education curriculum in Korea follows the standard curriculum determined by the Ministry of Education, and all students take at least a certain degree of mathematics and science education.

Higher education institutions in Korea are divided into two-year junior colleges, four-year universities, and graduate schools. Vocational training is led by vocational high schools and two-year junior colleges with much focus on technical skills training. Between 1 to 3 public universities are established in each region and a large number of private universities are distributed throughout the country. Many of the universities in Korea are private schools. However, many private universities depend heavily on state financial support and abide by government guidelines. And there is not much fundamental difference between private and public universities in terms of academic operation. In other words, the government has a strong influence in the design and operations of the education system for nurturing a science and technology workforce, with an emphasis on proceeding onto higher education. As a result, the government was able to train a science and technology workforce at the desired scale through higher education institutions.

2.3 Governance for nurturing science and technology manpower

The science and technology workforce was divided into technicians, junior engineers, and scientist-engineers depending on the level of knowledge and skills from the early stage of industrialization. Although this division was not initially clear, it was defined from the <Second Science and Technology Promotion Plan> in 1966 as follows. Technicians are skilled

workers who are put into the field after short-term vocational training. A junior engineer is a person who has graduated from a two-year junior college or vocational high school and has field experience for a certain period of time. A scientist-engineer is a person who has acquired expertise through a four-year university or higher education course. In its early days of industrialization, technicians or specialists with university degrees were called science engineers, but with industrial development and R&D growth, graduates with master's and doctoral degrees are sometimes called advanced science engineers. As Korea's science and technology development stage ramped up, the science and technology workforce policy focused on manpower above the university level as the main policy target. When necessary to be distinguished in this chapter, technicians or specialists with a master's or doctoral degree are referred to as high-level, and among them, those with great achievements are referred to as members of a core science and technology workforce.

Korea's science and technology workforce nurturing system consists of educational institutions and training courses managed by the dedicated ministries for each of the above three categories. In the Korean government, the Ministry of Education is in charge of this²⁷. All educational policies are under the purview of the Ministry of Education and include 12-year elementary and secondary education, 2-year junior colleges, 4-year colleges and graduate schools. Until the early 1960s, technical vocational education was also managed by the Ministry of Education. However, as the demand for a science and technology workforce was subdivided with industrialization, the task of nurturing a science and technology workforce was also differentiated. The dedicated industry-promoting ministry was in charge of technicians and their training in the industrial field, the Ministry of Education was in charge of nurturing a science and technology workforce through the public education system, and the Ministry of Science and Technology established in 1967²⁸ was in

27 The ministry in charge of education was the Ministry of Culture and Education between 1948 and 1990 and it was reorganized into the Ministry of Education in 1990. It was integrated with the Ministry of Science and Technology between 2008 and 2013 to be renamed to the Ministry of Education, Science and Technology and became the Ministry of Education in 2013. In this chapter, the name is indicated as the Ministry of Culture and Education pre-1990 and the Ministry of Education afterwards.

28 It was established as the Ministry of Science and Technology in 1967 and was elevated to the Ministry of Science and Technology in 1998. It became the Ministry of Education, Science and Technology after integration with the Ministry of Education between 2009 and 2013 and renamed to the Ministry of Science, ICT and Future Planning in 2013. With the addition of information communications function in 2017, it was renamed to the Ministry of Science and ICT. In this paper, it is referred to as the Ministry

charge of advanced science and technology workforce policy mainly focusing on R&D personnel. The decentralized system was effective in implementing policies to nurture human resources for each field based on demand forecasts. This division of labor system was effective in implementing a training policy based on the demand forecast for manpower for each field.

With the advancement of industry, the science and technology workforce policy gradually shifted the focus on R&D workforce. As a result, higher education on science and technology in Korea is currently conducted in a two-track format by educational institutions managed by the Ministry of Education and Ministry of Science and Technology, respectively. Universities are established by the <Higher Education Act> and are under the purview of the Ministry of Education. As the R&D competency became more important as Korea transitioned to a post-catch-up phase, education policies to foster graduate schools were implemented, and some universities developed into research universities. Since the 2000s, universities have also emphasized patents, research achievements, and commercialization results as much as academic papers.

S&T-specific universities were established not according to the <Higher Education Act>, but rather according to laws specifically enacted for each institution, with the aim to foster industry-university cooperation and R&D workforce. S&T-specific universities consist only of departments in the field of science and technology and put much emphasis on research and research-oriented education for undergraduate students. Because they are not restricted by the <Higher Education Act>, independent academic management such as early graduation is allowed, and they enjoy special status such as receiving more state financial support than other universities. Graduates from S&T-specific universities have higher rates of employment in companies and venture start-ups than graduates of general universities. As a result, education and research are designed and implemented in alignment with the goal of industry-oriented education and research.

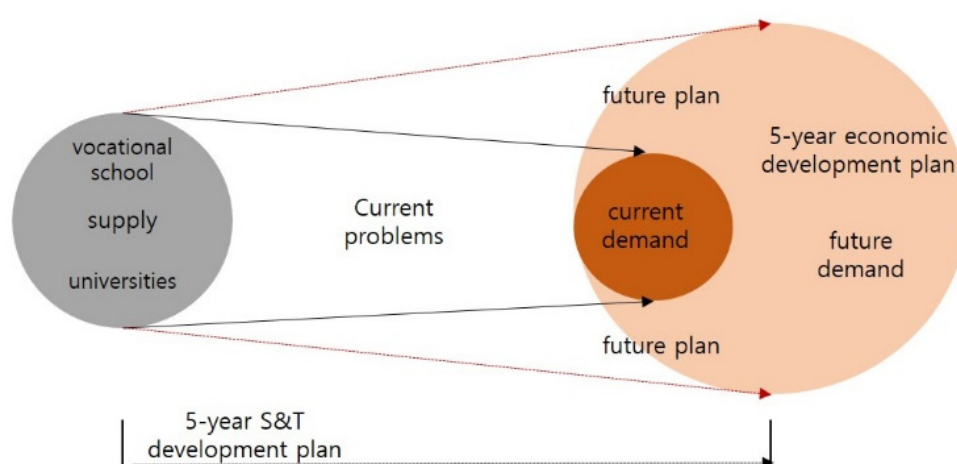
of Science and Technology pre-1998 and the Ministry of Science and Technology afterwards.

3.

In the 1960s: Laying the Foundation for Science and Technology Manpower Policy

From the early days of industrialization, the science and technology workforce policy was a part of science and technology policy and designed to support the comprehensive economic plan. The government established the Five-Year Economic Development Plan every five years since 1962 as a comprehensive economic plan. The Economic Planning Board, which oversees the Five-Year Economic Development Plan, was higher than other ministries in the hierarchy of state-run ministries. The science and technology policy was established in conjunction with it from 1962. At that time, science and technology policies were in charge of the Economic Planning Board's Technology Division since there was almost no science and technology base. The science and technology policy focused on the enactment for science and technology, system improvement, and utilization of technological resources and manpower. In other words, the science and technology policy and policy for nurturing a science and technology workforce in the 1960s marked a period of reorganizing the existing system and creating a new one.

[Figure 5-1] Supply vision of personnel for economic development



Source: HRST, *Human Resources in Science and Technology*, p.81

The comprehensive economic plan of the 1960s enabled to achieve high growth through rapid industrialization and modernization of industrial structure. The <First Five-Year Economic Development Plan (1962-1966)> sought to achieve high growth by expanding key industries such as fertilizer and oil refining, and focusing on the expansion of consumer goods manufacturing. According to the plan at the time, the average annual growth forecast for all industries for five years was 7.1 percent. In particular, the average annual growth forecast for the secondary industry was set at 14.8 percent, more than twice that. Numerous factories and industrial infrastructure were to be built starting from the absence of accumulated private capital during this period.

The government officials at the technology department of Economic Planning Board established the <1st Five-Year Plan for Technology Promotion> to support it. This plan focused on securing human resources in the technical field, and included promoting technology introduction and institutional reforms. At that time, the current status of the science and technology workforce in Korea was investigated and used as the basic data, while the demand for highly skilled workforce was predicted based on industrial growth forecasts. According to it, the current science and technology workforce in all industries in 1961 tallied about 300,000 people, but in 1966, the demand for a science and technology workforce was predicted to tally about 600,000 people. According to the plan, an additional science and technology workforce of 300,000 people would need to be supplied within five years. According to this prediction, the government pushed ahead with the additionally needed manpower training project.

The <1st Five-Year Plan for Technology Promotion> established a nurturing plan of science and technology manpower into three categories of scientist-engineer, junior engineer and technician based on the degree of expertise. Firstly, science engineers are experts who has a university degree or a certificate or license recognized by the government, such as architects, engineers, and science & technology researchers. Their duties include planning, design, supervision, and R&D. Secondly, junior engineers are personnel with field experience after graduating from a two-year college or a vocational high school. They are in charge of production method planning according to design, preparation of specifications,

and product testing, and thus supervise production sites and production processes. Thirdly, technicians are skilled workers who receive appropriate vocational training and are in charge of actual production activities. In this way, it can be seen that the science and technology workforce was categorized by level and the science and technology workforce policy was systematically implemented according to the industrialization plan at the time. Over time, these categories were redefined, and science and technology workforce policies were gradually narrowed down to scientists and engineers.²⁹

Initially, the Ministry of Culture and Education was in charge of nurturing a science and technology workforce but the Labor Administration was established to respond to industrial demand for manpower, mainly technicians. The Ministry of Culture and Education enacted the <Industrial Education Promotion Act> and expanded educational institutions at each level based on forecasts for manpower demand. For junior colleges and universities, the quota was adjusted for each required department according to the demand forecasts. But there was a problem due to a difficulty to quickly respond to the additional demand of 300,000 people with a method such as a new school. In addition, it was difficult to train the middle and lower-class skilled workforce required by the industry like technicians on a large scale through the public education system. Therefore, the Labor Administration was established in 1963 in charge of vocational training focused on nurturing technicians. The forecasting of demand for a science and technology workforce and a system for nurturing based on it were established and stabilized during the first five years of industrialization, while achieving a balance between supply and demand. This is because the human resources were produced according to the plan, and the high-growth industry hired most of them.

In the latter half of the 1960s, during the <Second Five-Year Economic Development Plan (1967-1972)>, a science and technology policy department was newly established, and policies for nurturing a science and technology workforce were differentiated and specialized. <The 2nd Five-Year Economic Development Plan> aimed to modernize the industrial structure and focused on laying the groundworks for advancing the industrial

29 It was defined systematically in the <Second Five-year Science and Technology Promotion Plan>(1966).

structure. The <Second Five-Year Science and Technology Promotion Plan> established in alignment with these goals was more systematic than the previous one. For the second plan, domestic and expatriate science and technology experts were consulted and the scope of science and technology was expanded to include R&D. In particular, the goal was to efficiently introduce advanced science and technology for industrial modernization and enhance scientific and technological capabilities through technology learning and research activities. Accordingly, the roles of junior engineers and scientist-engineers were emphasized compared to the previous period.

Foremost, the <2nd Five-Year Science and Technology Promotion Plan>, which emphasized high-tech learning and research activities, actively led the reform of the system to the extent that it was called a 'science and technology boom'. In 1966, the Korea Institute of Science & Technology (KIST), Korea's first government-funded research institute (GRI) was established to support industrial research. The GRI is a unique public research institute system in Korea. The government establishes it and provides a contribution for basic operations every year, but the main research activities are to be carried out in the form of external service research projects. This system was introduced to operate KIST according to the purpose of responding to the technological research demand of the Korean industry. At that time, KIST was practically the only research institute equipped with scientific and technological research facilities in Korea.³⁰

The Ministry of Science and Technology was established in 1967 under the recognition that a dedicated ministry was needed to enhance scientific and technological capabilities. The operation of KIST, which opened a year ago, is under the management of the Ministry of Science and Technology. As the Ministry of Science and Technology took full charge of science and technology policies, the nurturing of a science and technology workforce, centered on top professional scientists and engineers became the responsibility of the Ministry of Science and Technology, instead of the Ministry of Education. However, since

30 Moon, Manyong, "Science and Technology Boom' in the 1960s: The Formation of Science and Technology System in Korea", The Korean Journal for the History of Science, vol.29 no.1, 2007, 69-98, for the process of forming Korea's modern science and technology system and "Change in the R&D Activities of KIST, 1960-1980", The Korean Journal for the History of Science, vol.28 no.1, 2006, 81-115 for the process of establishing and operating KIST.

higher education institutions such as universities were under the purview of the Ministry of Education, there were many restrictions on the manpower training project related to the existing school system. Therefore, the Ministry of Science and Technology's manpower training had only indirect and limited means of research support for science and engineering professors, leveraging experimental facilities, and having access to research funds for graduate school education. Even though the budget was small, the effect of the support was not great. As a result, the Ministry of Science and Technology supported the operation of a new KIST rather than directly nurturing it, and focused on the policy of 'utilizing' excellent human resources centered on KIST.

The science and technology workforce policy of the Ministry of Science and Technology was not to foster but rather to secure top-notch expatriate scientists and engineers, and its representative project was attracting them. Korea's university education in the 1960s was poor in quality despite quantitative growth. There were not many well-qualified faculty, and the education budget was small, so there were almost no laboratories for science and engineering education. Therefore, there was a severe 'brain drain' in which excellent human resources who graduated from domestic universities did not return after studying abroad in developed countries. Through this project, the Ministry of Science and Technology created an opportunity for scientists and engineers working abroad to return to their home country. This project also saved the time it takes to train highly skilled scientists and engineers. In other words, R&D specialists were secured through 'recruiting' rather than 'nurturing'. Technically, it was possible to utilize the R&D achievements and know-how that the recruited scientists and engineers had accumulated in advanced countries. If possible, in order to match the local research environment and skill level of the recruited scientists, KIST was equipped with the best research facilities in Korea, guaranteed research autonomy to a large extent, and provided exceptional perks such as much higher salaries than other scientific engineers such as university professors. Thanks to the research achievements of these scientists at KIST, the level of science and technology R&D in Korea at that time ramped up in a short period of time.

It should be emphasized that the project to recruit top-notch overseas scientists and

engineers has become a catalyst for nurturing the next generation of advanced scientists and engineers. In fact, it is difficult to see the project attracting top-notch expatriate scientists and engineers as a human resources training project in itself. In addition, the Ministry of Science and Technology put its limited resources for the human resources policy to this project, therefore the research activities at domestic universities and support for graduate students had difficulty getting financial support from the Ministry of Science and Technology. Nevertheless, the existence of talented scientists and engineers in the high-tech field who repatriated through this project made it possible for top-talent students to choose science and technology, and the special perks they received are a kind of investigation for the development.

4.

In the 1970s: Establishment of Two-Track Manpower Nurturing System based on Sophistication of Industrial Structure

Korea's comprehensive economic policy in the 1970s aimed to lower the dependence on imports of capital goods through fostering heavy and chemical industrialization and to upgrade the industrial structure by increasing the ability to develop technologies autonomously. At the end of the <4th Five-Year Economic Development Plan> in 1981, export volume of USD 10 billion, per capita national income of USD 1,000, and 51 percent of the proportion of heavy and chemical industrialization in the industry were set as goals. According to the 3rd and 4th Five-Year Economic Development Plans, the government created huge industrial complexes for the industries to be nurtured nationwide, aiming for production efficiency and investment effects through concentration in the relevant industrial fields. Steel, oil refining & petrochemicals, non-ferrous metals, shipbuilding, general machinery, and electronics were the target businesses.

Accordingly, the <3rd Five-Year Science and Technology Promotion Plan> and the < 4th Five-Year Economic Development Plan: Science and Technology Sector Plan> focus on the two pillars of nurturing a science and technology workforce based on manpower to be deployed to industrial sites and manpower to be in charge of R&D. In the case of the heavy and chemical industry and brain-intensive industries, the level of technology was high, and a highly skilled science and technology workforce was needed to localize and improve the introduced technologies. Accordingly, the focus of nurturing a science and technology workforce changed from technicians & junior engineers in the 1960s to junior & scientist engineers. In particular, training highly skilled scientists and engineers to engage in R&D has emerged as a new policy task.

4.1 Track 1: Achieving quantitative supply through expansion of junior colleges and universities

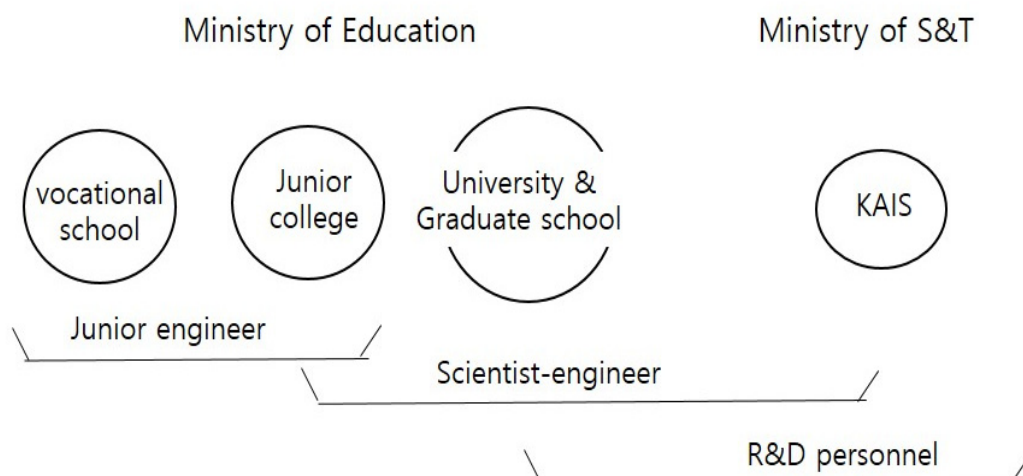
In order to solve these two kinds of tasks for nurturing science and technology manpower, the Ministry of Culture and Education and the Ministry of Science and Technology shared the system. The role of the Ministry of Science and Technology has expanded from the previous utilization of scientists and engineers to fostering. In particular, both ministries have adopted specialization strategies to meet the demands of the newly emerging heavy and chemical industries and electrical and electronic industries. First, let's take a look at the specialization strategy of the Ministry of Culture and Education, which was in charge of nurturing technicians and scientists. The Ministry of Culture and Education increased the entrance quota of departments that fit for the heavy and chemical industry at junior colleges and universities. In the heavy and chemical industry, compared to the light industry of consumer goods, the level of manpower in the field had to be raised, so various types of vocational education institutions were converted into junior colleges. The necessary departments were installed at new born colleges to increase the number of students. New curriculums were introduced and managed to expand the training scale, and to raise the educational level. By the end of the 1970s, there were 127 junior colleges and more than 78,000 students nationwide. At the same time, the number of science and engineering graduates at four-year universities was about 17,000. The result showed that a large number of junior engineers were educated enough to meet the demand of the heavy and chemical industry in the 1970s.

The entrance quota of universities also increased in mechanical, metal, electronic, and chemical engineering fields to prepare for the newly emerging demand for scientists and engineers. In addition to the expansion of the overall training scale, the Ministry of Culture and Education designated specialized departments universities adjacent to industrial complexes, which were tailored for the regional strategic industries.

For example, Kyungpook National University near Gumi Electronics Industrial Complex designated electronic engineering as a specialized field, and Chonnam National University near Yeosu Petrochemical Industrial Complex designated chemical engineering as a

specialized field, and each department operated a bachelor's degree on a large scale at the college level.

[Figure 5-2] Two track frame of S&T manpower training policy in Korea



4.2 Track 2: Establishment of a human resource development institution with special status

The Ministry of Science and Technology focused on the demand for a new and advanced, world-class science and technology workforce that emerged in tandem with the vitalization of a heavy and chemical industry and robust industrial structure. Since technology localization and technological development were emphasized, the demand for industrial technology R&D was predicted in all areas of the key nurturing industries. The project to recruit top-notch expatriate scientists and engineers continued steadily during the 1970s, but it was unable to meet all of the explosively increased demand for research manpower. Therefore, the need for domestic training of master's and doctoral degree-level scientific engineers to engage in R&D activities has increased. The Ministry of Science and Technology did not try to solve this problem by nurturing graduate schools in existing universities. Instead, they planned to establish a separate institution outside of the existing higher education system. The rationale was that it is necessary to nurture human resources with a strong ties to the domestic industry. By adopting this method, it was possible to directly

promote and support the science and technology workforce training project without encroaching on the work scope under the purview of the Ministry of Culture and Education.

The Korea Advanced Institute of Science (KAIS), which opened in 1971, was the first institution established through this initiative. KAIS benchmarked research universities in the U.S. to promote industry-university cooperation, and provided excellent domestic human resources with the highest level of research environment to become an alternative to studying abroad. KAIS was a government-funded institution established under the <Korea Advanced Institute of Science Act>. KAIS was a kind of graduate school that could award master's and doctoral degrees under this law. At the same time, it was a specialized graduate school in science and engineering that was not restricted by the <Higher Education Act> and set departments only in science and engineering. The established major fields were machinery, industry, biology, physics, materials, electricity, computing, chemistry, and chemical engineering. From this, it is clear that KAIS was an institution to support the heavy and chemical industry. World-class scientists and engineers from KIST joined the KAIS faculty or conducted research in connection with KAIS. KAIS is equipped with the highest level of research facilities, and all students are given state-funded scholarships. After completing their master's degree, students were supposed to perform research activities as technical research personnel at a designated institution in lieu of their mandatory military service. Since this special military service was a system that only KAIS graduates could enjoy, it was one of the important motivations for top-talent students to choose KAIS. From the government's point of view, it was possible to expect the dual effects. One is preventing the brain drain of top-talent manpower, which was painstakingly nurtured through going to study abroad after graduation. The other is allowing them to participate in the development of domestic industrial technology at research institutes. Since no graduate school in Korea enjoyed such support in the early 1970s, science and engineering professors at major universities opposed the establishment of KAIS or criticized the support given to KAIS as a preferential treatment.

<Table 5-1>Trend of employment of KAIS master program graduates, 1975-1980

	Graduates (industry- university)		Education institution	Research institution	Government institution	Companies	Doctorial program	Others
1975	92	(25)	23.9%	45.7%	10.9%	16.3%	2.2%	1.1%
1976	145	(36)	20.7%	37.2%	12.4%	26.9%	1.4%	1.4%
1977	139	(42)	12.2%	39.6%	6.5%	32.4%	8.6%	1.4%
1978	142	(25)	14.8%	38.7%	7.7%	26.1%	12.0%	0.7%
1979	225	(53)	21.3%	35.6%	6.7%	23.1%	13.3%	0.0%
1980	306	(104)	18.6%	22.9%	2.6%	35.6%	19.6%	0.3%

Source: <20 Year History of KAIST >

Now, let's examine the career paths of KAIS graduates to evaluate the performance of industry-oriented education at KAIS. KAIS operated the contract education program in which companies paid the tuition for participant students who already got the state-funded scholarships. KAIS produced a total of 1,049 master's graduates from 1975 to 1980. Looking at the career paths of graduates, it can be seen that the education and research at KAIS was industry-oriented. As the heavy and chemical industrialization progressed and corporate R&D activities became more active, the demand for advanced R&D personnel with doctoral level was rising. Looking at the institutions employing master's graduates, in the first half of the 1970s, when corporate research activities were weak, government institutions and government-funded research institutes accounted for more than 50% of the total employment. Then, as the 1980s progressed, the ratio of corporate employment and PhD programs increased.³¹

Despite the criticism from existing universities, KAIS was established for two reasons. One is because the government had a strong will to foster capital and technology-intensive industries like heavy and chemical industries. The other is because there was a great need to nurture science and technology manpower with the capability to support it. In fact, through the 1970s, the government sequentially established government-funded research

31 20 Year History of KAIST』 (1992)

institutes to support technology development in each of the key industries. For example, the ship research institute affiliated with KIST and the Korea Electronics Technology Research Center were separated and became independent laboratories, and the Machinery and Metals Research Center, the Korea Chemical Research Institute, and the Korean Standards Research Institute were newly established. These government-funded research institutes took KIST as their operating model and became key players in industrial technology R&D activities in the 1970s, when corporate research institutes were weak. In particular, after the mid-1970s, many new government-funded research institutes were located in the newly created research complex in Daejeon, which formed the first research cluster in Korea. In these government-funded research institutes, KAIS graduates, along with KIST scientists and scientists newly recruited from abroad, formed the main pillar of the R&D workforce. From the perspective of manpower supply and demand, KAIS was the starting point. It began to nurture excellent scientists and engineers in Korea. It was followed by specialized government-funded research institutes with excellent research environments to employ them and finally to make balance between the quantity and quality of supply and demand. KAIS was a new form 'invented' by the government for this effect, a Science and Technology centric(hereafter S&T centric) graduate school.³²

In summary, the policy of nurturing science and technology manpower in the 1970s tried to balance quantitative and qualitative supply and demand by adopting a specialization strategy to keep pace with the strategic development plan for the heavy and chemical industry and capital technology-intensive industries. A two-track policy was promoted: the expansion of the higher education system under the Ministry of Culture and Education for quantitative supply and the establishment of specialized graduate schools for science and engineering under the Ministry of Science and Technology for qualitative supply. And through the 1970s, the policy of nurturing science and technology manpower by the Ministry of Science and Technology expanded from the 'utilization' to the extent of 'training and higher education'.

32 Moon, Manyong, "From KIST to Daeduk Science Town: The Creation and Reproduction of Government-supported Research Institute during the Era of Park Chung-hee", *Critical Review of History*, vol.85, 2008, 262-289

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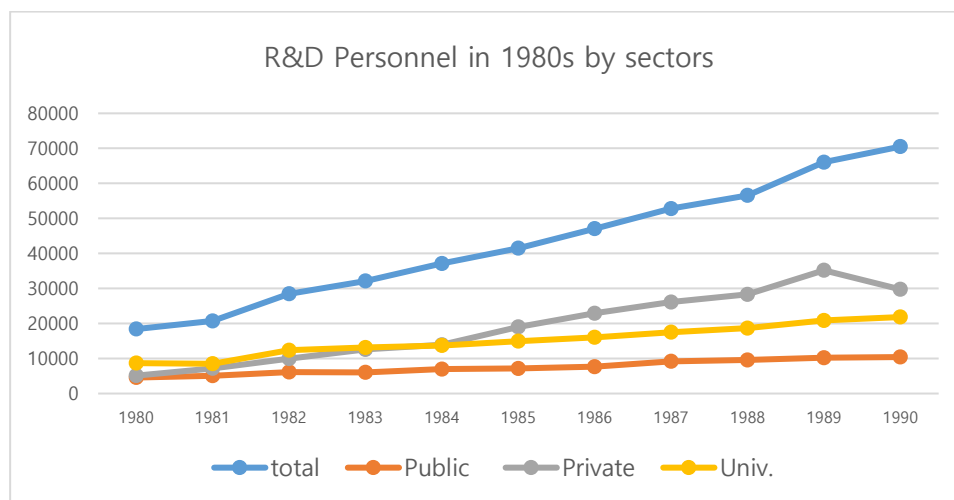
In the 1980s: Reorganization of R&D System and Establishment of Advanced Scientific and Technological Manpower Development System

5.1 ‘Technology Drive’ policy

The basis of the economic policy of the 1980s enabled to secure technologies for the stabilization of the heavy and chemical industry and development of high-tech industries. Since the start of industrialization in earnest, Korea's economy has grown rapidly, achieving the so-called ‘miracle on the Han River’, achieving the export target of USD 10 billion in 1977 earlier than planned with per capita income in 1980 surpassing USD 1,600. In order to maintain this growth, it was necessary to increase the value added of exports and turn to the high-tech industry. In the petrochemical, electronics, shipbuilding, and automobile industries, which had established a foundation to some extent during the 1970s, the main industries faced the challenges of technology localization and technology concentration. In addition, new challenges such as semiconductors, computers, information communication, and fine chemistry had to be pursued in a short period of time. But there was difficulty in introducing technology from advanced countries compared to the past due to the rise of technology patent protectionism in advanced countries and concerns about the rapid growth of the Korean economy. To solve this problem, the science and technology policy focused on promoting homegrown technological innovation through R&D and increased investment in R&D compared to before. As a result, the policy focus was placed on R&D personnel with a university degree or higher rather than the former approach to nurture a science and technology workforce.

In the background, there were general social factors such as an increase in the demand for higher education due to economic development and income-level increase, and an increase in professional jobs. With regard to science and technology, investment in R&D by companies has increased, resulting in the demand for R&D workforce. While government-funded research institutes were the main actors in R&D in the 1970s, companies and

universities in Korea showed relatively rapid growth in R&D activities in the 1980s. For instance, in 1981, there were about 50 corporate research institutes and about 7,000 researchers. The growth was driven by the industry's widespread recognition of the need for technological development activities, and the government's strong 'technology drive' policies based on technological development fund support, tariff reduction, and tax support.



5.2 Track 1: Growth of domestic universities and nurturing graduate schools

Even in the 1980s, the two-track structure of the Ministry of Culture and Education and the Ministry of Science and Technology, general higher education and specialized education in science and engineering, was maintained. Higher education, especially university (graduate school) policies under the jurisdiction of the Ministry of Culture and Education, was an expansionary policy. Through the 1980s, the number of graduates in all majors at universities and graduate schools increased sharply, especially in engineering majors. In 1980, about 10,000 bachelors, 690 masters, and 40 doctoral students were produced in engineering. However, by 1990, about 30,000 bachelors, 3,800 masters, and 450 doctoral students were produced. The increase in R&D investment by companies created a demand for scientists with bachelor's degree or higher. The forecast of the demand for manpower based on the 10-year long-term outlook requested the government to expand universities

and graduate schools in the early 1980s and made an answer to the demand. It can be said that the manpower demand forecast based on the government's long-term economic policy plan and the manpower training system in preparation for it were balanced to a considerable degree until the 1980s.

Since the 1980s, the government has expanded special military service exemptions as a policy tool to promote the cultivation and utilization of science and technology manpower. First, Bachelor's personnel who have graduated from universities can also substitute military service by working as specialized research personnel at a company-affiliated research institute designated as an alternative service organization. Second, a new special military service system was introduced to shorten service for master's graduates. It began in 1982 and ended in 1991. In the system, those who got master's degree from a domestic graduate school and passed the test spent six months of training and military service and then were recognized as having completed their compulsory service. They were commissioned as second lieutenants and discharged from service. That's why this system was called 'master-lieutenant'. In the case of active duty enlistment at that time, the service period was 30 to 33 months, and compared to the three years of service as a professional researcher, it had the advantage of being able to complete military service in a much shorter period. Master's graduates of all majors were able to apply for this system. However, the largest beneficiary of this system was the science and engineering graduate students, as the number of science and engineering fields increased the most in the graduate school. Of course, there were constraints such as having to pass the test and enlisting in the military if you fail the test. Along with the growth of corporate research institutes, however, this system became an important factor in the increase to choose domestic science and engineering graduate schools³³

In addition to the military service exemption, the government's expansion of R&D investment resulted in the improvement of the research environment of science and engineering graduate schools and finally made a positive effect on qualitative growth. The government started the Specific R&D Project in 1982 to promote the independence of

33 Choi Seong-woo et al. New Military Service System in the Science and Technology Sector (STEPI, 2003)

industrial technology. This project was the first large-scale national R&D project in Korea. As the name suggests, it aimed to develop mid- to long-term strategic technology necessary to achieve national development goals. At the same time, industry-university-research cooperative R&D was emphasized in the promotion method. Through this project, professors and graduate students of major universities were able to gain experience in high-tech science and technology R&D and industry-university cooperation. Professors and graduate students who participated in this research project experienced industry-university collaboration, which was rare at the time. Based on their research experience and achievements, they grew into first-generation researchers in the related high-tech field. Some of major achievements of the Specific R&D Projects are localization of 8-bit small computers, time-sharing digital electronic switching system TDX development, and 4M DRAM development.³⁴

5.3 Track 2: Establishment of a consistent system for specialized science and technology education

On the other hand, the special manpower training system for outstanding students with special talents in science and technology outside the general higher education system became more systematic than in the 1970s. The Ministry of Science and Technology expanded the scope of its intervention in science and technology education policies while emphasizing the importance of gifted education in science and technology. It was intended to expand science and technology specialized education to high schools and universities. To this end, science high school was first established in 1983 as a special purpose high school. A special purpose high school refers to a type of school that can operate a curriculum focusing on special fields such as arts and sports. As a special purpose high school, science high schools were able to operate a special system unique to science high schools. They also provide a curriculum centered on mathematics and science. For example, unlike all other high schools, they adopted an early graduation system that allows students to graduate in

34 Yoo, Sangun, "National R&D as Development of Weapons?: The Origin and Characteristic of Specific R&D Project, 1980-1991", Conference Proceedings of Korean Association of Science and Technology Studies, 2016.05, 129-145

two years. Additional science high schools continued to be established through the 1980s, and seven were in operation across the nation by the end of the 1980s.

A science and technology specialized education system was also established for students who graduated from science high school. Founded in 1971, KAIS, was merged with KIST in 1981 to become Korea Advanced Institute of S&T (KAIST). KAIST was integrated with the Korea Institute of Technology (KIT), which was established as a special science and engineering university in 1986, to operate the bachelor's program. Undergraduate courses at KAIST were also based on an independent law called the Korea Institute of Science and Technology Act, not the Higher Education Act, and it was under the jurisdiction of the Ministry of Science and Technology. KAIST is an educational institution equivalent to a university that grants bachelor's degrees, but unlike other universities, KAIST could have its own bachelor's programs, such as early graduation. KAIST was equipped with an unconventional scholarship support system. In addition, since KAIST was operated in conjunction with science high schools, science high school graduates were able to enter KAIST without national exams. The purpose of the entrance examination free was to allow science gifted students to study creatively and enter college early without the burden of entrance examination. All other universities could admit students only by means of national entrance examination under the Higher Education Act and all graduates from high schools could apply for KAIST only with the score of national entrance examination. Therefore, there were some complaints about the privilege for KAIST by universities. Since the establishment of KAIS, there has been continuous preferential treatment for educational institutions specializing in science and technology supervised by the Ministry of Science and Technology. Nevertheless, by emphasizing the national importance of science and technology, all these disputes over preferential treatment were allayed and the project was promoted.

KAIST's graduate programs continued to grow through the 1980s. After producing two PhDs in 1975, the number of established departments and the number of students for master's and doctoral programs continued to grow. As of February 1990, there were 372 bachelor's graduates, 532 master's graduates, and 168 doctoral graduates. In the same year, KAIST

alone produced approximately one-third of 468 PhDs in science and engineering from all universities in Korea. In addition, when looking at the number of students enrolled in the graduate school as of December 1991, the doctoral program had 1,536 students and the master's program numbered 1,174, showing an unusual composition, much more in doctoral programs.³⁵ Although general universities have a pyramid-shaped distribution of students in undergraduate, master's, and doctoral programs, KAIST shows a structure with a larger number of students as they go up to the upper level, revealing the aspect of a research-oriented university. Among KAIST graduates, there are many of the first generation of Korean venture startups in the 1990s as well as many leading personnels in corporate research institutes. This can be seen as a result of high research capacity, industry-university research, emphasis on entrepreneurship, and research related to cutting-edge technology. KAIST was positively evaluated as a pioneering model for research-oriented universities. As a result, the science and technology specialized education system for nurturing advanced science and technology manpower under the jurisdiction of the Ministry of Science and Technology, which leads to science high school, KAIST undergraduate program, and graduate school, was successfully completed.

In summary, even in the 1980s, the basic framework of Korea's science and technology manpower policy remained the same as before, and a balance between supply and demand was expected. Nevertheless, the role of companies in industrial development and technology development has increased, and the role of the government and government-funded research institutes have been relatively reduced. In particular, in the 1980s, as companies' R&D investment increased rapidly and technology became more advanced, companies created the largest demand for high-quality science and technology manpower. Therefore, unlike before, it became difficult to predict the demand for science and technology manpower by sector and level, and it became increasingly difficult to train and produce manpower accordingly. Nevertheless, thanks to the still high rate of economic growth, the industry continued to grow and the demand for manpower continued to rise accordingly, so the overall expansion of higher education was able to balance supply and

³⁵ 20 Year History of KAIST

demand to some extent. However, as the demand for science and technology manpower of companies, especially the demand for advanced R&D manpower with master's and doctoral level, is segmented by field and level, and there are many variables in business activities, forecasting becomes increasingly difficult. Unlike craftsmen or technicians, who can be trained in a short period of time, it takes several years to train advanced science and technology manpower at the master's and doctoral level, so it becomes difficult to respond to changes in demand forecasts. In the 1990s, these changes began to appear in earnest.

6.

In the 1990s: Nurturing Research-Oriented Universities and Fostering Excellent R&D Personnel

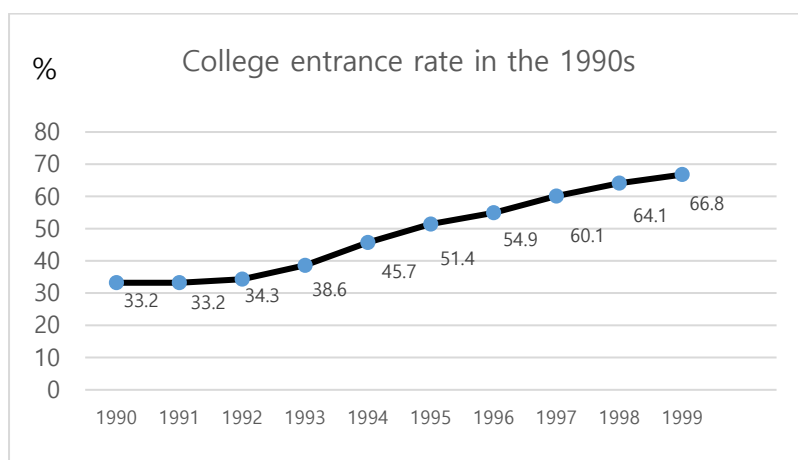
The 1990s marked a period of transition in Korea, while there has been a big worldwide wave of internationalization and informatization. Notably, informatization triggered industrial restructuring on a global scale. The global transition period of the industry serves as a time when a window of opportunity is open for developing countries in science and technology. This is because, unlike in existing industries, the technological divide between advanced and late-developed countries and the gap in accumulation of related capital and knowledge are relatively small. Korea was able to reach a level where it could compete with advanced countries by boldly seizing this opportunity in several fields such as semiconductors, information communication, and display. Accordingly, the science and technology policy of the 1990s was characterized by emphasizing R&D activities to advance into the high-tech field. The industry's motto of "We were late-comers for industrialization but are forging ahead with informatization" clearly expresses the spirit of science and technology policy in the 1990s.

Science and technology policies in the 1990s are broadly divided into large-scale national R&D projects and public R&D ones. Large-scale national R&D warranted establishing a cooperative system with companies and infrastructure for leading innovation in new industries, whereas R&D of a public nature required long-term investment. Through the 1980s, the company's R&D scale has grown significantly, and in some fields, such as memory semiconductors, it has secured a level of competitiveness that is leading the technology-chasing technology push. Therefore, the government's R&D investment focused on strategically promoting the R&D necessary to secure technological prowess in the emerging high value-added industry. Toward this end, a mid- to long-term technology prediction survey was conducted and a large-scale national R&D project was planned based on the results. The Leading Technology Development Project, which was launched in 1992, set R&D goals at the same level as seven advanced countries in high-tech industries such as

biotechnology and information & communication. In the emerging industrial field, it is impossible for companies to imitate or introduce technology. Therefore, in order to conduct homegrown technological development research and secure source technology from a long-term perspective, government-funded research institutes and universities had to undertake the public R&D projects and basic researches. In the science and technology workforce policy, under the principle of selection and concentration, the main task was to foster and support graduate schools for nurturing R&D workforce.

6.1 Track 1: Promoting the nurturing of graduate schools

In the 1990s, undergraduate and graduate programs still achieved rapid quantitative growth. As the regulation for establishing private universities were reduced, many new universities were established, and the number of university graduates increased accordingly. Underlying factors include an increase in the demand for higher education due to economic growth, a traditional passion for education, decrease in the gender gap in higher education, and increase in the demand for scientists and engineers rather than technicians in the labor market. Notably, the college entrance rate for high school graduates was about 30 percent in 1990, and has risen to about 67 percent by 1999. On the other hand, since the economic growth rate decreased compared to the high growth period of the 1970s and 1980s, the supply of undergraduate workforce was generally expected to be sufficient. Therefore, the main interest in nurturing a science and technology workforce through higher education institutions has changed to fostering graduate schools or improving undergraduate education. There may be some supply shortages depending on fields, but in most fields, it was no longer necessary to pay attention to the supply of scientists and engineers with bachelor's degrees.

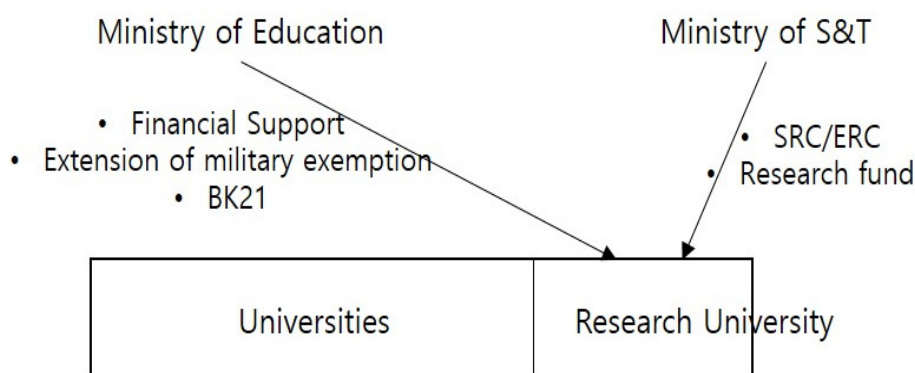


Domestic policies for fostering graduate schools were actively pursued. The Ministry of Education has already promoted policies to foster graduate schools during the 1980s. However, it became more serious in the 1990s. Leading universities in Korea have paid attention to fostering research universities as graduate schools. A lot of resources are required to operate and maintain a research university, and thus the principle of selection and concentration was introduced in the graduate school nurturing project. The Ministry of Education implemented several policies to support graduate schools, but the largest among them was the Brain Korea 21 (BK21) project. BK21 aimed to foster research universities with a focus on graduate schools for each academic field. Therefore, rather than a research project, BK21 is a support project to make the research environment of graduate schools equivalent to the level of advanced countries, and direct support for students participating in the project took up a large proportion. In other words, similar to graduate schools of research universities in the U.S., it provided fellowships for master's and doctoral degree students and post-doctoral researchers, so that they can complete the graduate program in a stable condition. This project was intended to prevent top-notch human resources from giving up or choosing to study abroad due to economic problems and to revitalize domestic graduate schools. However, since BK21 was started just in 1999, its performance would materialize in earnest in the 2000s.

Military service exemptions for R&D personnel were also reorganized to accommodate graduate school development. The 'master's degree lieutenant' system was abolished, and a new specialized research human resources system was introduced. It replaced military

service by attending a doctoral program at a designated graduate school or working at a designated university research institute after completing a master's degree. The system to exempt military service for those with master's degrees working in designated private and public research institutes was working already in the 1980s. It was reformed in the early 1990s to approve domestic graduate schools and university research institutes as designated. In other words, master's degree holders could attend a doctoral program and PhD degree holders can work as technical research personnel at a research institute affiliated with a university. In the case of excellent science and engineering graduate schools, students can expect research infrastructure, fellowships, research grants, research achievements, and military service exemptions. As a result, top-talent students, especially male students, had a strong motive for entering a domestic science and engineering graduate school.

[Figure 5-3] 1990s Two track policy for research university in Korea



6.2 Track 2: Launching a large-scale long-term R&D project to foster graduate schools

The Ministry of Science and Technology has chosen the principle of selection and concentration to effectively use its limited R&D resources. Rather than nurturing all graduate schools evenly, the ministry has chosen the option of intensively supporting the best-qualified groups for each major research topic. Through this, it was expected that the selected Korean universities would grow to a level where they could compete with research

universities in advanced countries. This project was carried out in two ways of increasing short-term research funding for individuals or small research teams, and supporting large-scale long-term group research. The small-scale research project allowed many research projects to be carried out at universities, thus creating a grassroots foundation for university research and broadly forming the minimum material basis for guiding master's and doctoral students.

However, it was the large-scale, long-term group research project that promoted research and development of universities in a more long-term, fundamental way. A prime example is the Excellent Research Center Project. This project was conducted in two ways through the Science Research Center (SRC), which aims for in-depth basic research, and the Engineering Research Center (ERC), which aims to promote technological development in the high-tech field. The method was to intensively support research funds for 9 years, while several universities were able to form a consortium to participate in SRC and ERC. Thanks to this, the hosting university was able to establish a research foundation by equipping research facilities comparable to those of world-class universities and expanding faculty, and other participating universities were able to conduct research by leveraging shared infrastructure.

The achievements of the joint SRC/ ERC project in the 1990s are described as follows. Starting with 13 in 1992, and by 2002, 43 SRCs and 57 ERCs, a total of 100 excellent research centers were designated. The average annual cost of each center was about KRW 1.2 billion. A total of 38,308 papers were published through this project for 10 years until 2001, of which 19,493 were published in overseas journals, and SCI papers accounted for about 47 percent of the total. There were 2,606 patent applications and 914 patent registrations. 23,788 master's and 17,584 doctoral students participated in this project, while 14,229 master's and 3,804 doctoral students were produced through this project. This project nurtured a master's and doctoral degree-level workforce and also served as a source of demand for utilizing them. The number of industry-university joint research has also grown to 3,167 cases.³⁶ The joint SRC/ ERC project also served as a policy tool for the Ministry of

36 An Analysis on the Performance of SRC/ERC Phased-out, Kistep, 2001.

Science and Technology to indirectly intervene in fostering a top-notch science and technology workforce in domestic graduate schools by means of research projects.

In summary, in the 1990s, the science and technology workforce policy shifted away from the previous quantitative supply-demand balance and focused on the qualitative aspect. While the scale of human resources fostering at junior colleges and universities has grown sufficiently due to the intensive support until the 1980s, the production of advanced manpower with master's and doctoral degrees for new technologies has been insufficient. It was after the mid-1980s that KAIST, which had the best research conditions among domestic universities, began to promote doctoral programs. In the 1990s, by benchmarking research universities in the U.S., the Ministry of Science and Technology devised a strategy to grow graduate schools through R&D investment in universities under the principle of selection and concentration. Through a large-scale long-term R&D project such as the ERC/SRC project, the effects of university's research infrastructure, fellowship, for master's and doctoral students and researchers, forming a research group, and excellent research results were intensively produced in a short period of time. Military service, a unique factor among Korea's science and engineering manpower support systems, was also reorganized aligned with these graduate school-oriented policies. Since then, the importance of nurturing an undergraduate workforce in Korea's science and technology workforce policy has sharply declined, especially in the policies promoted by the Ministry of Science and Technology.

7.

Paradigm shift in the 2000s: from quantity to quality, from nurturing to utilization

7.1 New tasks for science and technology manpower policy in transition to post catch-up period

Korea's science and technology policy in the 2000s was influenced by the global environment as well as the unique experiences of Korean society. In the global environment, neoliberalism emerged, technological innovation and R&D were emphasized in connection with the transition to a knowledge-based economy, global workforce mobility and access to information increased, and emerging technology-chasing countries such as BRICs grew rapidly. At the same time, in the process of overcoming the financial crisis of the late 1990s, Korean society accelerated market reforms, while going through industrial restructuring, increasing labor market flexibility, and expanding the roles of companies and private sector. In addition, as it entered the stage of establishing a new national innovation system suitable for the post-catch-up system, it began to redefine the roles of the government, public research institutes, companies, and universities.

In this environment, science and technology workforce policy has a different task than before. Firstly, the effectiveness of the training policy based on the demand forecast for manpower fell. This was because the role and proportion of companies in industrial development and investment in science and technology R&D has increased. With the industrial structure being reorganized at a rapid pace, the number of variables affecting the demand forecast by sector and level has increased and the accuracy has decreased. In addition, as a result of expanding higher education until the 1990s, it was predicted that the overall scale of nurturing scientists and engineers would be slightly larger or balanced between supply and demand.³⁷ This is because the college entrance rate had already reached 70 percent in 2000, and has risen to over 80 percent through the 2010s. However,

37 Long-term Forecast of Supply and Demand of Human Resources in S&T(2005-2014) and Preliminary Survey on the Status of Graduation in Science and Engineering (KISTEP 2005)

according to the 10-year long-term supply and demand forecast established in 2005, a manpower shortage for master's and doctoral degree holders in some rapidly growing fields is expected to occur.

Secondly, it was important to promote industry demand-oriented education with a focus on quality rather than quantity as the training scale was expected to meet the demand. Notably, from the late 1990s, even though the training scale was already sufficient, companies began to experience a qualitative discrepancy in which they had difficulties in recruiting. This was due to the fact that the departments and curriculums installed in universities did not change dynamically to the extent of industrial restructuring and as a result, the demand for manpower in the new technology field increased rapidly.

Thirdly, it has become important to nurture or secure world-class science and technology workforce with creativity and global competitiveness to meet the R&D demands of the post-catch-up phase. Fourthly, as the aging population gradually accelerates, the need to effectively utilize the capabilities of the national science and technology workforce in the long-term perspective has been raised. A highly skilled science and technology workforce requires a lot of time and investment to nurture. Therefore, it is necessary to support each career path stage to allow their careers to be uninterrupted and to pay policy attention to the female science and technology workforce, who have relatively less progress in the science and technology field.

7.2 Track 1: University's science and technology education innovation policy

Firstly, the key to S&T-oriented policies, in which the policy perspective has changed from quantity to quality, is to match the demands of companies or industries with the capabilities of human resources nurtured by universities. Toward this end, various educational innovation support projects are being promoted. The introduction of the Accreditation Board for Engineering Education of Korea is one of the policy attempts to change the university's major curriculum to be company and industry demand-oriented. The

engineering education accreditation system has benchmarked the U.S. Accreditation Board for Engineering and Technology. It is a system in which technical skills training courses are operated and accredited, so that when they enter the industrial field after graduation, they can be put into practice without additional training. The engineering education accreditation system was introduced in Korea as a pilot program in 2001 and has since been expanded to many universities. Although the Accreditation Board for Engineering Education of Korea is a private institution, it is an accreditation institution designated by the government and receives budget subsidies.

The Ministry of Education promoted a policy to strengthen industry-education sector cooperation in order to innovate science and technology education more broadly. A representative project is the Leaders in Industry-University Cooperation (LINC). The purpose of the project was to develop various industry-academic cooperation leading models and through this, the education and culture of universities were reorganized into an industry-academic close cooperation structure, thereby promoting the win-win development of local universities and local industries. In consideration of the industrial characteristics of universities and colleges distributed across the country, they were divided into two types specified as technological innovation and site-based types.

Since the technological innovation type focuses on the development and commercialization of source innovative technology, the business is organized centered on the undergraduate school, but the graduate school is operated in a combined form. On the other hand, the site-based type aims to nurture a highly skilled workforce tailored to the field manager and provides support mainly for undergraduate students. In the site-based type, the practice system linked with the company is activated, and in the case of long-term practice, the demands of the industrial field are reflected in education and networks. This project is supported every five years, and the competition for financial expansion of universities is fierce due to the large scale of support. The LINC project is being divided into more subdivided types after the first project has been completed. It has not been accurately analyzed whether the LINC project has positively increased the employment rate of graduates as a result of strengthening industry-university cooperation. However, through

this project, it is pointed out that each university has the effect of building an infrastructure and platform, such as building a human network and developing a convergence curriculum that can bolster win-win tie-ups with companies.³⁸

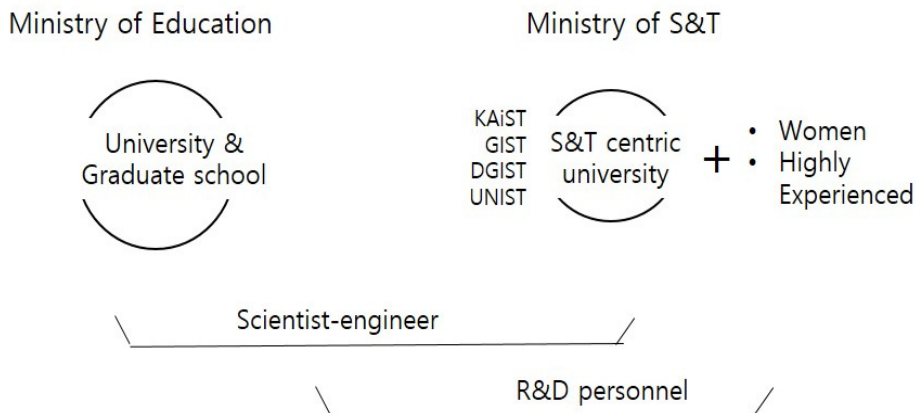
7.3 Track 2-1: Expansion of science and technology specialized universities

In order to nurture creative, high-level science and technology manpower that is industrial-oriented and required in the post catch-up stage, the university specialized in science and engineering, modeled on KAIST, has been expanded. Education in the age of catch-up focused on learning existing technologies and developing problem-solving abilities. This is because the main goal was to increase the ability to learn and digest introducing technologies developed and established in developed countries. However, in the post catch-up stage, independent technological innovation is required, so problem-setting ability is newly required. In other words, the ability to discover what to learn and the research capacity by oneself is additionally needed. To this end, the government additionally established a university specializing in science and technology that was modeled on KAIST but linked with the local science and technology innovation policy. In 1995, the Gwangju Institute of Science & Technology (GIST) was established as a graduate university, and an undergraduate program was added in 2010. Daegu Gyeongbuk Institute of S&T (DGIST) started out as a government-funded research institute, but strengthened its educational function by establishing undergraduate programs in 2008 and master's and doctoral programs in 2011. Ulsan Institute of S&T started as the National University of Science and Technology, Ulsan National University of Science and Technology, in 2009, but was converted to UNIST in 2015. Like KAIST, it operates a degree program like a university, but operates a characteristic curriculum such as R&D education and entrepreneurship orientation that emphasizes industry-university cooperation and problem-setting ability for undergraduate students. However, compared to KAIST, it is composed mainly of specialized

38 Lee, Sangmi et al., "The Analysis of the Effectiveness of LINC Projects", Korean Journal of Policy Analysis and Evaluation, vol.26no.4, 27-49.

majors related to the local industry near the university. The research facilities are excellent, financial support such as scholarships is abundant, and the graduate school is a designated institution for specialized research personnel, so it has good conditions to secure military service exemption, so it is highly preferred by excellent students.

[Figure 5-4] Two track frame of S&T manpower training policy in Korea



7.4 Track 2-2: Expanding the target of science and technology manpower policy

One of the new social agendas that emerged in Korea in the 2000s was to prepare for the rapid aging population. It is a common phenomenon that life expectancy increases and the fertility rate decreases with economic development. However, in the case of Korea, the aging of the population progressed compressively and rapidly as much as the compressed growth in the economy and industrialization. In 1955, before industrialization, the proportion of the population aged 65 and over was 3.3%, but it steadily increased to 3.9% in 1985, 5.12% in 1990, and reached 7% in 1999, becoming an aging society. Already in the early 2000s, Korea had to prepare for the possibility of a shortage of science and technology manpower in the long term, though not immediately, as it was predicted that the elderly population would reach 14.4% in 2019. So, instead of establishing the manpower nurturing policy in the past, the framework of the policy turned to nurturing and utilizing scientists and engineers throughout the entire career cycle. A support system has been established to prevent career interruptions that occur after completing the training course and before

finding a stable job in the R&D field. In form of research grant, It supported labor costs for doctoral and postdoctoral researchers, expanded training plans, and support young researchers in the early stages of their careers. Also, in preparation for the retirement of a large number of scientists and engineers of the baby boom generation (born between 1955 and 1974) in Korea, a support project was created so that the experience and know-how of highly experienced scientists and engineers could continue to be utilized. Projects such as consulting, education, and mentoring for small and medium-sized enterprises were promoted.

Another new policy for science and technology manpower in the 2000s was a support policy aimed at promoting the cultivation and utilization of female scientists and engineers. In the early days of industrialization, female students received the primary education at most, and the rate of entry into secondary education was lower than that of male students. Along with economic growth and the decrease in the number of children, the educational level of women rose rapidly. By the end of the 1980s, the female high school graduation rate was over 80%, but the college entrance rate was around 30%, which was 5-7% lower than that of male students. Since the 1990s, the female college entrance rate has grown rapidly, and the gap between male and female college entrance rates has also narrowed. In relation to science and technology, the problem of female workers appeared in two aspects. First, compared to the rate of women entering higher education, the rate of entering science and engineering, especially engineering with high demand for manpower, was low. Second, the employment rate of women in science and engineering fields was lower than that of men, and there were many career interruptions. This is a waste of investment in education both for individual and the national point of view. In addition, there were demands from Korean civil society in 2000 to increase women's rights and expand social advancement.³⁹

Since 2004, the Korean government has been promoting a policy of fostering women to expand their entry into the science and technology and support each career stage to prevent career interruption. Its main projects are support measures to promote more

39 Lee, Eunkyong, "Boundary Agenda between Gender Equality and Human Resource: The Establishment of Policy for Women in Science and Technology in Korea", *Asian Women*(2009), 25-2, 29-47.

women into fields with high demand for manpower, a recruitment target system to lower entry barriers to R&D sites, support for work-family balance to prevent career interruptions, and return of career-interrupted female scientists and engineers. As the result of the policies, there has been some advancement in the number of women in science and engineering, especially in the engineering, and the increase in the ratio of women in the R&D workforce.

7.4.1 Remaining tasks

Despite many new attempts and achievements, Korea's science and technology manpower policy has several challenges or weaknesses. First, science and technology manpower is being nurtured appropriately in quantity, but in terms of quality, more progress is needed considering the level required for the post catch-up stage. Despite policies such as selection and concentration, fostering of research universities, establishment of universities specializing in science and technology, and strengthening of industry-university cooperation, the international competitiveness of leading domestic universities is in the lower group among OECD countries. Second, policies from the global perspective should be promoted more actively. As the global workforce moves more actively, Korea faces the possibility of another brain drain. During the technology catch-up phase, there were many kinds of special support for science and technology manpower. However, since the 2000s, in a situation where global competition to secure excellent science and technology manpower has intensified, the former advantages are not likely as much effective as before. As the mandatory service period continues to be shortened, the merits of military service exemption like Technical Research Personnel for three years. It can be benefit only if master's graduates enter a doctoral program. The fact that the compensation for domestic science and technology manpower is relatively low compared to other competitive professions such as finance and medicine also makes it difficult to secure excellent manpower in the field of science and technology. In terms of policies for attracting, nurturing, and utilizing excellent foreign students and R&D personnel, Korea seems to be less competitive compared to other Asian countries such as Singapore..

Lastly, there are still many challenges in the area of converting the entire higher education

curriculum established under technology catch-up period to system suitable for transition toward post catch-up period. Since the college entrance examination focuses on selecting students, elementary and secondary education are regarded as the preparation for concern the entrance examination. So, during the K-12 education, students get used to solve the problems with right answer following the established process. By the way, the higher education at colleges is trying to transform into a new kind of education to stimulate and encourage students' creative challenges. However, it is not easy task to achieve yet.

8.

Implications for the Policy of Fostering Scientific and Technological Manpower in Lao PDR

Korea's science and technology workforce policy had different paradigms for each period of catch-up and transition to post catch-up. During the catch-up phase, Korea's policy to nurture a science and technology workforce was a policy to balance supply and demand based on the development plan. Because it was state-led economic development, demand forecasting was relatively easy. In addition, supply adjustment was possible because all educational institutions were under the government's support and control. It was enough to learn and acquire established technologies rather than creating new ones. Because there was a high demand for a science and technology workforce with low skill levels, it was possible to foster and supply large-scale manpower in a short period of time. This is one of the important factors that made it possible to promote compressed industrialization and economic growth during the catch-up phase.

It is important to note that the public responded to the government's supply-demand balance policy actively to make it effective. This is because the government's strategy to leverage Korea's strong value placed on education and top-talent manpower being drawn to the science and technology field. Among them, there was the specialization strategy that provided unprecedented support to institutions established outside the higher education system rather than support for the general higher education system. Attracting top-notch expatriate scientists and engineers, high salaries, a good research environment, and military service exemptions were key elements of the specialization strategy. These factors made the whole nation share the power of science and technology for the Korea's national development.

The current situation in Lao PDR has something in common with Korea since it has been through the catch-up phase. Lao PDR is an agricultural country with poor infrastructure for industry, science and technology, and education. The Lao government can play a leading role in overall economic policy and science & technology policy, and the government and people

have a strong drive to achieve industrialization and science & technology advancement. These common factors can be advantageous when the Lao government in the catch-up phase benchmarks Korea's national science and technology-oriented policy. On the other hand, it is not clear whether Lao PDR also has factors that facilitated the development of Korea, such as public's strong value placed on education, social expectations of science and technology as the ticket to economic prosperity, and military service exemptions that lured top-talent students to the science and technology field. If there are similar factors, Lao PDR can avail itself of them, and if not, Lao PDR needs to identify its own localized social factors.

Based on these assumptions, it is possible to identify following policy implications from experience of Korea.

Firstly, during the catch-up phase, science and technology workforce with high demand can be produced on a large scale through general public education. If the basic education infrastructure is established and foundation for elementary and secondary education is solidified, it can be used as a basis for nurturing a science and technology workforce.

Secondly, it is necessary to train and utilize the small-scale elite science and technology workforce ICT, energy, health care, and national strategic industries. Toward this end, Lao society can utilize meaningful and socially symbolic elements and use unconventional incentives and support strategies. Then It can provide expertise in establishing national science and technology policies and planning activities, lay the foundation for nurturing.



CHAPTER 6

STI Higher Education and Universities in Lao PDR

National Master Plan for STI Research & Development
and Application in Lao PDR 2020-2022

Chapter 6. STI Higher Education and Universities in Lao PDR

1. The Role of the University for National Economic Development based on STI in Lao PDR

1.1 Overview: STI & National Development

The science and technology such as big data, IoT (Internet of things), and AI (artificial intelligence), are important factors of economic and social development in the 21st century. The 1st Industrial Revolution occurred through the steam engine and brought about tremendous changes in the economy and society. Since then, it has brought about rapid changes and development through electricity energy and computer, and it is now called the 4th Industrial Revolution. Currently, exchanges and cooperation of technology and knowledge are active through trade and international cooperation. Anyone can use science and technology as much as possible. National development has been made easier by science and technology. The socio-economic development of nations is fundamentally an education process, in which people learn to create new institutions, utilize new technologies, cope with their environment, and alter their patterns of behavior (Adams, D, 2002). One of the most important factors of the development of science and technology is human resources development (HRD). Human resources are nurtured through research institutes and industries based on higher education. In particular, universities play an important role as suppliers. It is important to educate the workforce in certain fields with certain curriculums of universities.

Education in a broad sense improves the capabilities of individuals and capacity of institutions, and becomes a catalyst for closely interrelated economic, social, cultural, and demographic changes that represent the course of national development (Adams, D, 2002).

From this point of view, education, science and technology are very closely linked to national development. Most of the human resources development in the science and technology sector takes place at universities. The development of science and technology has a very important relationship with the role of universities. And basic scientific research and technological development have played a crucial role in the modern economic growth experience of contemporary developed countries (Todaro, M. et, 2011).

OECD points out the universities play an important role in national development. In the globalizing knowledge economy institutions of higher education and research have a pivotal role to play. They are now portrayed as vital sources of knowledge, seedbeds of innovation, and engines of growth, making major contributions to the economic, social, and cultural development of their societies (OECD, 2005). In particular, OECD emphasized that university-led knowledge creation and technology transfers are essential for enhancing the competitiveness of both regions and countries.

In the role of universities in the 21st century, UNESCO approaches in terms of building cooperative partnerships between universities and local industries. As a partner of industry-university cooperation, universities lead regional and national development, which is a result of internationalization. Changes in economy and technology increase the need for learning (UNESCO, 2005). In conclusion, it increases the importance of knowledge, while education for the knowledge economy is conducted through universities. Since 2000, the paradigm of the knowledge economy was introduced, in which it is more important for the role of universities to educate for acquiring, managing, and applying knowledge.

According to the World Bank (2002), knowledge accumulation and application have become major factors in economic development and are increasingly at the core of a country's competitive advantage in the global economy. The innovations of science and technology raise national productivity, and most of these innovations are the product of research and development (R&D) activities undertaken in universities. Progress of the environment, health & wellbeing, and industry sectors, in particular, is mainly dependent on continuing innovations. Tertiary education institutions have a crucial role in supporting knowledge-driven economic growth strategies and building well-functioning democratic, socially

cohesive societies (Teferra, D, 2016). The roles of universities in the 21st century are defined as 1) response to changing education and training needs, 2) adaptation to rapid change of higher education from internationalization, and 3) contribution to regional economy in located universities. This concept has been valid since the beginning of the knowledge economy society. In particular, recently, the World Bank is taking a new look at the role of the university in the prolonged pandemic. Higher education is instrumental in fostering growth, reducing poverty, and boosting shared prosperity. In a word, higher education is recognized as an indispensable element for economic and social development. Higher education is a fundamental requirement for employment in the industries and provides opportunities for individual development. A tertiary education system plays role of national economy for providing the goods and commodity in a society. Also, tertiary education offers human capitals for social mobility and economic development through professional skills and research⁴⁰.

Teaching and learning activities in university are shifting from classroom-based knowledge transfers to interactive experiential, task-oriented learning activities. Moreover, with pandemic restrictions, a new normal has emerged with online learning. It is enabling non-face-to-face learning, discussions, and evaluations online through real-time video conferencing and lectures. This is a learning transformation in higher education and also contributes to resolve the digital divide as various online contents. Moreover, there are various changes such as student mobility, exchange programs, and transferring credits through internationalization of higher education.

Tertiary institutions often constitute the backbone of a national information infrastructure in their role as repositories and conduits of information (through libraries, databases, etc.), computer network hosts, and internet service providers (Van Weert, Tom J, Kendall, Mike, 2004). In addition, the norms, values, attitudes, and ethics that tertiary institutions impart to students are the foundation of the human capital necessary for constructing sound civil societies and cohesive cultures—the very basis of good governance and democratic political

40 <https://www.worldbank.org/en/topic/tertiaryeducation#1>

systems (Harrison & Huntington, 2000)

Universities are devoted to assisting their respective country's challenges, such as research and development. The research provides invaluable knowledge, insights, and local recommendations for policy formulation, as well as developing and turning out the relevant skills and knowledge (F-K, Eric, 2021) for the times. In the global arena, it changes the demands of social life and develops the science and technology field. The university contributes to national development by nurturing human resources in science and technology. Responding to the impact of internationalization on delivering education and critical adaptability to the needs of economic and social life, higher education institutions (HEI) effectiveness depends on the use of technology and contribution to national development (Magzan, Masa et, 2011).

The purpose of this paper seeks the role of university for the national development of Lao PDR. Toward this end, the role of university was reviewed for a focus on national development driven by science, technology, and innovation (STI). Lao PDR has multiple features of industrial structure, export items, and population composition. About 70 percent of the Lao population is still engaged in agriculture. The main export industries are mining and hydropower, mainly foreign-owned with many of the high-skilled positions filled by foreigners. It is very important for the role of university to develop science and technology. Lao PDR has five public universities, but there is a shortage of research manpower. A key reason for the low quality of higher education in Lao PDR is the lack of qualified academic experts with advanced degrees.

For this study, it reviewed the status of economic social development, science, technology, & innovation, and higher education institutes. Also, it summarized the case of Korea's development experience in terms of economic development and science & technology development. Finally, from the teachable lessons of Korea's experience, it proposes the development of science & technology, national development, and the role of universities. However, there was notably a limitation of the fields of data collection & analysis and science & technology in Lao PDR.

2. The Analysis of Status of Lao PDR

2.1 National Development Plan in Lao PDR

Lao PDR has established a five-year national development plan aligned with the direction of the 11th Party Congress and 9th Five-Year National Socio-Economic Development Plan (NSEDP). The one of the priority tasks in the 9th NSEDP is related to human resources. It is very important to develop human resources and management for national development. Lao PDR has established and implemented a national manpower development strategy for economic social development. National socio-economic development of Lao PDR is implemented with the 'Vision for 2030' as the basic direction. Based on the national characteristics and situation along with the international environment and national development priorities, the 'Vision for 2030' contains elements of the main themes of 'upper-middle income', 'innovative, green and sustainable economic growth', 'strong basic infrastructure system', 'socialist market economy', 'social justice, peace, & order', 'people's livelihoods improvement and solidarity promotion', 'improvement of disparities between urban and rural segments', 'improvement human development', 'people's rights protection under the effective rule of law', 'the administrative system enhancement by the 3-builds directive', 'environmental protection', 'political stability and strength', 'toward regional and international integration' (Lao PDR, 2016). Lao PDR aims to achieve middle-income country status by 2030 through political stability, development of economy and society. Toward this end, she has carried out human resources development, bridging the divide between an urban and rural areas and promoting natural resource protection & international cooperation.

The Lao government set up its '10-year socio-economic development strategy (2016–2025)' for achieving its 'Vision for 2030'. Its 10-year socio-economic development strategy (2016–2025) consists of seven strategies as follows:

- i. Strategy on quality, inclusive, stable, sustainable and green economic growth
- ii. Strategy on LDC graduation by 2020 and progress on the SDGs

- iii. Strategy on human development
- iv. Strategy on sustainable and green environment with effective and efficient use of the natural resources
- v. Strategy to enhance government's role in social management under the effective rule of law
- vi. Strategy on regional and international integration
- vii. Strategy on industrialization and modernization.

One of 7 strategies is human resource development. National HRD strategy in Lao PDR has carried out since 2015. It called 'The Human Resource Development (HRD) Strategy to 2025'. It should be revised to make it up-to-date, practical and in line with the new phase of political duties as well as the National Socio-Economic Development Plan so that it can be used as the key reference for a harmonized human resource development nationwide. Likewise, it is covering all aspects of HRD in large, complex sectors such as education and health. And it was to focus on two areas: (1) HRD interventions that require multi-sectoral cooperation, and (2) cross-cutting themes that support HRD in multiple sectors. The scope of the draft strategic framework covers the six areas as follows:

- i. Political/ideological development
- ii. Civil service
- iii. Education and training
- iv. Health and nutrition
- v. Labor and social welfare
- vi. Information, culture, tourism and sport

Currently, Lao PDR is carrying out detailed strategies for education and training, health and nutrition, and information and technology closely related to science and technology.

2.2 STI (Science, Technology and Innovation)

Recently, STI in Lao PDR has changed in terms of governance. Originally, there was a research institute on science and technology under the Prime Minister's Office, and then

the Ministry of Science and Technology was established by combining this institute and several functions of each Ministry. In 2021, the government has restructured the organizations of ministries, Ministry of Science and Technology (MOST) has relocated with several ministries. Ministry of Technology and Communications (MTC) is a governmental institution at the central level, acts as a secretary to the government and executes at a macro level on the post, telecom, information communication technology, technology and innovation throughout the country. MTC played a role of main body of technology and innovation (T&I). The Department of Technology and Innovation (DTI) under the MTC, which has been recognized, is responsible for managing T & I in Lao PDR. And the Ministry of Education & Sports (MOES) took science and R & D. MOES is responsible for the implementation of the policies on science, R&D in terms of long-term strategy. National Science Council (NSC) is under the MOES, has a mandate for advising and managing R&D institutes. NSC might discuss long-term strategies in terms of research and HRD in science sector.

Lao PDR have five universities, such as National University of Lao PDR (NUOL) in Vientiane, capital city in Lao PDR, Souphanouvong University (SU) in Luang Prabang, Champasak University (CU) in Pakse, Savanakheth University (SKU) in Kaysone Phomviharn, and University of Health Science (UHS) in Vientiane. Only UHS belongs to Ministry of Health, and others belong to Ministry of Education & Sports. Line Ministries is established institutes / centers/ council relating to the research, development, application and management of science, technology and innovation for serving into their own activities. For example, National Agriculture & Forestry Research Institute (NAFRI) belongs to Ministry of Agriculture & Forestry. NAFRI is mandated to undertake integrated agriculture, forestry and fisheries research in order to provide technical information, norms and results which help to formulate strategy in accordance with government policies. NAFRI has four main functions including: carrying out adaptive research, developing methods, tools and information packages, providing policy feedback, and coordinating and managing research. Task that NAFRI is involved in including land-use planning and assessments of changes in forest cover

and biomass⁴¹.

National Strategy for STI Development in Lao PDR is to strengthen the national information system (NIS), to promote R&D, technology, innovation and creativity. The STI policy outlined consisted of five main goals (Lee, JH, Maliphol, S, HW Kang, 2014): i) to improve R&D to ensure the mobilization of STI resources, ii) to upgrade higher education for STI knowledge aimed to serve industrial, agricultural and services development contributing to the poverty reduction and to the modernization and the industrialization of the country, iii) to strengthen the cooperation and coordination between sectors and research institutions to ensure the participation of the production sectors and scientific community for maximizing the country's benefit from S&T, iv) to upgrade and strengthen the public's STI and engineering knowledge, v) To promote essentially the adaptation and technological innovation system.

2.3 Higher Education and Human Resources in Science & Technology

The human resources in universities, mainly, consist of bachelor and master degree, so it is low quality of research and teaching. Even though the target of degree in university is to Ph.D. 10: Master 40: Bachelor 50, it could not achieve. In 2015, 45.2% of staff in the National University of Lao PDR (NUOL) had a postgraduate degree (6.7% had doctoral degree and 38.5% had master's degree) and the remaining 54.8% had bachelor's degree.

In the academic year 2018-2019, MOES still provides scholarships to talented and marginalized students by 20% share of the total enrollment, whereas 80% have to pass the entrance exam and pay the tuition fee. There were 12,258 students enrolled at the four universities under MOES and one university under Ministry of Public Health adding to a total number of 39,497 students in 2018-2019. Based on UNESCO's classification, the new entrants were grouped as follows: 1) Education and Sports 1,126; 2) Language and

⁴¹ <https://www.landportal.org/organization/national-agriculture-and-forestry-research-institute-lao>

Humanity 1.050; 3) Social Sciences 768; 4) Business, Management and Laws 3.570; 5) IT 118; 6) Sciences 593; 7) Engineering 2.653; 8) Agriculture and Forestry 1.677; 9) Health and Social Welfare 592; and 10) Services 111. In the meantime, based on the same classification model the number of graduates in 2018- 2019 were grouped as follows: 1) Education and Sports 1.165; 2) Language and Humanity 592; 3) Social Sciences 685; 4) Business, Management and Laws 2.601; 5) IT 74; 6) Sciences 480; 7) Engineering 1.890; 8) Agriculture and Forestry 985; 9) Health and Social Welfare 219; and 10) Services 89.⁴²

As above, there are five universities in Lao PDR. The National University of Lao PDR (NUOL) came about as a result of the national policy to merge all higher educational institutions together in order to standardize its higher educational component in 1995. It now consists of 13 faculties, 11 offices, two institutes and five centers. The numbers of students enrolled and graduated have significantly increased since its foundation. Currently, there are a total of 23,861 students with 11,425 female students and 758 foreign students. It is well known for its diversity programs in 153 Bachelor Programs, 66 Master Programs and 6 Doctoral Programs. The total number of staff is 1,762 with 823 females, teaching staff is 1,444 with 636 females, and supporting staff is 318 with 187 females. Among teaching staff has 186 Ph.D. with 41 females, 965 masters with 408 female, 568 bachelors with 343 female as below table. The ratio of Ph.D. master, and bachelor is 11:56:33.

<Table 6-1> Number of Degree of Staff in NUOL

Content	PhD	Master	Associate Degree	Bachelor	Professor	Assoc. Prof.
Teaching Staff	186 (41)	956 (408)	4 (0)	568 (343)	7 (1)	89 (11)

Souphanovong University (SU) has established in 2003, located in Luang Prabang, Lao PDR, and the University is named after Prince Souphanouvong, who was the 1st President of the Lao PDR. SU is one national university of Lao PDR to serve higher education for 8 provinces

⁴² Ministry of Education and Sport, Education and Sports Sector Performance Annual Report 2018-2019 And Development Plan for 2019-2020

in northern part of Lao PDR. SU seems to be a symbolic institution of Korea in Lao PDR, because Lao PDR had received an ODA fund from the Korean government to establish the new campus in 2007. Specially, SU is symbol of closely relationship between Lao PDR and Korea, in 25th anniversary diplomatic relations. The organization structure of SU is one president and 2 Vice presidents. The 9 office's administrative compose of the offices of general administration, finance, academic affair, audit, organization and personnel, student affairs, construction and service, international cooperation, research and development services, and library and ICT. SU has 6 faculties and 34 programs in faculties including 2 master degree programs, about 3,500 students (34 students are Master degree) and 300 lectures. Ph.D. has 10, master degree got 161 staffs and lectures, and others have bachelor and high diploma. The ration of Ph.D., master, and bachelor is to 3:53:44.

<Table 6-2> Lecturers by Academic Degree and Faculties at SU (2018-2019)

No.	Faculties	Doctoral		Master		Bachelor		Diploma	
		T	F	T	F	T	F	T	F
1.	Education	1	0	29	4	39	18	0	0
2.	Agriculture and Forestry	3	0	38	8	17	7	0	0
3.	Economic and Tourism	2	1	34	11	23	6	0	0
4.	Architecture	1	0	14	2	18	6	0	0
5.	Engineering	2	0	27	2	20	3	1	0
6.	Languages	1	0	19	8	19	7	0	0
Total		10	1	161	35	136	47	1	0

Source: SU (AY: 2018-2019)

The 6 faculties consist of education, 6 bachelor degree programs: Lao language and literacy, English, mathematics, physical, political science, and chemistry, the faculty of economics and tourism has 8 programs: economics, international business management, general business management, IT business management, tourism management, hotel management, finance and banking, and accounting, the faculty of agriculture & forest resources offers 5 programs: plant science, animal science, forest resources, food science and technology, and ecotourism, the faculty of engineering has 6 programs: electrical engineering, computer

engineering, civil engineering, material engineering, logistic technology, and electronic and communication, the faculty of architecture offers 3 programs: architecture, interior design and Lao architecture, the faculty of languages has 4 programs: Lao language, general English language, Korean language, Vietnamese language. SU also has some institute and centers such as Confucian institute, Korean language center and Lao-Korea business incubating center.

Champassak university (CU) has established in 2002, located in Pakse, Champassak province, southern of Lao PDR. CU has Bachelor's degree has a 4-year course and 27 programs. The Associate degree has a 3-year course and 7 programs. The continuing program for bachelor's degree has a 2-year course and 13 programs. Ph.D. has 20, master degree got 153 staffs and lectures, and 258 have bachelor and high diploma. The ration of Ph.D., master, and bachelor is to 5:35:60.

<Table 6-3> Status on Staff in Champasak University

No.	Department/Faculties	Doctoral		Master		Bachelor		Total	
		T	F	T	F	T	F	T	F
1.	Cabinet	5	2	25	10	77	38	107	50
2.	Education	3	0	34	12	39	16	76	28
3.	Agriculture and Forestry	5	3	23	5	36	17	64	25
4.	Economic and Business	1	1	24	6	38	20	63	27
5.	Natural Science	2	0	20	7	23	9	45	16
6.	Engineering	4	0	14	2	14	3	32	5
7.	Law and Political Science	0	0	13	4	31	12	44	16
Total		20	6	153	46	258	115	431	167

Source: CU (AY: 2018-2019)

Champassak University has focused on agriculture and will be a center of excellence in Agriculture. Importantly, it also has 3 demonstration centers for agricultural experiments with a total land area of 65.8 hectares, such as cattle, pigs, goat, poultry, agronomy and wild-life. And the Centre Laboratory of the University is playing a vital role for scientific

examination on food contamination as well as agricultural product contamination, for instances pesticide residues and chemical residues. The Faculty of Economics is also famous for the evaluation of environmental and recreational services (UNESCO, 2018).

Savannakhet University (SKU) has established in 2009. SKU has academic and administration staff: 321 persons (148 women) and training staff: 43 persons (20 women). The distribution according to the educational level and gender is the following: 4 persons with Ph.D. (0 women); 23 persons with M.S. 23 (8 women); 133 persons with B.S. (44 women); 53 persons with Diploma (43 women), 16 persons with Middle Diploma (15 women) and 7 persons with First state levels (6 women). The total number of students is 3,442 persons (1,820 women). SKU has 6 faculties (Agriculture; Business Administration; Linguistics; Food Science; Natural Sciences, and Education), 1 Information Technology Center and a Secondary School for Ethnic and Talent Students (UNESCO, 2018).

There are short of doctor degree as researchers and professors. The academic research capacity of Lao PDR universities is limited; the bulk of the research that is conducted is basic and supports education and teaching, rather than the development of science and technology disciplines as found in research universities. Among the four public universities, only NUOL has a sufficient number of teaching staff with the ability to conduct research, as a lack of qualified academic staff and limited research facilities constrain the three other universities. Some of the Ph.D. holders are working at government agencies, from high ranking level to middle level officer.

Currently, university education is highly favored over technical-vocational education by students and parents, and most university students prefer to study business, medicine, education, IT or the social sciences rather than agriculture, science, math or engineering. Informal procedures are typically used to fill many types of jobs. For example, most wage and salary employees report that they obtained their current jobs through social networks rather than through formal mechanisms.⁴³ It is frequently reported that job applicants have

⁴³ For example, in the 2012 World Bank STEP household survey, 48% of surveyed wage and salary workers indicate that they got their job through social networks (friends/relatives), compared to 23% from employment agencies (public and private), 2% through university/school career offices, and only 0.3% through an internship or apprenticeship.

to pay fees to obtain jobs in both the public and private sectors. When less qualified people are hired for jobs, the effect is to lower productivity and growth⁴⁴.

Asia Development Bank (ADB) is the most systematic contributions in higher education in Lao PDR. In 2009, the Strengthening Higher Education Project (SHEP) was designed to support the Ministry of Education and Sports (MOES) of the Lao PDR to produce university graduates with skills to match labor market demands. The 1st SHEP envisaged improving the country's higher education system and making it more equitable, targeting three public universities: Champasak University; National University of Lao PDR (NUOL); and Souphanouvong University. The total project cost was \$28.8 million, of which \$24.8 million was financed by Asian Development Bank, \$2.0 million by the government of the Lao PDR, and \$2.0 million by foreign partner universities. The most of cost used to establish the new campus of Champasak University⁴⁵. Under a matching scholarship program,¹⁹ the project provided funds for 45 staff from Champasak University, NUOL, Souphanouvong University, and DHE (of which 14, or 31%, were women) to undertake master's and doctoral degrees abroad.²⁰ At project completion, 30 scholarship recipients had completed their studies and returned to their universities to teach, 12 recipients were in the process of completing their studies, and 3 recipients were unable to complete their studies. The three universities planned to strengthen their research capacities and increase the annual output of research publications in international journals. By project completion, 44 research reports were completed and approved by the universities' research committee members, of which 2 reports were published in Thai and ASEAN journals. The universities are expected to be increasingly involved in research activities in the future, as 904 faculty staff (of which 315, or 35%, were women) had received training on research planning and design.

The 2nd SHEP strengthened the linkage between higher education (HE) services and socioeconomic needs by improving HE quality, enhancing access to modern HE programs, and strengthening governance and management of HE institutions⁴⁶. The target of the

⁴⁴ Human Resource Development Strategy to 2025

⁴⁵ ADB (2016), Lao People's Democratic Republic: Strengthen Higher Education Project: Project Completion Report.

⁴⁶ ADB (2016), Lao PDR: Second Strengthen Higher Education Project, Project Administration Manual.

project focused on four universities including Savanakheth University (SKU). The project has a total estimated cost of \$45.67 million from 2017 to 2021. 51.2% of the base cost is to civil and related works (including in establishment of new campus of SKU), 9.4% is to equipment and vehicles, 17.7% is to human resource capacity building, and 9.4% is consulting services for project implementation, supervision and monitoring, as well as technical supervision of the civil works contracts. SHEP improved access to higher education by upgrading the learning environment, particularly at Champasack University, and a similar approach will be undertaken under the project for Savannakhet University. The project will consolidate and sustain the gains of SHEP in reforming the higher education sub-sector by providing further support to enhance the capacity of universities to deliver higher quality programs and services.

3. The Case of Korea

Korea's economic development is a remarkable case in the world. Among the countries that joined UCTAD in 1964, it is the only case in which the status of a developing country was changed from that of a development country. Despite the pandemic crisis, it continues to grow and has a stable and systematic health and welfare. This paper examines the role of science and technology for national development, and the relationship between industry and HRD through the case of Korea. Specially, it draws some implications for national development from three perspectives, such as export promotion for economy, HRD and development of science and technology.

3.1 Economic-Socio Development and Export

Korea had an insufficient foundation for economic development in early stage. As a newly independent country, foreign aid was a catalyst that revitalized the economy and society. However, the Korean War broke out before basic foundation of a state. It was huge damage on infrastructure and human resources. More than 60-70% of industrial facilities and plants were destroyed, farmland was devastated, and many houses and buildings collapsed. After that, large-scale reconstruction began in each sector, such as urban and rural, factory and schools, houses and buildings, road and bridge, etc. Government policies, foreign aid, and corporate participation coexisted and cooperated with aspirations of the people.

It has grown little by little since the establishment of the 5-year economic development plan in 1962. The exports increased, and the economy developed remarkably through foreign aid, loans and financing, and policies. There are various opinions about the success factors of national development in Korea. Even though the lack of nature resources, one opinion of them is qualified human resources have laid the foundation for development. There were a number of complex factors, but it regarded as best practice of success. As it shown in the table, there are growths in population, GDP, GDP for capital, etc., but the most notable is the size of exports. Exports in 1960 were 33 million, and 512, 850 million in 2020, more than

15,000 times.

<Table 6-4. Population, GDP, Exports in Korea(1960-2020)

	1960	1970	1980	1990	2000	2010	2020
Population(1,000)	24,989	30,882	37,436	43,411	46,136	48,580	51,780
GDP (US\$, B)	2.0	8.2	65.4	283.3	576.4	1,143.9	1,631
GDP per capita (US\$)	79	253	1,714	6,608	12,261	23,083	31,489
Exports (US\$, M)	33	835	15,015	59,861	169,524	463,835	512,850

Republic of Korea established a government in 1948, after independence from Japan in 1945. At that time, it was a very difficult situation, so it was supported by the USA and other agencies. In the chaos after liberation, Korea barely built a national framework for politics, economy and social culture etc.

However, as above, Korea made efforts for national reconstruction after the Korean War. One of the most important characteristics of the reconstruction was the use of foreign aid. It occupied a very high proportion in the national budget. Without foreign aid, there would be no foreign exchange reserves. From 1945 to 1999, Korea received about \$13.9 billion in ODA (official development assistance). The public non-concessional loan amount to about \$31.1 billion, and total amount to about \$45 billion including ODA and non-concessional loans.

<Table 6-5> Status on External Assistance in Korea (1945-1999)

Contents	ODA(1945-1999)		Non-Concessional loan(1964-1999)
	ODA	NGO	
	127.76 B\$	12 B\$	
Sub total	139.76 B\$		311.96 B\$
Total	451.72 B\$		

It was mainly used for humanitarian aid and emergency relief. After liberation, USA installed GARIOA (Government Appropriation for Relief Occupied Area, 1945-1948) and provided

with food, clothes and medical supplies. During Korea War, CRIK(Civil Relief in Korea, 1950-1954) served secondhand clothes, flour, blanket, bed, rice, salt, buckwheat, sanitary box, rubber shoes, caramel etc. UNKRA(UN Korea Reconstruction Agency, 1953-1958) supported the infrastructure except for food, clothes and medical supplies, such as construction of flat glass plant, cement plant and fertilizer plant. These plants provided for materials as well as fertilizer of industry and agriculture. It was not just a factory plant construction, but also learned know-how such as technology, engineering, and knowledge, which spread to other industries. For example, many talents and qualified engineers employed in cement factory, which learned from A to Z about cement manufacturing process. Several of knowledge was accumulated and extended to other industries.

According to USAID's internal report in the 1950's, Korea is like filling a bottomless vessel. Reader's Digest mentioned expecting democracy out of Korea would be like looking for a rose among trash. In the 1960's, GNP per capita is only \$82, 40% of populations was extreme poverty, and 71.1% of amount of import and 42.9% of national budget depended on foreign aid.

After the 5-year economic development plan in 1962, Korea received in concessional loans and public loans (non-concessional loans) from developed countries, such as the USA, Japan and others etc. These loans put into various infrastructure projects such as purchase of diesel locomotive, water supply system, apartment construction etc. Korea started to reorganize its institution with a focus on industrialization as one of the main goals of its economic development plan. USA and other international agencies said that it was a reckless challenge about the national plan. However, despite all kinds of difficulties, Korean government carried out policies according to plan, and 2nd and 3rd round by 1976.

Korea reformed the structure of the government organization. It installed Ministry of Reconstruction for restarting, and then focused on restructure and change of infrastructure. After a certain period of time, some functions of the Ministry of Reconstruction transferred to the Ministry of Construction. And the Economy Planning Board (EPB) established in 1961, in order to strengthen the planning function for national economic development. EPB played a very important role in economic development of Korea during 1960-1980. At the

same time, it enacted in 'foreign capital act' and installed in 'Department of foreign capital' under EPB. One of the important things in science and technology was the establishment of 'Department of Technology Management' under EPB in 1962.

In Korea, it is very important for 5-year economic plan to grow national economy. The plan started from 1962, laid the basic foundation for 15 years, such as the 1st plan(1962-1966), the 2nd plan(1967-1971), and the 3rd plan(1972-1976), and gave an opportunity for rapid economic growth.

Korea has adopted an export-led economic policy. In addition, the President conducted an export meeting every month and executed export promotion as a top priority. At that time, Korea government launched 'Monthly National Trade Promotion Meetings'. The first National Export Promotion Meeting, later to be called the 'National Trade Promotion Meeting', was held in 1965. The monthly meetings have since provided a forum for exchanges of information between firms and agencies for the export drive. Chaired by the President, the monthly trade promotion meetings are select the ministers and top bureaucrats responsible for trade, such as the chief executives of export associations, research organizations, and universities, and representatives of a few firms, general trading companies and other large firms. When Korea's exports passed \$100 million on 30 November 1964, the milestone was marked by declaring 'Export Day', which has been celebrated every year. The focus of the celebration is the award of prizes at public gathering, a prize that the representatives of firms (Rhee, Y.H. ed. 2011). Initially, Korea exported light industry goods such as wigs and socks and was processing trade. It was the only option for Korea, which lacks raw materials and original technology. The processing trade implemented successfully, increased exports by more than 40% every year. In the 1970s, it expanded to the heavy and chemical industry. Korea recognized that there is a limit to export and economic growth only with light industry goods and decided that heavy and chemical industry should be strengthened for epoch-making growth. Some peoples have promoted the view that Korean incentives have been close enough to being right much of the time to enable Korean producers to exploit the country's comparative advantage in export production.

There are several systems and policies that have driven economic development in Korea. First, as mentioned above, EPB, which take the overall responsibility on national development, was established in 1961. The Minister of EPB and Deputy Prime Minister had full authority over economic development and generalize all Ministries. The President entrust Deputy Prime Minister for economy with full authority and Minister carried out policies with strong leadership. Also, government established in KOTRA (Korea Trade Investment Promotion Agency), which took full charge of export, in 1962. Although exports were only about \$100 million, Korea established an agency for entering global market and conduction market survey. There were various policies for promotion of export, such as 'Monthly National Export Promotion Meetings', 'Export Promotion Fund', and 'Technical Assistance for Small Medium Enterprises'. It appeared that 'General Trade Company', which is responsible for international trade and domestic distribution, such as Daewoo International Corporation, Samsung Corporation, Hyundai Corporation, LG Corporation, Hyosung Corporation and Dongyang Corporation etc. In line with the government's export-led policy, the private sector also contributed to the expansion of exports. Also, large-scale infrastructure has carried out, which was a turning point of economy in Korea, such as Gyeongbu(Seoul-Busan)Expressway(1967-1970), Soyang multipurpose Dam(1967-1973), Pohang Steel Company (1970-1973) etc. At the same time, It established national think tank for research & development, such as Korea Institute for Science & Technology (KIST, 1966), Korea Development Institute (KDI, 1971), and Korea Education Development Institute (KEDI) etc. The foundation for national development completed in the early 1960s and 1970s.

3.2 Human Resources Development: Higher Education & HRD for Science and Technology.

One of the important factors in economic development in Korea is human resource development. After liberation, it established numerous universities as an open-door policy, and as a result, this led to an increase in the number of universities. Although there was a decline in quality due to quantitative expansion, the number of universities increased from only 19 to 42 during 1945-1948. Even during the Korean War, total 6,405 students studied at United Universities, which consist of 31 public and private universities in Seoul. United

University was established by special law during the war. That is why Korea had a high desire for education.

At the end of the Korean War, UNKRA commissioned the Nathan Association to establish the 'Reconstruction Plan of Korea (1952.12)'. In particular, this report focused on Korea's human resources and mentioned HRD would be a driving force for national prosperity. It emphasized on labor market, development of technician and skilled worker and role of government for HRD. First of all, government restored the facilities and equipment of university for nurturing talents, and more concentrated on science and engineering department. In, addition, the government introduced a 'Peabody Program (1956-1961)' to cultivate high-quality teacher. That is, teachers studied in 'George Peabody College for Teachers' on teacher training and library education.

The University of Minnesota, USA and Seoul National University have signed a memorandum of understanding (MOU) on the exchange and education program for human resources development. This MOU program focused medicine, agriculture and engineering etc. As a result, a large number of lectures and professor in Seoul National University studied abroad to University of Minnesota. In particular, currently, medicine has been revived through 'Lee Jong-Wook Fellowship' between Korea and Lao PDR, contributing to the improvement of health care system in Lao PDR.

At that same time, active study abroad implemented after the Korean War, and there were 3,602 students for 7 years until 1958. After they returned to Korea, it was in progress the 5-year economic development plan and they worked in the right place. Based on this foundation, the human resources development plan started together with the economic development plan from 1962. The Korean government established the 'Long-term Manpower Supply-Demand Plan'. In according to this plan, it launched a 'Human Resources Development Committee' at EPB. Chairperson is the President, EPB supervised the long-term perspective, total national budget, and economic development plan and Ministry of Science and Technology supervised human resources development plan and investment plan.

In 1963, it enacted the 'Professional Engineers Law', and established KDI (Korea Development Institute) in 1971 and did KEDI (Korea Educational Development Institute in 1972. These two national research institutes contributed to economic development and educational development. In 1970, Korea wanted to transform from an agricultural-oriented to an industrial-oriented, and it invested human resources as well as industry infrastructure. In result, it established a high technology & science university in 1974, it called KIST (Korea Institute of Science and Technology), and KIST contributed to development of science and technology. And also, it enacted 'National Qualification System law' for the quality control of technician and engineer in 1973. New science park located in Daeduck, Daejeon City, and established Daeduck Research Park (1974-1983), which concentrated on resources and facilities, and established 'National Vocational Training Agency' in 1982. Regarding industry, as above mentioned, it constructed glass plant, cement plant, steel plant, expressway and mega multipurpose dam etc.

In the 1970s, it was human resources development through the policy for specialization of the universities and establishment of research institutes. It provided human resources for export promotion to economic plan and domestic economic growth. Simultaneously, all universities opened a new department or expanded the major for needs and demands of the industry. As a result, good talented students graduated from universities, and some moved to industries and companies, and others studied abroad or went to graduate schools. The former contributed to development of industries, the latter did improvement of science and technology in universities as well as research institutes. In the 1980s, it expanded an entrance quota of the universities and opened the research institutes in accordance with explosive population growth and social demand. The development of higher education in Korea opened the new departments and cultivated human resources for needs and demand of industry age.

3.3 Science and Technology Development

It also surprised in the development of science and technology in Korea. In 2019, Korean status in the world is as follows; GERD (total R&D expenditure) is 5th, number of researchers

are 5th, GERD/GDP ratio is 2nd, international academic paper is 12th(2.45%), patent grants is 3rd in world. Most of the indicators had great take-off comparing with 1960s.

<Table 6-6> Basic STI Data in Korea (1963-2019)

	1963	1970	1980	1990	2000	2010	2019
GERD (US\$, M)	4	32	321	4,676	12,249	37,935	76,403
Public vs. Private (%)	97:3	71:29	64:36	19:81	28:72	28:72	21:77
R&D/GDP(%)	0.24	0.39	0.56	1.72	2.39	3.47	4.64
Researchers	1,750	5,628	18,434	70,503	108,370	264,118	430,690
Papers		27	159	1,587	12,472	41,991	69,618
US Patent		3	9	290	3,472	12,508	21,684

The growth of science and technology started from ‘Master Plan for Promotion of Science and Technology’ suitable for ‘5-Year Economic Development Plan’ since 1962. Even though a developing country, Korea thought national economic growth should be based on the development of science and technology. First, it consisted of 40 experts on ‘Advisory Committee for Sciences and Technology Policy’. It made a drastically decision considering on the future prospect. At that time, Korea engaged in agriculture population, 67.5% of the total population, \$ 80 per capita national income only, \$ 40 million export, and the industrial structure was 40.3% of primary industry, 20.6 of secondary industry, and 39.1% of tertiary industry. Under these circumstances, it was difficult to invest in science and technology. Nevertheless, the Korean government accepted the advisory committee’s recommendation, and first, the priority of national policy was science and technology, and it provided the budget and administration support. Second, the chief executive of state, the President should lead the way to get the civil servants and people working. The basic directions of science and technology are follows; i) Human resources were the most important available resources, so the 1st technology promotion plan considered not only economic development plan but also human resources in other sector, such as science, vocational education and training etc., ii) It actively introduced and applied overseas advanced science

and technology, iii) It reorganized research institutions, invited talented people working abroad, and formed a research-oriented scientific community, iv) It established an administrative body in charge of science and technology and supported related laws and policies, v) It created a scientific climate and transcended it into a nationwide momentum.

From 1980 to 2001, it cultivated 15,780 people in science and technology, and there were 9,780 in graduate of science and engineering, 4,000 in research institute for science and technology, 2,000 in overseas human resources, and 3,780 in post-doctoral students. It formed the industry-university cooperation committee and expanded the human resources policy deliberation committee. It made linkages and cooperation in the process of establishing a legal system, nurturing human resources, establishing government organizations and research complex, and resulted in remarkable growth.

3.4 Implications

The implications of Korea's development experiences are as follows. First, it established the economic development plan as well as human resources development plan at the same time. In particular, it established a system of technician, engineer, and professional engineer for human resources development of science and technology and allowed to cultivate human resources in vocational schools, colleges and universities. This human resource development plan took into account the needs of the industry and made it possible for human resources to be used immediately. It executed strong leadership and legal system. Lao PDR is also preparing an economic development plan as well as human resource development. The best practice of Korea will be helpful in implementing these plans. Second, the government used foreign aid appropriately. Because it was not enough for the national budget, so, foreign aid covered the most of the budget. It revitalized the market through foreign aid. At that same time, it developed manufacturing industry, which focused on export, for securing the national finances. Third, it invested in basic infrastructure, such as road and express, multipurpose dam, plant etc., and it put in human resources. Also, it made investment in higher educational institutions as well as vocational training institutions. Government put human resources into a large-scale infrastructure to learn knowledge and

skill, and then expanded to other fields. Lao PDR is undertaking large-scale infrastructure such as dams and roads. Recently, a high-speed railway is under construction supported by China. It is very important for human resources of Lao PDR to be put into these infrastructure projects to work and learn knowledge and skill. As the case of Korea, in carrying out large-scale infrastructure, it is necessary to investigate the supply and demand of manpower, and to produce the human resources through TVET or universities. Forth, it made the most of specialization strategy for human resources development. Considering the specialization of the region and university, it has fostered human resources suitable for each field and sectors. These make application of Lao PDR, she had 4 universities, which located in capital city, southern, central, and northern. The universities should be able to cultivate human resource in line with characteristics of region and government policy. In particular, it should strengthen the role of national research institutions for the foundation of basic science. It should conduct the sustainable research through biodiversity and the development of herb medicine. It is essential for this to cooperate with foreign countries. Furthermore, it is necessary to closely cooperate and put appropriate human resources in the process from preparation to practice and performance of research. Finally, governance played a very important role in the case of Korea. The EPB, a government executive body, was established, and implemented the Committee for Human Resources Policy and the Science and Technology Advisory Committee. It established a legal and policy system for leading strong leadership and achievement of goals.

4.

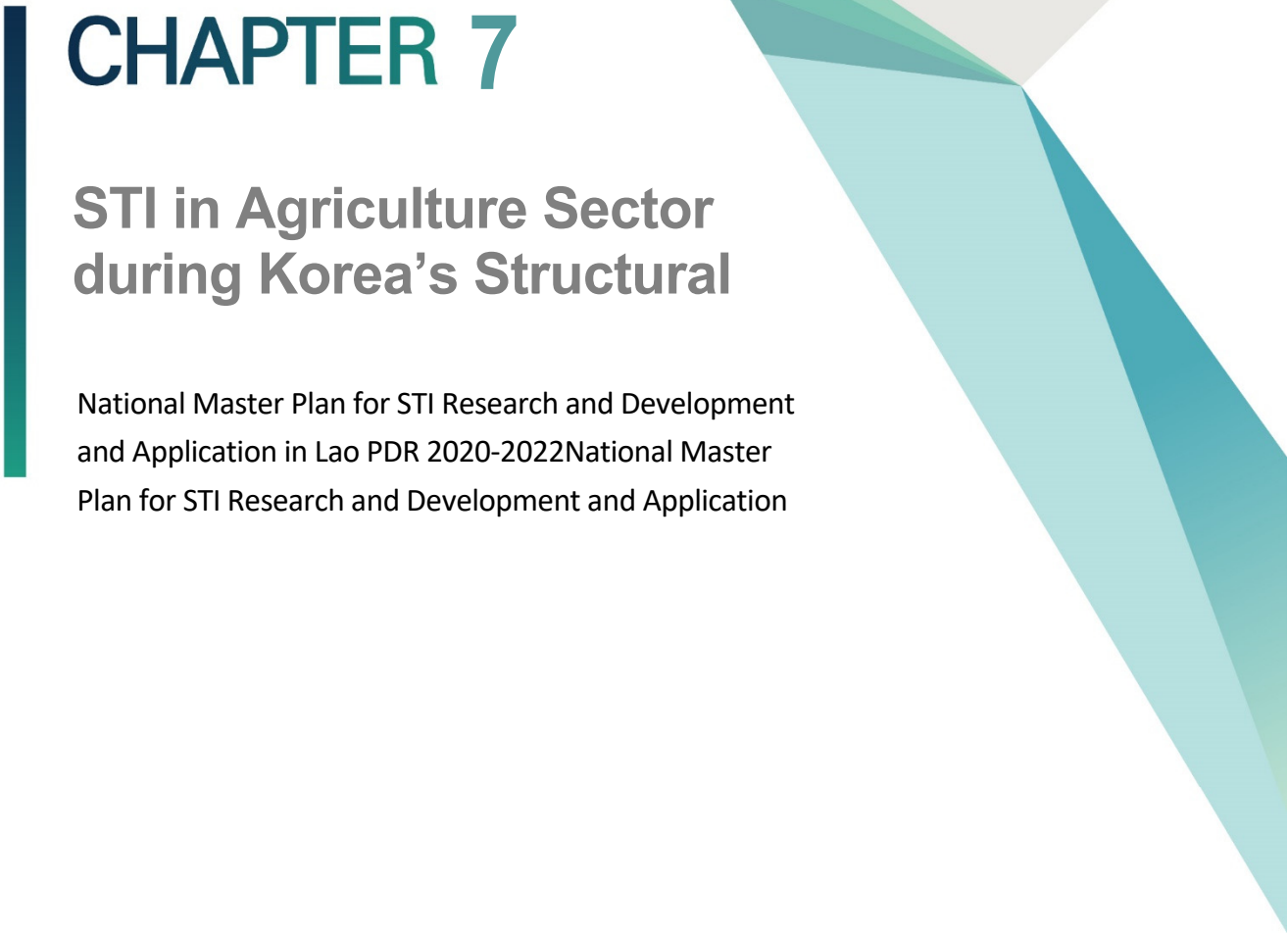
The Role of the University for National Development

As mentioned above, university plays an important role as institution for cultivating high-level human resources for national development. In particular, all countries in the world are trying to do national development through innovation and development in 21st century. Korea promoted human resource development and industrial development in line with its economic development plan. To this end, Korea implemented the support of facility and infrastructure, policy execution to the plan, and securing finance. It achieved polices and performances of comprehensive and consistent as a best practice of national development. The shortage of natural resources made up for human resources development. It accomplished the industrial development through export expansion police. And the insufficient finance supplemented appropriate foreign aid. By using foreign aid as leverage, it received foreign investment and revitalized the economy.

Lao PDR and Korea are in different situations and systems. Lao PDR has a small population compared to its resources and land size. Moreover, there is a shortage of human resources development. Lao PDR has an important turning point from the least developed country to the middle-income country. It aims to cultivate core human resources through establishing a higher education system. To this end, it has successfully carried out the SHEP (Strengthen Higher Education Project) by ADB (Asia Development Bank). Through reform of government structure in the field of science and technology, it is trying to harmonize industry and technology as well as education and R & D. It is important for national development, which is based on science, technology and innovation to cultivate human resources, universities are at the center. The 5 universities are trying to balance the development of central government and local government in Lao PDR. It is worth trying to develop the country through selection and concentration of cultivation high-level human resources considering the circumstance of Lao PDR. To this end, the role of university as follows;

First, it should establish a system for cultivation and management of high qualified human resources in university. It may play a role in the members of the university so that function

of education and research, in addition it shall provide sufficient financial support to secure doctor degree holders. Also, it is necessary to collect database of researchers by fields so that it may share and utilize information and data in connection with state, university, and research institute. Second, it should strengthen the capacity and function of the research in university. In the national innovation system, it expands the research role of universities, but R&D expenses are at low level in Lao PDR. Therefore, it should reserve the research funds and continue education and research in special areas and fields. These make it possible to secure and utilize qualified human resources. Third, it should promote industry-university-research cooperation, and enhance the filed suitability of the curriculum. It may introduce customized education, industry-university-research cooperation education, and interdisciplinary and convergence course etc. It will contribute to the supply of human resources of science and engineering based on demands of the market and industry. Forth, considering on the regional characteristics of Lao PDR, regional universities should develop and improve into specialization and professionalization as hubs of regional innovation. It is necessary to develop into competitive universities by discovering areas of strength in consideration of the research capacity and the characteristics of the community. It should also carry out activities as a think tank such as supporting the creation of new business and operating body for regional development. Fifth, it is necessary to promote the internationalization and informatization of university in response to the circumstance in the era of globalization. It may facilitate cooperation with joint research and exchange programs with foreign universities and research institutions, and also focus on efforts to secure competitiveness through international cooperation such as joint curriculum and joint degree with foreign universities. Sixth, university should be faithful to the function of nurturing human resources, which are required for the 21st century, such as new problem-solution ability, creative thinking, and data analysis ability etc. While responding to changes in the circumstance of the global era, it should nurture qualified human resources who can demonstrate their abilities in the industrial field even after graduation. Finally, it is very important for the support of the government to develop and improve the universities, such as partnership with industry-academy-research-government, improvement of infrastructure on education and research for universities, financial support for research and development in national budget etc.

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CHAPTER 7

STI in Agriculture Sector during Korea's Structural

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022National Master
Plan for STI Research and Development and Application

Chapter 7. STI in Agriculture Sector during Korea's Structural Transformation Period

1. Historical Trends in the Development of Agriculture Sector in Korea

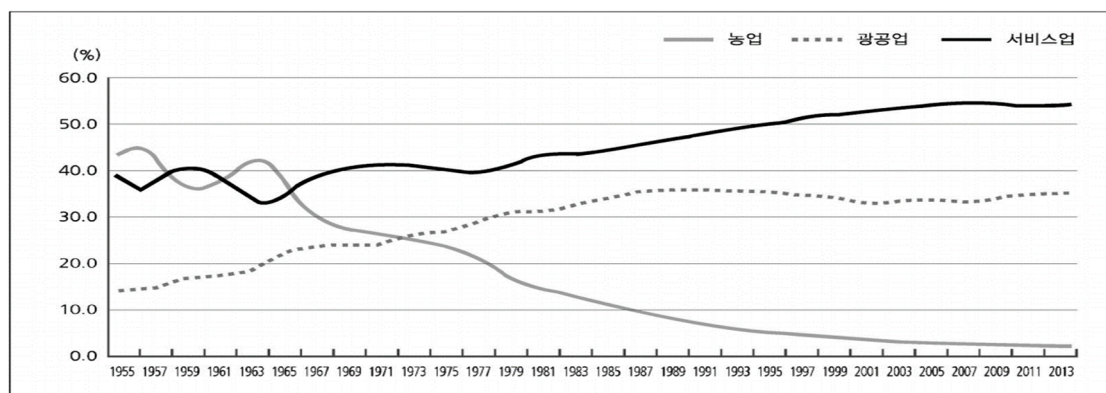
1.1 Structural transformation of agriculture in Korea⁴⁷

1.1.1 Reduced proportion of agriculture sector in Korean industry

The proportion of South Korea's gross domestic product (GDP) from agriculture showed a sharp decrease from the 1960s due to rapid progress of industrialization and stellar growth of the non-agricultural sectors. Since the mid-1960s, the growth rate of the non-agricultural sector increased to 10 percent, indicating a sharp decline in GDP by the agriculture sector. The percentage of agricultural output in South Korean GDP was at least 40 percent in the early 1960s, but this fell sharply to less than 20 percent in 1979 and less than 10 percent in 1988. At that time, this rate of decrease was 2.5 to 6 times faster than that of developed countries.

⁴⁷ Main reference: Korea Rural Economic Institute(2015), Ministry of Agriculture, Food and Rural Affairs(2020), statistical data of FAO(Food and Agriculture Organization) (accessed on 2021.10)

[Figure 7-1] Trend of GDP by Sector (in %) of Agriculture, Mining and Manufacturing Industry, Service Industry in Korea



Source: Korea Rural Economic Institute (2015)

Compared with the manufacturing industry between 1970 to 2000, the GDP of agriculture, forestry, and fisheries industries grew at a compound annual rate of 7.7 percent, but in terms of percentage in total GDP, it showed a sharp decrease from 26.5 to 3.9 percent, whereas the GDP of the manufacturing industry increased at a compound annual rate of 16.5 percent, resulting in an increase in the GDP share from 17.4 to 26.4 percent.

<Table 7-1> Trends in GDP of agriculture/forestry/fisheries industry and manufacturing industry in Korea, 1970-2000

Unit: 1 million dollars

	1970	1975	1980	1985	1990	1995	2000	CAGR
GDP	9,005	21,784	65,398	101,296	283,366	566,581	576,179	14.9%
Agriculture, forestry and fisheries industry GDP	2,385	5,334	9,332	11,906	21,554	30,191	22,228	7.7%
% of total GDP	26.5	24.5	14.3	11.8	7.6	5.3	3.9	
Manufacturing industry GDP	1,563	4,386	14,378	24,542	70,895	146,186	152,390	16.5%
% of total GDP	17.4	20.1	22.0	24.2	25.0	25.8	26.4	

Source: FAO(2021a)

The sharp decline of agricultural production in the national GDP at that time indicates that the labor force in farms and farmland underwent fast changes from the agricultural sector to non-agricultural sectors. The actual number of farm households decreased from the end of the 1960s and showed a decrease by 44.3 percent in the number of households from 2,483,000 to 1,383,000 people between 1970 to 2000. In terms of proportion against the total number of households, this change corresponds to a sharp reduction from 42.4 to 9.6 percent. The rate of decrease in the farm population was even faster and showed a decrease by 72.0 percent in the number of population from 14,422,000 to 4,031,000 people between 1970 to 2000. In terms of proportion against the total population, this change corresponds to a drastic reduction from 44.7 to 8.6 percent. The number of employed persons in agriculture also decreased by 54.4 percent from 4,916,000 to 2,243,000 people between 1970 to 2000, and the proportion against the total employed fell sharply from 50.4 to 10.6 percent.

<Table 7-2> Trends in Korean farm households, 1970-2000

Unit: 1,000 households, %

	1970	1975	1980	1985	1990	1995	2000	Ratio of change
Total households	5,857	6,754	7,969	9,571	11,355	12,958	14,391	▲ 145.7
Farm households	2,483	2,379	2,155	1,926	1,767	1,501	1,383	▼ 44.3
Percentage	42.4%	35.2%	27.0%	20.1%	15.6%	11.6%	9.6%	

Source: Ministry of Agriculture, Food and Rural Affairs(2020)

<Table 7-3> Trends in Korean farm population, 1970-2000

Unit: 1,000 ppl, %

	1970	1975	1980	1985	1990	1995	2000	Ratio of change
Total Population	32,241	35,281	38,124	40,806	42,869	45,093	47,008	▲ 45.8
Farm population	14,422	13,244	10,827	8,521	6,661	4,851	4,031	▼ 72.0
Percentage	44.7%	37.5%	28.4%	20.9%	15.5%	10.8%	8.6%	

Source: Ministry of Agriculture, Food and Rural Affairs (2020)

<Table 7-4> Trends in economically active population in Korean agriculture, 1970-2000

Unit: 1,000 ppl

	1970	1975	1980	1985	1990	1995	2000	Ratio of change
Employed persons in agriculture	4,916	5,425	4,654	3,733	3,237	2,403	2,243	▼ 54.4
Employed persons in agriculture Percentage (%)	50.4	45.7	34.0	24.9	17.9	11.8	10.6	

Source: FAO(2021b)

The area of farmland also showed a steady decline after peaking between 1965 to 1970. Until the 1980s, there were cases of the increase in farmland from reclamation projects but after that period, the farmland area continued to decrease with the progress of industrialization and urbanization. The area of farmland decreased by 17.8 percent from 2,298,000 hectares in 1970 to 1,889,000 hectares in 2000. In this period, the percentage of farmland from total national land decreased from 23.3 to 19.0 percent. As for the composition of the farmland, the area of dry paddies showed a more rapid decrease between 1970 to 2000, and thus the proportion of the rice paddies increased from 55.4 to 60.8 percent, and that of dry paddies decreased from 44.6 to 39.2 percent.

<Table 7-5> Trends in the area of farmland and composition in Korea, 1970-2000

Unit: 1,000 ha, %

	1970	1975	1980	1985	1990	1995	2000	Ratio of change(%)
Area of national land (A)	9,848	9,881	9,899	9,914	9,927	9,927	9,946	▲ 1.0
Area of farmland (B)	2,298	2,240	2,196	2,144	2,109	1,985	1,889	▼ 17.8
Percentage (B/A)	23.3%	22.7%	22.2%	21.6%	21.2%	20.0%	19.0%	
Area of rice paddies (C)	1,273	1,277	1,307	1,325	1,345	1,206	1,149	▼ 9.7
Percentage (C/B)	55.4%	57.0%	59.5%	61.8%	63.8%	60.8%	60.8%	
Area of dry paddies (D)	1,025	963	889	819	764	779	740	▼ 27.8
Percentage (D/B)	44.6%	43.0%	40.5%	38.2%	36.2%	39.2%	39.2%	

Source: Ministry of Agriculture, Food and Rural Affairs (2020)

1.1.2 Structural changes in the agricultural production

The decrease in farm households resulted in an increase in the area of cultivated land per farm household. Also, due to the increase in the farmland prices caused by the increase in the conversion of the land from farmland to non-agricultural use and the agricultural mechanization, the type of farmland converted to the leased land type. First, the area of cultivated land per farm household increased by 47.6% from 0.93 ha to 1.37 ha between 1970 and 2000.

<Table 7-6> Trends in the area of cultivated land per farm household in Korea, 1970-2000

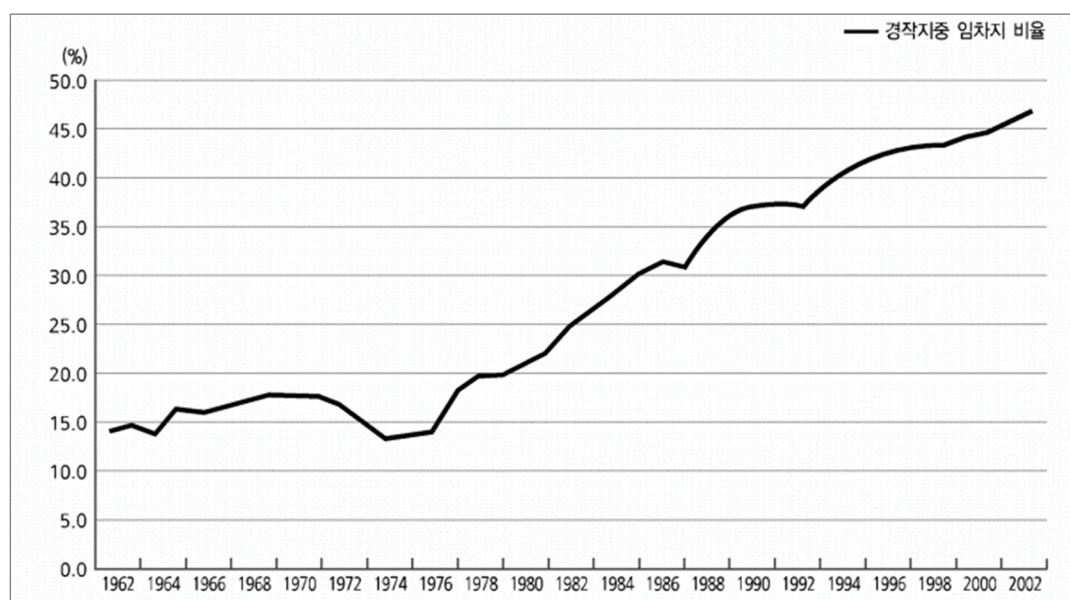
Unit: ha, %

	1970	1975	1980	1985	1990	1995	2000	Ratio of change
Area of cultivated land per farm household	0.93	0.94	1.02	1.11	1.19	1.32	1.37	▲47.6

Source: Ministry of Agriculture, Food and Rural Affairs(2020)

Until the 1980s, the supply of farmland increased with the decrease in the number of farm households. However, conversion of farmland to non-agricultural use also increased, leading to an increase in farmland prices. In this period, the relocated farm households tended to retain their property assets, and thus the proportion of leased land increased out of the farmland. Also, with a decrease in the supply of agricultural labor force and the increase in the wage, the existing large farms adjusted the size of their farmland, increasing the proportion of midsize farms.

[Figure 7-2] Trend in the Proportion of Leased Land out of Total Agricultural Land in Korea



Source: Korea Rural Economic Institute(2015)

Since the 1990s, as the mechanization of agricultural work progressed, the land burden capacity of large farmers improved, and as large farmers through leases increased, the concentration of large farmers increased. Between 1980 and 2000, the number of farm households decreased by 35.8% from 2.16 million to 1.38 million households, but the number of farms with 2.0ha or larger increased, and the number of farm households with 3.0ha or larger increased by 171.8%. The proportion of farm households by size of cultivated land was 63.9% for the area of 0.5-2.0 ha and 6.4% for the area larger than 2.0 ha in 1980. In 1990, the proportion was 61.5% for the area of 0.5-2.0 ha and 9.8% for the area larger than 2.0 ha. In 2002, the proportion was 52.8% for the area of 0.5-2.0 ha and 14.3% for the area larger than 2.0 ha.⁴⁸

<Table 7-7> Trends of the farm households by size of the cultivated land in Korea, 1980-2000

Unit: household, %

	Total	No land	Less than 0.5ha	0.5-1.0	1.0-2.0	2.0-3.0	3.0-5.0	5.0ha or more
1980	2,155,073	27,877	611,698	747,579	629,197	107,559	31,163	-
	100.0%	1.3%	28.4%	34.7%	29.2%	5.0%	1.4%	0.0%
1990	1,767,033	23,803	482,703	544,457	543,027	129,510	43,533	-
	100.0%	1.3%	27.3%	30.8%	30.7%	7.3%	2.5%	0.0%
1995	1,500,745	23,918	432,982	432,107	417,960	123,333	54,896	15,549
	100.0%	1.6%	28.9%	28.8%	27.9%	8.2%	3.7%	1.0%
2000	1,383,468	14,170	440,605	378,655	351,534	113,790	61,068	23,646
	100.0%	1.0%	31.8%	27.4%	25.4%	8.2%	4.4%	1.7%
Ratio of change(%)	▼35.8	▼49.2	▼28.0	▼49.3	▼44.1	▲5.8	▲171.8*	

※ Note: * indicates a change in farm households with not less than 3.0ha or more

Source: Park, Seok Doo , Hwang, Euisik(2002)

⁴⁸ Park, Seok Doo , Hwang, Euisik(2002)

From the 1970s, the rate of agricultural mechanization showed a rapid progress with the progress of agricultural infrastructure preparations, which contributed to resolving the problem of shortage of agricultural labor force. The number of agricultural machines showed a rapid increase from about 420,000 in 1975 to about 3 million in 1995.

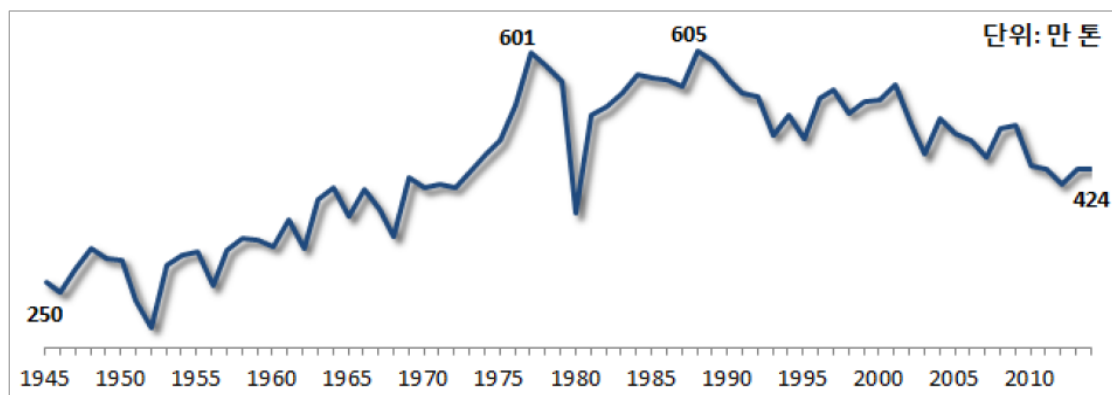
<Table 7-8> Trends in the number of major agricultural machines in Korea

	1975	1985	1995
Power tiller	85,722	588,962	868,870
Tractor	564	12,389	100,412
Rice-transplanter	16	42,138	248,009
Combine	56	11,667	72,268
Grain dryer	694	5,437	28,408
Speed sprayer	137,698	517,530	682,675
Agriculture water pump	65,993	286,298	384,900
Total	417,937	1,795,260	3,042,872

Source: Ministry of Agriculture, Food and Rural Affairs(2020)

Agricultural production, which mainly consisted of food crops, shifted to vegetables, fruits, and livestock products due to the improvement in dietary style and changes in profitability by agricultural products after achieving self-sufficiency in staple food grain in the late 1970s. First, in the 1970s, rice production dramatically increased through the development of Tong-il rice cultivar, which is resistant to insects pests and diseases and the application of cultivation technique for high yield.

[Figure 7-3] Trends in Production of Rice (Milled Grain) in Korea

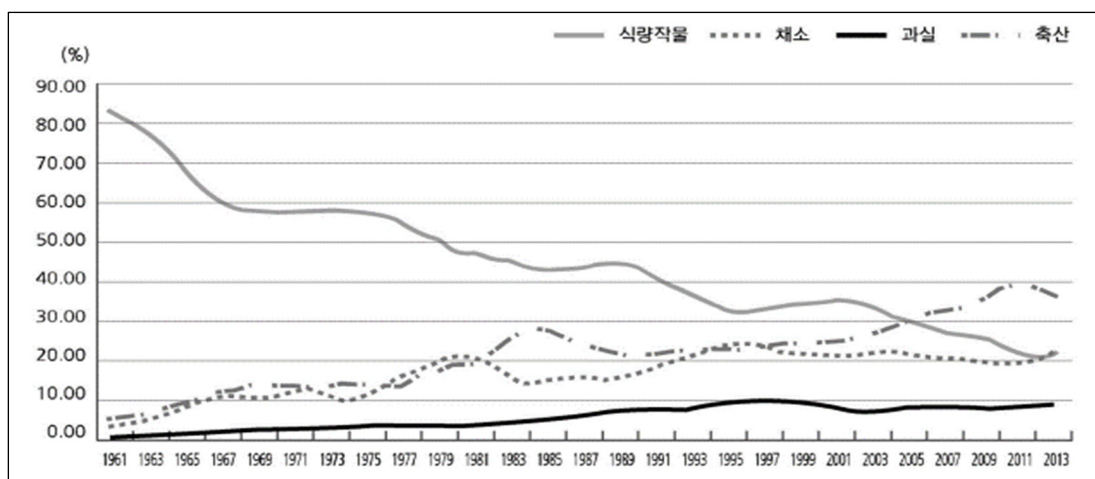


Unit: 10,000 tons, Note: based on milled rice/brown rice ratio of 92.9%

Source: Song, Mi-ryeong, et al.(2015)

As employment in the non-agricultural sector increased with industrialization, the opportunity cost of labor rose, and the agricultural products with high value-added productivity of land were substituted with those with high value-added productivity of labor. Examining the proportion of value of agricultural production by sector to gross value of agricultural production, food crops decreased from 80% in the early 1960s to 30% in the early 1990s, whereas the proportion of vegetables increased from 4% to 20%, fruits from 1% to almost 10%, and livestock from 5% to 20%.

[Figure 7-4] Trends in Composition of Agricultural Products in Korea



Food crops, vegetables, fruits, livestock

Source: Korea Rural Economic Institute (2015)

As for the composition of agricultural products in terms of production value by commodity, between 1980 and 2000, the production of rice grew at an compound annual rate of 8.2%, and the proportion decreased from 34.1% to 32.9%, while fruit production showed a rapid increase at an compound annual rate of 12.4% from KRW 247 billion to KRW 2.581 trillion, which corresponds to an increase in proportion from 3.9% to 8.1%. The livestock products showed a growth at an compound annual rate of 9.9%, resulting in an increase in the proportion from 19.1% to 25.3%.

<Table 7-9> Trends in the value and percentage of agricultural production by commodity in Korea

Unit: 1 billion KRW, %

	1980		1990		2000		CAGR	Ratio of change
Total agricultural production	6,415	100.0%	17,860	100.0%	31,968	100.0%	8.4%	▲ 398.3
Rice	2,188	34.1%	6,538	36.6%	10,505	32.9%	8.2%	▲ 380.1
Vegetable	1,449	22.6%	3,433	19.2%	6,739	21.1%	8.0%	▲ 365.1
Fruits	247	3.9%	1,310	7.3%	2,581	8.1%	12.4%	▲ 944.9
Livestock farming	1,227	19.1%	3,923	22.0%	8,082	25.3%	9.9%	▲ 558.7

Source: Korea Rural Economic Institute (2015)

The cultivated area of food crops decreased by 51.3% from 2.706 million hectares in 1970 to 1.317 million hectares in 2000. Among different types of food crops, the cultivated area of barley decreased by 91.8% and other grains by 79.7%.

Rice: paddy rice and field rice, Barley: hulled barley, naked barley, wheat and rye, Other grains: millet, sorghum, corn, buckwheat, Pulses: bean, red bean, mung bean, Millets: sweet potato and potato

<Table 7-10> Trends in the composition of food crop cultivation area, 1970-2000

Unit: 1,000 ha, %

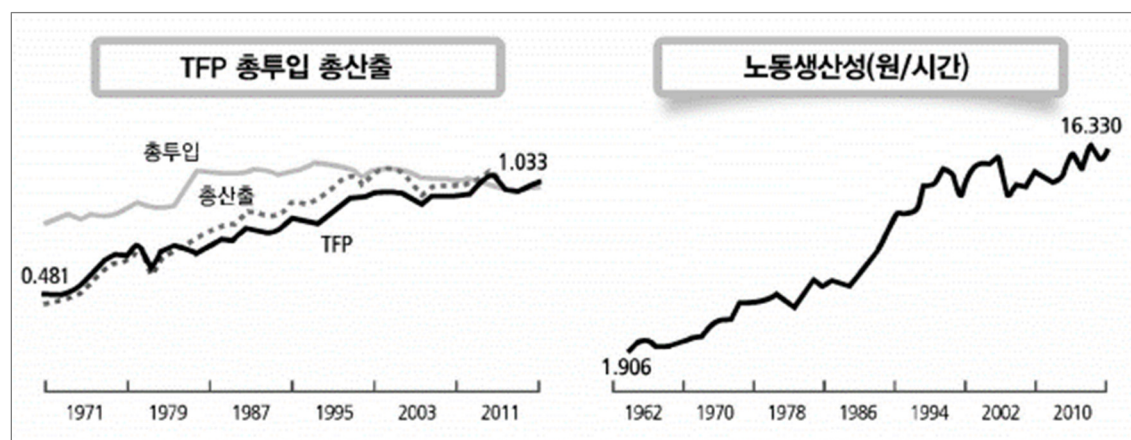
	1970	1975	1980	1985	1990	1995	2000	Ratio of change
Food crops	2,706	2,531	1,994	1,780	1,669	1,346	1,317	▼51.3
Rice	1,203	1,218	1,233	1,237	1,244	1,056	1,072	▼10.9
	44.5%	48.1%	61.8%	69.5%	74.5%	78.5%	81.4%	
Barley	834	761	360	242	160	90	68	▼91.8
	30.8%	30.1%	18.1%	13.6%	9.6%	6.7%	5.2%	
Pulses	365	333	256	196	188	132	107	▼70.7
	13.5%	13.2%	12.8%	11.0%	11.3%	9.8%	8.1%	
Millets	181	146	92	65	40	40	44	▼75.7
	6.7%	5.8%	4.6%	3.7%	2.4%	3.0%	3.3%	
Other grains	123	73	53	40	37	28	25	▼79.7
	4.5%	2.9%	2.7%	2.2%	2.2%	2.1%	1.9%	

Source: Ministry of Agriculture, Food and Rural Affairs (2020)

1.1.3 Increase in agricultural productivity

In Korean agriculture, there was a rapid increase in total factor productivity (TFP), which represents production compared to input factors, from the 1970s to the 1990s. Although there was a shift of land and labor from agricultural to non-agricultural sectors, agricultural production still expanded through increased capital and use of intermediate goods. Up to the 1990s, output increased faster than input, resulting in a remarkable increase in labor productivity.

[Figure 7-5] Trends in Agricultural Total Factor Productivity (TFP) and Labor Productivity in Korea



*Total Input, Total Output, Labor productivity (KRW/hour)

Source: Ministry of Agriculture, Food and Rural Affairs(2020)

1.1.4 Increase in the agricultural trade

From 1970 to 2000, imports and exports of Korean agricultural products increased at an compound annual rate of 10%, indicating a significant increase in trade volume. Between 1970 and 2000, exports grew from \$78.48 million to \$1.53 billion, and imports increased from \$440 million to \$8.30 billion. This is the result of a change in the agricultural trade policy stance from restriction of imports to gradual liberalization of imports and the opening up of the agricultural products market following the implementation of the Uruguay Round commitments of World Trade Organization (WTO) in 1995. The trade deficit rose sharply from \$360 million in 1970 to over \$8 billion in 1995, and recorded around \$6.76 billion as of 2000.

<Table 7-11> Trends in Korean agricultural trade, 1970-2000

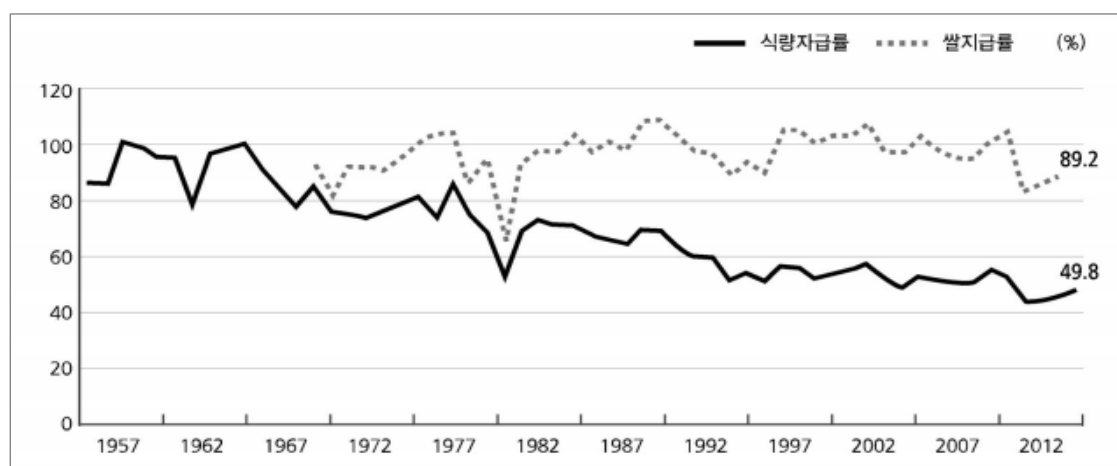
Unit: 1,000 USD

	1970	1975	1980	1985	1990	1995	2000	CAGR
Export	78,479	352,948	640,823	502,347	1,145,283	1,650,989	1,533,348	10.4%
Import	438,503	1,425,958	3,300,101	3,074,718	6,459,489	9,674,506	8,298,246	10.3%
Balance	-360,024	-1,073,010	-2,659,278	-2,572,371	-5,314,206	-8,023,517	-6,764,898	10.3%

Source: FAO(2021c)

With the progressive opening up of the agricultural products market, the food self-sufficiency rate of Korea fell below 60% since the 1990s. Only the rice has maintained a high self-sufficiency rate of over 80%, and the overall food self-sufficiency rate has been on the decline since the late 1970s, with pulses, fruits, meat and dairy products greatly affected by the market opening.

[Figure 7-6] Trends in Food and Rice Self-Sufficiency Rates in Korea



Note: food self-sufficiency rate (excluding crops for livestock feed) = (crop production/domestic consumption) x 100, the value of year 2014 is a tentative value.

Source: Korea Rural Economic Institute (2015)

1.2 Major Cases of Development of Agricultural Technology in Korea ⁴⁹

1.2.1 Basic infrastructure of agriculture

In the national soil survey project (1964-1999) that started in the 1960s, a detailed soil map with 1:5000 scale was produced, and active R&D on weeds and weed control has started from the 1970s. In the 1960s, the sericulture industry grew with the implementation of the sericulture production promotion plan, resulting in the industry accounting for 3.4-8.8% of the total export value.

⁴⁹ Main reference: Science & Technology Policy Institute(2017)

In the 1970s, a power tiller that could replace animal power was developed and quickly distributed to nationwide farms. In the 1980s, rice transplanting using a machine became a common practice, and for the entire paddy field, the farm work from tillage to sowing and transplanting, was handled by machines. Likewise, in the horticulture and livestock sectors, mechanical equipment was developed and disseminated, and the agricultural mechanization was rapidly promoted.

From the 1960s, as crop breeding projects showed an active development, awareness of the necessity of genetic resources increased, and the number of genetic resources increased to 12,131 in total in 1970 from 2,014 in 1951, amounting to a six-fold increase.

1.2.2 Food crops

From 1964 to 1976, the Tongil rice cultivation project with the aim of cultivating and distributing new high-yield varieties that are less prone to falling over, disease-resistant, and have strong fertilizer tolerance, was implemented, opening the era of the Korean Green Revolution. A technology of rice variety development that overcomes the cross sterility between ecotypes of different lineages by three-way cross was applied, and the shuttle breeding technique that enabled selection and generation advancement with different ecological conditions was introduced, thus establishing a foreign seed propagation system.

The positive effects of Tongil rice development in terms of breeding include expansion of genetic resources, shortening of breeding period, establishment of selection system through development of characterization test method, and diversification of breeding methods for pyramiding of blight resistance genes. The average income per farm household (based on 0.3ha) increased from 24,000 won in 1972 to 128,000 won in 1977 owing to the improvement of yield and cultivation safety, because remarkably high yield and reduction of insect control cost of the new variety. As Tong-il rice was developed in 1971 and several Tongil-type rice varieties were developed subsequently, domestic rice production, which was 3.5 million tons in 1965, increased to over 6 million tons in 1977, achieving food self-sufficiency.

<Table 7-12> Trends in rice yields with development of Tongil-type rice varieties

Unit: 1,000 USD

	Rice yields (kg/10a)	Range
'72~75	491	449~554
'76~80	510	439~581
'81~86	562	496~605

Source: Korea Rural Economic Institute (2015)

The significance of the development of Tongil rice lies not only in the new variety of rice but also in the dissemination of scientific cultivation technology and establishment of the overall system with agricultural R&D and dissemination. The development of Tongil Rice resulted in shifting away from the conventional cultivation methods and provided cultivation techniques of scientific farming such as seedling, transplanting, fertilization, pest control, and water management to farmers. The system of development of new varieties with continuous improvement and that of fast dissemination in farms were established, thus laying the foundation for the development of new varieties of other crops and advancement of cultivation technology. The development of breeding technology later leads to the development of related industries such as agricultural machinery and vinyl, which serves as the starting point of the White Revolution that supplies fresh fruits and vegetables across all seasons.

1.2.3 Horticultural crops

The cultivation of first-filial generation (F1) of vegetables, which Dr. Woo Jang-chun started in the 1950s, further led to the development of horticultural variety No. 1 and 2 of cabbage and onion in 1960 after his death. After the superiority of F1 hybrid, which exhibits a clear heterosis, was verified, the F1 hybrids were developed for almost all vegetables in the mid to late 1970s.

The consumption of vegetables per capita in Korea nearly doubled from 60 kg in the late 1970s to 120 kg in the early 1980s, which was made possible by the White Revolution. The White Revolution refers to the rapidly expanded area of protected cultivation covering the

ground surface in winter by the white color of the greenhouses. White Revolution was possible because of a considerable increase in productivity in the of greenhouse due to the domestic production of polyethylene film for agricultural use and the increase in the area of protected cultivation from the reduction of logistics time owing to the opening of the Gyeongbu Expressway in 1970, and the change in the frame materials of the greenhouse from bamboo to steel pipes following the completion of Pohang Steel Mill in 1973. The area of greenhouse per capita in Korea was approximately 10m² from 1996, which is the world-highest level.

Fruit breeding was time-consuming and costly, and thus it was implemented by a governmental agency and the effect was insignificant. Since the late 1960s, new varieties of fruits such as pears, peaches, and apples have been developed and distributed. Following spread of new varieties of fruits and imports, fruit consumption per capita in the 1970s, which was only 15-20 kg per year, increased to 65 kg in the 2000s.

1.2.4 Livestock farming technology

Since the 1960s, with the progress of agricultural mechanization, the role of domestic cattle for ploughing and other farming work was reduced, which led to the increase in research on the improvement of the domestic cattle for meat production, and the feeding strategy for the cattle. The Korean Hanwoo Improvement Association, founded in 1960, discussed the goals and methods of improvement of Hanwoo, the standards of assessment, measurement methods and establishment of feeding standards. In 1978, the direction of Hanwoo improvement was categorized into purebred improvement and crossbred improvement.

As for the breeding of Hanwoo, various methods were developed such as the utilization of frozen-thawed spermatozoa in 1970s, identification of an appropriate thawing method of the frozen spermatozoa in 1980s, and technology development on improving the viability of spermatozoa in the 1990s. Thereafter, the development of a deep uterine artificial insemination device served as the opportunity that Korean Hanwoo raising farms can familiarize themselves with the artificial insemination.

2. Key Policies and Systems in Agriculture Sector in Korea⁵⁰

2.1 Major policies and systems during the period of the 1st and 2nd 5-Year Economic Development Plans(1962 ~ 1971)

During the period of the 1st 5-Year Economic Development Plans (1962-1966), the main goals of the agricultural sector were to increase agricultural production and modernize the production process. The plan was focused on the food grain production for food self-sufficiency. Reclamation projects for expansion of farmland, agricultural land consolidation and sewer installation projects for reform of agricultural production infrastructure, and research, guidance and dissemination projects were carried out for the increase in the agricultural productivity.

During the period of the 2nd 5-Year Economic Development Plans (1967-1971), agricultural administration advocated uniform advancement of agriculture and industry, aiming to increase farm household income and modernize the national agriculture along with achieving the food self-sufficiency through increased production. As a result of the imbalance between the development of industrial sectors due to the governmental support of industrialization through the low-grain price policy, the policy of uniform advancement of agriculture and industry was promoted that aim for the promotion and development of both agriculture and industry, and the government attempted to link the increased agricultural product to the increase of the farm household income. Above all, the idea of the uniform advancement of agriculture and industry was based on processing of agricultural products to turn them into industrial products, and through domestic sales and exports to overseas market, both the industrial development and increase in the income of farmers and fishermen, who are raw material producers, could be achieved. In addition, to increase food production, farmland expansion and consolidation, expansion of irrigation facilities, seed improvement, land fertility enhancement, disaster countermeasures, and

⁵⁰ Main reference: Korea Rural Economic Institute(2015) and Song, Mi-ryeong et al.(2015)

mechanization were implemented. As part of the uniform advancement of agriculture and industry, Korean Rural Community Development Corporation was established, which is an agency for development of processing and storage of agricultural products, and furthermore, various projects such as groundwater development project, large-scale integrated agricultural development project, and price support policy were introduced.

In accordance with the discussion that for promotion of agricultural cooperatives, they should be integrated with the agricultural bank and thus as of on July 29, 1961, the "Agricultural Cooperatives Act" was promulgated. As of August 15 in the same year, an integrated agricultural cooperative in charge of credit-based loan business, economic business, and guidance projects was launched. In addition, for unification of agricultural administrative system, the establishment of new external agency of the Ministry of Agriculture and Forestry and integration of regional agencies were discussed, which led to the enactment of "Agricultural Community Development Promotion Act" as of March 21, 1962, and establishment of Rural Development Administration as of April 1 of the same year. Rural Development Administration was established as an external agency of the Ministry of Agriculture and Forestry and the affiliated organizations included provincial rural development agencies for each province and agricultural service center, thereby establishing the unification of the agricultural administrative system.

As one of the pillars for the uniform advancement of agriculture and industry, the Korean Rural Community Development Corporation was established in December 1967 for development of processing and storage of agricultural products. As another pillar to achieve the uniform advancement of agriculture and industry, the First Special Project on Rural Income Increase was implemented from 1968 to 1971. This was a comprehensive agricultural administration for regional development that aimed for transformation from subsistence farming to commercial farming by focusing on the production cash crops and export items, centering on the building of the main production complex.

At the same time, a price support policy was implemented for agricultural products. In order to increase the production of rice, high rice price policy was implemented starting from the 17% increase compared to previous year in the purchase price of rice in 1968, leading to an

annual average increase of 25.1% in rice purchase price between 1968 and 1971. In order to increase production and promote consumption of barley, a dual barley price system was implemented from 1969, which led to the lowering of governmental release price of barley compared to the purchase price, and since 1972, the purchase price of barley has been indicated and unlimited purchases have been made.

2.2 Major policies and systems during the period of the 3rd and 4th 5-Year Economic Development Plans(1972 ~ 1981)

During the period of the 3rd 5-Year Economic Development Plans (1972-1976), the main goal of the agricultural sector was achieving self-sufficiency in staple food grain. From the viewpoint of the total development of national industries, this plan was based on the recognition of structural problems such as aggravated disparities between urban and rural areas, between industries and between regions due to rapid economic development of the last 10 years. From the perspective of the agricultural sector, the top priority was still the increase in the food production and achieving the self-sufficiency in staple food grain. Along with the development of agricultural production infrastructure through improvement in irrigation and land reforms, there were a variety of projects such as development and distribution of Tongil rice, continued implementation of the price support policy for agricultural products, and policy of increased production of cash crops, livestock farming and sericulture industry as part of the Special Project on Rural Income Increase, and the rural Saemaul Undong (also known as the new village movement of Korea to facilitate the modernization of rural economy).

During the period of the 4th 5-Year Economic Development Plans (1977-1981), the main goals in the agricultural sector were stable food supply and an increase in the income of farmers and fishermen. The 4th Five-Year Plan was oriented toward private sector-led growth and market opening, focusing on the implementation of self-growth structures and achieving technological innovation. In the agricultural sector, government-led development and domestic market protection policies were maintained to start with but since the late 1970s, changes in the policy stance started such as the withdrawal of the high rice price

policy. In terms of the agricultural sector, the main focus was on sustaining growth, stable food supply, increasing incomes of farmers and fishermen, stabilizing prices of agricultural products, modernizing distribution, and improving living conditions in rural areas.

For the large-scale integrated agricultural development project, the limitations of the groundwater development project were recognized and the direction was changed to securing large-scale agricultural water from 1971, and the integrated development project for the river basin of the four major rivers (Han River, Geum River, Yeongsan River, Nakdong River) was implemented with the target of completion by 1981. At that time, the basin areas of the four major rivers accounted for 64% of the total national land and 54% of the total agricultural land. In this project, there were programs on improvement on irrigation and land reforms such as the development of water for agricultural use, farmland consolidation, drainage improvement, and land reclamation, development programs such as agricultural improvement, rural area transformation, and environmental improvement. These programs were implemented by dividing into large-scale unit for each river basin, and achieved the 31,533 ha of farmland development by 1976 and increased production of rice by 63,000 tons. The large-scale integrated agricultural development project, which was more actively carried out from 1977 with the signing of the loan agreements, led to the construction of Sapgyocheon district, Mihocheon district, Nam River district, estuary bank in the phase II project of Yeongsan River, Nakdong River district, and Nonsan district.

The development and distribution of high-yielding new varieties of rice, which began in the 1960s, came to fruition. Due to the depression in the rice production, from the late 1960s, through hybrid crossbreeding between remote varieties, a new variety that is resistant to disease and pest and lodging was developed and this new variety was distributed with the name of "Tongil rice" to nationwide farms in 1971. The Tongil rice significantly contributed to achieving the self-sufficiency in staple food grain through a drastic increase in rice production, and with the application of the high-yielding technique, it also contributed to the increase in the yield per unit area of general rice variety, thereby leading the Korean Green Revolution. Between 1971 and 1976, total production of food grains increased from 6.78 million tons to 8.18 million tons, while food grain consumption decreased slightly

according to the governmental policy of reducing food grain consumption. Therefore, the self-sufficiency rate of rice increased from 82.5% to 100.5%, and the self-sufficiency rate of barley increased from 91.8% to 97.9%, thus finally achieving the self-sufficiency in staple food grains.

In order to resolve the problem of urban-rural disparity and overcome the economic downturn, Saemaul Undong was implemented. Saemaul Undong achieved results such as improvement of living environment, reform of agricultural production infrastructure, increase in the income, and education of the rural public to raise awareness on key issues. Saemaul Undong was operated in a mixture of top-down and bottom-up approaches, in which the government allocated resources and the rural village residents contributed in terms of labor. It proceeded as a kind of public awareness movement emphasizing diligence/self-help/cooperation, while fostering change facilitators called Saemaul Leaders. Between 1971 and 1982, through Saemaul Undong, the progress was focused on the reform of agricultural production infrastructure such as farm road construction, road pavement, construction of a village community center, village warehouse, and communal work center and the improvement of living environment of rural villages such as housing improvement, village restructuring, and installation of sewer system.

<Table 7-13> Major Achievements of Saemaul Undong, 1972-1982

Project Description	Unit	Output ('72~'82)
Farm road construction	km	64,686
Saemaul road pavement	km	6,187
Village community center	Dong	39,231
Village warehouse	Dong	22,468
Communal work center	Dong	6,323
Rural housing improvement	1,000 Dongs	258
Village restructuring	Village	3,047
Sewer system installation	km	17,202
Small stream improvements	km	21,562
Cash crop development	Complex	1,616

Source: Korea Rural Economic Institute(2015)

As part of the creation of the agricultural production infrastructure, the “Agricultural Mechanization Promotion Act” was enacted in 1978 and related financial support to farm households was expanded with the establishment of Agricultural Mechanization Promotion Fund and the National Investment Fund. In addition, agricultural mechanization projects were actively carried out such as the expansion of shared use of agricultural machinery and reinforcement of boarding training. Furthermore, with the completion of the 7th Fertilizer Factory, the largest in Korea at the time, in 1977, the supply capacity of fertilizers improved, satisfying the required volume for domestic consumption as well as exports of the fertilizers.

The production of cash crops increased as the demand for fruits has surged along with the advancement of dietary lifestyle. A complex for fruit production was established, collaboration in cultivation of fruits was promoted, and funds were provided to expand storage facilities in production areas. In January 1977, a total of 92 vegetable main production complexes by city and county (465 eup-myeons, 37,945 ha in area) were designated, and in 1979, a total of 141 vegetable production complexes by items (690 eup-myeons, 49,996 ha in area) were designated by the Notice of the Ministry of Agriculture and Fisheries.

The implementation of the policy for improvement of the distribution structure of agricultural products, which has significantly contributed to advancing the commercialization of Korean agriculture, began during this period. At that time, in Korea, the economic growth led to a significant increase in the consumption of agricultural products per capita as well as the surging per capita consumption of fruits and vegetables and livestock products, leading to a remarkable change in the patterns of food consumption of the consumers. However, since the infrastructure for distribution of agricultural products was not established, leading to repeated problems of unstable supply/demand of agricultural products and severely varying prices. Therefore, a necessity of public wholesale markets was proposed, which can facilitate the effective resource distribution by agreeing on an optimal price in the agricultural products market, and enhance the distribution efficiency through transparency of transactions, fairness in price formation and disclosure of market information. Thus, policy was implemented to transform the distribution

structure from the previous structure centered on private wholesale market to a new structure centered on public wholesale market.

In December 1976, “Act On Distribution And Price Stabilization of Agricultural And Fishery Products” was enacted, thus establishing a legal basis for the formation and operation of the wholesale market of agricultural products and achieving stability in supply and demand for agricultural products and price. In 1977, a policy of establishing a large scale agricultural and fishery products wholesale market was determined, and a plan for construction of Seoul (Garak) agricultural and fishery products wholesale market was confirmed in 1980, which was the first case in Korea. The construction started in April 1982 and the wholesale market opened in June 1985. Thereafter, a total of 33 construction projects of the public wholesale markets for agricultural and fishery products were carried out over 29 years, and the latest one was the opening of the Busan International Seafood Wholesale Market in September 2008.

Prior to the establishment of the distribution network of the public wholesale markets, there were 7 complex stages of distribution, and due to the unnecessary duplication of distribution functions by many middlemen on the distribution route, distribution costs (gross margin) increased and profit share for the farm households decreased. In the quasi-wholesale markets operated by contracted wholesalers centered on large-scale traditional markets, 90% of the distribution of the total agricultural products was carried out, and the functions of a real wholesale market were performed in these markets. However, in the quasi-wholesale markets without legal basis, prices were formed in a secretive and unfair manner due to undisclosed activities of middle merchants.

With the establishment of the public wholesale markets, the following effects were achieved. 1) First, as the number of public wholesale markets increased, the quantity of the agricultural products in those public wholesale markets increased, which led to a rapid change in the market that involved the replacement of the existing quasi-wholesale markets. From the opening of the Seoul wholesale market in 1985, for the period of 26 years up to 2012, the trading volume in the public wholesale market rapidly increased at an average annual rate of 7.3%, and the proportion of trading in the quasi-wholesale markets decreased

sharply. 2) A transparent and fair practice was established for trading of agricultural products and a stable trading basis was established accordingly. By adopting auction as the basic trading method, open and fair price formation became possible, and as producers' options for place of shipment were increased, the bargaining power of producers increased significantly. The producers' distrust in the price formation process of wholesale markets was resolved, and the auction also provided incentives to improve product quality by inducing producer ratings and standardization since prices were formed according to grades and quality. 3) Costs were reduced due to the reduction of the agricultural products margin rate in the wholesale stage. From the existing low-efficiency and high-cost trading structure between small-scale shippers and contracted merchants, efficiency in trading and logistics was enhanced due to various factors such as an increase in the proportion of joint shipments, an increase in logistics efficiency, a shortening of trading time through rapid trading process, and a decrease in the amount of loss during distribution process.

<Table 7-14> Changes in the distribution/retail margin rate of fruits and vegetables before and after the opening of public wholesale market

Unit: %

Year	Apple	Pear	Water melon	Tomato	Autumn Radish	Garlic	Dried chili/pepper	All types of vegetables	All types of fruits
1982	20.6	20.7	26.1	15.8	28.8	27.0	46.2	36.0	19.5
1990	13.4	9.1	20.3	10.9	16.6	15.5	38.5	25.7	15.9
2000	10.0	8.5	10.9	7.4	14.6	17.2	13.2	15.9	11.1

Source: Korea Rural Economic Institute (2015)

In addition, 4) the range of price fluctuations of agricultural products have also been alleviated. As the ratio of individual shipments decreased and the ratio of joint shipments increased, the shipper's market bargaining power and control has strengthened. For example, if oversupply occurs due to an increase in production, price fluctuations can be mitigated as producers jointly dispose of or store production before shipment to prevent a sharp drop in wholesale market prices. 5) Farm household income has increased. Among

the total production of agricultural product, the ratio of market sales, that is, the commercialization rate, has increased sharply, and farm household income growth is estimated to have increased relatively due to the increase in the proportion of shipments to the public wholesale market. 6) The commercial value and distribution efficiency of agricultural products have been greatly improved. Previously, agricultural products were shipped in the form of non-grade and unpackaged products, so it was impossible to form a price that reflects the supply and demand situation by product grade. After the opening of the public wholesale market, an appropriate price for each grade was set thanks to the rapid increase in the shipment volume of standard products. This contributed to the added value of products and increase in farmer household income. In addition, as packaging standardization has progressed significantly due to distribution standardization policy, distribution efficiency such as transportation, storage, and loading and unloading has been enhanced. Furthermore, the improvement of the entry method into the wholesale market reduces the auction time and the loading and unloading costs thanks to mechanization of loading and unloading operations. Therefore, it reduces the overall distribution costs in the wholesale market.

Most of all, 7) the bargaining power/skills of small farmers in trading has improved significantly. In 1:1 trade between individual farmers and merchants, it was difficult to collect and analyze market information, so farmers' bargaining power was very low. However, as the proportion of joint shipments of standardized agricultural products has increased, their influence on bargaining power and price determination has increased.

<Table 7-15> Changes in the proportion of each shipment type in the wholesale market before and after the opening of public wholesale market

Unit: %

Year	Individual Shipment	Producer joint system of National Agricultural Cooperative Federation	Local Assemblers	Others	Total
1982	65.0	22.4	12.6	0	100.0
1990	47.5	35.2	12.4	4.9	100.0
2000	37.7	49.5	10.7	2.1	100.0

Source: Korea Rural Economic Institute (2015)

2.3 Major policies and systems during the 5th and 6th 5-Year Economic Development Plans(1982 ~ 1991)

During the 5th Five Year Economic and Social Development Plan (1982-1986), the basic goals of the agricultural sector were to make self-sufficiency in grains possible, create a foundation for modernization of agriculture, increase income for farmers, and create a foundation for welfare farming and fishing villages. Since the late 1970s, along with the shift in economic policy, the advocacy of food policy change and the abolition of agricultural protection emerged. Meanwhile, an open farming theory was raised that agricultural prices should be stabilized through control of purchase price of grain, promotion of an industry with a comparative advantage, and imports of insufficient agricultural products. The direction of agricultural policy has shifted to stabilizing prices by expanding agricultural imports and reducing price support, increasing non-farm income by building rural industries, and increasing agricultural income through income crops other than staple grain/crops. Control the purchase price of grain and foster an industry with a comparative advantage

The direction of agriculture during the 6th Five-Year Economic and Social Development Plan (1987-1991) was to improve productivity and increase income in rural areas through improved agricultural structure, and various measures were taken regarding the followings: difficulties of the rural economy, cattle price crisis, and liberalization of import. Difficulties in the rural economy were intensifying, in particular, there was a dramatic decrease in cattle price in 1984 after the encouraged buying of live cattle in farmhouses in the early 80s, and liberalization of imports of agricultural, fishery products was implemented due to pressure from the U.S. in the late 1980s. Consequently, various measures were taken, such as promotion of agricultural and fishing villages and debt alleviation policy. Under the 6th Five-Year Plan, the basic goals of the agricultural sector were to foster efficient and productive agriculture, forestry and fisheries industries, foster farmers and fishermen as professional managers, and improve convenience facilities and welfare in rural areas. The strategies to reach the goals were as follows: 1) adjustment of the production system in response to changes in food consumption structure, 2) improvement of agricultural and fishery structure and productivity, 3) Improving the efficiency of supply and distribution of agricultural and fishery products, 4) increase of income of farmers and fishermen and

improvement of rural living environment, 5) enhancement of agricultural and fishing manpower, development of new technology and 6) strengthening the support system for the activation of the rural economy.

After the basic policy of liberalizing agricultural products was finalized in February 1978, the import liberalization policy of agricultural products began as import liberalization measures were implemented three times in May, September, and January of the following year. The import liberalization rate of agricultural products expanded in the 1980s by liberalizing imports mainly on items that have little impact on domestic farms, such as primary hair, wood, raw sugar, coffee beans, cocoa beans, and feed. From the 1978 import liberalization measure to 1988, the import liberalization rate expanded to 72.8% for agricultural products, 93.4% for forest products, and 54.2% for fishery products, and the average tariff rate for agricultural and fishery products decreased from 43.2% to 25.2% during the same period. Since 1983, the need for import liberalization measures based on comparative advantage that liberalization of imports can stabilize prices and increase exports in the long run through reduced manufacturing costs was raised. And, the "import liberalization advance notice" was adopted in 1984, which selects and announces items subject to import liberalization in advance. In 1988, as pressure from the U.S. to open the agricultural product market increased, the "Three-Year Plan for Import Liberalization Advance Notice of Agricultural, Forestry and Fisheries Products" from 1989 to 1991 was announced and import liberalization was implemented step by step for a total of 243 items. Meanwhile, Korea was able to restrict imports of major agricultural and fishery products under the so-called Balance of Payments (BOP) clause for developing countries until the late 1980s, but as it recorded a three-year trade surplus from 1986, it was no longer able to restrict imports of 273 items and had to liberalize their import annually.

The Fostering Project for Agricultural and Fishery Successors was first introduced in 1981 to foster successors who will succeed in agricultural management as a way of improving the structure of farming and fishing villages as well as responding to the shortage of agricultural manpower. The project was based on the Agricultural and Fishery Successors Fund Act enacted in 1980 and its Enforcement Decree promulgated in 1981. The qualifications for

successors are defined as those who have completed military service or are exempted from the service. The age limit was raised from under the age of 35 in the early stages of the project to under the age of 40 in 1992. The fund for fostering successors was a full state loan, and the amount of support was differentially paid according to the applicant's farming design. An average of 4 million won was provided in 1981, and since 1995, when the WTO system was launched, the per capita support limit has been raised to 20 million won to 50 million won per item. Gradually, the repayment conditions of the principal were eased and interest rates were lowered.

The project, which took effect in 1981, provided KRW 189,835 million to 28,013 successors of farmers and KRW 20,932 million to 1,931 successors of fishermen by 1986. The proportion of farmers' successors by production item was 35.9% for livestock, 31.7% for cultivation, 28.6% for complex farming, 1.1% for fruit trees, and 2.8% for special crops. The use of funds was limited to the purchase of farmland and agricultural machinery, the installation of facilities, the building of livestock, the creation of grassland, and the purchase of seedlings. Since then, the measures to improve the structure of rural areas announced in 1991 included a project plan related to fostering successors for the next 10 years. Based on the 42 trillion won investment loan plan, the measures were planned to foster 10,000 successors of farmers and fishermen a year, and the Korea Federation of Agricultural and Fishery Successors was founded as a national organization of successors of farmers and fishermen.

Also, efforts to stabilize agricultural prices were materialized. Through the revision of the Rural Development Act in October 1982, the Rural Development Corporation, which was in charge of stockpiling and releasing agricultural products purchased by the National Agricultural Cooperative Federation, imported and stockpiled agricultural products with insufficient supply and released when their prices rose. Funds for low-temperature storage of onions and fruits, which have large seasonal price fluctuations and fluctuations in production by year, were provided to merchants and processors, while funds for vegetables that are difficult to store were provided to producers or producer organizations to control of amount and timing of agricultural products shipment. A joint shipping organization was fostered for 24 items, including vegetables, cucumbers, peppers, and apples, allowing farmers to autonomously control production and shipments, while wholesale markets,

storage, and processing facilities were expanded.

From the mid-1980s, measures continued to reduce the debt of farming and fishing households, which had increased due to factors such as the cattle price crisis in the early 1980s, and improve the structure of farming and fishing villages. First of all, on March 5, 1986, the "Comprehensive Rural Measures" was announced in the basic direction of introducing rural industries, diversifying income sources, and improving the structure of agriculture and fishery as a countermeasure against losses resulting from the cattle price crisis and increased farm debt.

<Table 7-16> Comprehensive Measures and Burden Reduction Measures of Comprehensive Rural Measures

1. Comprehensive measures.	2. Immediate measures to reduce the burden.
1) Support for taxation of factories in rural areas and designation of 100 agricultural and industrial districts by 1991, etc. 2) New establishment of mid- to long-term low-interest farmland loans and increase of loans for successors of farmers and fishermen. 3) Improving living conditions in rural areas. 4) Supply of mid- to long-term low-interest funds through the establishment of a special fund for rural development of KRW 500 billion by 1988. 5) Improvement of the system, such as the name of the Agricultural Promotion Corporation as the Rural Development Corporation and the Rural Development Corporation as the Agricultural and Fishery Products Distribution Corporation.	1) Extension of interest period and interest rate reduction of various funds related to the funds of buying of cattle in '83 and '84. 2) Converting loans related to water supply facilities into subsidies. 3) 10% exemption from special consumption tax for dairy products and juice drinks.

Source: Korea Rural Economic Institute (2015)

Since the comprehensive measures in 1986, debt reduction measures have continued to cope with the immediate debt problem in rural areas. The "Rural Household Debt Reduction Measures," announced on March 16, 1987, included replacing existing high-interest rural

bonds with low-interest funds, lowering mid- to long-term interest rates and extending repayment periods. The "Comprehensive Measures to Revitalize the Rural Economy," announced on December 9, 1987, included expanding the size of funds for agriculture and fishery, further cuts in interest rates on existing funds, extending repayment periods, and expanding government subsidies for water supply facilities.

Judging that the previous three measures lacked "fundamental resolution or self-reliance" related to rural issues, the "Comprehensive Rural Development Measures" was announced on April 28, 1989, and the legal and institutional device "Rural Household Debt Reduction Special Measures Act" was promulgated on December 3, 1989. The "Comprehensive Rural Development Measures" stipulated the development of farming associations and consigned farming companies as a collaborative effort to overcome the small scale of small farming structures.

<Table 7-17> Basic Direction and Blueprint of Comprehensive Rural Development Measures

1. Basic direction.	2. Rural blueprints in the 2000s.
1) Establishing mid- to long-term plans for rural development. 2) To find a solution to the current problem by reducing the burden and expanding support. 3) As a legal and institutional device, the enactment of the Special Measures Act, the establishment of the Rural Corporation, and the preparation of institutional support and promotion system for occupational change such as vocational training.	1) Management on an efficient scale centered on commercial full-time farmers and fishermen. 2) Expansion of non-farm income sources by creating an economic zone in rural areas where agriculture and industry coexist. 3) A farming and fishing village as a comfortable and convenient settlement space.

Source: Korea Rural Economic Institute (2015)

The previous "Comprehensive Measures for Rural Development" did not establish structural improvement in rural areas in the basic direction, so there was a problem that financial resources for this were not supported. Accordingly, in July 1991, the "Rural Structure Improvement Measure" was announced to promote the structural improvement project in earnest. The Rural Structure Improvement Measure secured financial resources for the

project and suggested directions and policy measures for 10-year agricultural administration by establishing a funding investment plan of a total of 42 trillion won over the 10 years from 1992 to 2001, specifically 35.5 trillion won in enhancing competitiveness and 6.2 trillion won in boosting rural vitality,. In particular, the policy to improve the distribution function of production areas through the promotion of main production complexes by item and the support for processing/distribution infrastructure expanded the proportion of joint production and shipments of production areas and served as an opportunity to enhance the power of transaction bargaining and market dominance of production areas.

2.4 Major policies and systems during the 5-Year New Economy Plan (1993-1997)

The 5-Year New Agricultural Administration Plan adopted the transition from the existing policy focusing on increased production and price to a policy focusing on structural improvement and the improvement of producer autonomy as basic strategies to enhance competitiveness and self-sustainability of the agriculture, forestry and fisheries industry. As of July 2, 1993, the "5-Year New Agricultural Administration Plan" was finalized as an action plan in the agriculture, forestry and fisheries sector in harmony with the civil government's 5-Year New Economy Plan. New Agricultural Administration was a concept suggested at a time when the domestic agricultural market was experiencing changes in the global agricultural environment and nearing market opening, and aimed at achieving new agriculture that fundamentally changes the direction of agriculture. Heo Shin-haeng, the then Minister of Agriculture, Forestry and Fisheries suggested the direction of new agriculture as technological agriculture, high-quality agriculture, sustainable agriculture, and export agriculture⁵¹.

⁵¹ Heo Shin-haeng (1999); Korea Rural Economic Research Institute (2015)

<Table 7-18> Idea of New Agriculture from Heo Shin-haeng, the then Minister of Agriculture, Forestry and Fisheries

<Global Changes in Agricultural Environment>

- 1) The central axis of the economy shifts from resources to capital or technology.
- 2) The transition from a time when food was scarce to a time when food was left.
- 3) The transition in agricultural consumption from low-grade products to high-end products.
- 4) The center of economic power moves to Northeast Asia while entering an era of infinite competition.

<Direction of New Agriculture>

- 1) technological agriculture 2) high-quality agriculture 3) sustainable agriculture 4) export agriculture

Source: Heo Shin-haeng(1999)

<Table 7-19> The Key Contents of the Five Year Plan of New Agricultural Administration

< Objective >

- 1) Fostering the technology agriculture, forestry and fisheries industry with competitiveness and self-sustainability
- 2) Training professional agricultural managers who respond to market opening.
- 3) Preparation of farming and fishing villages with abundant industrial and living spaces.

< Basic Strategy >

- 1) A shift in policy focus from increased production and price to structural improvement to improve productivity.
- 2) Improvement of autonomy that the government is in charge of infrastructure such as production infrastructure preparation, and producers lead production and market activities according to market functions. i.e. specifically, the production and shipment control of producers' organizations by item and the lead in market and distribution

<Preliminary task>

- 1) Reorganization of the autonomous system for farmers and fishermen.
- 2) Preparation and reinforcement of the agricultural, forestry and fisheries support organization.
- 3) Selective intensive investment through adjustment of investment priorities.

<Key policy measures>

- 1) Improvement of the system for promoting new agricultural administration, including the grain management system, farmland system, manpower development, and technological innovation system.
- 2) Improving the structure to enhance the competitiveness of each field, such as the rice industry and livestock industry, fisheries industry, and forestry industry.
- 3) Producer-led market and distribution innovation (Strengthening the government's function of building market and distribution substructure and fostering distribution, such as fostering producer organizations by item for autonomous production and shipment control, wholesale market construction, and agricultural product shipment standards)

Source: Heo Shin-haeng(1999)

The "5-Year New Agricultural Administration Plan" made production infrastructure preparation and technology development, which are the core projects of structural improvement, a priority for investment. In addition, by converting the project promotion system to an autonomous method of farmers and fishermen, the government presented investment plans and guidelines of the project, while allowing local governments to establish regional development plans and producers to select, plan and execute projects on their own.

In February 1994, the Rural Development Committee was formed as a presidential advisory body to propose comprehensive measures, and based on this, the Rural Development and Agricultural Reform Promotion Plan (Nongbal Measures) was established on June 14. The main contents were the selection of 10 core processes to strengthen the competitiveness of the farming and fishing industry, reform of related systems, improve living conditions in farming and fishing villages, implement welfare promotion projects, and arrange financial resources. The Plan was characterized by a shift in agricultural administration methods from the protection of small scale farmers to a selective support centered on full-time farmers and agricultural corporations. In addition, the role of cities and counties was emphasized by suggesting a shift from top-down agricultural management to bottom-up autonomous management, while the autonomy of farmers, producers organizations, and local governments was respected that the "rural development council" composed of residents

representatives deliberated their projects and budgets.

Meanwhile, with the establishment of the "5-Year New Agricultural Administration Plan" in 1993, the completion of the Rural Structure Improvement Project (a plan to invest 42 trillion won from 1992 to 2001), which was announced in July 1991, was advanced to 1998. Under the 1994 national agreement, a "special rural tax" was newly established to invest 15 trillion won in agriculture and rural areas for 10 years from 1995 to 2004, providing funds to reduce damage to agriculture and fishing caused by UR negotiations and realize major agricultural measures.

Technology development was emphasized as the core of structural improvement during this period, and since 1994, the Ministry of Agriculture, Forestry and Fisheries has promoted the "Agriculture, Livestock and Food Technology Development Project." The project was to overcome the difficulties of Korean agriculture and rural areas due to market opening and intensifying competition by developing the agriculture, livestock and food industries into the next-generation growth engine industry. The government budget provided a total of 508.3 billion won for 12 years from 1995 to 2006. The detailed projects until 2004 were "High-tech Development Project," "On-site Difficulty Technology Development Project," and "Venture-type SME Technology Development Project."

3.

Major Universities in Agriculture Sector in Korea

3.1 (An example/ a case) Seoul National University⁵²

3.1.1 Composition of the department

The case of Seoul National University is when the demand for agricultural and rural research and manpower training increases and various departments are newly established in line with changes that differentiate into several branches. With the establishment of Seoul National University in 1946, the College of Agriculture was newly started, and it consisted of the Department of Agriculture, Department of Forestry, the Department of Livestock, the Department of Agricultural Engineering, the Department of Agricultural Economics, and the Department of Agricultural Biology. Since then, in response to social changes such as raising agricultural added value and rural development, the Department of Damsa, the Department of Agriculture, the Department of Agriculture, the Department of Horticulture, the Department of Food Engineering, the Department of Pregnant Processing, and the Landscape were additionally established.

<Table 7-20> newly established department in the College of Agriculture in Seoul National University between 1960s and 1970s.

The year of establishment	Name of the department	The purpose of establishment
1962	Department of Agricultural Education.	Focusing on fostering agricultural instructors and rural instructors who will strive for social development in rural areas.
1967	Department of horticulture.	Vegetables, fruits, flowers, and ornamental trees are breeding, breeding, cultivation, and management tasks, and nutritious vegetables and fruits.

⁵² Mainly refer to the 50-year history compilation committee (1996a) of Seoul National University and the 50-year history compilation committee (1996b) of Seoul National University

The year of establishment	Name of the department	The purpose of establishment
1967	Food Processing major.	A study of basic theory and expertise necessary to process food based on agricultural products, livestock products, and marine products.
1969	Department of Forest Product Processing.	Comprehensive and systematic food-related education conducted individually by the Department of Agriculture and Chemistry, Livestock, Agriculture, Farm Administration, and Food and Nutrition at Home University.
1972	Department of Landscape Architecture	A field in which forest products are reasonably processed and used as useful products separately from the forestry department.

Source: It's written by referring to the 50-year history compilation committee of Seoul National University

In the 1960s, when the veterinary department included in the College of Agriculture became independent as a veterinary school in 1974, the College of Agriculture consisted of 14 departments, which were maintained until the introduction of an undergraduate system that integrated similar departments in 1997. In the 1990s, the college name was changed from the College of Agriculture to the College of Agriculture and Life Sciences, and several departments changed their names from traditional names to more engineering and industrial vibe.

<Table 7-21> The composition of College of Agriculture at Seoul National University, 1962-1996

1962	1976	1984	1996
Department of Agriculture.	Department of Agriculture.	Department of Agriculture.	Department of Agriculture
Department of forestry.	Department of forestry.	Department of forestry.	Department of forestry.
Department of Agricultural Engineering.	Department of Agricultural Engineering.	Department of Agricultural Engineering.	Department of Agricultural Engineering.
Department of Livestock.	Department of Livestock.	Department of Livestock.	Department of Livestock.

1962	1976	1984	1996
Department of Agricultural Chemistry.	Department of Agricultural Chemistry.	Department of Agricultural Chemistry.	Department of Agricultural Chemistry.
Department of Agricultural Economics.	Department of Agricultural Economics.	Department of Agricultural Economics.	Department of Agricultural Economics.
Department of Agricultural Biology.	Department of Agricultural Biology.	Department of Agricultural Biology.	Department of Agricultural Biology.
Department of Sericulture.	Department of Sericulture	Department of Sericulture	Department of natural fiber
Department of farmhouse	Department of farmhouse	Department of farmhouse	Department of farmhouse
Department of Agricultural Education.	Department of Agricultural education.	Department of Agricultural education.	Department of Agricultural education.
Department of horticulture.	Department of horticulture.	Department of horticulture.	Department of horticulture.
Department of Food Processing.	Department of Food engineering.	Department of Food engineering.	Department of Food engineering.
Department of forest product processing	Department of forest product processing	Department of forest product processing	Department of forest product processing
Department of landscape architecture	Department of landscape architecture	Department of landscape architecture	Department of landscape architecture
Department of Veterinary medicine.			

※ Yellow: Revised name of the department

Source: It's written by referring to the 50-year history compilation committee of Seoul National University

3.1.2 Size of the college

The number of teachers and students in the College of Agriculture has increased overall, but there are times when the number of teachers and students has decreased sharply due to changes in policies such as adjusting the quota of Seoul National University as a whole. In the 1960s, the number of students in the College of Agriculture decreased slightly while the number of teachers increased significantly, reducing the ratio of teachers to students from 17 to 11, and the ratio of the college was better than that of the College of Engineering in

the 1960s and 1970s. It can be estimated that the increase in the number of teachers at the College of Agriculture in 1963-73 was influenced by the revitalization of research in the agricultural sector according to the policy of increasing production at the time. On the other hand, the decrease in the number of students at the same time and the surge in the number of students at the College of Engineering seem to be due to the focus of fostering manpower or the shift of students' desired fields from agriculture to industry.

<Table 7-22> Number of instructors and students in the college of engineering and college of agriculture at Seoul National University 1963-1993

Unit: People

Category		1963	1968	1973	1978	1983	1988	1993	1963-1973	1973-1978	1978-1993
Number of teachers (including TA)	College of engineering	86	123	132	134	187	217	203	▲ 53.5	▲ 1.5	▲ 51.5
	College of Agriculture	83	98	119	90	117	125	132	▲ 43.4	▼ 24.4	▲ 46.7
Number of students	College of engineering	1,766	2,119	3,383	2,621	3,745	3,449	3,931	▲ 91.6	▼ 22.5	▲ 50.0
	College of Agriculture	1,415	1,145	1,352	1,072	2,366	2,387	2,283	▼ 4.5	▼ 20.7	▲ 113.0
Ratio of teachers to students	College of engineering	20.5	17.2	25.6	19.6	20.0	15.9	19.4			
	College of Agriculture	17.0	11.7	11.4	11.9	20.2	19.1	17.3			

Source: It's written by referring to the 50-year history compilation committee of Seoul National University (1996b)

3.1.3 Trends in faculty and research activities.

Since 1962, overseas trained faculty and facilities have been distributed to the College of Agriculture, and at the same time, the relationship with the Rural Development Administration has become close, and research activities by agricultural faculty have become active. First of all, research activities were promoted as faculty trained at the University of Minnesota and facilities were distributed and research subsidies were paid. In 1966, more than 50 professors from agricultural colleges were appointed as standing

members of the Rural Development Administration's Agricultural Test Research Guidance and participated in guidance and joint research to improve the qualities of test researchers. Following the promulgation of the Industry-Academic Cooperation Act in 1971, the connection between universities and test research institutes became more concrete and the quantity and quality of research projects could be improved.

From the early 1960s to the mid-1970s, the research field mainly focused on food production by crop cultivation, and excellent varieties were fostered as a major task in crop science. In the field of cultivation and physiology, studies have also been conducted on improving the cultivation environment, fertilizers, controlling pests by new pesticides, and controlling weeds by herbicides.

According to the status of the final degree-acquiring countries of 100 faculty members in 1995, 27 of the 100 faculty members were Korea, 54 were the United States, 10 were Japan, 3 were Germany, 2 were Britain, 1 was France, 1 was Australia, and 1 was the Philippines, 3/4 were overseas.

3.1.4 Research Institutes

In the 1960s, there were two affiliated research institutes in the College of Agriculture. In March 1967, the Tropical Agricultural Research Institute was established to study tropical agriculture and seek overseas expansion through agriculture, and in August 1967, the Agricultural Science Research Institute was established to promote domestic agricultural research. In May 1972, the Agricultural Science Research Institute absorbed and integrated the Tropical Agricultural Research Institute to contribute to agricultural and rural development through more organized and systematic research activities. After moving to Seoul National University campus in 1975, the Agricultural Science Research Institute was promoted to a legal research institute affiliated with the College of Agriculture and renamed the Agricultural Development Research Institute. In 1976, the Division of Basic, Applied and Agricultural Development and the Operation Office were established, and in 1978, the Research Department and the Administrative Office were established. As of the mid-1990s, the Research Division of Production, the Research Division of Biological Engineering, the

Research Division of Agricultural Engineering, the Research Division of Animal and Plant Conservation, the Research Division of Agriculture and Forest Product Processing, the Research Division of Natural Conservation, the Research Division of Rural Development, and the Research Division of Overseas Agriculture were established.

Between 1968 and 1985, 228 studies were conducted, and 46 studies were conducted in 1984, indicating that the number of research projects increased in the 1980s. The research fields were agricultural production infrastructure and mechanization research, agricultural production research, agricultural production processing research, and rural development research. Between 1986 and 1994, 281 academic seminars were held, and 443 service studies were conducted. The themes of the annual International and Domestic Symposium from 1989 to 1994 were agricultural biotechnology, environmental pollution and agriculture, the course of Korean agriculture after UR agricultural negotiations, sustainable agricultural and environmental conservation, and agricultural structure improvement strategies in the era of internationalization.

3.2 (Case/example) Korea University.⁵³

3.2.1 Composition of departments.

Korea University's case is a case in which the boundaries between existing colleges and departments have been broken and reorganized as technological changes have been reflected in the academic system due to the development of biotechnology since the 1970s.

Founded in 1953, Korea University's College of Agriculture consisted of the Department of Agriculture, Department of Forestry, Department of Agricultural Chemistry, and Department of Agricultural Management as of 1960, followed by the Department of Horticulture and Department of Livestock in 1963, the Department of Food Engineering in 1982, and Department of Genetic Engineering in 1983. In 1992, the name of the college was changed from College of Agriculture to College of Natural Resources, and as of 1994, the

⁵³ Mainly referred to Korea University (2008a), Korea University (2008b), and Korea University (2011).

College of Natural Resources was composed of Agricultural Biology (Plant Conservation), Agricultural Economics, Agricultural Chemistry, Forest Resources (Forestry), Food Resources(Agriculture), Food Engineering, Horticultural Science(Horticulture), Genetic Engineering, and Applied Animal Science(Livestock).

In 1995, some professors from the College of Natural Resources, led by those from the Department of Genetic Engineering and Food Engineering, and professors of the Department of Biology from the College of Science formed a new graduate school organization called the Graduate School of Biotechnology and were selected for Priority Promotion of Science and Engineering Graduate Programs (PPSEGP) by the Ministry of Education to foster research universities. Professors majoring in the latest biotechnology fields such as molecular biology and biotechnology within the College of Natural Resources and professors in the field of biology, in other words, have integrated the field of agriculture and biology, and basic and applied fields into the category of "biotechnology." This organization was completely separated from the existing colleges in 2000 and became an independent college with its undergraduate program.

3.2.2 West German government aid and Establishment of research infrastructure.

Behind the formation of the biotechnology-based organization beyond the boundaries of colleges and departments in the mid-1990s was Korea University's special situation, where the latest biotechnology field flowed into rather than to the Department of biology.

In 1973, the College of Agriculture established and implemented a long-term development plan with free education aid from the West German government. The development plan aimed to establish a "cooperative system of pure science and applied science" to raise the level of agriculture, and to this end, research infrastructure was mainly built, including the expansion of facilities, equipment, books, and the dispatch of faculty and Ph.D. students to West Germany. Through the investment of Korea University and the aid, a new building for the College of Agriculture with 2,738 pyeong of ground floor and 4 stories above ground was completed at the end of 1976, and research equipment and instruments needed to

conduct the latest research were expanded along with the construction of new building. Dr. Lee Se-young of the Korea Atomic Energy Research Institute, the first molecular biologist in Korea in the late 1970s, was appointed to the College of Agriculture, and it is said that the latest research facilities of the College of Agriculture was one of the most important reasons for him to take his position in the College.

In 1977, Korea's first special graduate school of agriculture, Food Development Graduate School, was also opened with the free aid to the College of Agriculture by the West German government. Along with academic research on food issues, the goal was to foster practical agricultural workers, and master's and researchers' courses were established to advance agricultural technology. There were four fields related to food processing in food chemistry, food microbiology, food processing, and fermentation chemistry, and three fields related to agriculture in horticulture, food economics, and food resources. The faculty consisted of 30 full-time faculty members and 15 instructors, and the total capacity was 70.

3.2.3 Research Institutes.

In 1964, the International Agricultural Resources Research Institute was established as an affiliated institution of the College of Agriculture, and aimed at introducing and researching advanced overseas agricultural technologies, improving the quality of agricultural engineers, and carrying out subsidiary projects. In the 1960s, the institute was conducted research related to agricultural management and economy, and later renamed the Korea Agricultural Research Institute when it was officially launched as a research institute affiliated with Korea University in 1971 due to internal operational difficulties. The purpose of the institute was to contribute to the modernization of Korean agriculture by investigating and researching agricultural theories and technologies, and seven sectors were agriculture, forestry, agricultural chemistry, agricultural economics and management, horticulture, livestock, and food processing.

4.

Major Research Institutes in Agriculture Sector in Korea

4.1 (Case/example) Rural Development Administration

4.1.1 Establishment.

Through the enactment of the Rural Development Act (1962.3.21) and the Ordinance of the Rural Development Administration (1962.3.29), the Rural Development Administration was established by integrating the existing Central Agricultural Technology Institute, Farming Institute, Department of Regional Society at the Ministry of Agriculture and Forestry, and Training Institute. As a result, the unification of the farming test research and technology distribution system and differentiation of functions by specialized field were achieved. Specifically, by establishing rural promotion centers in each province to integrate test research and training projects in each province, and by establishing rural guidance centers in each city and county to combine research and guidance functions at various levels, organizations of agricultural research and guidance were systematized. In addition, by installing research institutes and test sites by field, test research organizations have been specialized and systematized⁵⁴. At that time, there was a plan to integrate the guidance project into the Ministry of Agriculture and Forestry, or general administrative organizations, but based on the perception that technological innovation cannot be expected with authoritative way of guidance, and that linking connection between test research and technology distribution are essential⁵⁵. It was finally decided to install an institution that integrates test research and guidance functions as an external office of an administrative organization.

⁵⁴ Agricultural administration research center(2012); Rural economic research center (2015)

⁵⁵ Donghee Kim(1999a)

4.1.2 Changes in policy stance, business concept, and performance.

As industrialization and economic growth policies were promoted from the 1960s, the performance of the Rural Development Administration changed along with the agricultural and political stance. In the 1960s, the early stages of the implementation of the economic development plan, the Rural Development Administration established an agricultural research and guidance system, and in the 1970s, the period of high growth and when income growth was attempted through complex farming, it realized increased production through the supply of new varieties and production technologies. In the 1980s, when the economic structure changed and agricultural competitiveness improved, it established a year-round production system through the development and distribution of greenhouse agriculture, and in the 1990s, when the agricultural market opened, high-quality, it developed low-cost production technology and promoted mechanization⁵⁶.

<Table 7-23> Core achievements of the Rural Development Administration according to changes in the internal and external environment and policies.

Time Period	Environmental and Policy changes	Key outcome/Achievements
1960s	The beginning of industrialization.	Establishing a research guidance system.
	- The 1 st and 2 nd Economic Development Plan.	- Farmers' education and the spread of new technologies.
	- Rural Modernization Promotion Act	
1970s	Economic development and compressed growth.	Evacuation of barley head (Green Revolution)
	- Enter the industrial society.	- Development of food production technology.
	- Multi-farming test project.	▶ Promote and distribute unified rice.
1980s	Advanced industrialization.	Overcoming the Seasonality of Agriculture (White Revolution)
	- Changes in economic structure.	- Establishing a production system throughout the year.

⁵⁶ Agricultural administration research center(2007); agricultural research center2012)

Time Period	Environmental and Policy changes	Key outcome/Achievements
	- Comprehensive Measures for Rural Development	► Developing a vinyl farming method.
1990s	Localization, internationalization, and informatization.	Improving competitiveness in agricultural technology.
	- The launch of the WTO system.	- High quality, low cost production technology.
	- Basic Act on Agriculture and Rural Affairs.	- Mechanization and automation.

Source: Agricultural Research Center(2007)

The Rural Development Administration selected the top 50 technologies from the 1960s to the 1990s in 2012. They were 1) development of Tongil-type rice varieties, 2) development of large-scale cabbage varieties, 3) introduction of agricultural machinery and establishment of farmland infrastructure, 4) development of silkworms and mulberry varieties, and 5) corn hybrid development for the 1960s and 1970s; 6) green house and mulching cultivation technology, 7) overcoming *pyricularia grisea*, 8) rice integrated machine technology development, 9) high-quality Korean beef production technology, 10) fruit tree bag cultivation and unforeseen cultivation technology for the 1980s, and 11) rice varieties were replaced, 12) Publication of Soil genealogy-Outline of Korea Soil, 13) Comprehensive Management IPM Technology, 14) Fruit Plantation-Apple Reservoir and pear Y-massed Cultivation, 15) Korean Seed Pig Development, 16) Fruit Non-destructive quality determination technology, 17) Development of popular greenhouse model, 18) New mushroom variety development, 19) Vegetable factory-like production of seedling technology, 20) Cultivation of new domestic varieties of fruit trees, 21) disease-free seed potato production technology, 22) use of natural enemies for pest control of vegetable crops, 23) Korean hydroponics technology, 24) start of flower breeding and industrialization.

The purpose of the Rural Development Administration and the concept of research, guidance, and education projects have changed from the time of market opening. The purpose of the institution has shifted from farming technology and living guidance for rural families to agricultural development through the promotion of agricultural science and

technology. The concept of each project was changed. For the test research project, the range of agricultural technology was expanded and function of establishing standards, and quality management were included, and the unit of agricultural management was expanded to producers' organizations. For rural guidance project, the content changed from centering on the spread of farming technology and the improvement of living in rural areas into the development and distribution of field difficulties technology and the strengthening of the organization of rural workers such as successors and rural women. For the education and training project, the scope of the target was expanded to rural women and agricultural students, and education and training for farmers' successors and rural women were strengthened.

▪ ¹¹ **Agricultural research center (2012)**

<Table 7-24> Changes in the purpose and concept of research and guidance and education projects of the Rural Development Administration.

Category	1962~1994	Changes starting from 1995
Purpose	- Promoting the welfare of farmers through test research, enlightenment guidance, dissemination of technology, and training of leaders for the promotion and development of rural areas.	- Development of agriculture and improvement of farmers' welfare through test and research projects, rural guidance projects, and education and training projects for agricultural people to promote agricultural science and technology.
Test Research Business	- ÿ Agricultural technology (including agricultural and livestock product use, processing, and agricultural technology) Improvement test research on rural life (clothing and shelter) - Production of high-quality crops, high-quality livestock, and prevention of livestock epidemics - Research and Research for Agricultural Management and Rural Development	- A test study for the development of agricultural science and technology (including technologies for improving agricultural materials such as agricultural machinery, pesticides, fertilizers, etc.) - Experimental research on advanced technology development and agricultural environment conservation for agricultural and biological industries - Production of high-quality crops, high-quality livestock, useful microorganisms, etc. and development of preventive measures for livestock infectious diseases

Category	1962~1994	Changes starting from 1995
		- Setting standards and quality control for agricultural materials such as agricultural machinery, pesticides, and fertilizers - Investigation and research on proper scale and efficiency of agricultural management and improvement of management of agricultural producer organizations
Business for fostering rural leaders	- Production of blue-chip crops, blue-chip livestock, and prevention measures for livestock infectious diseases. - Dissemination of scientific knowledge and technology related to agriculture and living improvement - Rural side business knowledge and technology dissemination - Education of farmers on the conservation and use of natural resources useful for agricultural development and peasant life - Fostering rural organizations for the purpose of improving agriculture and peasant living - Establishment of a pilot rural construction project	- Fostering agricultural organizations for the purpose of enhancing agricultural productivity and improving the lives of farmers - Fostering agricultural successors such as rural youth and agricultural successors - Supply of high quality seeds and livestock for agricultural products - Propagation of test research project development technology - Development and dissemination of local agricultural development and farmers' difficulties in the field - Scientific observation of crop diseases and pests, spread of control information and technical guidance in preparation for meteorological disasters - Guidance for improving the quality and dignity of crops - Guidance on quarantine technology for the prevention of livestock diseases A test study on the development of advanced technologies and preservation of agricultural environment for the agricultural bio-industry.
Business of educational training	- Research for Agricultural Management and Rural Development - Ministry of Agriculture and Forestry-related technical civil servant training	- Production of high-quality crops, high-quality livestock, useful microorganisms, etc. and development of measures to prevent livestock infectious diseases.

Category	1962~1994	Changes starting from 1995
	- Agricultural research, training for agricultural guidance business workers - Executive training for executives and staff of organizations under the Ministry of Agriculture and Forestry - Training of rural resource leaders	- Education and training for pilot research project, rural guidance project and education and training project - Agricultural technology education and training for teachers and students of agricultural schools through cooperation between agricultural industry, academia, relations, and research institutes - Education and training for farmers, rural youth, rural women and members of related organizations - Farmer side job training and job placement - Education and training for nurturing professional agricultural manpower such as full-time farmers and agricultural successors

Source: Agricultural Research Center (2012)

4.1.3 Changes in an organization and budget

<Table 7-25> Organizational change of Rural Development Administration, 1962-1994

Time period	Main history	Details
1962	Rural Development Administration was launched.	2 bureau, 11 institutions
1963	Reorganization: Training center → Agricultural Public Officials Training Center	
1965	Reorganization: Irijijang → Breeding Research Institute.	
1966	New planning and management office.	
1967	Capital Six-Breeding Research Institute → Honam Crop Test Center Reorganization	
1969	Yeongnam Crop Test Center was established.	
1973	Forest Service transferred.	3 bureau 12 department 7 officers in charge
1977	Establishment of Agricultural Management Research Institute	
1978	New sports ground.	

Time period	Main history	Details
1979	Agricultural Management Research Institute → Transferred to the Ministry of Agriculture and Forestry	
1981	Plantation → Transferred to the Ministry of Agriculture and Forestry.	
1983	New Office for Technical Distribution and Technical Information	3bureau 13departments 12officers in charge
1991	Institute of Plant and Environment → Institute of Agricultural Technology.	3bureau 13departments 13officers in charge
1994	Tropical Agricultural Center → Technology Cooperation Center Reorganization Examination bureau → Reorganization of Research and Management Bureau. Integration of Guidance Bureau and Technology Distribution Bureau → Reorganization of Technical Guidance Bureau → Agricultural Materials Quality Management Function Integration of Agricultural Technology Research Institute, Agricultural Genetic Engineering Research Institute, Agricultural Pharmaceutical Research Institute, and National Agricultural Materials Inspection Office (Ministry of Agriculture and Forestry) → Agricultural Science and Technology Institute reorganization → Horticultural test site and fruit research institute integrated → Horticultural research institute reorganization → Submarine Testing Laboratory, National Office of Agriculture and Forestry (Ministry of Agriculture and Forestry) Integrated → Submarine Insect Research Institute Reorganization Livestock test site.	

Source: Agricultural Research Center (2007)

Since its launch, the Rural Development Administration has been reorganized, such as the establishment of a new research institute or the abolition and integration of existing research institutes according to changes in the trend of the times. In the 1960s and 1970s, when food production, self-sufficiency, and nutrition improvement were urgent, barley research institutes and rural nutrition improvement research institutes were established, pesticide research institutes were established in the 1980s and agricultural genetic engineering research institutes in the 1990s. As the importance of horticulture and livestock grew, organizational changes such as the establishment of an orchard research institute, the subsequent integration with the horticultural test site, and the reorganization of the livestock test site took place in the 1990s.

The number of people in the Rural Development Administration started from 781 in 1962, increased sharply to 1,651 between the late 70s and early 80s, and in 1991, a large number of 680 people, including research workers, were added to reach 2,289. In 1994, it reached 2,713 due to changes in the organization that organizations and 520 employees belonging to the Ministry of Agriculture and Forestry were transferred.

The Rural Development Administration's budget started at 230 million won in 1962 and grew rapidly from the mid-1970s to 50.2 billion won in 1990. It increased very rapidly throughout the 1990s when technology development investment was emphasized amid the crisis of opening the agricultural market, and reached 235.5 billion won in 1996..

4.2 (Case) Korea Rural Economic Institute⁵⁷

4.2.1 Establishment.

The Korea Rural Economic Institute was founded in 1967 as the Agricultural Management Institute at the Rural Development Administration, which was reorganized into the Agricultural Economy Research Institute under the direct control of the Ministry of Agriculture and Forestry in 1970 and the National Agricultural Economy Research Institute in 1973. The Rural Development Administration's Agricultural Management Institute, which

⁵⁷ Mainly referenced to Kim Dong-hee (1999b) and Korea Rural Economic Research Institute (2008).

was in charge of economic analysis of technology development by the Rural Development Administration, conducted research on agricultural management, including agricultural statistics, agricultural observation, and grain policies, survey and analysis of agricultural distribution, and research on agricultural administration development such as research of long-term agricultural development strategies. At the same time, in response to the serious shortage of talent, it hired overseas Ph.D. professors as contract officials to entrust research tasks, while strengthening the overseas dispatch training of researchers to the extent that it once dispatched up to a third of its research personnel.

On April 1, 1978, the Korea Rural Economic Institute was established as a government-funded research institute dealing with the agricultural and fishing village economy, and the "Korea Rural Economic Institute Promotion Act" was enacted the following year. At the time of establishment, the purpose was to "contribute to the development of the national economy and national welfare by comprehensively investigating and researching the development of the agricultural and fisheries economy and rural society." It can be seen that the ultimate goal was set beyond agriculture to national economic development and aimed for comprehensive research that encompasses not only the economic aspect but also the social development aspect. The functions of the Korea Rural Economic Institute were largely classified into four categories: 1) rural economic research, 2) rural social research, 3) rural poll, 4) education and training⁵⁸.

4.2.2 Development of research institute activities by period.

In the early days of 1978-1985, a systematic and comprehensive research and support system were established to effectively cope with agricultural administration during the transition period. Research tasks such as long-term agricultural development strategy, agricultural distribution structure, food supply and demand, farmland system, agricultural mechanization, non-farm income, and rural settlement development were carried out. Along with short-term research projects requested by the government, it suggested directions of agricultural administration and improvement measures by holding policy

⁵⁸ Donghee Kim(1999b))

associations and conducting polls on major agricultural administration issues, and tried to improve the quality of research projects with close research management through the Research Advisory Committee and evaluation committee. In 1978, Korean academic journal "Rural Economy" and the English journal "JRD (Journal of Rural Development)" was created to advance academic research. Since there were only a few Ph.D.-level personnel, they actively promoted overseas trips and training and held various seminars and special lectures, especially international seminars inviting overseas scholars, while actively securing new personnel through public recruitment of Ph.D. holders and master's degree holders at home and abroad. Accordingly, the number of research staff increased from 50 at the time of opening to 112 at the end of 1985. While promoting joint research and large-scale international symposiums with the Asia-Africa Rural Development and Reconstruction Organization (AARRO) and the Asia-Pacific Development Center (APDC), it conducted regional development research by establishing rural regional development research team, established agricultural sector plans for national economic and social development plans such as the national long-term development plan for 2000 and the 6th 5-Year Plan, and collected public opinions.

From 1986 to the early 1990s, in addition to research studies, the Institute led the establishment and advice of policies such as joint work with the government, various committee activities, and the operation of a presidential advisory body in the process of preparing countermeasures for the Uruguay Round negotiations (UR). First of all, he led policies such as restructuring to improve agricultural competitiveness and easing the debt burden on farmers in response to changes in trade conditions that expand import liberalization of agricultural products in the mid-to-late 1980s. While conducting research to prepare measures to negotiate the Uruguay Round, Vice President Choi Yang Bu was appointed as an advisor to the Minister of Agriculture, Forestry and Fisheries in 1991 to form and operate a "UR Task Force" within the Institute. Agricultural structures, farmland ownership and lease, and farm income and debt issues were actively studied, which were the basis for restructuring agriculture and establishing measures for urgent debt issues, including "Comprehensive Rural Measures" in 1986, "Farm and Fishery Debt Reduction Measures" in 1987, and "Rural Economy Vitalizing Measures" in 1989. In 1993, he led the

civil government to establish a new agricultural administration and prepare countermeasures for the settlement of Uruguay Round negotiations (UR). In the civil government founded in 1993, Heo Shin-haeng was selected as the first Minister of Agriculture, Forestry and Fisheries to lead the new agricultural administration with a special report to the Blue House, including reform measures expanding farmland ownership limits and allowing seasonal amplitude of rice prices. Meanwhile, Dr. Choi Yang-bu was selected as senior secretary for agriculture, forestry and marine affairs at the Blue House and served for four years. The secretariat of the Rural Development Committee to respond to the settlement of the Uruguay Round negotiations was established within the Institute and operated for six months in 1994. Chung Young-il, then director and Ph.D. personnel actively participated in the Committee.

4.2.3 Organization and budget.

At the time of its establishment in 1978, the number of personnel was 87, including one chairman and director, and it was increased to 135 by 1980 and maintained until 1984. Since the mid-1980s, as discussions on market opening have become active, the demand for research and the resulting demand for research personnel have soared. The quota reached 195 in 1988 through steady increase and conversion to regular workers, but the quota was adjusted to 188 in 1994 due to deepening manpower outflow.

The total revenue budget consists of government contributions and self-income and increased rapidly from KRW 5.8 billion in 1978 to KRW 3.6 billion in 1988. Government contributions increased from 500 million won in 1978 to 3 billion won in 1988 and 5.4 billion won in 1996, and its own income, including service income, increased from 80 million won in 1978 to 600 million won in 1988 and 5.3 billion won in 1999.

5. Implications of STI in Agriculture Sector during Korea's Structural Transformation Period

Since the 1960s, Korea has undergone a decrease in agricultural labor, an increase in leased land farming, a change in the composition of agricultural products due to changes in food consumption patterns and profitability, and agricultural productivity has improved significantly through variety improvement and mechanization.

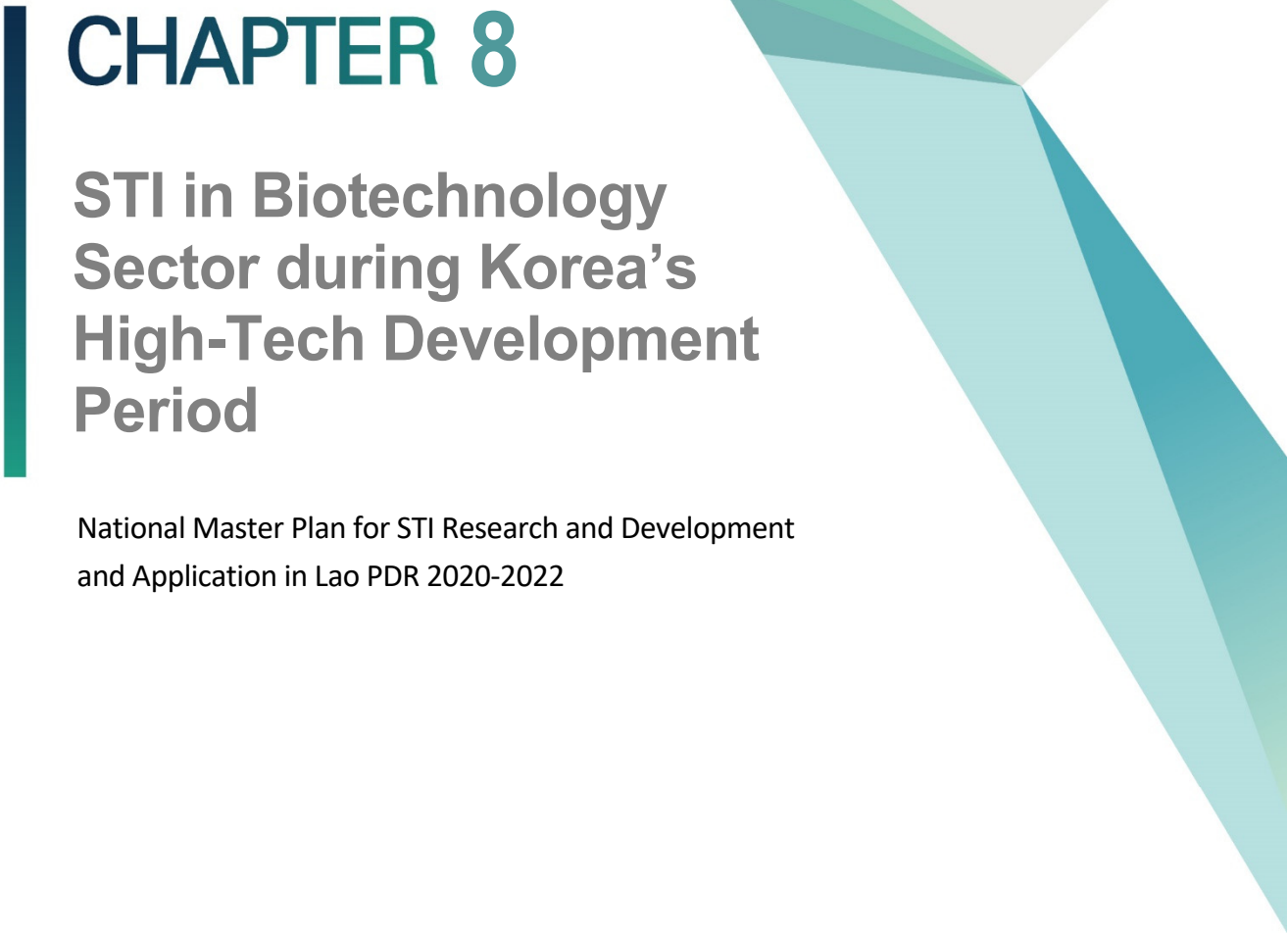
By the mid-70s, the government focused on food production increase and food self-sufficiency, and efforts to develop technologies represented by Tongil-type rice paid off, while making efforts to change the structure of agriculture and rural areas in response to changes in food consumption and agricultural market opening. In response to the increasing difficulties of agriculture and rural areas such as farm debt, it provided various support, promoted high value-added agriculture through reform of the distribution market and technology development.

Researchers at universities in the agricultural field had opportunities for research activities and contributing to food growth through cooperation with the Rural Development Administration, such as joint research and research guidance, and have advanced agricultural technology development with foreign aids. The Rural Development Administration established an efficient system outside the administrative agency at a time when it was urgent to develop and quickly distribute varieties and cultivation technologies and supported technology development in line with changes in social needs. Through agricultural/farm survey, research, and information provision, the Korea Rural Economic Institute established a development system for agricultural policies. In response to the problem of lack of research manpower and competency, universities and research institutes promoted training through overseas training and attracting overseas Ph.D. personnel in the short term, and training through university and graduate education in the long term.

Korea's agricultural technology development has shifted its role from increasing production to increasing productivity and increasing added value. In the process, the development of

agricultural technologies such as variety development, vinyl greenhouses, mechanization and automation, processing and distribution technology, alleviated the impact of industrial structure transformation, and supported the transition of agricultural production and sales system.

It appears that policies to increase the autonomy and market dominance of farmers and farmers were effective at the rural and farm level. Distribution network innovation through the establishment of a public wholesale market not only lowered the distribution margin rate but also increased the trading bargaining power of producers, and similarly, innovation in production areas distribution functions increased market dominance through joint production and shipment at production sites.

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in shades of teal and light blue, also pointing downwards.

CHAPTER 8

STI in Biotechnology Sector during Korea's High-Tech Development Period

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022



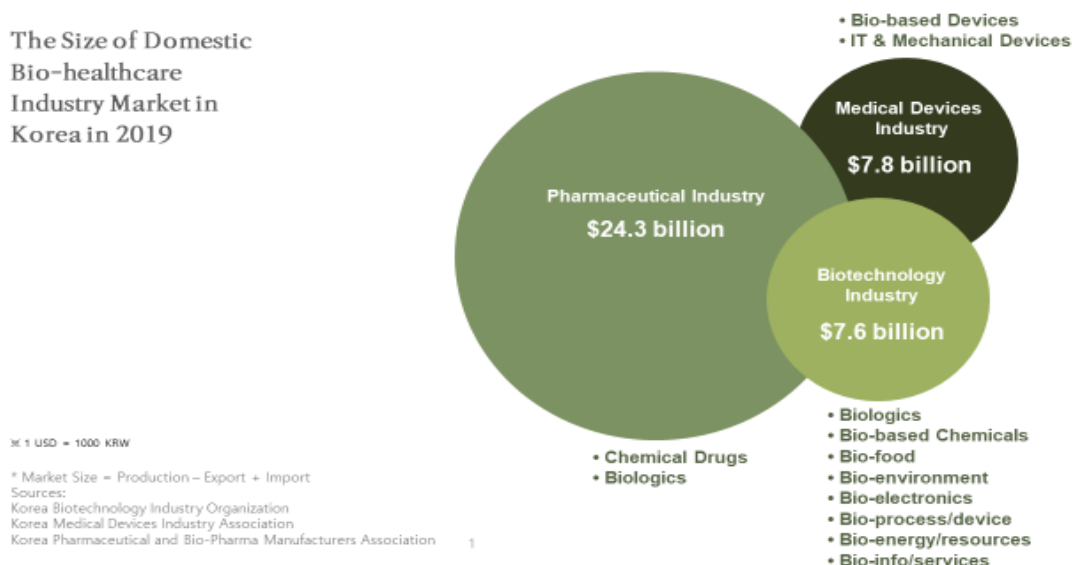
Chapter 8. STI in Biotechnology Sector during Korea's High-Tech Development Period

1. Korea's Biotechnology Sector

1.1 The current status of the biotechnology sector in Korea.

The biotech industry is defined as a field that "includes industries that have been reborn through innovative techniques using biotechnology along with new industries created by using biotechnology as innovative tools based on biological resources." Based on the basic science of biology or life science, biotechnology is widely applied to medicine, agriculture, food, chemistry, environment, and energy. In particular, biotechnology is very actively used in the health and medical fields, so the biotech industry combined with the pharmaceutical and medical devices industries is sometimes referred to as the bio-health industry. As of 2019, Korea's bio-health industry is approximately 39.7 trillion won, combining the pharmaceutical industry worth 24.3 trillion won, the medical device industry worth 7.8 trillion won, and the biotech industry worth 7.6 trillion won.

[Figure 8-1] The size of domestic bio-health industry market in Korea in 2019



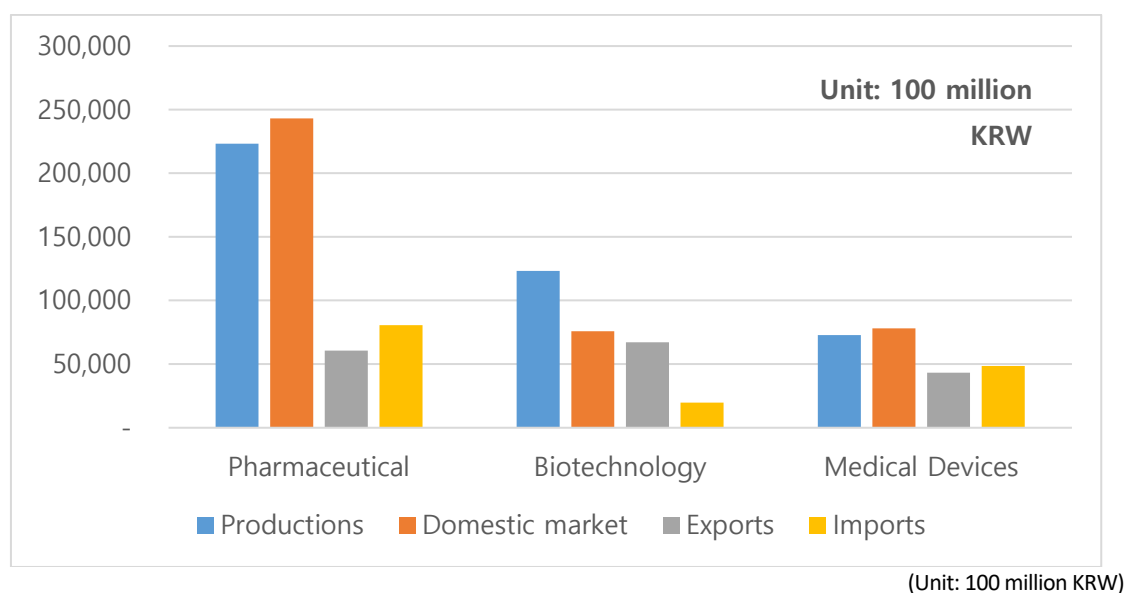
Sources: Korea Biotechnology Industry Organization (KoreaBIO), Korea Medical Devices Industry Association (KMDIA), Korea Pharmaceutical and Bio-Pharma Manufacturers Association (KPBMA)

The biotech industry has been supporting the growth of the pharmaceutical and medical device industries. Currently, companies and startups based on biotechnology which are called biotech companies, are leading innovation in the pharmaceutical and medical device industries. As can be seen from the rapidly growing cases of biologics and in-vitro diagnostics such as COVID-19 test kits, the biotechnology sector is driving the growth of the pharmaceutical and medical device industries.

In Korea, the biotech industry shows that the scale of production is much larger than that of the domestic market, and in particular, the export is three times larger than import at the present time, showing its export-oriented industrial structure. On the other hand, the pharmaceutical industry shows bigger scale in domestic market than the production, and the size of its imports exceeds that of exports, indicating the industrial structure centered on the domestic market. Such differences well reflect the characteristics of Korea's biotech industry, which has grown with new technology-based products and services, and the pharmaceutical industry, which has grown centered on domestic demand, especially generic drugs. Currently, while the biotech industry has a domestic market size of 1/3

compared to the pharmaceutical industry, the production is half size and the export scale is larger.

[Figure 8-2] Status of pharmaceutical, biotech, and medical device industries in Korea in 2019



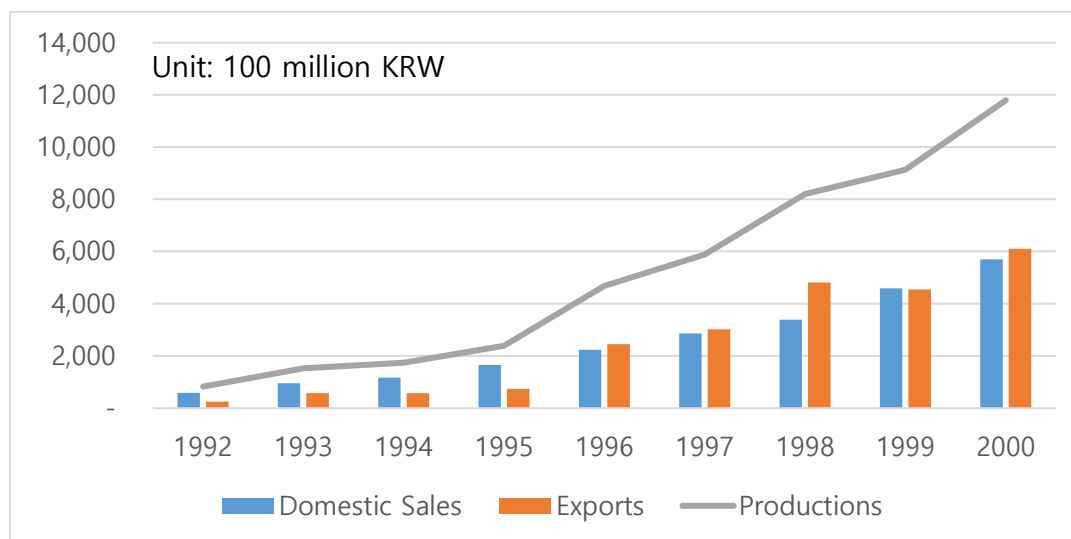
	Production	Domestic market	Export	Import
Pharmaceutical	223,132	243,100	60,581	80,549
Biotech	123,235	75,756	67,124	19,644
Medical devices	72,794	78,039	43,245	48,490

Sources: KPBMA (2020); KoreaBIO (2020); KMDIA (2021)

1.2 Growth in the biotechnology sector in Korea

The scale of Korea's biotech industry production rapidly grew at a compound annual rate of 39.4% from ₩82.7 billion in 1992, when statistics were first identified, reaching ₩1.1795 trillion in 2000. Since the mid-1990s, the export scale has been similar to or larger than the domestic sales, as the proportion of exports in production scale has increased.

[Figure 8-3] Statistics on bioindustry production scale in Korea, 1992-2000

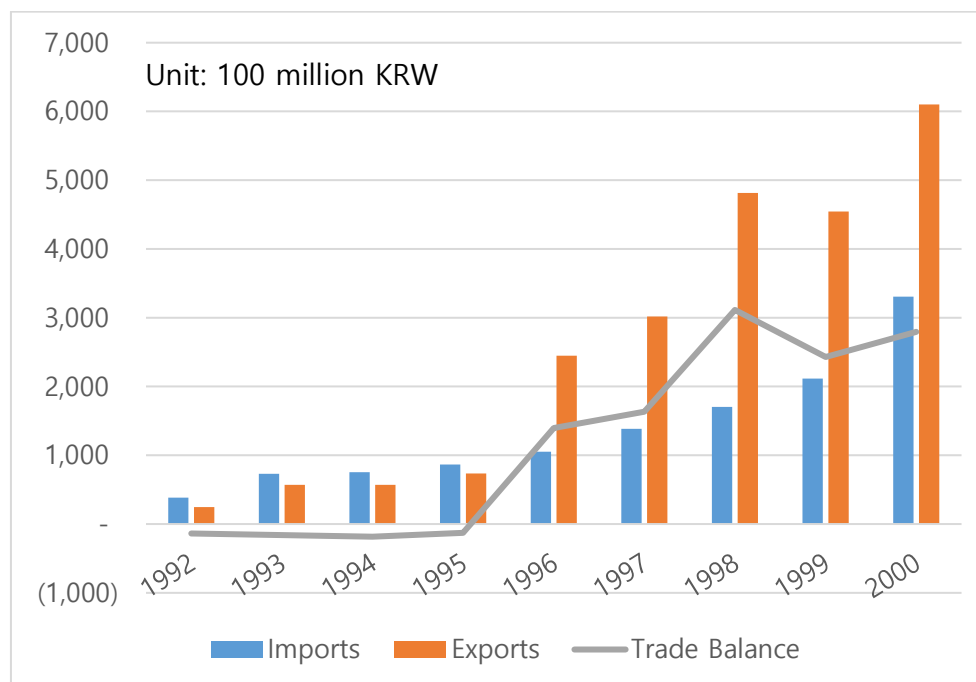


	1992	1993	1994	1995	1996	1997	1998	1999	2000	CAGR
Domestic sales	581	952	1,166	1,652	2,233	2,861	3,383	4,587	5,694	33.0%
Exports	246	569	569	735	2,448	3,018	4,815	4,543	6,101	49.4%
Production	827	1,521	1,735	2,387	4,681	5,879	8,198	9,130	11,795	39.4%

Source: KoreaBIO (2012)

Between 1992 and 2000, Korea's biotech industry import scale increased from ₩38.4 billion to ₩330.6 billion and exports from ₩24.6 billion to ₩610.1 billion. Exports grew faster than imports at a compound annual rate of 49.4%, and the trade balance turned into a surplus in 1996, reaching ₩279.5 billion as of 2000.

[Figure 8-4] Trend of bioindustry trade in Korea, 1992-2000

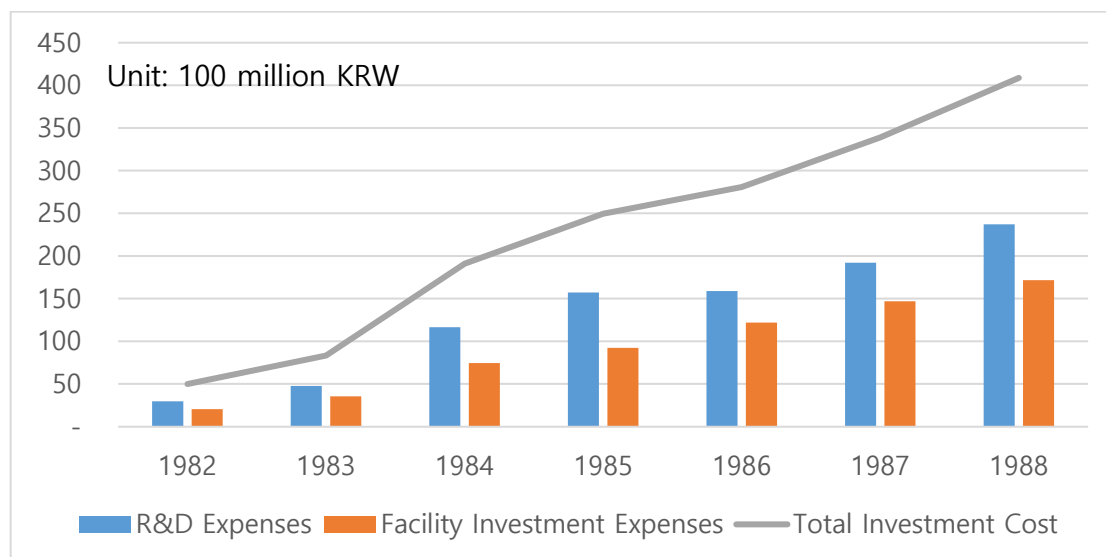


	1992	1993	1994	1995	1996	1997	1998	1999	2000	CAGR
Imports	384	731	753	864	1,052	1,384	1,702	2,114	3,306	30.9%
Exports	246	569	569	735	2,448	3,018	4,815	4,543	6,101	49.4%
Trade balance	- 138	- 162	- 184	- 129	1,396	1,634	3,113	2,429	2,795	

Source: KoreaBIO (2012)

The rapid growth of Korea's biotech industry in production and export scale in the 1990s can be regarded as a result of corporate investment in facilities and R&D that has begun in the 1980s. Investment in the bio sector has begun in 1982 by the industry, when the field of biotechnology was fostered under the narrower category of genetic engineering focused on genetic modification. In the 1980s, investment in the genetic engineering field increased from ₩5 billion in 1982 at a compound annual rate of 41.9%, reaching ₩40.9 billion in 1988, and R&D investment was larger than facility investment.

[Figure 8-5] Investment trends in genetic engineering field by industry in Korea, 1982-1988



	1982	1983	1984	1985	1986	1987	1988	CAGR
R&D expenses	30	48	117	157	159	192	237	41.5%
Facility investment expenses	20	36	74	92	122	147	172	42.6%
Total investment cost	50	83	191	249	281	339	409	41.9%

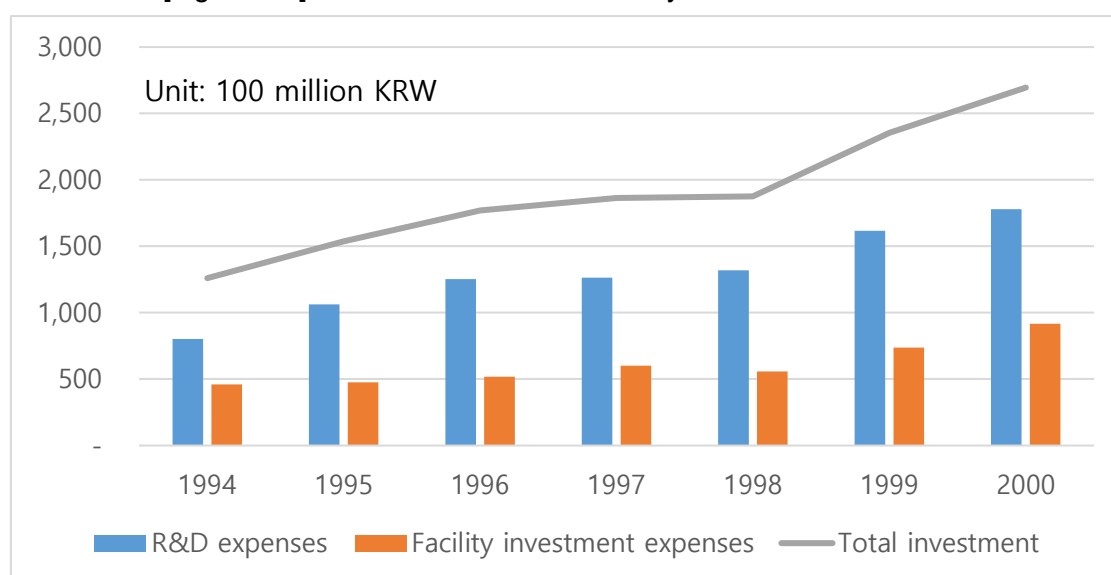
Source: KoreaBIO (2012)

At the time, while companies were aware that the genetic engineering field required long-term investment, they were struggling in the commercialization process with clinical expenses accounting for more than 50% of the total development cost as well as insufficient product development capabilities. Accordingly, some companies set up research institutes overseas or promoted joint research with foreign companies as a countermeasure to acquire advanced technologies and improve R&D capabilities. In the midst of these difficulties, companies have continued to invest, such as attracting doctorate holders or

building research facilities.⁵⁹

Continued to increase since the late 1980s, the scale of investment in the biotech industry has tripled for six years between 1988 and 1994, reaching ₩126 billion. The investment scale continued to increase steadily in the 1990s, and the total investment cost exceeded ₩200 billion in 1999, and as of 2000, it reached approximately ₩270 billion with the R&D expenses of ₩177.9 billion and the facility investment expenses of ₩91.6 billion.

[Figure 8-6] Trends of Korea's bioindustry investment, 1994-2000



	1994	1995	1996	1997	1998	1999	2000	CAGR
R&D expenses	801	1,062	1,252	1,263	1,318	1,616	1,779	14.2%
Facility investment expenses	459	475	517	600	557	737	916	12.2%
Total investment cost	1,260	1,537	1,769	1,863	1,875	2,353	2,695	13.5%

Source: KoreaBIO (2012)

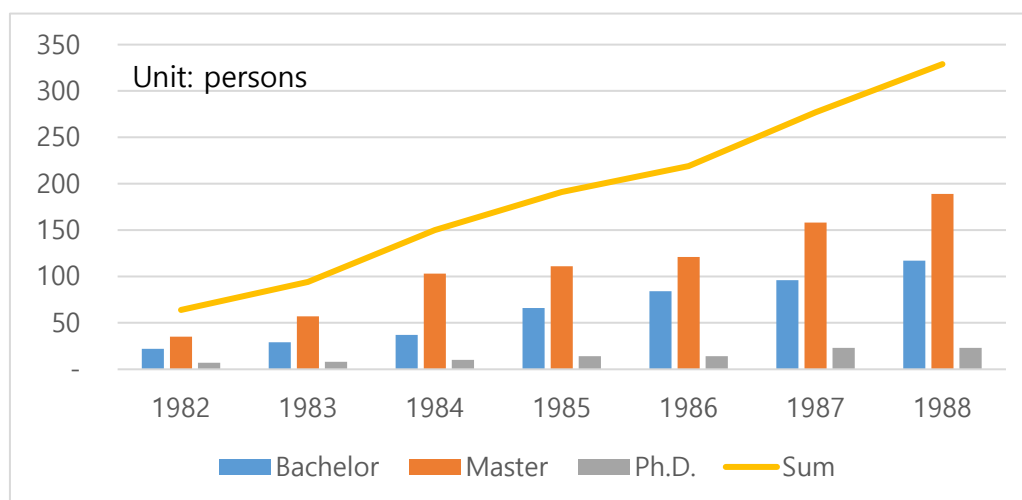
The lack of research manpower was the most serious problem faced by Korean companies

⁵⁹ Shin Hyang Sook (2013)

sought to conduct business in the biotech field in the 1980s. In response, companies have made great efforts to expand research manpower in genetic engineering field by attracting Korean scientists in genetic engineering field from abroad and promoting domestic and overseas training for existing researchers.⁶⁰

While the number of researchers in genetic engineering field across the industry was only 64 in 1982, it increased at a compound annual rate of 32%, showing rapid growth to 117 in 1988. However, compared to the growth rate of the bachelor's and master's manpower, the growth rate of the doctoral manpower was shown to be low, indicating that the conditions for the industry to lead R&D on its own have not yet been created.

[Figure 8-7] Trends in research manpower in the genetic engineering field in Korea's industry, 1982-1988



	1982	1983	1984	1985	1986	1987	1988	CAGR
Bachelor	22	29	37	66	84	96	117	32%
Master	35	57	103	111	121	158	189	32%
Ph. D.	7	8	10	14	14	23	23	22%
Sum	64	94	150	191	219	277	329	31%

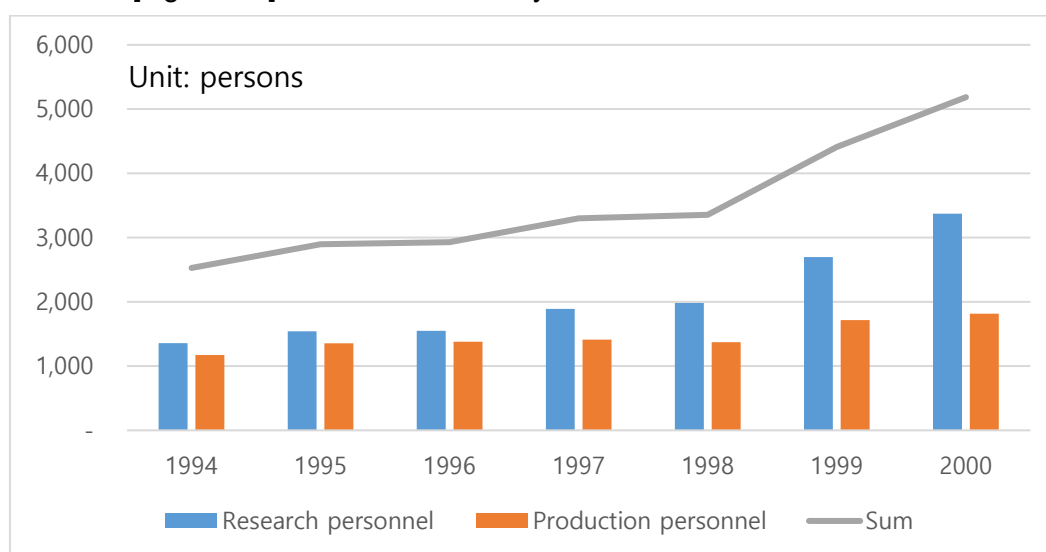
Source: KoreaBIO (2012)

The manpower of the biotech industry has been rapidly expanded since the late 1980s.

⁶⁰ Shin Hyang Sook (2013)

Between 1988 and 1994, the number of research personnel in the biotech industry quadrupled to 1,357 in six years. The number increased by at a compound annual rate of 16.4% between 1994 and 2000, reaching 3,371 in 2000. In particular, 1,389 researchers were added over the two years between 1998 and 2000. It is considered to have resulted from nurturing human resources in the biotech sector that began in the mid-1980s, and the expansion of investments by companies. In addition, the number of production personnel as well as research personnel has been counted since the 1990s. The number of production workers in the biotech sector in Korea grew from 1,171 in 1994 to 1,815 in 2000 with a compound annual rate of 7.6%, accounting for 35 to 47% of the total workforce during the same period.

[Figure 8-8] Trends of bioindustry workforce in Korea, 1994-2000

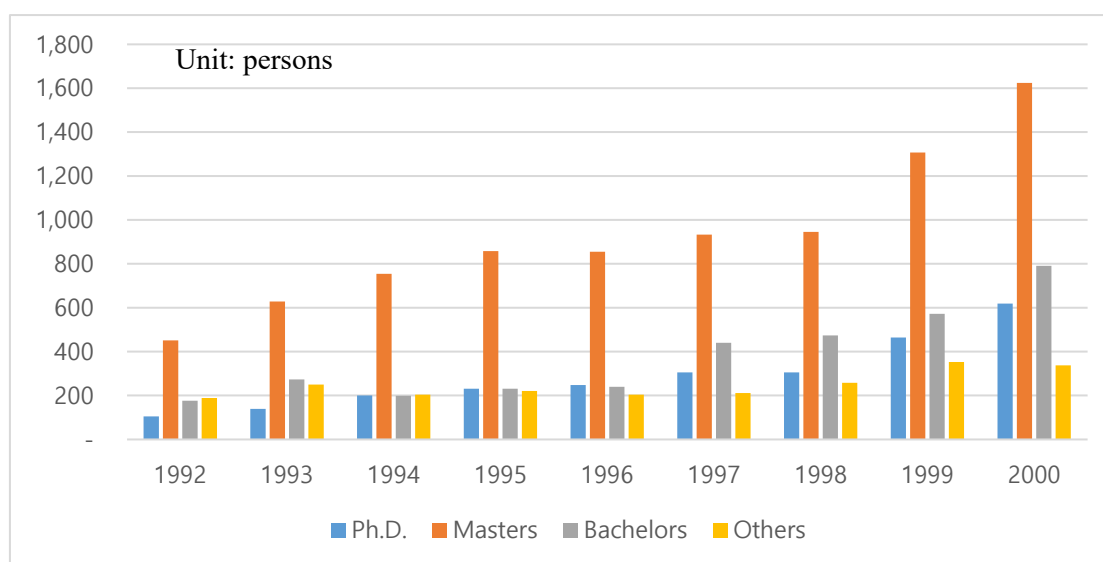


	1994	1995	1996	1997	1998	1999	2000	CAGR
Sum	2,528	2,895	2,927	3,300	3,354	4,411	5,186	12.7%
Research personnel	1,357	1,541	1,547	1,889	1,982	2,696	3,371	16.4%
	53.7%	53.2%	52.9%	57.2%	59.1%	61.1%	65.0%	
Production personnel	1,171	1,354	1,380	1,411	1,372	1,715	1,815	7.6%
	46.3%	46.8%	47.1%	42.8%	40.9%	38.9%	35.0%	

Source: KoreaBIO (2012)

As for the composition of research manpower in the biotech sector in the 1990s by degree, master's workforce accounted for the most, whose proportion in the total research manpower rose from 49.0% in 1992 to 55.7% in 1995, and then decreased again to 48.2% in 2000. The expansion of Ph.D. manpower is notable. The Ph.D. workforce has grown the fastest, whose proportion in the total research workforce has increased significantly from 11.4% in 1992 to 18.4% in 2000.

[Figure 8-9] Trends of research manpower in Korea's bioindustry, 1992-2000



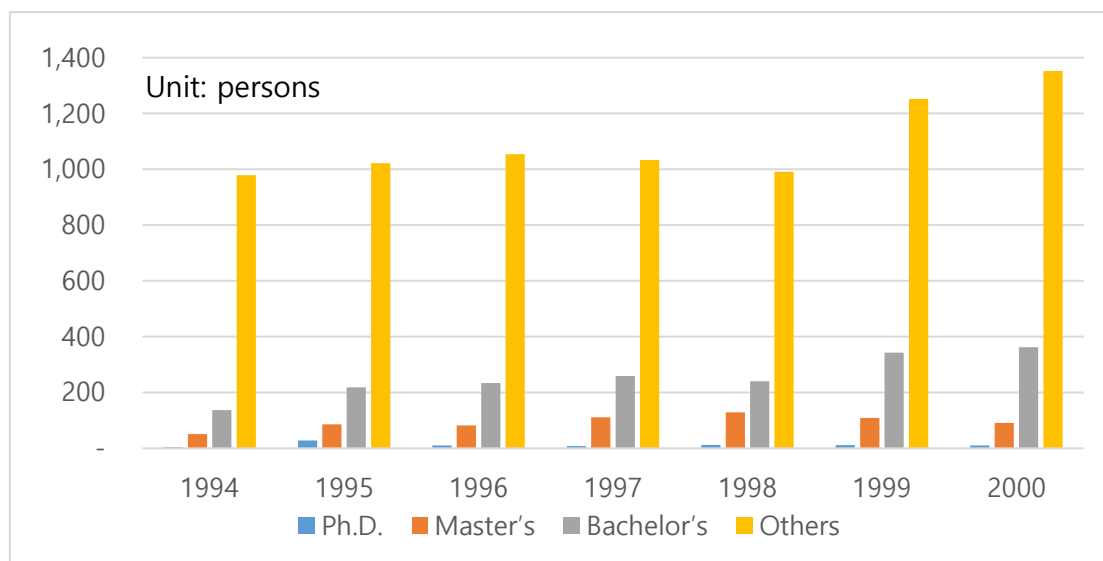
	1992	1993	1994	1995	1996	1997	1998	1999	2000	CAGR
Ph. D.	105	139	200	231	248	305	305	464	619	24.8%
Master's	451	628	754	858	855	933	945	1,307	1,624	17.4%
Bachelor's	176	273	199	231	240	440	474	572	791	20.7%
Other	189	250	204	221	204	211	258	353	337	7.5%
Sum	921	1,290	1,357	1,541	1,547	1,889	1,982	2,696	3,371	17.6%

Source: KoreaBIO

As for the composition of production manpower in the biotech sector in the 1990s by degree, other manpower was the largest with 70% or more. The number of bachelor's manpower increased from 137 in 1992 to 362 in 2000, whose proportion in the total workforce grew from 11.7% to 20.0%. It is regarded that the production manpower has

been reinforced mainly with the bachelor's manpower.

[Figure 8-10] Trends of production manpower in Korea's industry, 1994-2000



	1994	1995	1996	1997	1998	1999	2000	CAGR
Ph. D.	4	28	10	8	12	11	10	16.5%
Master's	51	86	82	111	129	109	91	10.1%
Bachelor's	137	218	234	259	240	343	362	17.6%
Other	979	1,022	1,054	1,033	991	1,252	1,352	5.5%
Sum	1,171	1,354	1,380	1,411	1,372	1,715	1,815	7.6%

Source: KoreaBIO (2012)

1.3 Major cases of first-generation venture companies in the Korea's biotech sector ⁶¹

In 1992, Korea Biotechnology Co., Ltd. (now Bioneer Corporation), focusing on “gene manipulation technology, protein synthesis, and development of reagents for biotechnology research,” was established as the first bio venture company in Korea. It was established with a capital of 80 million KRW by a joint investment of 12 researchers from

⁶¹ The Science & Technology Policy Institute (STEPI: 2017) was mainly referred

the Korea Institute of Science and Technology (KIST), and initially focused on gene amplification and reagents. The company developed PCR equipment for the first time in Korea in 1995 and a gene amplification device in 1998. At that time, it was the only technology in the world to manufacture a gene amplification reagent that works stably at room temperature, which was patented in the United States. In 1997, the company name was changed to Bioneer Co., Ltd., with sales growing rapidly from ₩550 million in 1994 to ₩1.15 billion the following year. Bioneer was listed on the KOSDAQ in 2005 and surpassed ₩1 trillion in market cap in August 2021. Currently, the company is developing and providing not only gene analysis services such as PCR and NGS, but also various molecular diagnostic devices related to diseases such as AIDS, hepatitis and H1N1, and probiotic products through its subsidiary company. It is also developing drugs based on RNAi (RNA interference), which is attracting attention as a next-generation drug development technology.

Macrogen, an on-campus venture company established by Professor Seo Jeong-Sun of the Gene Transplant Research Institute at Seoul National University College of Medicine in 1996, started out as a company that provided laboratory mice for early gene transfer research. Started with a capital of ₩420 million and grew up through venture capital investment, Macrogen became the first biotech venture to be listed on the KOSDAQ in February 2000 and caused a boom in biotech ventures. Since then, the company, which has focused on the gene business, is currently developing and providing gene analysis services and clinical diagnosis services such as cancer genome and rare disease tests. Currently, the scope of services provided has been expanded to include personal genome analysis service and intestinal microbiome analysis service. Macrogen recorded sales of ₩112.6 billion, operating profit of ₩7.2 billion, and net profit of ₩90.8 billion in 2020 due to increased demand for COVID-19 testing.

2.

Major Policies and Systems in Biotechnology Sector in Korea

2.1 Establishment of the foundation for fostering, the 1980s⁶²

Until the 1970s, although 'genetic engineering' was introduced to the media as a novel cutting-edge technology, there was little awareness of industrial application and the necessity of nurturing strategies for the field of biology or life sciences in Korea. However, the "5th Five-Year Economic and Social Development Plan (1982-1986)" established in 1981 served as an important opportunity for change. In early 1981, Dr. Han Moon-Hee of the Department of Biological Engineering at the Korea Advanced Institute of Science and Technology (KAIST) led the report for the biological field in the Economic and Social Development Plan, "Studies on the Technology Assessment and R&D Policy for the Development of Life Science and Biotechnology". The report informed the industrial applicability of the high-tech bio (biotechnology) field and urged a policy to foster it, which led the government and the media to recognize the biotechnology sector as a field with great industrial value and an urgent need for strategic development. However, since the field of biotechnology has been popularized under the concept of 'genetic engineering', policies and legal systems to foster biotechnology in the 1980s were mostly promoted under the name of 'genetic engineering'.

The "5th Five-Year Economic and Social Development Plan (1982-1986)" established in 1981 recognized genetic engineering as an important basic research field for the development of core industrial technology and contained measures to foster through national R&D projects for domestic indigenization as a core strategic technology. Based on this, on January 4, 1982, genetic engineering was selected as a research field for the Designated National R&D Program, Korea's first national R&D program. Between 1982 and 1989, a total of ₩23.26 billion of the Designated National R&D Program was invested in the field of genetic engineering, and the proportion of genetic engineering in the total Designated National

⁶²Shin Hyang Sook (2013) was mainly referred

R&D Program cost was about 4% on average. Government's investment was ₩14.202 billion, 1.6 times higher than corporate investment of ₩9.059 billion. While investment for the Designated National R&D Program was less than 10% of their total R&D investment, the program provided an interest and motivation for companies to continue investing in R&D.⁶³

< Box. The Designated National R&D Program >

The Designated National R&D Program was “the first national R&D program in which the government directly supported large-scale research funds for the advancement of science and industrial technology, and a goal-oriented R&D program to narrow the technological gap with advanced countries in a short period of time according to mid- to long-term science and technology development plans and strategies. The Designated National R&D Program was divided into government-led and private-led programs, and universities, government-funded research institutes, and companies were required to share their roles according to the technological development stages of basic, applied, and industrialized. Such implementation strategy means that the target of government's R&D support has expanded from government-funded research institutes to industries and academia. Since the Designated National R&D Program started with the national-led R&D program (later, the national-led research program) and the company-led technology development program (later, the government-private joint research program, the targeted basic research program, the new technology commercialization and development research program, and the International joint research program have been added later.⁶⁴

In 1983, the ‘Genetic Engineering Promotion Act (Act No. 3718)’, Korea's first law to foster a specific industry, was enacted. The Genetic Engineering Promotion Act provides a legal basis for promoting various policies necessary to foster genetic engineering, including manpower training, establishment of research institutes, and research fund raising, which allowed not only the Ministry of Science and Technology, but also various government

⁶³ Shin Hyang Sook (2013)

⁶⁴ Shin Hyang Sook (2013)

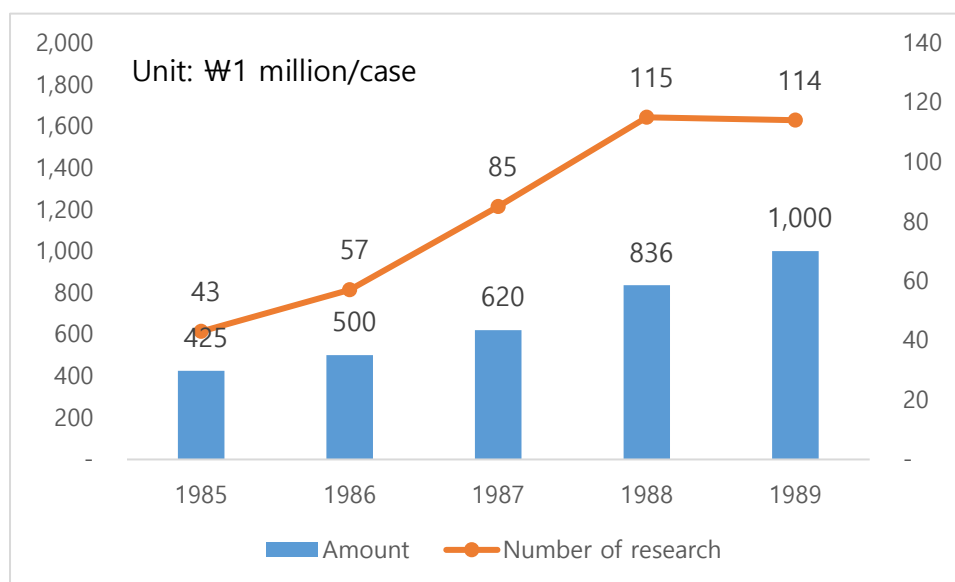
ministries such as the Ministry of Education and Culture, the Ministry of Agriculture and Fisheries, the Ministry of Commerce and Industry, and the Ministry of Health and Social Affairs to actively participate in fostering genetic engineering.. Based on this law, on February 1, 1985, the Genetic Engineering Center affiliated with the KAIST (now 'Korea Research Institute of Bioscience and Biotechnology') was established.

The university's biotechnology field has also developed in accordance with the genetic engineering fostering policy. First of all, funding support for researchers in the biotechnology field at universities has begun. In 1983, through the Targeted Basic Research Program, which started as a part of the Designated National R&D Program, researchers at universities were able to receive research grants in the field of genetic engineering. The purpose of the Targeted Basic Research Program was to support basic research by devoting a certain amount of the Designated National R&D expenses, which intended to support problem-oriented research unlike research funding for universities to support pure academic research until then. Accordingly, genetic engineering was supported as a fundamental field for industrial technology development. In 1983, the first year of the Targeted Basic Research Program, ₩250 million was provided for 79 projects in the field of genetic engineering.

In addition, since 1985, the Ministry of Culture & Education has allocated a part of the science and technology basic research funds among the academic research funds to the fields of semiconductors and genetic engineering, and funds allocated to the field of genetic engineering were named genetic engineering research funds. This was the first case in which the Ministry of Culture & Education has allocated science and technology basic research funds to a specific field. Genetic engineering research funds were paid to the university-affiliated genetic engineering research institute or related research institutes at each university, through which the research system in genetic engineering field has been established. Many universities have set up university-affiliated genetic engineering research institutes to receive genetic engineering research funds. The number of university research institutes that received the fund increased from 6 in 1985 to 22 in 1989. Meanwhile, the review plan prepared by the Genetic Engineering Review and Assessment Committee,

which was formed to review research funding, suggested the direction of the university's nurturing of the genetic engineering field. The Genetic Engineering Review and Assessment Committee suggested the development of technologies, enhancement of research capabilities, and the cultivation of excellent research manpower for the promotion of the field of genetic engineering as the purpose of research funding. Funding guidelines were roughly set up as 1) studies on basic and applied fields of genetic engineering using genetic engineering-related technologies, 2) outstanding ongoing studies or excellent basic research with the potential to grow to an international level in the near future, and 3) studies that are currently weak, but needs support or expected to contribute to the development of the industry. Genetic engineering research funds more than doubled in five years from about ₩ 420 million in the first year to ₩ 1 billion in 1989.

[Figure 8-11] Status of genetic engineering research funding by the Ministry of Culture & Education, 1985-1989



Source: Organized by referring to Shin Hyang Sook (2013)

As the government's support for the biotechnology field was strengthened due to the policy to foster genetic engineering, the growing demand for manpower training in the biotechnology field has led to the active establishments of new departments advocating biotechnology in universities. The departments were named Genetic Engineering, Gene

Science, Biotechnology, and Molecular Biology, and 16 related departments were newly established over four years from 1983 to 1987.

Meanwhile, the introduction of the substance patent system in 1987 became both a crisis and an opportunity for Korean pharmaceutical companies. Until then, inventions of substances themselves were excluded from patents in Korea, and domestic pharmaceutical companies could develop generic drugs without infringing on the patents of advanced foreign companies if they obtained a manufacturing method patent through improved manufacturing method. However, the material patent system was put into effect in 1987 under the pressure of introducing the material patent system by developed countries including the United States, which forced domestic companies to obtain the approval of the material patent holder in order to utilize the patented technology even if they obtained a manufacturing method patent. As the introduction of the substance patent system made it difficult to introduce foreign products and technologies, domestic pharmaceutical companies pursued restructuring through strategic alliances and mergers and acquisitions in the short term while investing in R&D with the goal of expanding independent R&D capabilities and developing new drugs in the long term. With government-level support, the Ministry of Science and Technology has launched a global new drug research project as part of the Designated National R&D Program. The Ministry of Science and Technology formed drug development consortium consisting of domestic pharmaceutical companies, biotech companies, government-funded research institutes, and universities to support the development of five major treatments: antibiotics, anticancer drugs, cardiovascular disease drugs, antiulcer drugs, and antifungals.⁶⁵

2.2 Establishment of the fostering system, the 1990s

From 1992 to 2001, the “Highly Advanced National project (G7 Project)” was implemented by the Ministry of Science and Technology. The G7 project was a 10-year R&D project to secure world-class technology in specific products and technologies with the goal of

⁶⁵ KPBMA(2015);STEP(2017)

'entering the science and technology level of seven advanced countries'. Regarding the field of biotechnology, a new functional biological material technology development project (1992-2002), a new drug and agrochemical development project (1992-1997), and a medical engineering technology development project (1995-2001) were promoted. The new functional biological material technology development project, which was overseen by the Ministry of Science and Technology with the cooperation of the Ministry of Agriculture, Forestry and Fisheries, provided a total of ₩ 210 billion to 33 institutions and 600 companies to develop biodegradable polymer materials, recombinant proteins, and new food materials. The new drug and agrochemical development project, which was overseen by the Ministry of Science and Technology with the cooperation of the Ministry of Health and Welfare, provided a total of ₩152.2 billion to 55 institutions and 215 companies to develop new drugs and new agrochemical products. The medical engineering technology development project, overseen by the Ministry of Health and Welfare with the cooperation of the Ministry of Science and Technology and the Ministry of Commerce, Industry and Energy, provided a total of ₩123.1 billion to 40 institutions and 472 companies for the development of imaging diagnosis and measurement/treatment equipment, rehabilitation equipment, artificial organs, and materials for treatment.

In December 1993, the 1st Biotechnology Promotion Master Plan (Biotech 2000, 1994-2006) was deliberated and resolved by government. The goal of the master plan was to "nurture our technologies in the biotechnology field into the level of seven advanced countries by the year of 2000 to become a strategic export industry of the 21st century" and to "take a 5% share of the global market by challenging the vast market worth \$100 billion in the world and ₩4 trillion in Korea with our technology in the 2000s."⁶⁶ To this end, three basic directions and ten action plans were presented.

⁶⁶ Ministry of Science and Technology etc. (1994)

<Table 8-1> Basic direction and action plans of the 1st Biotechnology Promotion Master Plan

< Three basic directions >

I . Strategic promotion of biotechnology source technology and advancement of technology level

II. Establishing a biotechnology R&D system and strengthening support

III. Enhancing the international competitiveness of the domestic biological industry and promoting entry into the global market

< Ten Action plans >

[Establishment of promotion system]

1. Establishing a nationwide biotechnology R&D system

2. Strategic selection and intensive investment in key areas

[Expanding the base of creative R&D]

3. Strategic promotion of medium-tech development projects

4. Goal-oriented promotion of advanced technology development projects

5. Continuous fostering of basic bio sciences

[Fostering technology development entities and support systems]

6. Expanding the basis for nurturing professional manpower

7. Establishing a domestic biotechnology belt network (Bio Techno-Belt)

8. Reinforcing public-based biotechnology support system

9. Promoting a bold internationalization strategy

[Institutional support]

10. Supplementation of legal and institutional devices for fostering biotechnology

Source: Ministry of Science and Technology, etc. (1994)

The biological materials, health care, agriculture, forestry and fisheries and food, environment/safety management and conservation and use of biological resources, alternative energy, and basic life science were selected as key area of promotion according to the second plan of the ten major action plans. For fields other than basic life sciences, ten key R&D tasks were also presented.⁶⁷

⁶⁷ Kim Seok Gwan et al. (2013)

<Table 8-2> National strategic areas and key R&D tasks of the 1st Biotechnology Promotion Master Plan

Technical field related to biological materials
1. Technology development for new functional biological materials
2. Technology development for industrial use of biological functions Technical field related to Health care
3. Development of biomedical engineering technology
4. Molecular biological study of biological functions
5. Technology development for genome analysis and use Technical field related to agriculture, forestry and fisheries and food
6. Development of genetic engineering breeding and propagation technology of biological resources
7. Development of food biotechnology Technical field related to environment/safety management and conservation and use of biological resources
8. Development of biodiversity and environmental conservation technology
9. Development of environmental and safety evaluation technology Technical field related to alternative energy
10. Technology development for biological alternative energy Basic Life Sciences

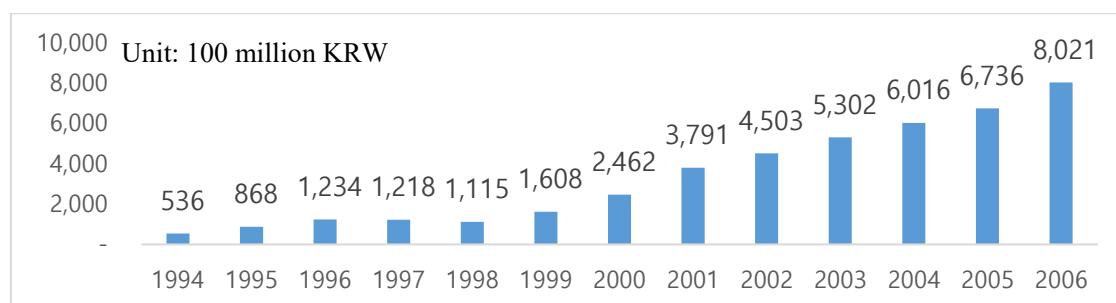
Source: Ministry of Science and Technology, etc. (1994)

The plan to build the infrastructure to support R&D included to expand the size of the gene bank of the Genetic Engineering Research Center at the time, and joint use of industry-academia-research institutes of the Bio Pilot Plant, strengthen the safety evaluation function of each institution's biotechnology products, establish a center for preclinical evaluation, strengthen the clinical trial function of domestic medical schools and medical institutions, and establish a center for clinical trial evaluation.

The 1st Biotechnology Promotion Master Plan was implemented by dividing the promotion period into three stages and analyzing and evaluating the performance of the previous stage when implementing the stage to supplement the problems and establishing a plan for the next stage. The goals for each stage are as follows: 'Establishment of a technology base to secure essential technology for the domestic biotechnology industry' in stage 1(1994-1997), 'Strengthening of biotechnology base and establishment of application base' in stage 2

(1998-2002), 'Advancement of biotechnology and achievement of export strategy industrialization' in stage 3 (2003-2007). The achieving goal of technological level by stage was that 'Securing production technology base to the level of advanced countries' in stage 1, 'Securing source fundamental technology to the level of advanced countries' in in stage 2, and the 'Industrialization of export strategy for biotechnology industry' in stage 3.⁶⁸ The government's investment amount by stage to implement the 1st Biotechnology Promotion Master Plan was ₩385.6 billion in stage 1, ₩897.6 billion in stage 2, and ₩3.0578 trillion in stage 3, for a total of ₩4.3410 trillion. The investment amount by ministry consisted of ₩1.7873 trillion (41.2%) by the Ministry of Science & Technology, ₩780 billion (18%) by the Ministry of Commerce, Industry and Energy, ₩718.4 billion (16.5%) by the Ministry of Health & Welfare, and ₩555.6 billion (12.8%) by the Ministry of Agriculture and Forestry.⁶⁹

[Figure 8-12] Government's investment by stage of the 1st Biotechnology Promotion Master Plan



Source: Kim Seok Gwan et al. (2013)

In addition to the growth of the bio industry (production and market scales) introduced above, the achievements of the 1st Biotechnology Promotion Master plan are also reflected in the achievements of papers and patents. The number of papers published in SCIE increased from 420 in 1994 to 4,089 in 2005, and in particular, the number of papers published in NSC (Nature, Science, Cell) gradually increased to 6 in the 1st stage, 25 in the

⁶⁸ Ministry of Science & Technology etc. (1994); Ministry of Science & Technology etc. (1998); Ministry of Science & Technology etc. (2001)

⁶⁹ Kim Seok Gwan et al. (2013); Ministry of Education etc. (2006)

2nd stage, and 55 in the 3rd stage, showing growth not only in terms of quantity but also in terms of quality. The number of domestic patent applications increased from 535 in 1994 to 1,856 in 2005, and from the middle of the second stage, the number of domestic patent applications began to surpass that of foreigners. The number of registered patents in the United States increased as the phase progressed, recording a total of 207 over the three stages.⁷⁰

⁷⁰ Ministry of Science & Technology (2008)

3.

Major Universities in Biotechnology Sector in Korea

3.1 (Case) Seoul National University⁷¹

In the case of Seoul National University (SNU), basic academic fields of biotechnology, such as microbiology and molecular biology, were introduced into the existing biology departments. While most other universities introduced the cutting-edge field of biotechnology through the establishment of departments such as the Department of Genetic Engineering, Biotechnology, or Molecular Biology in the wake of the genetic engineering promotion policies in the 1980s, SNU established the new department of Microbiology relatively early in 1970 and introduced cutting-edge fields through changes within the existing Department of Botany and Zoology.

Until the 1960s, morphology and taxonomy were the main fields in the Department of Botany and Zoology in the College of Natural Sciences. Although a professor of physiology and microbiology was appointed in the late 1960s, it was difficult to access the latest academic fields due to poor research funds and facility environments, and professors who had accessed to cutting-edge fields such as molecular biology and genetic engineering in North America and Europe were also rare.

The change began in 1970 when the Department of Microbiology was established as a department for education in the basic fields for fermentation industry and plant pathology related to agriculture by order of the Ministry of Education. The Department of Microbiology newly recruited professors from the field of Genetics (Princeton Uni.) and the field of Biochemistry (MIT) with existing professors of microbiology from the Department of Botany as its founding members. The entrance exam cut-off for the Department of Microbiology was much higher than that of the Department of Zoology and Botany, reflecting students' high interest in novel and cutting-edge fields. Meanwhile, the Korea

⁷¹ Kim Ji Hyun (2016) was mainly referred

Advanced Institute of Science (KAIS) established in 1971, which advocated applied fields and research, newly established the Department of Biological Science and Engineering focusing on molecular biology and bioengineering. This series of changes facilitated the subsequent changes in the Department of Botany and Zoology.

In the 1980s, not only the Department of Microbiology, but the Department of Botany and Zoology introduced various latest fields of biology by recruiting professors who studied in North America and Europe. The majors of professors newly appointed by the departments of biology at SNU in the 1980s included fermentation, biophysics, immunology, and molecular biology in the Department of Microbiology, and molecular genetics, molecular biology, biochemistry, and neuroendocrine in the Department of Botany and Zoology. In common, the fields were concentrated in the latest and experimental fields of biology, especially molecular biology.

<Table 8-3> Status of professors in departments of biology field at SNU in the 1980s

Department	Major	University where doctoral degree acquired	Year of appointment
Zoology	Molecular Genetics	Florida State University	1980
	Biochemistry	University of Alabama	1983
	Neuroendocrinology	University of Illinois	1985
	Systematics	Florida State University	1989
Botany	Plant Physiology	Universität Freiburg	1980
	Plant Molecular Genetics	Oregon State University	1981
	Molecular Biology	University of Tennessee	1983
	Plant Molecular Biology	McGill University	1986
	Plant Systematics	Cornell University	1990
Microbiology	Fermentology	University of New South Wales	1983
	Biophysics	Justus-Liebig-Universität Giessen	1984
	Microbial Ecology	Christian-Albrechts-Universität zu Kiel	1983
	Immunology	Cornell University	1986
	Molecular Biology	University of Wisconsin-Madison	1986
	Mycology	University of Tennessee	1989

Source: Kim Ji Hyun (2016)

In 1991, the Department of Zoology was renamed the Department of Molecular Biology, and the Department of Botany the Department of Biology. Although there was some exchange of professors between the two departments, the reorganization at the time was not for the adjustment of departments according to the type of field, but rather a change due to the trend toward the latest field of molecular biology. While the contacts between the three departments centering on molecular biology were already gradually expanding in the late 1990s, the three departments underwent an integrated reorganization and launched the Department of Life Sciences in the year 2000 in the process of applying for a government program to support research-centered and graduate school-centered reorganization.

From the establishment of the Department of Microbiology in 1970 to the integration of the three departments in 1999, the process of reorganizing the field of biology at SNU into a field centered on molecular biology and experiments was accompanied by a change in the introduction and reinforcement of the research management system. The professors in the field of biology at SNU have led the introduction of the research management system during the establishment of the Natural Sciences Research Institute in the 1970s, or the introduction of the selection criteria of “Excellence” in the planning process of “Science and Technology Basic Research Focused Research Fund” project by the Ministry of Education in the 1980s. In other words, the professors of biology field at SNU intended to increase the credibility of support for research activities by actively adopting systematic research management, especially research management systems such as selection and evaluation, and consequently lead to the expansion of support.

3.2 (Case) Korea Advanced Institute of Science and Technology⁷²

In the case of the Korea Advanced Institute of Science and Technology (KAIST) established as a research-oriented as well as an applied field-oriented university, molecular biology, which is a basic science and has great applicability, was introduced and expanded. In 1971, the Korea

⁷² Kim Ji Hyun(2016) was mainly referred

Advanced Institute of Science (KAIS) was established as a graduate university specialized in science and technology as well as an educational institution that advocates application fields to support industrialization, where the Department of Biological Science and Engineering was established. Claiming to advocate an applied field, the Department of Biological Science and Engineering at the KAIS focused on applied fields such as biochemistry, applied microbiology, food biochemistry, food processing engineering, and microbial physiology.

KAIS was merged with the Korea Institute of Science and Technology (KIST), a research institute, in 1980 under the government order, to become the Korea Advanced Institute of Science and Technology (KAIST). As the Korean government promoted the promotion of science and technology for the advancement of industrial technology in the 1980s, support for basic fields besides applied fields was strengthened. Accordingly, Department of Biotechnology of KAIST has more expanded the fields related to molecular biology such as biophysics, molecular biology, and molecular physiology.

<Table 8-4> Status of professors in Department of Biological Engineering at KAIS in the 1980s

Year of appointment	Major	University where doctoral degree acquired
1980	Biophysics	Clark University
1980	Biological Engineering	Université de Technologie de Compiègne
1982	Molecular Biology	University of Chicago
1986	Molecular Physiology	Washington State University
1986	Biochemistry	Columbia University
1988	Biomolecular Engineering	Université de Technologie de Compiègne

Source: Kim Ji Hyun (2016)

Meanwhile, in 1984, the Korea Institute of Technology (KIT) was established as an undergraduate university specialized in science and technology, where the biology major was opened. While KIT advocated applied fields and engineering fields like the KAIS for the purpose of nurturing industrial manpower, at the same time, education in basic field had to be provided due to the nature of an undergraduate university. Since its establishment, professors from basic

fields centered on molecular biology such as molecular biology, biochemistry, cell biology, and molecular genetics were appointed in the major of biology in the 1980s.

<Table 8-5> Status of professors in Biology at KIT in the 1980s

Year of appointment	Major	University where doctoral degree acquired
1984	Environmental Microbiology	Technical University of Berlin
1985	Toxicology/Biochemistry	University of Wisconsin
1985	Plant Molecular Biology	University of Tokyo
1985	Biochemistry/Cell Biology	North Texas State University
1986	Molecular Genetics	Indiana University at Bloomington
1987	Genetics	North Carolina State University
1988	Molecular Virology	UCLA

Source: Kim Ji Hyun (2016)

In 1989, the graduate school organization of KAIST and KIT were merged to establish a new educational institution with both undergraduate and graduate schools by order of the government. The two organizations, whose basic fields had been strengthened with a focus on molecular biology over the 1980s, were briefly divided into the Department of Life Sciences, centered on groups that advocated basic fields, and the Department of Biotechnology, centered on groups that advocated applied fields, shortly after the integration. In 1995, the two organizations were reunited to establish the Department of Biological Sciences, providing a departmental system in which education and research in basic and applied fields coexist. In the 1990s, under the background of strengthening support for basic research and scientific research, the Department of Biological Sciences continued to expand the basic fields centered on molecular biology.

4.

Major Research Institutes in Biotechnology Sector in Korea

Korea Research Institute of Bioscience & Biotechnology⁷³

4.1 Establishment

In 1981, when the Department of Biological Engineering of the KAIST prepared a report “Studies on the Technology Assessment and R&D Policy for the Development of Life Science and Biotechnology”, the need for a “Life Science Research Institute”, a specialized research institute to promote the field of biotechnology was first raised. This report drew the government’s interest in fostering the field of biotechnology and supporting measures, and led social attention toward the field of biotechnology under the name of 'genetic engineering' through the media. Afterwards, the legal basis for the establishment of a genetic engineering research institute was prepared with the enactment of the 'Genetic Engineering Promotion Act (Act No. 3718)' in 1983, and on February 1, 1985, the Genetic Engineering Center affiliated with the KAIST (now ‘KRIBB’) was officially established.⁷⁴

Securing its location within the Daedeok Research Complex in accordance with the Enforcement Decree of the Specific Research Institutions Promotion Act in 1985, the Genetic Engineering Center purchased a land of about 86,000 m² (research building of about 63,000 m²) to construct a new building, and moved in 1990. In order to attract and secure excellent research manpower, the Genetic Engineering Center secured manpower centering on the existing research personnel of the Faculty of Biological Engineering at the KAIST, while actively recruiting overseas Korean scientists who have acquired knowledge and field experience from advanced countries such as North America, Europe and Japan. In addition, the center secured essential research equipment through the Oversea Economic

⁷³ KRIBB(2015) was mainly referred

⁷⁴ Shin Hyang Sook (1993),

Cooperation Fund (OECF) and loans from the (International Bank for Reconstruction and Development (IBRD).

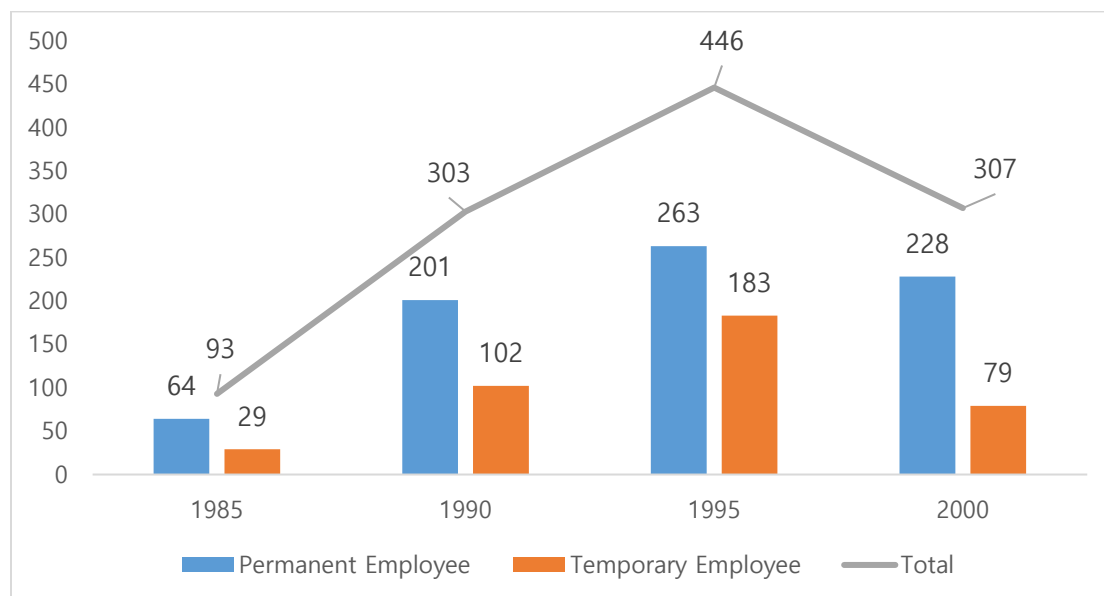
4.2 Organization and personnel, budget, and publication performances

When the KIST was separated by the reorganization of the KAIST in 1989, the Genetic Engineering Center was changed to an affiliated institution of the KIST. After the center was renamed Genetic Engineering Research Institute (GERI) affiliated with the KIST in December 1990 the following year, it was changed to Korea Research Institute of Bioscience and Biotechnology (KRIBB) in 1995. It became an independent corporation belonging to the Korea Research Council of Fundamental Science Technology in 1999, and then renamed its Korean name without change in its English name the KRIBB in 2001.

When the Genetic Engineering Center was established in 1985, the research organization started with six labs under the research department: Molecular Biology, Enzyme Engineering, Cell Biology, Biocontrol, Microbial Resources, and Microbial Engineering, and, in 1992, expanded to 21 laboratories under two research departments, Genetic Engineering and Applied Biological Process. In 1993, eight research groups were established with the introduction of senior research director system, and in 1998, a reorganization was carried out to greatly simplify the project team to strengthen the competitiveness of the institution.

The scale of personnel has increased from 93 in 1985 to 446 in 1995. After the reorganization carried out to strengthen the competitiveness of the institution in 1998, it reached 307 in 2000.

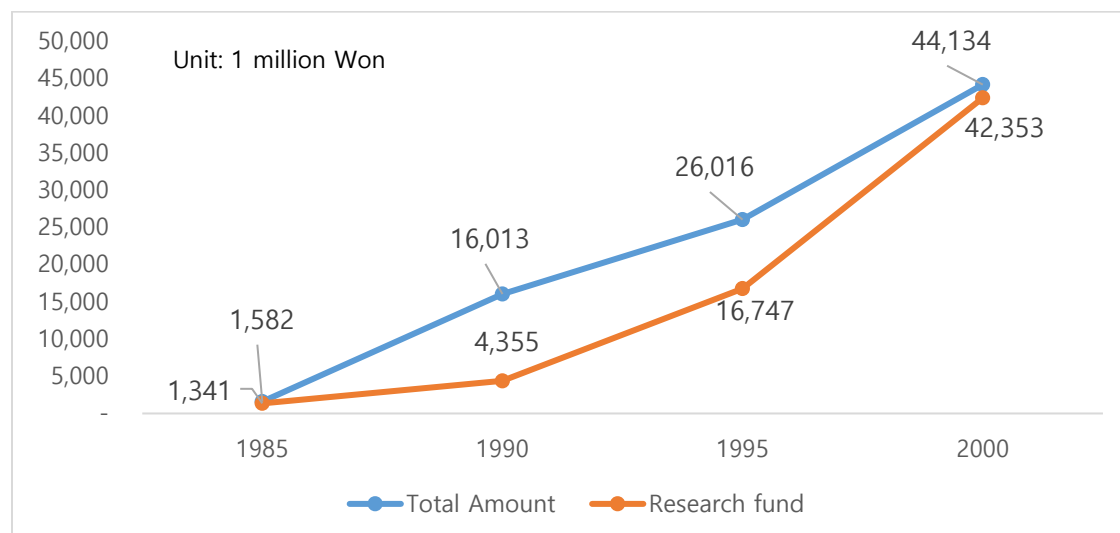
[Figure 8-13] Human Resources Trend of Korea Research Institute of Bioscience and Biotechnology (KRIBB), 1985-2000



Source: KRIBB(2015)

The institution's budget grew from 1.58 billion won in 1995 at a compound annual rate of 24.8%, reaching 44.11 billion won in 2000. During the same period, research expenses increased more than 30 times in 15 years from 1.34 billion won to 42.35 billion won. Looking at the composition of research funds, the scale of government-trusted research funds was the largest, and showed the rapid increase from KRW 1.17 billion in 1985 to KRW 31.66 billion in 2000, and private-trusted research conducted on a full scale from the late 1990s.

[Figure 8-14] The progress of KRIBB's total budget and research funds, 1985-2000



Source: KRIBB (2015)

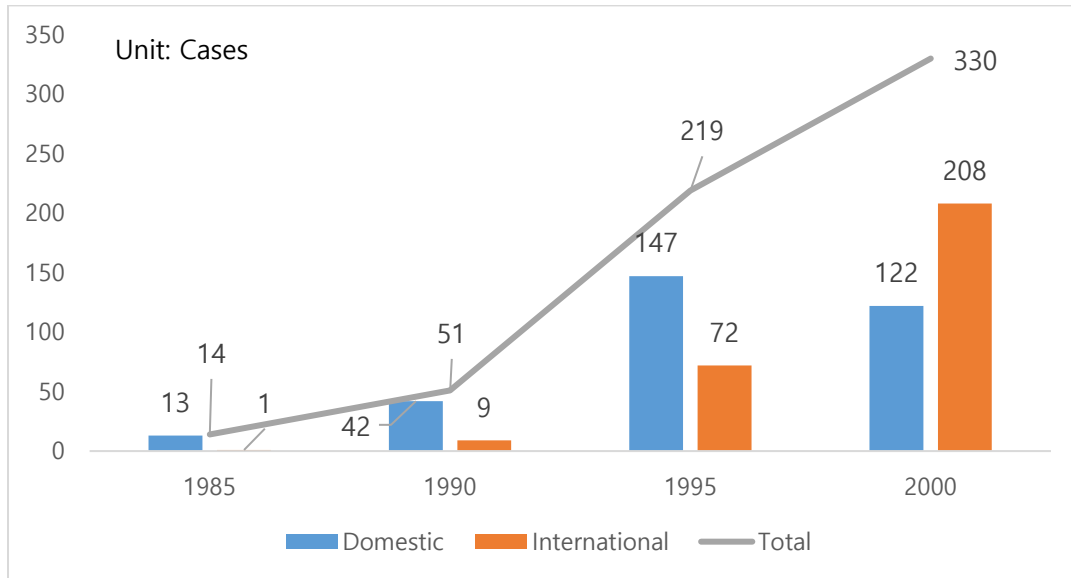
<Table 8-6> Composition of KRIBB's Research Funds, 1985-2000

	1985	1990	1995	2000
Main Business	92	-	530	9,025
Government Entrusted	1,173	4,315	15,530	31,657
Private Entrusted	76	40	661	1,626
Total Research Funds	1,341	4,355	16,747	42,353

Source: KRIBB (2015)

The publication performance of the Korea Research Institute of Bioscience and Biotechnology has also increased sharply since 1990 due to the investment in research funds. In 1985, there were only 14 publications in total, but in 1990, the number of publications increased to 51, then it increased to 330 in 2000. Between 1990 and 1995, the number of domestic journal publications increased from 42 to 147, whereas between 1995 and 2000, the number of domestic journal publications decreased to 122, and the number of international journal publications increased from 72 to 208. It can be seen that the research capability of researchers has advanced after 1990s.

[Figure 8-15] Trends of KRIBB's domestic and international journal publications performance, 1985-2000



Source: KRIBB (2015)

5.

Implications of STI in Biotechnology Sector during Korea's High-tech Development Period

Korea's biotech industry exports three times the size of imports, and it has a huge ripple effect as it is applied to various industrial fields including pharmaceuticals and medical devices. Two biologics companies are included in the top 15 by market capitalization of the KOSPI, and bio/healthcare is the sector where venture capital investment is the most. This seems to be the result of pursuing advanced technology development and development of high value-added products and services from the time strategic development for the biotech industry was planned.

Since the early 1980s, R&D and facility investment in the biotech industry and the scale of research manpower have increased rapidly. In the 1990s, the proportion of R&D investment increased, and the recruitment of people with Ph.D. was faster than that of people with master's degree. Since the mid-1990s, exports out of total production have tended to exceed domestic sales, and exports greatly outpaced imports.

In the 1980s, against the backdrop of the establishment of policies for technological development and industrial advancement, the biotech sector emerged as an urgent field for strategic promotion. As the biotech field has a very high industrialization value and R&D capabilities are recognized as the key, Korea's first law to foster a specific industrial fields was enacted. In order to strengthen R&D capabilities and advanced manpower training, support for R&D activities by those encompassing industry, research institutes, and university and special support for education at universities were provided. From the 1990s, the Biotechnology Promotion Master Plan, a cross-ministerial strategy for the biotech sector, was established to develop the biotech industry as a future strategic export industry, and R&D capabilities in the biotech sector have grown significantly.

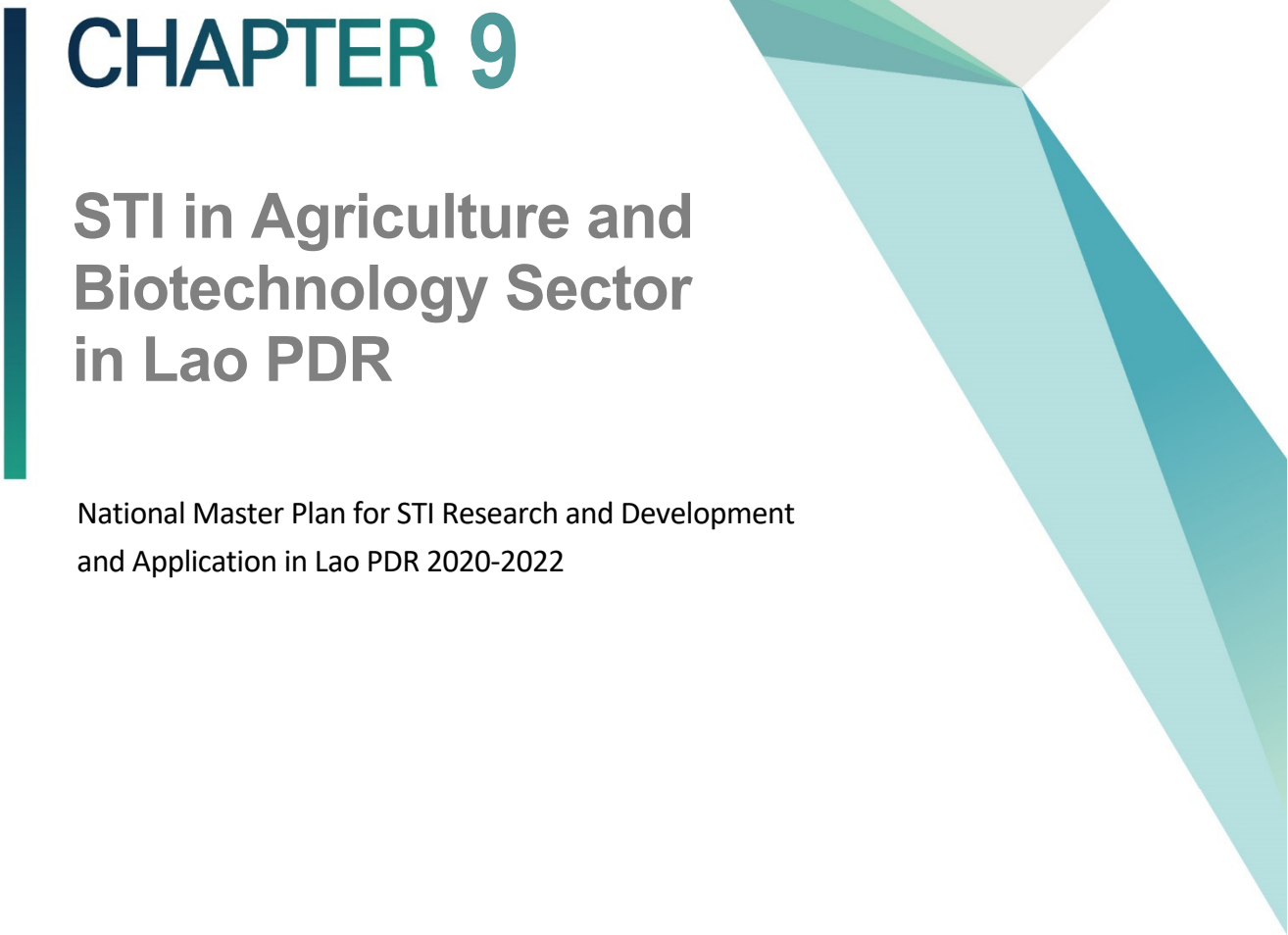
From the beginning, the biotech sector promotion policy emphasized the enhancement of research capacity and the cultivation of high-quality research manpower. In response to the expansion of research funding support and increased demand for manpower, universities

actively expanded education and research activities in molecular biology-related fields, which are the basic fields for the biotech industry, and introduced a system for improving research capabilities such as research management and excellence evaluation., The Korea Research Institute of Bioscience and Biotechnology which was established on the basis of the Promotion Act, started developing technologies in the biotech field by concentrating public investment such as foreign loans in the early stages, and supported universities and companies that lacked R&D infrastructure and carried out joint research.

In Korea, the biotech industry has established itself as one of the main industries. Under the vision of advancing industry and fostering the future industry, the long-term strategy of investing in fostering high-quality human resources and enhancing research capabilities and promoting the development of new technologies from the beginning is working out.

¹ Footnote: Mainly referred to Se-won Kim, Dong-hwan Ahn, Tae-yun Kim (2017), Heo Jang et al. (2017)

¹ Heo Jang, Mina Lee(2020).

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CHAPTER 9

STI in Agriculture and Biotechnology Sector in Lao PDR

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 9. STI in Agriculture and Biotechnology Sectors in Lao PDR

1. STI in Agriculture Sector in Lao PDR

1.1 Current Status of Agriculture Sector in Lao PDR

1.1.1 Overview of Current Status

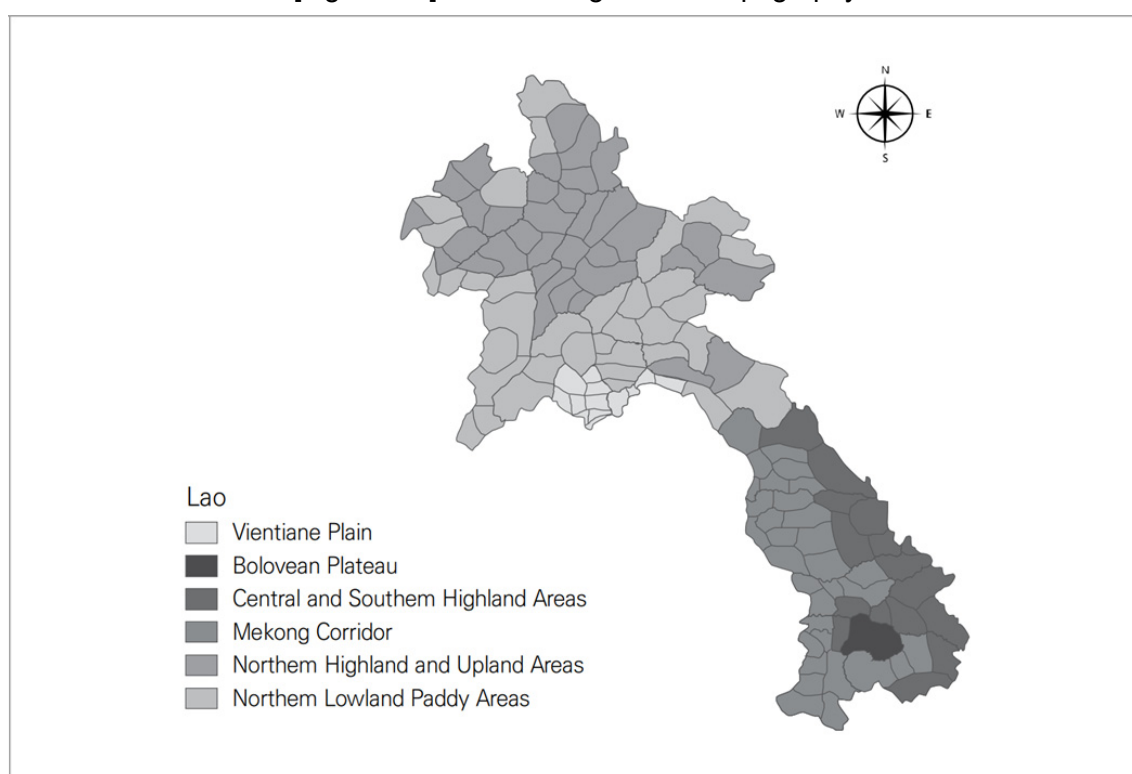
Lao PDR has a tropical monsoon climate, with the plains showing tropical climates, and the northern and mountainous regions showing subtropical climates. It is divided into rainy season from May to October and dry season from November to April, with an average annual rainfall of about 2,045mm, and little rain falls in December and January during the dry season. The average annual rainfall is 1,200 to 2,200mm in the plains and 3,000mm in the northern mountains (Jang and Lee, 2020).

The entire topography of Lao PDR consists of mountainous topography in the northern and eastern regions bordering China and Vietnam, the central plain of the Mekong River basin, and the plains extending to the south. The agricultural topography consists of six regions: the Vientiane Plain, the Bolovan Plateau, the Central/Southern Highlands, the Mekong Corridor, the Northern Highlands, and the Northern Lowlands. The Vientiane Plain is located at an altitude about 500 to 1,000m above sea level in Lao PDR, with the highest annual average rainfall of 2,500 to 3,000mm. The Bolaven Plateau is located in grasslands and forest areas 500 to 1,500m above sea level, with an average annual rainfall of 2,500 to 3,000mm.

The central and southern highlands are located in the upper valleys and highlands at an altitude of 200 to 500 m above sea level, and the average annual rainfall is 2,000 to 3,000

mm. The Mekong River basin is located in the lowland plains and gentle slopes at an altitude of 100 to 200 m above sea level. The average annual rainfall is 1,500 to 2,000 mm. The northern highlands are located in steep mountainous terrain at an altitude of 1,500 to 2,500 m above sea level, and the average annual rainfall is 1,300 to 2,500 mm. The northern lowlands are located in the mountains at an altitude of 500 to 1,500 m above sea level, and the average annual rainfall is 1,500 to 2,000 mm.

[Figure 9-1] Lao PDR' agricultural topography



Source: Kim Se-won et al. (2017)

In Lao PDR, about 70% of the land is composed of mountainous areas, and although the agricultural area has increased, it accounts for about 10% of the total land as of 2019. As of 2019, the agricultural land area of Lao PDR was about 2.39 million hectares, and the crop cultivation area was about 1.72 million hectares. Between 1998 and 2019, the proportion of agricultural area in the total land area increased from 7.6% to 10.4%, and the area under cultivation for crops in agricultural area increased from 53.3% to 71.8%.

<Table 9-1> Current status of farmland and crop cultivation area in Lao PDR

Unit: 1,000 ha

	1998	2001	2004	2007	2010	2013	2016	2019
Land area (A)	23,080	23,080	23,080	23,080	23,080	23,080	23,080	23,080
Agricultural area (B)	1,765	1,841	1,946	2,063	2,220	2,335	2,494	2,394
Crop cultivation area (C)	940	1,051	1,186	1,338	1,530	1,658	1,819	1,719
Agricultural area % (B/A) of land area	7.6%	8.0%	8.4%	8.9%	9.6%	10.1%	10.8%	10.4%
Crop cultivation area % of agricultural area (C/B)	53.3%	57.1%	60.9%	64.9%	68.9%	71.0%	72.9%	71.8%

Source: FAO(2021d)

Lao PDR is a traditional agricultural country, and the proportion of agricultural workers is the largest among the five Mekong countries. The proportion of agricultural workers in Lao PDR was more than 80% in 1998, but it steadily decreased as industrialization progressed, dropping to 61.4% in 2019. However, it is still higher than other Mekong countries, which are below 50% in 2019. In Lao PDR, more than half of the agricultural population is said to be self-sufficient farmers, earning a living on less than \$300 per year.

Foot note: WorldBank ; Heo Jang et al. (2017).

Heo Jang, Lee Mina (2020).

<Table 9-2> Proportion of Agricultural Workers in Five Mekong Countries, 1998-2019

Unit: %

	1998	2001	2004	2007	2010	2013	2016	2019
Lao PDR	83.3	81.1	78.3	75.1	71.5	67.8	64.7	61.4
Myanmar	62.8	60.5	56.8	54.7	53.1	52.4	51.2	48.8
Vietnam	64.8	64.0	57.9	49.3	48.7	46.8	41.9	37.2
Cambodia	76.8	68.9	63.9	58.1	57.3	49.5	37.5	34.5
Thailand	51.3	46.0	39.3	39.5	38.3	41.8	31.2	31.4

Source: WorldBank (20212021.10.28.)

1.1.2 Agricultural production scale

Although the proportion of agricultural workers in Lao PDR is over 60%, the proportion of agriculture, forestry and fisheries in GDP is less than 20% due to low productivity. The share of agriculture, forestry and fisheries in the total GDP is decreasing, and although it is affected by changes in the industrial structure, the low productivity of the agriculture, forestry and fishery industry is considered to be the major cause. Although the Lao government is implementing policies and investment incentives to foster the food processing industry as a measure to increase farm households and national income, the effect is insufficient due to the lack of manufacturing infrastructure. Compared with other Mekong countries, the manufacturing share of Lao PDR has decreased significantly compared to 2010, and it is the lowest figure of less than 10% as of 2019. The decrease in the proportion of GDP in agriculture, forestry and fisheries does not appear to be related to the development of the manufacturing sector.

Foot note: Kim, Dong-Kwan (2017); FAO (2021.10.22.).

Heo Jang et al. (2017).

<Table 9-3> Agriculture, forestry, fisheries and manufacturing GDP of five Mekong countries in 2010 and 2019

Unit: one million dollars

	Lao PDR		Myanmar		Vietnam		Cambodia		Thailand	
	2010	2019	2010	2019	2010	2019	2010	2019	2010	2019
GDP	7,313	18,823	44,847	76,785	115,932	261,921	11,242	27,098	341,105	542,017
agriculture, forestry and fisheries	1,611	2,878	15,438	17,731	21,307	36,555	3,809	5,612	35,892	44,966
% of total GDP	22.0	15.3	34.4	23.1	18.4	14.0	33.9	20.7	10.5	8.3
Manufacturing	791	1,407	8,425	18,422	15,009	43,172	1,652	4,411	105,505	146,860
% of total GDP	10.8	7.5	18.8	24.0	12.9	16.5	14.7	16.3	30.9	27.1

Source: FAO (2021a2021.10.22.)

Lao PDR' Gross Production Value in 2018 was \$5.27592 billion based on the nominal price. Since 1998, the proportion of grains has decreased and the proportion of vegetables and fruits has increased. Gross agricultural output nearly six fold increased from \$909.59 million in 1998 to \$5.27592 billion in 2018, at a compound annual growth rate of 9.2%. During the same period, the total production value of vegetables and fruits increased the fastest at a compound annual rate of 9.7%, and its share increased from 35.4% to 38.6%, while the total production value of cereals increased relatively slowly at a compound annual rate of 7.8%, and its share decreased from 33.9% to 26.3%.

<Table 9-4> Lao PDR' Gross Production Value (GPV) (based on nominal prices), 1998-2019

Unit: one million dollar

	1998	2001	2004	2007	2010	2013	2016	2018	CAGR
Agriculture	909.59	745.18	964.68	1,764.73	3,501.54	4,097.09	6,041.22	5,275.92	9.2%
	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	
Cereals, Total	308.50	243.96	295.24	551.98	1,032.28	1,301.11	1,758.46	1,388.52	7.8%
	33.9%	32.7%	30.6%	31.3%	29.5%	31.8%	29.1%	26.3%	
Vegetables and Fruit Primary	322.11	345.54	434.54	735.39	1,390.64	1,526.66	2,243.79	2,038.02	9.7%
	35.4%	46.4%	45.0%	41.7%	39.7%	37.3%	37.1%	38.6%	
Livestock	123.69	77.51	131.98	281.56	509.42	540.58	705.07	711.25	9.1%
	13.6%	10.4%	13.7%	16.0%	14.5%	13.2%	11.7%	13.5%	

Source: FAO (2021e2021.10.22.)

Looking at the top agricultural items in terms of production value in Lao PDR in 2018, rice and vegetables were ranked first and second with more than USD 1 billion. Other crops included bananas, sugarcane, cassava, corn, tobacco and watermelon in the top ten, while pork and chicken made it into the top ten for livestock products.

<Table 9-5> Top Agricultural Items by Gross Production Value in Lao PDR in 2018 (Based on Nominal Price)

Unit: Thousand dollars

Ranking	List of items	Value
1	Rice, paddy	1,203,308
2	Vegetables, freshness (the vegetables that are not included in the list)	1,197,479
3	Bananas	430,932
4	Sugar cane	357,183
5	Meat indigenous, pig	318,928
6	Cassava	295,233
7	Maize	185,211
8	Meat indigenous, chicken	137,633
9	Tobacco, unmanufactured	137,620
10	Watermelons	120,560
11	Oranges	91,856
12	Coffee, green	88,191
13	Tangerines, mandarins, clementines, satsumas	87,429
14	Meat indigenous, buffalo	85,607
15	Eggs, hen, in shell	72,615
16	Sweet potatoes	62,684
17	Groundnuts, with shell	61,551
18	Meat indigenous, cattle	51,079
19	Chillies and peppers, dry	49,793
20	Sesame seed	30,044

Source: FAO (2021e2021.10.22.)

1.1.3 Production status of major crops⁷⁵

The per capita consumption of rice in Lao PDR is about 160 kg per year, which is the highest in the world. Rice is staple in the diet of the Lao people and affects food security. Lao people generally prefer glutinous rice compared to non-glutinous rice or other varieties, and most of the glutinous rice production is consumed domestically. The production of rice, the main crop of Lao PDR, is mainly concentrated in the central region, and rice production has maintained more than 3 million tons since 2010. About 70% of the total rice production in Lao PDR comes from Savannakhet, Bolikamxai, Khammouan, Vientiane and Xieng Khouang provinces. Savannakhet produces 22% of the total rice production. Owing to the recent increase in investment in agricultural technology and research, the production of high-quality rice has increased, and as of 2016, about 60,000 tons of high-quality rice were produced in the central region. Farm households using chemical and organic fertilizers account for about 40% of the total farm households, and as the use of agricultural machinery increases, the number of farm households using small-sized standing tractors reaches 64% of the total farm households. Rice production was less than 2 million tons in 1998, but it increased rapidly and maintained more than 3 million tons after 2010, and the rice production area reached about 780,000 hectares as of 2019. Comparing 2004 and 2019, production compared to production area has significantly increased.

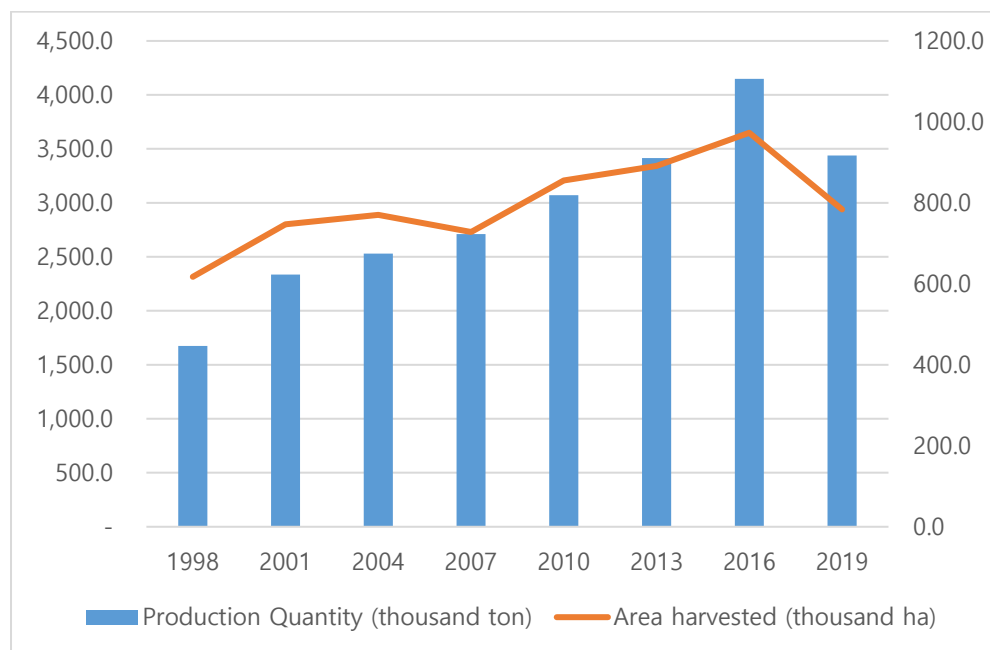
⁷⁵ Foot note: Chang-ho Shin. "Status of Rice Industry in Lao PDR" (unpublished data)

Se-Won Kim, Dong-Hwan An, and Tae-Yun Kim (2017); FAO

Se-Won Kim, Dong-Hwan An, and Tae-Yun Kim (2017); FAO

Heo Jang, Mina Lee (2020)

[Figure 9-2] Trend of Rice Production and Production Area in Lao PDR, 1998-2019

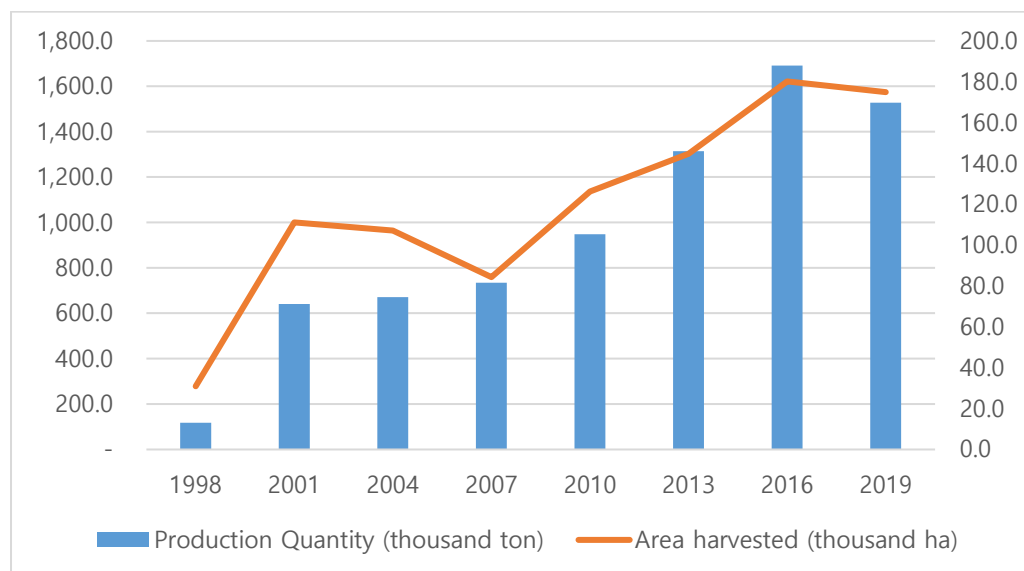


	1998	2001	2004	2007	2010	2013	2016	2019	CAGR
Production Quantity (thousand ton)	1,674.5	2,334.7	2,529.0	2,710.1	3,070.6	3,414.6	4,148.8	3,438.0	3.5%
Area harvested (thousand ha)	617.5	746.8	770.3	727.9	855.1	891.2	973.3	783.8	1.1%

Source: FAO (2021f 2021.10.22.)

Vegetables are one of the main products and exports of Lao PDR, and production has increased rapidly since the late 2000s. In irrigated agricultural areas, vegetables are grown mainly in urban areas during the dry season, while vegetables are grown for household consumption in highland areas. Vegetable production was at the level of 117 thousand tons in 1998, but it increased rapidly and reached 1,527 thousand tons as of 2019.

[Figure 9-3] Trends in vegetable production and production area in Lao PDR, 1998-2019

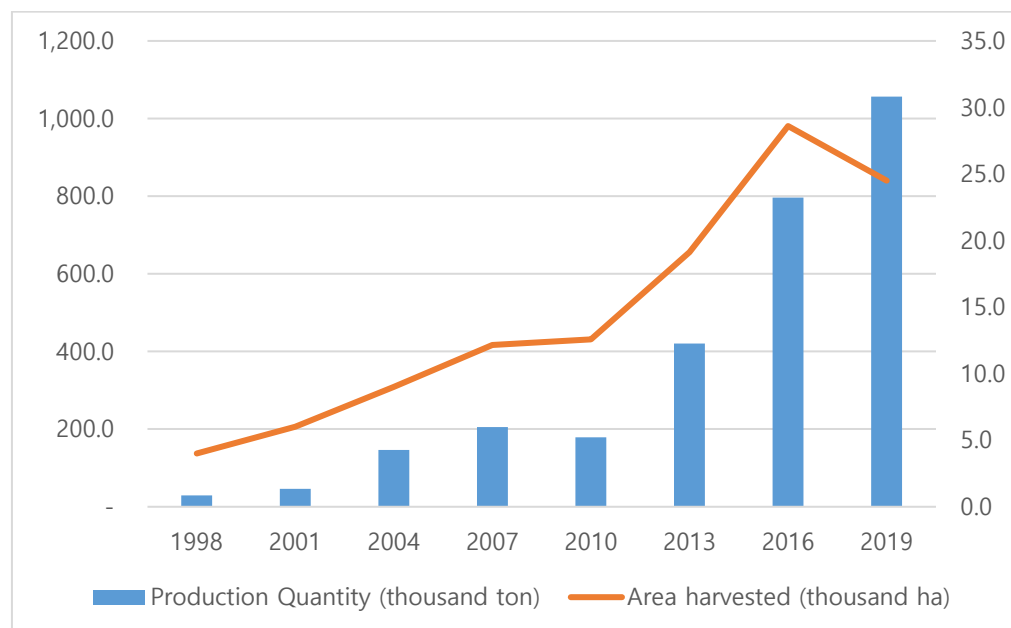


	1998	2001	2004	2007	2010	2013	2016	2019	CAGR
Production Quantity (thousand ton)	117.3	640.7	670.5	734.4	947.7	1,313.7	1,690.9	1,527.2	13.0%
Area harvested (thousand ha)	30.9	111.2	107.2	84.3	126.3	144.7	180.2	174.9	8.6%

Source: FAO (2021f2021.10.22.)

Lao PDR has a relatively undeveloped fruit industry. Lao PDR has been importing tropical fruits from neighboring countries such as Thailand and Vietnam because the production of tropical fruits cannot meet market demand due to lack of technology and funds. However, the production of bananas and watermelons is increasing rapidly. In particular, banana production has increased from 29,000 tons in 1998 at a compound annual rate of 18.7%, reaching 1.057 million tons in 2019. As of 2018, the production value of bananas was 430 million dollars, ranking third in terms of production value.

[Figure 9-4] Trend of Banana Production and Production Area in Lao PDR, 1998-2019



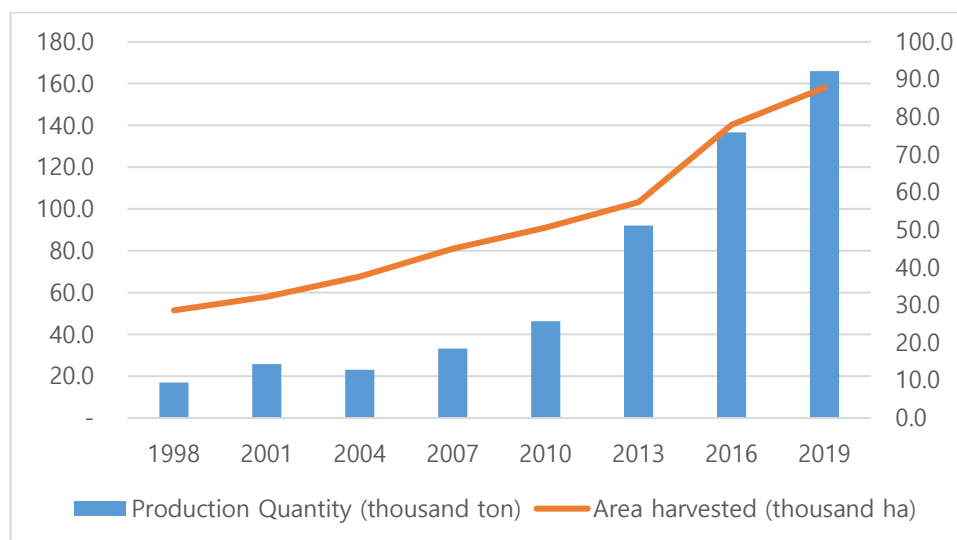
	1998	2001	2004	2007	2010	2013	2016	2019	CAGR
Production Quantity (thousand ton)	29.0	46.0	146.0	205.0	178.6	420.4	796.2	1,056.5	18.7%
Area harvested (thousand ha)	4.0	6.0	9.0	12.2	12.6	19.1	28.6	24.5	9.0%

Source: FAO (2021f2021.10.22.)

Coffee is being encouraged by the Lao government as a part of revitalizing the national economy, and production has steadily increased, reaching 166,000 tons as of 2019. Coffee is produced in Bolaven Plateau in the southern Champasak Province with a cool climate and high altitude, Tongeng in Pakxong Province, and Laongam in Saraban Province. The Lao government is implementing policies to support coffee farms, such as the 2+3 policy, in which farmers provide land and labor and investors provide funds, technology and trading markets. Coffee production was less than 17,000 tons in 1998, but has grown at a compound annual rate of 11.5%, reaching 166,000 tons as of 2019.

Foot note: Heo Jang, Mina Lee (2020)

[Figure 9-5] Trends in coffee production and production area in Lao PDR, 1998-2019

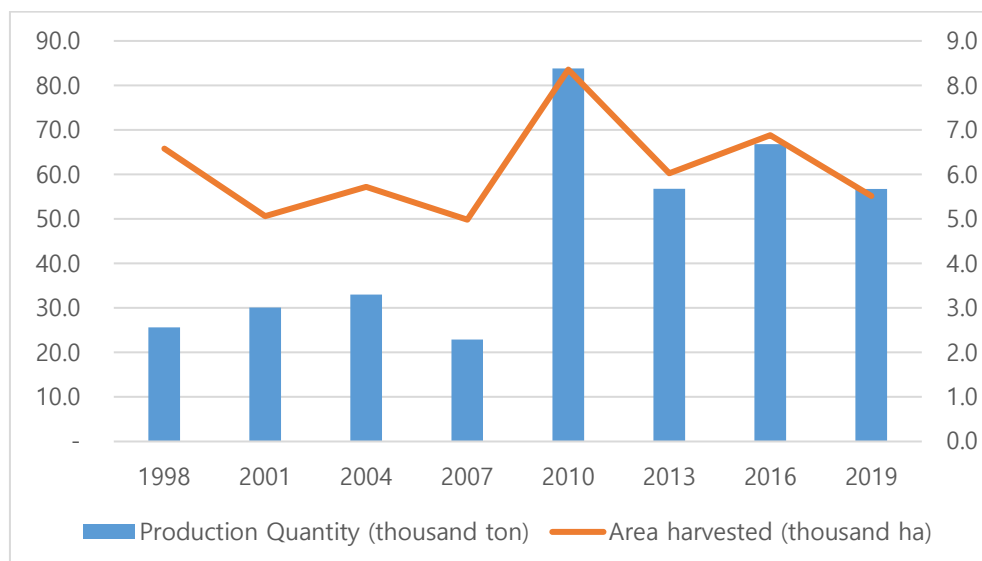


	1998	2001	2004	2007	2010	2013	2016	2019	CAGR
Production Quantity (thousand ton)	17.0	25.8	23.1	33.2	46.3	92.0	136.6	165.9	11.5%
Area harvested (thousand ha)	28.6	32.2	37.6	45.0	50.6	57.3	77.9	87.8	5.5%

Source: FAO (2021f2021.10.22.)

Cash crops in Lao PDR have been encouraged to protect forests and generate income in mountainous regions, and tobacco production has been maintained at over 50,000 tons since 2010. As slash-and-burn agriculture in the mountainous regions acted as a cause of poor soil quality and reduction of tropical forests, the Lao government has encouraged the cultivation of cash crops. Tobacco is a crop with abundant production experience and potential in mountainous and highland regions. Tobacco production was less than 30,000 tons in 1998, but has been maintained at more than 50,000 tons since 2010, and is about 57,000 tons as of 2019.

[Figure 9-6] Trends in Tobacco Production and Production Area in Lao PDR, 1998-2019



	1998	2001	2004	2007	2010	2013	2016	2019	CAGR
Production Quantity (thousand ton)	25.6	30.1	33.0	22.9	83.8	56.8	66.8	56.7	3.9%
Area harvested (thousand ha)	6.6	5.1	5.7	5.0	8.4	6.0	6.9	5.5	-0.8%

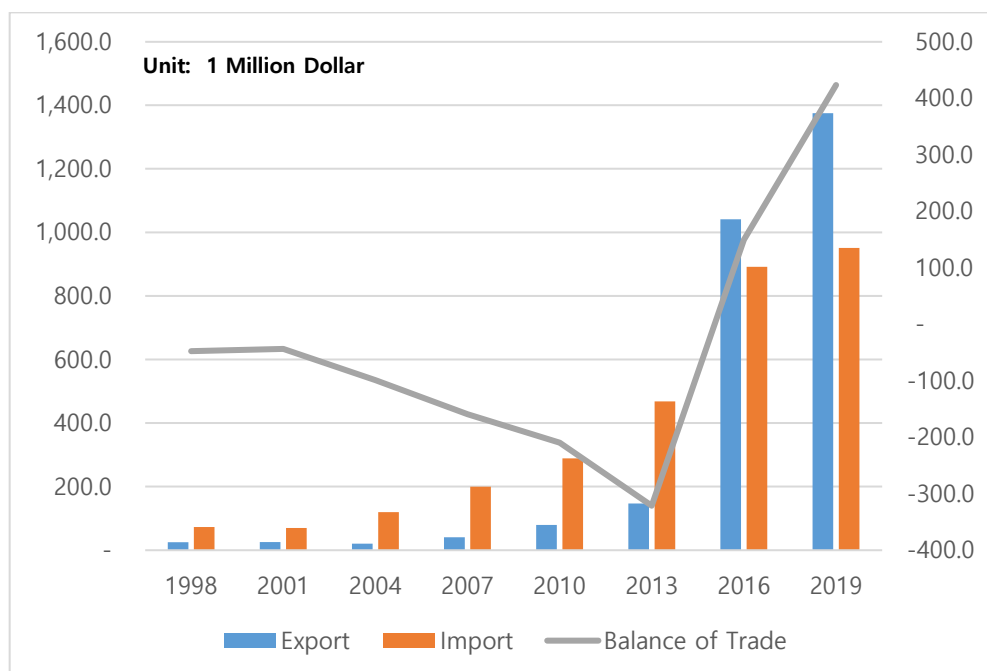
Source: FAO (2021f2021.10.22.)

1.1.4 agricultural trade

As of 2019, Lao PDR' total agricultural and food exports amounted to \$1.374 billion, and total imports were \$951.2 million, recording a surplus of \$423.5 million. Looking at the trend since 1998, it appears that there is a shift from agriculture centered on the domestic market to the expansion of entry into the trade market. Lao PDR' agricultural and food exports jumped from \$146.7 million in 2013 to \$1.0409 billion in 2016, and the trade balance for agro-food turned into a surplus during the same period. Lao PDR' agricultural and food imports approximately doubled from \$467.9 million in 2013 to 2019, but the size of the trade surplus is increasing due to the

rapid increase in exports.

[Figure 9-7] Agricultural products trade in Lao PDR, 1998-2019



	1998	2001	2004	2007	2010	2013	2016	2019	CAGR
Export	24.9	25.7	20.6	40.4	79.2	146.7	1,040.9	1,374.7	21.0%
Import	72.7	69.6	119.7	199.8	289.0	467.9	891.6	951.2	13.0%
Balance of Trade	-47.8	-43.9	-99.2	-159.5	-209.7	-321.1	149.3	423.5	

Source: FAO (2021g2021.10.22.)

As of 2019, the top agricultural export items in Lao PDR were non-alcoholic beverages and rubber, cattle, refined sugar, and buffalo. In 2019 compared to 2013, the ranking of coffee, maize and fresh fruit was lowered, and non-alcoholic beverages took the first place and cattle took third place.

<Table 9-6> Top 10 export items in Lao PDR, 2013, 2016, 2019

Unit: thousand dollars

	2013		2016		2019	
	List of items	Value	List of items	Value	List of items	Value
1	Coffee, green	72,589	Bananas	197,813	nonalcoholic beverage	229,164
2	Maize	42,590	nonalcoholic beverage	155,823	Rubber, Rubber natural dry	165,145
3	Crude materials	4,894	Cigarettes	72,279	Cattle	143,337
4	Fresh fruits	4,411	Rubber, Rubber natural dry	67,930	Refined sugar	102,166
5	Bananas	3,106	Coffee, green	63,317	Buffaloes	83,342
6	Beer of barley	1,672	Cassava	62,203	Dried Cassava	74,098
7	Sesame seed	1,328	Dried Cassava	50,373	Cigarettes	65,791
8	Fruits, prepared	1,311	Cattle	37,601	Coffee, green	54,767
9	Soybeans	94	Cereal preparations	37,006	rubber natural	52,341
10	Dried fruits	38	rice, paddy (rice milled equivalent)	34,897	corn	31,177

Source: FAO (2021g2021.10.22.)

As of 2019, the highest agricultural imports in Lao PDR were non-alcoholic beverages and cattle, refined sugar, and buffalo. Compared to 2013, the ranking of refined sugar, rice and cigarettes in 2019 was higher.

<Table 9-7> Top 10 imported items in Lao PDR, 2013, 2016, 2019

Unit: thousand dollars

	2013		2016		2019	
	List of items	Value	List of items	Value	List of items	Value
1	Food prep nes	67,152	nonalcoholic Beverages	177,077	nonalcoholic Beverages	207,226
2	Cattle	61,996	meat, chicken	127,369	Cattle	142,895
3	nonalcoholic beverage	57,718	rice, paddy (rice milled equivalent)	78,674	Refined sugar	88,307
4	Food wastes	41,154	rice, milled	75,856	Buffaloes	85,989
5	Coffee, extracts	28,716	Food prep nets	75,153	Food wastes	37,403
6	Tabaco products nets	21,721	Refined sugar	44,395	Rice, paddy (rice milled equivalent)	25,016
7	distilled alcoholic beverage	17,585	Coffee, extracts	34,641	Cigarettes	24,145
8	Rubber	16,566	Cattle	30,261	Coffee, extracts	22,897
9	Pigs	16,513	Food wastes	28,286	distilled alcoholic beverage	22,572
10	Refined sugar	16,334	Cigarettes	22,387	Tobacco unmanufactured	21,192

Source: FAO (2021g2021.10.22.)

1.1.5 Food distribution

Lao PDR has poor food manufacturing and distribution facilities. As agricultural products manufacturing facilities and logistics are not well developed, the distribution of self-produced agricultural products is also not smooth, so investment in refrigeration and refrigeration warehouse facilities and the development of logistics systems are required at the government

level. Due to the nature of the landlocked country without a sea, the food culture is not diverse, and in the case of seafood, there are no large seafood import companies or large refrigerated warehouses, so it appears that they are distributed in the market through small merchants in Thailand and Vietnam.

■ **Foot note: Dongkwan Kim (2017)**

Food imports from neighboring countries are increasing as Lao PDR is free to enter and exit from neighboring countries and has eased or abolished customs and import/export regulations. Because it is a landlocked country, trade with other countries uses Bangkok, Thailand or Hai Phong, Vietnam. The import of food produced and distributed in neighboring countries such as Vietnam, Thailand and China is expected to increase in the future owing to the poor situation in the country and the direction of the trade policy.

Lao PDR' distribution market is being transformed into a modern distribution structure, but the traditional distribution market still dominates. Traditional markets and supermarkets are the main ones, there are few large-scale distribution stores, and imports or wholesale and retail distribution systems have not been developed. There is no import or wholesale/retail distribution system for most of the items, and there are many cases where importers also engage in wholesale distribution or even retail distribution. Large marts are delaying their entry into the market due to the small market size and lack of development of fresh food producers, product manufacturers, and logistics companies. As income rises, many consumers who are not satisfied with the retail market in Lao PDR are visiting large marts in Thailand. Lao consumers go shopping to Thailand for leisure and entertainment facilities for several reasons such as low prices, a variety of products, and high-end restaurants. In the case of major cities such as Vientiane, Pakse, and Savannakhet, it is a 1-1.5 hour drive across the border to access large Thai retail stores such as Tesco, Big-C, and Makro in nearby cities. Therefore, rather than operating stores in Lao PDR, large marts are operating stores in a Thai city near the Lao PDR border.

In 2019, the offline market grew by 9.5% compared to the previous year to reach 10.45 trillion kip (about 1.22 trillion won in Korean Won). The proportion of traditional distribution channels is 96.7% and that of modern distribution channels is 3.3%. Although the size of the online food

market is not officially counted, it is expected to have high growth potential in the future due to the increase in Internet and mobile phone penetration.

1.2 Agricultural Policy in Lao PDR

1.2.1 Policy Promotion Agency

The central unit organizational system of the Ministry of Agriculture and Forestry (MAF), which is in charge of agricultural policy, consists of 11 departments and the National Agriculture and Forestry Research Institute (NAFRI), an institution in charge of agricultural research. Eleven departments are specialized in fields and policy areas such as agriculture, forestry, livestock, fisheries, irrigation, land management, planning, and technology expansion. The NAFRI, which has the same status as each department under the Ministry of Agriculture and Forestry, was established in 1999 as an organization that coordinates research and development in agriculture and forestry.

Foot note: “Lao PDR” agricultural and food export information, search date: 2021-10-23.
<https://www.kati.net/nation/basisInfo.do?lcdCode=MD620>

Se-Won Kim, Dong-Hwan An, Tae-Yun Kim (2017) Se-Won Kim et al. (2017); Heo Jang et al. (2017)

Ministry of Agriculture and Forestry website, retrieved from: 2021-10-04.
<https://www.maf.gov.la/>

< Table 9-8> 11 countries of the Lao Ministry of Agriculture and Forestry (MAF)

-
- Department of Agriculture (DOA)
 - Department of Forestry (DOF)
 - Department of Irrigation (DOI)
 - Department of Forestry Inspection (DOFI)
 - Department of Agricultural Land Management (DALam)
 - Department of Planning and Finance (DOPC)
 - Department of Technical Extension and Agro-Processing (DTEAP)
 - Department of Livestock and Fisheries (DLF)
 - Department of Organization and Personnel (DOAP)
 - Department of Rural Development and Cooperatives (DRDC)
 - Department of Policy and Legal Affairs (DOPLA)
-

Source: Ministry of Agriculture and Forestry website

The Ministry of Agriculture and Forestry has regional branch organizations, and currently has 18 Provincial Agriculture and Forestry Office (PAFO), 145 District Agriculture and Forestry Office (DAFO), and 268 Technical Training Centers. The NAFRI has 14 research centers in addition to its headquarter.

1.2.2 Policy

1) System

Currently, Lao PDR' agricultural and rural development policy is centered on the larger national socio-economic development plan (NSED) and the agriculture development strategy (ADS), which are specialized strategies for agriculture. The National Socio-Economic Development Plan (NSED) is a comprehensive development strategy for Lao PDR' social policy system and includes agriculture as a major component. Agriculture Development Strategy (ADS) contains mid- to long-term goals and strategies for agricultural and rural development.

2) National Socio-Economic Development Plan (NSED)

The National Social and Economic Development Plan (NSED) is a five-year social and

economic development plan that directly or indirectly links agricultural and rural development with the output of the performance goal. In 2016, the Lao government announced and implemented the 8th Five-Year National Socio-economic Development Plan (2016-2020), hereinafter referred to as the 8th NSEDP. Recently, the 9th Five-Year National Socio-economic Development Plan (2021–2025, hereafter referred to as 9th NSEDP) was approved by the National Assembly.

In 2016, the Lao government announced the 8th NSEDP, which set three macro performance goals to break out of LDC status. 8th NSEDP set three macro performance goals related to 1) economic growth, 2) human resource development, and 3) protection and utilization of natural resources and the environment. The agricultural and rural development policies in the 8th NSEDP consist of “Sustainable development of agriculture and forestry, allocation of farmland and production quantity for food security and commercial production, increase in production of local specialties through organic agriculture, increase in processing facilities for high added value of agricultural products, and formation of farmers’ producers’ organizations, management and protection of forest resources”.

■ < agricultural production >

- Setting targets for rice production. Ex) By 2020, about 2.5 million tons of rice will be produced in a total of 600,000 hectares of agricultural land in 10 major rice-producing regions
- Set income crop and livestock and aquaculture production goals. For example, about 230,000 tons of corn, about 300,000 tons of potato and root crops, about 800,000 tons of fruits, about 1.5 million tons of vegetables

Establishment of agricultural production plans for each major agricultural production area in consideration of topographical characteristics. Ex) In the great plains, rice production is promoted through the maintenance of irrigation facilities, fruit and vegetables, livestock industry, forestry development, and in the highlands and mountainous areas, the plains focus on the production of fruits and vegetables and cold flowers.

Setting goals and measures to improve national nutrition. Ex) Increase the per capita annual caloric intake to 2,600-2,700 kcal and produce about 258,000 tons of meat and eggs and about 229,000 tons of fish and aquatic products per capita of eggs, fish, and eggs per capita annually Guaranteed to be 65 kg

■ < High value added of agri-food >

- Creating employment in the agricultural sector by supporting the diversification of farm households
- Expansion of organic agricultural products market
- Improve post-harvest management and processing technology
- Establish a modern rice milling facility and expand food production infrastructure that can meet international standards
- Transition from natural livestock to livestock industry through the introduction of modern livestock technology

Set development goals reflecting the geographical and climatic specificities of each region. Ex) In the northern part, production and processing facilities and livestock and fishery products center were established by item, in the central part, forest protection and improvement of the quality of non-timber forest products (NTFPs), and food quarantine standards were strengthened, and in the southern part, the cultivation of commodity crops was spread, and the livestock industry and fishery industry were developed. , expanding production of non-timber forest products (NTFPs)

■ < Rural Development >

- Expand the land area equipped with irrigation facilities from about 320,000 ha in 2016 to about 480,000 ha by 2020
- Minimize the instability of agricultural production due to natural disasters by improving the quality of agricultural water and building a hydroelectric power plant in the agricultural center.

- Resolving the problem of infringement of land use rights of Lao farmers and mobile arable land by mountain dwellers. Ex) Expansion of land allocation survey and soil survey, issue of farmland ownership to family farms in agricultural priority areas by 2020

■ < Agricultural education and agricultural research >

- Strengthen support for research on Lao native seeds and increase access to agricultural information and technology for farmers' organizations
- Research on high-quality agricultural products and livestock breeding and providing agricultural technology services to farmers
- Establishment of research institutes and agricultural technology centers to strengthen competitiveness of agri-food

■ < Food market policy and international cooperation >

- Establish various credit and financial policies to improve access to credit for farmers, producers, and agro-food companies
- Establish an integrated production system between SMEs and large enterprises by applying policies on price control, tax and investment in the agri-food market, and strengthen the public-private partnership system to ensure that a sufficient amount of agricultural products can be continuously supplied to food processing companies
- Enforce policies on intellectual property rights, standards and measures to build consumer confidence in agricultural food and forestry products and ensure trademark rights;
- Continuous use of low-interest concessional loans from advanced countries, and strengthening South-South and trilateral cooperation
- The Asian Development Bank (ADB) expands international cooperation projects in the Greater Mekong Sub region through the use of the Association of Southeast Asian Nations (ASEAN)

<Table 9-9> Details of 8th NSEDP's Agricultural and Rural Policy

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Source: Composition with reference to Kim Se-won et al. (2017)

The agricultural production sector aims to improve national nutrition and achieve national food security through the production of a sufficient amount of agri-food, while the high value-added sector in agriculture seeks to foster the agro-food industry through improvement of post-harvest management, sales and distribution, and the agricultural value chain. The rural development sector promotes infrastructure maintenance, land reorganization, village development, and climate change response to resolve the development gap between urban and rural areas due to economic growth. The aim of agricultural education and agricultural research sector is to strengthen the capabilities of agricultural educational institutions for the transfer of agricultural technologies and the dissemination of seeds. The agri-food market policy and international cooperation sector seeks to develop the agri-food market and utilize the financial resources and support projects of international organizations and donor countries.

According to the Ministry of Agriculture and Forestry of Lao PDR, as a result of the implementation of the 8th NSEDP, there have been achievements in maintaining food self-sufficiency and improving national nutritional status, but there are still unresolved problems. Since the mid-1990s, food production in Lao PDR has soared, exceeding the current per capita food production target in all sectors except for aquatic products. The diversity of food consumption in urban areas has increased due to economic growth and regional integration. On the other hand, most of the food production is still for livelihood, poverty is high in rural areas, food distribution is unsatisfactory, and about 20% of the population suffers from food insecurity at the household level. Growth inhibition rate of children under the age of 5 nears

50% in some provinces, and growth inhibition rate exceeds 50% in some villages, particularly those with poor road access, ethnic minorities, and high-altitude villages.

In March 2021, the Lao Parliament approved the 9th NSEDP, which aims to break out of LDC status by 2025 and enter a middle to high-income country by 2030. The 9th NSEDP's five-year average annual GDP growth rate target is at least 4%, with sectoral growth rate targets of 2.5% for agriculture, 4.1% for industry and commerce, 6% for services and 5.8% for tariffs and taxes. 9th NSEDP focuses on 1) guarantee of continuous economic growth, 2) guarantee of sustainable development, 3) reinforcement of human resource capacity, 4) continuation of political stability, 5) effort to expand international cooperation, 6) national green growth strategy, setting the following four directions: 1) quality, 2) focused, 3) green, and 4) sustainable. The 9th NSEDP's six performance goals are 1) achieving continuous, high-quality, stable and sustainable economic growth, 2) improving the quality of human resources, 3) improving public welfare, 4) improving environmental protection and reducing disaster risk, 5) promoting regional and international cooperation and integration; 6) improving public governance and administration; and a society that is equal, fair and protected by legal norms.

Agricultural and rural development policies within the 9th NSEDP are found to contribute to 5 out of 6 goals, including food production, commercial production and agricultural business, domestic and foreign investment, investment, innovation, and technology in agriculture, rural industry and service livelihood, forestry and forest resource management, and harmonized production chain among large enterprises and small and medium enterprises.⁷⁶

⁷⁶ Se-Won Kim, Dong-Hwan An, Tae-Yun Kim (2017) Se-Won Kim et al. (2017); Heo Jang et al. (2017)

Se-Won Kim, Dong-Hwan An, Tae-Yun Kim (2017) Se-Won Kim et al. (2017)

<Table 9-10> Contribution of Agriculture, Livestock and Forestry Sector to 9th NSEDP Goals

9th NSEDP outcomes	Objective of agricultural and live stock, and forestry sector
Outcome 1	(Food production) - Ensure nutrition and food security – expand rice farming areas; provide water for irrigation; produce other crops, livestock, fisheries and NTFP for a diversified diet. (Commercial production/Agribusiness) - Produce more and higher commodities for sale in domestic export markets for fresh and processed items. (Domestic and foreign investment) - Facilitating environment to attract investment - inspection, assessment of agricultural and other projects in agricultural / rural areas
Outcome 2	(Investment, innovation, and technology in agriculture) - Building capacity for officials in enterprise registration in industry and commerce sector; etc.. - Application of modern technology and innovation in the agriculture - Youth and agriculture - Farmers and small holders
Outcome 3	(Food production) - Ensure nutrition and food security – expand rice farming areas; provide water for irrigation; produce other crops, livestock, fisheries and NTFP for a diversified diet. (Rural industries, services and livelihoods) - Promote value-addition goods production and services in agriculture. Livestock, fisheries and rural areas. - Irrigation
Outcome 4	(Forestry and forest resources management) - Support restoration of forestry resources for richness natural resources base and ecosystems. - Natural Resource and Disaster Risk Management - Disease control

9th NSEDP outcomes	Objective of agricultural and live stock, and forestry sector
Outcome 5	(Harmonized production chain among large enterprises and small and medium enterprises) - Technical assistance; cooperation and coordination with SME-supporting international organizations; research and development; finance; feasibility studies; etc.

Source: Inthichack(2021)

1.2.3 Agriculture Development Strategy (ADS)⁷⁷

Agriculture Development Strategy (ADS) presents various implementation measures and action plans along with mid to long-term goals for agriculture and rural development. In 2015, the Laotian Ministry of Agriculture and Forestry announced the Agriculture Development Strategy to 2025 and vision to the year 2030 (hereafter ADS 2025), which contains implementation goals and strategies by 2025 and vision by 2030. ADS 2025 presents comprehensive and mid to long-term goals and strategies and does not contain priorities or specific investment plans regarding the various policies presented.

ADS 2025 specifies goals and achievements for agriculture and agriculture production development from 2020 to 2025 while presenting its vision to 2030 as “ensuring food security, producing agricultural products with comparative competitiveness, developing clean, safe and sustainable agriculture, and gradually moving towards the modernization of the productive agricultural economy, but in connection with regional development that contributes to the establishment of an economic foundation”.⁷⁸ Goals and achievements set targets for the production of food and agricultural products, such as minimum nutritional targets, food composition ratio, rice production, and exports for each target period.

⁷⁷ Chang-Ho Shin (2020); Jang Heo, Mi-Na Lee (2020); MAF(2015)

⁷⁸ Chang-Ho Shin (2020)

<Table9- 11> ADS 2025 Goals and achievements by target period

Category	By 2020	By 2025
Agricultural GDP	Annual growth of 3.4% to contribute 19% to the national economic structure	
National nutrition	Minimum daily intake of 2600Kcal per person - Rice, starch: 62% - Meat, egg, fish: 10% - Vegetables, fruits, beans: 6% - Fat, sugar, milk: 22%	Minimum daily intake of 2600Kcal per person - Rice, starch: 54% - Meat, egg, fish: 13% - Vegetables, fruits, beans: 8% - Fat, sugar, milk: 25%
Food and agricultural products	- Rice(paddy): non-glutinous rice 30% included, production of 4.7 million tons - Clean rice and vegetable production are improved to meet GAP requirements - Commercial production of rice: at least 1 million tons for domestic and export - Corn for livestock feed: 1.3 million tons - Coffee: 120,000 tons - Sugar cane for sugar production: 2.15 million tons - Cassava: 1.5 million tons - Beans: 50,000 tons - Meat, egg: 262,500 tons - Livestock growth rate: 6% annually - Beef and carabeef exports: 10,000-15,000 tons	- Rice(paddy): non-glutinous rice 30% included, production of 5 million tons - Clean rice and vegetable production are improved to meet GAP requirements - Commercial production of rice: at least 1.5 million tons for domestic and export - corn for livestock feed: 1.4 million tons - Coffee: 280,000 tons - Sugar cane for sugar production: 2.4 million tons - Cassava; 1.6 million tons - Beans: 52,000 tons - Meat, egg: 414,000 tons - Livestock growth rate: 6% annually - Beef and carabeef exports: 15,000 tons

Source: Chang-Ho Shin (2020); MAF(2015)

ADS 2025 enumerates in detail the technical measures related to the production of food and agricultural products along with overall measures and draws actions plans and tasks to execute them. Overall measures include improvement in policies and laws, decision of

mediation bodies, enhancement of cooperation and investment, organizational reform, and human resource development in the agriculture and forestry sector. Specific technical measures of food crops and commercial crops, livestock products for consumption and sale, and food security and nutrition include crop production systems, livestock breeding technology development, farmland measurement and distribution, strengthening research guidance, expanding productive labor force development, and constructing irrigation and technical infrastructure. In addition, ADS 2025 has set a total of 16 action plans and 120 tasks to achieve food and agricultural product production, which in detail consists of 9 action plans and 62 tasks related to crop production, and 7 action plans and 58 tasks related to livestock and fisheries.

1.3 (At present) Agricultural sector tasks by 2023⁷⁹

According to the Laotian Ministry of Agriculture and Forestry, the agricultural sector in Lao PDR will focus on two major tasks by 2023: agriculture production and agricultural product production. The main tasks are 1) promoting agricultural production to replace imports and reduce foreign trade flows, and 2) promoting the production of agricultural products related to a commercial processing for export to attract foreign currency. The direction of the agriculture and forestry sector for three years from 2021 to 2023 is to maintain an annual average agricultural growth rate of 2.5% and an agriculture GDP share of 15.3%.

<Table 9-12> Key tasks for the agricultural sector in Lao PDR by 2023

Key task 1	Accelerate the export of agricultural products that have been signed with trading partner countries, focusing on the Chinese market, and at the same time speed up the negotiation of the signing of agricultural export agreements that have been identified in the list of crops and plant products that are priority for market opening negotiations with China and other countries.
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⁷⁹ Inthichack(2021).

Key task 2	Accelerate the improvement of standards and quality of agricultural products, including capacity building to inspect and certify the standards of agricultural products, such as Sanitary and phyto-sanitary (SPS), certification of product identification, certification of organic and good agricultural standards (GAP, GAHP GMP, HACCP); Expedite the dissemination of legislation, policies and standards related to the implementation of local standards (SPS) and Entrepreneurs to expand and reach targets; Accelerate the mechanism of planning, coordinating and exchanging production and market information for agricultural products between the centers. Central, local and entrepreneurship improved.
Key task 3	Accelerate the identification of list of agricultural products to replace imports and comprehensive promotion policies, along with the expansion and improvement of production capacity (production groups, agricultural cooperatives, associations, farmers' networks, etc.) and a wider and more practical production model.
Key task 4	Accelerate the improvement and development of production infrastructure and techniques to reduce production costs such as: dryers, seed sorting machines, drying racks, storage houses, laboratories and retail / wholesale markets.
Key task 5	Accelerate the prioritization and identification of agricultural investment zones, pay attention to monitoring and inspecting investment projects in the agricultural sector in accordance with the signed agreements; Improve investment mechanisms in the form of public-private partnerships, contract farming based agriculture production, such as 2 + 3, PPP, leases or state land concessions.

Source: Inthichack(2021)

1.4 Laotian Universities in Agriculture

1.4.1 Overview of the university system in agriculture

In Lao PDR, four out of five national universities have agricultural colleges. National universities with agricultural colleges are the National University of Lao PDR, Champasack University, Souphanouvong University, and Savannakhet University, with the National University of Lao PDR being the only case in which agriculture, forestry, livestock, and fisheries are independent at the college level. Departments in the forestry sector in Lao PDR

appear to be participating in projects to strengthen the forest-related higher education system in the Mekong region or Lao PDR under the funding of Erasmus+, a European education support program.

1.4.2 National University of Lao PDR⁸⁰

The National University of Lao PDR, located in Vientiane, the capital of Lao PDR, has the Faculty of Agriculture and Faculty of Forest Science as its agriculture college. The Faculty of Agriculture is on the Nabong Campus, located on the outskirts of Vientiane, and the fisheries program within the Faculty of Agriculture and the Faculty of Forest Science are on the Dongdok Campus, which is the main campus located inside the city of Vientiane.

<Table 9-13> Agricultural colleges and departments of National University of Lao PDR

Division	Department
Faculty of Agriculture	- Department of Plant Science
	- Department of Livestock and Fisheries
	- Department of Rural Economics and Food Technology
	- Department of Veterinary Medicine
Faculty of Forest Science	- Department of Forest Economics and Wood Processing Technology
	- Department of Forest Management
	- Department of Water Resources Management and Land Use
	- Department of Community Forestry and Rural Development
	- Department of Conservation Tourism

Source: Website of the National University of Lao PDR and each college

⁸⁰ Website of the National University of Lao PDR and each college (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. <https://www.nuol.edu.la/>; <https://fag.nuol.edu.la/index.php/en/>; <http://ffs.nuol.edu.la/>; <https://fes.nuol.edu.la/>

Founded in 1975, the Faculty of Agriculture manages bachelor's and master's degree programs related to the four departments and has associated research facilities. The vision is to have a group of excellent professionals in higher agricultural education following regional and international standards, and the goals are 1) to develop qualified agricultural professionals through the cultivation of professional knowledge, competence, ethics, and morals; 2) to promote and develop agricultural research following regional and international standards; and 3) to develop faculty in clean and green fields and expand modernized educational facilities. Currently, six bachelor's programs and four master's programs are in operation.⁸¹

<Table 9-14> National University of Lao PDR' Faculty of Agriculture Bachelor's and master's degree programs

1. Bachelor's degree program	2. Master's degree program
- Crop Production - Plant Protection - Livestock - Fisheries - Agricultural Economics and Food Technology - Veterinary Medicine	- Sustainable Agricultural Resource Management - Crop Science - Animal Science - Sustainable hydropower development

Source: National University of Lao PDR website

National University of Lao PDR' Faculty of Agriculture possesses various laboratories and pilot farms, and its research fields include plant production, animal production, management and use of agricultural resources, value chain analysis, animal breeding, fisheries and fish ecology, pests, and plant diseases. In addition, the Faculty of Agriculture provides technical services that are largely composed of education and consultant services.⁸²

⁸¹ National University of Lao PDR' Faculty of Agriculture website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. <https://fag.nuol.edu.la/index.php/en/>

⁸² National University of Lao PDR website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into

The fields of education and research of the Faculty of Forest Science are forest management, forest economics and wood processing technology, water resource management and land use planning, conservation and nature conservation tourism, local forest and rural development, NTFP management and marketing, climate change and natural resources, and environmental research, and the Faculty of Forest Science manages a doctoral degree program in addition to the bachelor's and master's degrees.⁸³

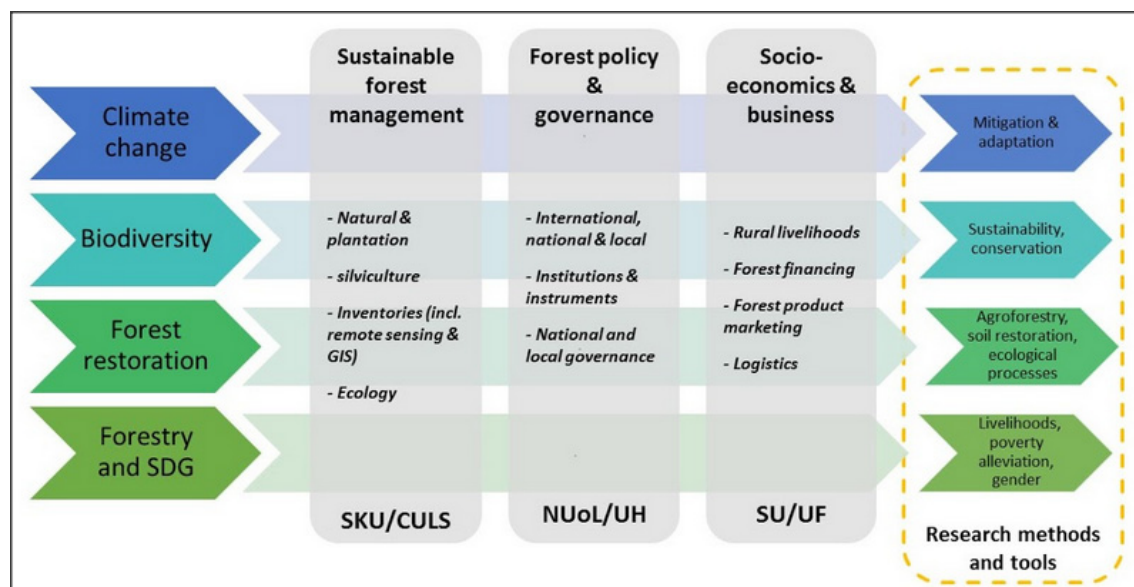
The Faculty of Forest Science appears to be participating in the ForHeal (Forestry Higher Education Advancement in Lao PDR) project, which strengthens Lao PDR' forest-related higher education system, organized by Europe. ForHeal is funded by Erasmus+, a European education support program and its purpose is "to strengthen the forest-related higher education system in Lao PDR to improve domestic capacity to sustainably manage forests for the benefit of the people and the environment, while at the same time enabling Lao PDR to effectively respond to the demands and opportunities of international processes such as the Agenda 2030 (SDGs) and the Paris Agreement on Climate Change".⁸⁴ Program institutions are the University of Helsinki in Finland, the Czech University of Life Sciences Prague in the Czech Republic, and the University of Freiburg in Germany, and the partner institutions in Lao PDR are the National University of Lao PDR, Savannakhet University, and Souphanouvong University. Concerning the project, the National University of Lao PDR' Faculty of Forest Science adopts forest policy and governance, Savannakhet University adopts sustainable forest management, and Souphanouvong University adopts socioeconomics and business as its academic focus areas.

English through Google Translate. <http://www.nuol.edu.la/index.php/%E0%BA%84%E0%BA%B0%E0%BA%99%E0%BA%B0%E0%BA%A7%E0%BA%B4%E0%BA%8A%E0%BA%B2/2020-12-01-04-27-39>

⁸³ National University of Lao PDR website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. <https://www.nuol.edu.la/index.php/%E0%BA%84%E0%BA%B0%E0%BA%99%E0%BA%B0%E0%BA%A7%E0%BA%B4%E0%BA%8A%E0%BA%B2/2020-12-01-04-29-53> ;
National University of Lao PDR' Faculty of Forest Science website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. <http://ffs.nuol.edu.la/>

⁸⁴ National University of Lao PDR' Faculty of Forest Science website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. <http://ffs.nuol.edu.la/>

[Figure 9-8] ForHeal project's academic focus areas and framework



Source: National University of Lao PDR' Faculty of Forest Science website

1.4.3 Champasack University⁸⁵

Champasack University, a national university located in Champasack Province, has the Faculty of Agriculture and Forestry as its agricultural college. The vision of the Faculty of Agriculture and Forestry is to establish the Faculty of Agriculture and forestry as the center of education, research, scientific experiments, technical service provision to society, and technical cooperation at the regional and international levels.⁸⁶

The Faculty of Agriculture and Forestry consists of two departments, the Department of Agriculture and the Department of Animal Husbandry, and manages bachelor's and master's degree programs and has associated research facilities. The degree program manages five bachelor's programs, three consecutive bachelor's programs, and one master's program. As of January 2021, there are a total of 64 (26 females) staff and

⁸⁵ Champasack University website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. (<http://www.cu.edu.la/index.php/en/>)

⁸⁶ National University of Lao PDR' Faculty of Agriculture website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. (<https://fag.nuol.edu.la/index.php/en/>)

professors, consisting of 31 bachelor's degree holders (16 females), 23 master's degree holders (6 females), and 10 doctoral degree holders (4 females).

<Table 9-15> Champasack University's Faculty of Agriculture and Forestry Bachelor's and master's degree programs

1. Bachelor's degree program
- Agriculture (4 years)
- Livestock (4 years)
- Food Technology (4 years)
- Forest Resources and Environment (4 years)
- Veterinary Medicine (5 years)
2. Consecutive Bachelor's degree program
- Forest Science (2 years)
- Agriculture (2 years)
- Livestock (2 years)
3. Master's degree program
- Sustainable Agriculture (2 years)

Source: Champasack University website

1.4.4 Souphanouvong University

Souphanouvong University, a national university located in Luang Prabang, has Faculty of Agriculture and Forest Resources as its agricultural college.⁸⁷ Faculty of Agriculture and Forest Resources consists of six majors – Animal Science, Food Science and Technology, Plant Science, Rural Development, Forest Resources, and Eco-tourism.⁸⁸

The Faculty of Agriculture and Forest Resources is participating in the FRAME (Forests, climate change mitigation and adaptation: Higher Education Cooperation in Mekong region) project, which strengthens higher education in the field of forestry and climate change

⁸⁷ It was built with the support of the Infrastructure Development Loan Project (Business Year: 2004, Approval amount: \$22.7 million) of Korea's Economic Development Cooperation Fund (EDCF).

⁸⁸ Souphanouvong University website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. (http://www.su.edu.la/?page_id=95)

response in the Mekong region, organized by Europe. FRAME is funded by Erasmus+, a European education support program, and its purpose is “to strengthen the forest-related higher education system in Lao PDR and Thailand by modernizing, improving accessibility and internationalization of partner HEIs curriculum by supporting the development of new learning and teaching tools (including MOOCs and open-source ICT tools) specifically in relation to sustainable forest management and forest restoration & rural livelihoods”.⁸⁹

Program institutions are the University of Helsinki in Finland and the Czech University of Life Sciences Prague in the Czech Republic, and partner institutions in the Mekong region are Savannakhet University and Souphanouvong University in Lao PDR, and Chiang Mai University and Kasetsart University in Thailand. The academic focus areas are sustainable forest management, forest recovery, and rural livelihood.

<Table 9-16> FRAME academic focus areas by partner institutions

< Sustainable Forest Management >
- Institutions: Savannakhet University, Kasetsart University, Czech University - Natural and Plantation - Biodiversity - Silviculture - Inventories (incl. remote sensing and GIS) - Forest Technology - Ecology - International, National, & Local Forest Policy & Governance
< Forest Restoration & Rural Livelihoods >
- Institutions: Souphanouvong University, Chiang Mai University, University of Helsinki - Climate Change Mitigation and Adaptation - Rural Livelihoods - Forest Financing - Forest Products Marketing - Livelihoods, Poverty Reduction & Gender - Agroforestry, Soil Restoration & Ecological Processes

Source: Souphanouvong University website (<https://www.su.edu.la/Higher-Education-Cooperation-in-Mekong-Region>)

⁸⁹ Souphanouvong University website <https://www.su.edu.la/Higher-Education-Cooperation-in-Mekong-Region>

1.4.5 Savannakhet University⁹⁰

Savannakhet University, founded by the Laotian Ministry of Education and Sports in 2009, has the Faculty of Agriculture and Environment as its agricultural college. Faculty of Education, Faculty of Natural Science, Faculty of Business Management, and Faculty of Linguistics and Humanity are installed in the first campus (Na Sang Campus), and the second campus (Nong Phue Campus), which opened in 2018, includes the Faculty of Agriculture and Environment and Faculty of Food Science. The second campus which is currently under construction with the support of the Asian Development Bank (ADB), will consist of the Faculty of Agriculture and Environment, Faculty of Linguistics and Humanity, Faculty of IT, Faculty of Food Science, and Faculty of Natural Science, and is scheduled to be completed in May 2022.⁹¹

Faculty of Agriculture and Environment has four departments – Crop Science, Animal Science, Environment Science and Forestry Resource. Currently, the Faculty of Agriculture and Environment has a total of 75 faculty members (29 female), including 3 doctoral degree holders and 29 master’s degree holders, and 434 students are enrolled.

1.5 Laotian Agricultural Research Institutes

National Agriculture and Forestry Research Institute of Lao PDR, hereinafter NAFRI is an institution under the Ministry of Agriculture and Forestry established in 1999 with the purpose of integrating national agricultural and forestry research activities and developing a coordinated research system.⁹² NAFRI has the same authority as each department under

⁹⁰ Chang-Ho Shin, Director of Lao PDR KOPIA Center; “Savannakhet University, Faculty of Agriculture and Environment” Development Aid (Accessed on 2021-10-04) <https://www.developmentaid.org/#!/organizations/view/226387/savannakhet-university-faculty-of-agriculture-and-environment>

“Member Universities-Savannakhet University, Lao PDR” ACNET-EngTech (Accessed on 2021-10-04) http://acnet-engtech.tju.edu.cn/member/asean_side/201511/t20151105_267695.htm

⁹¹ The Korean Embassy in Lao PDR, “Beginning construction of the Savannakhet University second campus” The Korean Embassy in Lao PDR, 2021-01-07.

⁹² National Agriculture and Forestry Research Institute of Lao PDR (NAFRI) website (Accessed on 2021-10-04). Information in Laotian reflects the content translated into English through Google Translate. http://www.nafri.org.la/?page_id=203

the Ministry of Agriculture and Forestry and includes agricultural, livestock, aquaculture, and forestry research institutes scattered across the country.⁹³

The purpose of NAFRI is to plan strategies that is in harmony with government policies by providing technical information, norms, and results through research in the fields of agriculture, forestry, and fisheries, and its functions consist of 1) performing adaptive research, 2) developing methods, means, and information package, 3) providing policy feedback, and 4) coordinating and managing research.

NAFRI's responsibilities include directly carrying out research activities along with the development of R&D projects and tasks, as well as interaction and information exchange with stakeholders in R&D in the agriculture and forestry industry. The details are as follows.

Develop specific programs and projects to implement priority programs of the Ministry of Agriculture and Forestry and the government

- Play a pivotal role in land-related research such as land-use planning, zoning, and land management
- Conduct research for genetic resource collection and screening, seed/cultivar improvement, proliferation, and other services
- Conduct research for effective and demand-based agricultural and forestry production technologies, methods, models, and systems
- Collect, process, manage, and exchange information and research results with partners related to agricultural and forestry development
- Harmonize with all relevant R&D partners
- Manage, monitor, and evaluate research activities in the field of charge
- In charge of the affairs of the Ministry of Agriculture and Forestry's Science Committee

⁹³ KREI, 2017;

NAFRI consists of two central departments, seven agricultural product-based research centers, two non-agricultural product-based research centers, and five regional research centers. The existing Agriculture Research Center has been divided into Maize and Cash Crop Research Center and Rice Research Center.⁹⁴

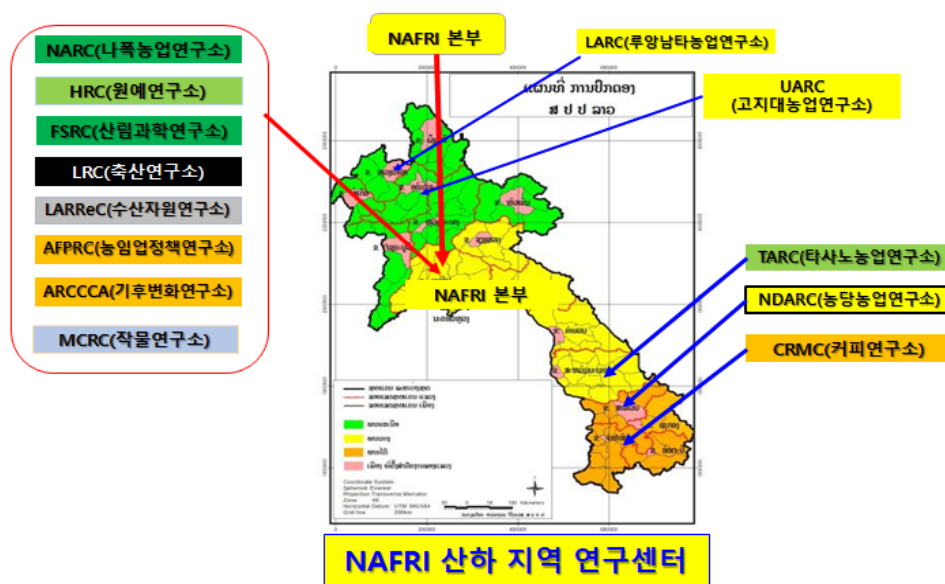
<Table 9-17> NAFRI's organization composition

< Central Department >	
- Administration and Personnel Division	
- Planning and Cooperation Division	
< Agricultural Product-Based Research Center >	
- Forestry Science Research Center	
- Horticulture Research Centre	
- Livestock Research Center	
- Living Aquatic Resources Research Centre	
- Maize and Cash Crop Research Center	
- Research Center to Climate Change Resilience in Agriculture	
- Rice Research Center	
< Non-agricultural Product-Based Research Center >	
- Agriculture and Forestry Policy Research Center	
- Center for Agriculture and Forestry Information and Communication	
< Regional Research Center >	
- Coffee Research and Multiplication Center	
- Luangnamtha Agriculture Research Center (Luangnamtha Province)	
- Nong Daeng Agriculture Research Center (Champasack Province)	
- Thasano Agriculture Research Center (Savannakhet Province)	
- Upland Agriculture Research Center (Luang Prabang Province)	

Source: NAFRI website and Chang-Ho Shin, Director of Lao PDR KOPIA Center

⁹⁴ Chang-Ho Shin, Director of Lao PDR KOPIA Center

[Figure 9-9] The location of NAFRI's headquarters and affiliated research centers (research institute)



Source: Chang-Ho Shin, Director of Lao PDR KOPIA Center

It is said that NAFRI has difficulties in conducting independent research projects due to a lack of budget and the problem of a lack of professional research personnel in each field.⁹⁵ Only two-thirds of NAFRI's employees receive salaries from the government, the rest are paid from projects supported by foreign countries, and the affiliated research centers are having difficulties conducting appropriate research suitable for its function due to a lack of research funds. Of the 340 employees, only about 20 of them are doctoral degree holders, and there are 0-2 doctoral degree holders at each affiliated research center. Many doctoral degree holders work in areas other than their majors, since they cannot recruit experts according to the function of the institution due to the lack of diversity in their majors.

⁹⁵ E-mail interview with Chang-Ho Shin, Director of Lao PDR KOPIA Center (2021.9.28.) Information obtained from conversations with NAFRI's director and executives.

This is the translation of the Korean text in [Figure 30] (from left to right)

NARC (Napok Agricultural Research Center), HRC (Horticulture Research Centre), FSRC (Forestry Science Research Center), LRC (Livestock Research Center), LARReC (Living Aquatic Resources Research Centre), AFPRC (Luangnamtha Agriculture Research Center), ARCCCA (Research Center to Climate Change Resilience in Agriculture), MCRC (Maize and Cash Crop Research Center), NAFRI headquarters, LARC (Luangnamtha Agriculture Research Center), UARC (Upland Agriculture Research Center), TARC (Thasano Agriculture Research Center), NDARC (Nong Daeng Agriculture Research Center), CRMC (Coffee Research and Multiplication Center), NAFRI's regional affiliated research centers

In Lao PDR, doctoral degrees are not conferred easily in its country, and currently, doctoral degree holders at NAFRI have received scholarships individually and have obtained their degrees abroad, with more than half of them in their 50s and only two younger doctoral degree holders in their 30s.

<Table 9-18> Distribution of NAFRI's doctoral degree personnel in headquarters and affiliated research centers

NAFRI's 22 doctoral degree holders (5 females)	
Position and affiliated research center	Field of study
DG (Director General), DDG (Deputy Director General)(4 employees)	Agronomic
	Forest Biological Science
	Agriculture Promotion
	Rice Improvement
Planning and Cooperation Division	Hydrology
	Unknown
Economic and Rural Development Research Center	Management and Development of Economic
	Regional and Rural development
Rice Research Center	Crops Cultivation
	Horticulture
	Horticulture
Maize and Economic Crops Research Center	Crops Cultivation
	Plant Improvement
Horticulture Research Center	Plant Genetic
Livestock Research Center	Feed and Raising system
	Unknown
Fisheries Research Center	Fisheries Management
Forestry Research Center	Forest Biological Science
Lao Rubber Research Center	Science Technology
Research Center For Climate Change in Agriculture	Agriculture
Upland Agriculture Research Center	System Agriculture
	Plant Genetic Improvement

Source: Chang-Ho Shin, Director of Lao PDR KOPIA Center

<Table 9-19> Distribution of NAFRI's doctoral degree personnel by age group

Age Group	Birth Range	Male	Female	Total
30	1982-1991	1	1	2
40	1972-1981	6	1	7
50	1962-1971	9	3	12
60	1952-1961	1	-	1
Total		17	5	22

Source: Chang-Ho Shin, Director of Lao PDR KOPIA Center

1.6 Agricultural sector research expenditure and current status of researchers in Lao PDR⁹⁶

1.6.1 Overview

As of 2017, Lao PDR's research expenditures and size of researchers in the agricultural sector appear to be small compared to other Mekong region countries. Based on the 2017 Purchasing Power Assessment Index, Lao PDR's agricultural research expenditure was \$19.3 million, less than 3% of Thailand's \$847.2 million, less than Cambodia (\$30.2 million) and Myanmar (\$46.6 million). The proportion of agricultural research expenditure to agricultural GDP is 0.26%, similar to that of Cambodia, higher than Myanmar, and lower than Thailand. The number of Full-Time Equivalent (FTE) researchers are 203 in Lao PDR, which is the smallest compared to Cambodia (319), Myanmar (657), and Thailand (2,911), with 47% of researchers with master's degree or higher, which is similar to Cambodia (45%) and Thailand (50%), but higher than Myanmar (37%).

Of the 203 Full-Time Equivalent (FTE) researchers in the agricultural sector, 79 researchers, or 39% are in universities, and 124 researchers, or 61% are working at NAFRI. As such, more than 70% of doctoral degree holders and 66% of bachelor's degree holders work at NAFRI. Lao PDR lacks R&D infrastructure in the agricultural sectors, including research funds, and

⁹⁶ ASTI(2019)

doctoral degree holders who can lead research activities are concentrated in NAFRI, whose research funding is supported by government and foreign partner organizations.

<Table 9-20> Distribution of Lao PDR' Agricultural Researchers by Institution and Degree as of 2017

Category	Universities		NAFRI		Total	
	Number of persons	%	# of persons	%	# of persons	%
PhD	6	27.3%	16	72.7%	22	100.0%
MSc	36	49.3%	37	50.7%	73	100.0%
BSc	37	34.3%	71	65.7%	108	100.0%
Total	79	38.9%	124	61.1%	203	100.0%

Source: ASTI(2019)

1.6.2 University

While the number of agricultural researchers at universities (higher educational institutions) in Lao PDR increased from a total of 14 in 2000 to 79 in 2017, the proportion of bachelor's degree holders has decreased, and the proportion of master's degree holders has increased. The proportion of doctoral degree holders rose to 8.8% in 2013 and then dropped to 7.6% in 2017, maintaining 7 to 8%. The proportion of master's degree holders remained similar from 35.7% in 2000 to 36.8% in 2013 but increased to 45.6% in 2017. The proportion of bachelor's degrees fell from 57.1% in 2000 to 46.8% in 2017, and the proportion of bachelor's degree holders decreased, and the proportion of master's degree holders increased. Due to the large proportion of master's degree holders and a small proportion of doctoral degree holders, universities appear to be in a difficult environment for high-quality research activities.

<Table 9-21> Distribution of Lao PDR' Agricultural Researchers in Universities, 2010, 2013, 2017

Category	2000		2013		2017	
	# of persons	%	# of persons	%	# of persons	%
PhD	1	7.1%	5	8.8%	6	7.6%
MSc	5	35.7%	21	36.8%	36	45.6%
BSc	8	57.1%	31	54.4%	37	46.8%
Total	14	100.0%	57	100.0%	79	100.0%

Source: ASTI (2019)

1.6.3 NAFRI

The number of agricultural researchers at NAFRI increased slightly from 98 in 2000 to 124 in 2017, while the proportion of master's degree or higher decreased. The proportion of doctoral degree holders rose from 6.1% in 2000 to 16.0% in 2013 and then dropped to 12.9% in 2017. The proportion of master's degree holders remained the same at 40.8% from 2000 to 2013, then decreased to 29.8% in 2017. The proportion of bachelor's degree holders dropped from 53.1% in 2000 to 43.2% in 2013, then rose to 57.3% in 2017. The proportion of those with a master's degree or higher increased from 46.9% in 2000 to 56.8% in 2013, then fell to 42.7% in 2017, which is lower than in 2000. Compared to 2000, it seems that the research capacity has increased somewhat as the master's degree holders were replaced with doctoral degree holders, while the proportion of both master's and doctoral degree holders has decreased since 2013, leading to insufficient expansion of advanced research personnel.

<Table 9-22> Distribution of NAFRI's Agricultural Researchers by degree, 2010, 2013, 2017

Category	2000		2013		2017	
	# of persons	%	# of persons	%	# of persons	%
PhD	6	6.1%	20	16.0%	16	12.9%
MSc	40	40.8%	51	40.8%	37	29.8%
BSc	52	53.1%	54	43.2%	71	57.3%
Total	98	100.0%	125	100.0%	124	100.0%

Source: ASTI (2019)

2. STI in Biotechnology Sector in Lao PDR

2.1 Current status of biotech sector in Lao PDR

Lao PDR has not developed a biotech industry that utilizes biological systems such as genes and cells due to the overall lack of technology-based industry development and this data research investigated the current status of the healthcare industry as a related industry.

Lao PDR appears to have a very poor medical system. As of 2018, the proportion of health expenditure to GDP was 2.246%, which is lower than 3.793% of Thailand and 4.791% of Myanmar.⁹⁷ In particular, Lao PDR's medical infrastructure appears to be suffering from significantly weak human capabilities. As of 2020, the status of medical facilities in Lao PDR is 1,056 public health centers which are primary care facilities, 135 municipal hospitals which are secondary care facilities, 17 provincial hospitals which are tertiary care facilities, and five general hospitals which are quaternary care facilities, which does not seem to be too short compared to the population of 7.4 million. However, only about 10% of the 200 doctors produced every year are put into hospitals, and it appears that the capabilities of human infrastructure is very weak, such as difficulty in using advanced equipment. According to buyers in the healthcare sector, medical facilities and medical information systems are being improved due to foreign grants and aids, but the effect of practical improvement is limited as the medical staffs are not familiar with cutting-edge equipment. Thailand's Kasemrad International Hospital, a modern hospital built in Vientiane last August, has 12 Thai doctors, but the cost is high, making it less accessible for the general Laotians.⁹⁸

The type of hospital use appears to have higher demands for trauma-focused emergency treatment than health promotion, and this seems to be due to the characteristics of the poorest countries. According to the patient details of major hospitals, 207,532 of total

⁹⁷ World Bank Data.

⁹⁸ Pil-Seong Kim (2021a)

263,038 patients were emergency patients, accounting for 77.7% of the total.⁹⁹

Meanwhile, Lao PDR's healthcare consumption is expected to grow at an annual growth rate of 8.2%, from \$522 million in 2019 to \$839 million in 2025, and in particular, pharmaceutical sales are expected to grow at an annual growth rate of 11.8%, from \$89 million in 2019, reaching \$174 million in 2025. The proportion of pharmaceutical sales to healthcare consumption is projected to increase from 17.0% to 20.7%.¹⁰⁰

The production capacity of the healthcare industry in Lao PDR appears to be very limited. All medicines produced in Lao PDR are for domestic use, and most of them are essential medicines with expired patents or have been exempted from patents for Least Development Countries under the WTO agreement.¹⁰¹ Lao PDR has manufacturing plants for pharmaceuticals and medical devices, but very few of them. Locally produced medicines are pill products produced using locally grown medical plants, which are mainly generic products such as cold medicine and diabetes-related medicines, and all prescription medicines rely on foreign imports.¹⁰²

<Table 9-23> Status of local medicines and medical devices manufacturers in Lao PDR

Company/factor name	Type of investor	Location	Product type/area	Note
Pharmaceutical Factory No. 2	Laotian government	Vientiane	Generic medicine	Domestic consumption
Pharmaceutical Factory No. 3	Laotian government	Vientiane	Generic medicine, Solution	Domestic consumption
CBF Pharma	Public institutions 2-8%	Pakse	Generic medicine, Solution	Domestic consumption
Kodupha	Vietnamese private company	Vientiane	Generic medicine	Domestic consumption, Export
KPN	Laotian private company	Vientiane	Generic medicine, Herbal medicine	Domestic consumption

⁹⁹ Pil-Seong Kim (2021a)

¹⁰⁰ Pil-Seong Kim (2021a)

¹⁰¹ EuroCham Lao PDR (2018)

¹⁰² Pil-Seong Kim (2021b)

Company/factor name	Type of investor	Location	Product type/area	Note
Pharmaceutical Factory 104	Military	Vientiane	Generic medicine	Domestic consumption
Tongmeng Pharmaceutical	Laotian-Chinese private company	Vientiane	Liver hematoma	Domestic consumption, Export
Chinchieng	Chinese private company	Luangnamtha	-	Domestic consumption, Export
VT Greater Pharma Co.	Laotian-Thai private company	Vientiane	Herbal medicine	Domestic consumption, Export
Primes Medical	Laotian-Canadian private company	Vientiane	Medical device	Domestic consumption, Export

Source: Pil-Seong Kim(2021a)

With limited local medicine production capabilities, Lao PDR' pharmaceutical market mostly relies on imports. As of 2017, a total of 1,542 medicines were registered in Lao PDR, with Thai products accounting for the largest share with 616 (39.3%), and local production in Lao PDR with 234 (15.2%), followed by European products with 186 (12.1%), Malaysian products with 148 (9.6%), and Indian products with 92 (6.0%).¹⁰³

<Table 9-24> The distribution of registered medicines in Lao PDR by country as if 2017

Ranking	Country of production	Type of medicines	Ranking	Country of production	Type of medicines
1	Thailand	616	6	Pakistan	77
2	Lao PDR	234	7	China	53
3	Europe	186	8	Vietnam	47
4	Malaysia	148	9	South Korea	41
5	India	92	10	Bangladesh	39
Total			1,542		

Source: EuroCham Lao PDR (2018)

¹⁰³ EuroCham Lao PDR (2018)

As of 2020, medicine imports amounted to \$31.68 million, slightly reduced compared with the previous year due to the impact of COVID-19 but have been showing an increasing trend over the past five years. The total amount of imports has grown at a compound annual growth rate of 16.4% over the past five years from 2016, and imports are mainly from Thailand, India, Denmark, Vietnam, Malaysia, Belgium, and Switzerland. Imports from South Korea have surged from 2019 and appear to have increased 184.2% compared to the previous year in 2020. As of 2020, countries with more than \$1 million imports are Thailand, India, Denmark, Vietnam, and Malaysia, and imports from Thailand in particular, amounted to \$17.58, accounting for more than 50% of the total.

<Table 9-25> The Size of Lao PDR' main medicine imports by country

Unit: \$

Ranking	Country	2016	2017	2018	2019	2020	Increase/decrease rate compared to the previous year (%)
	Total	14,826,030	21,492,498	28,228,825	31,843,241	31,681,469	▼0.5
1	Thailand	8,560,062	12,961,191	15,008,082	17,172,799	17,578,254	▲2.4
2	India	724,628	908,046	1,890,519	4,127,934	5,227,270	▲26.6
3	Denmark	15,250	230,127	1,043,133	430,102	1,932,920	▲349.4
4	Vietnam	626,928	694,005	725,263	1,164,567	1,206,998	▲3.6
5	Malaysia	589,486	769,096	1,140,050	950,372	1,002,756	▲5.5
6	Belgium	25,795	280,244	1,662,028	465,806	645,594	▲38.6
7	South Korea	83,326	53,800	79,309	219,667	624,310	▲184.2
8	Switzerland	1,183,467	999,803	1,038,576	996,715	544,405	▼45.4
9	United Kingdom	869,708	933,218	777,274	818,740	294,396	▼64.0
10	Pakistan	697,295	863,292	775,860	676,494	276,876	▼59.1

Source: Pil-Seong Kim (2021a)

As of 2020, Lao PDR' medical devices imports amounted to about \$11.66 million, showing a slight increase despite the impact of COVID-19. The sharp increase and sharp decrease in

the past five years make it difficult to know the trend of increase and decrease and 2020, in particular, is estimated to be largely affected by the import of diagnostic kits due to COVID-19. Total imports have grown at an compound annual rate of 25.5% over the past five years since 2016, imports are mainly from Thailand, China, and Vietnam. As of 2020, countries with more than \$1 million of imports are Thailand, China, Japan, and Vietnam, and imports from Thailand, the No. 1 country of import, are approximately \$3.38 million, accounting for 25.5% of the total.

<Table 9-26> The size of Lao PDR' main medical devices imports by country

Unit: \$

Rank ing	Country	2016	2017	2018	2019	2020	Increase/decrease rate compared to the previous year (%)
	Total	3,743,354	17,501,726	20,405,140	11,457,748	11,657,554	▲ 1.7
1	Thailand	1,508,135	1,914,562	3,198,747	2,825,140	3,384,272	▲ 19.8
2	China	315,483	230,964	5,775,716	653,653	2,728,063	▲ 317.4
3	Japan	345,569	125,927	160,998	358,575	1,602,955	▲ 347.0
4	Vietnam	538,770	310,579	500,608	1,405,824	902,493	▼ 35.8
5	South Korea	143,177	88,345	506,248	261,793	602,438	▲ 130.1
6	Singapore	87,012	92,771	505,701	382,896	378,555	▼ 1.1
7	Taiwan	136,499	172,905	217,512	160,953	336,606	▲ 109.1
8	France	54,985	72,756	95,566	359,262	200,351	▼ 44.2
9	United States	82,409	89,435	153,873	386,803	195,842	▼ 49.4
10	United Kingdom	0	4,508	10,949	280,916	70,751	▼ 74.8

Source: Pil-Seong Kim (2021c)

2.2 Lao PDR' biotech sector policy

Lao PDR is currently found to have no established policies focusing on biotechnology-based industries, and accordingly, this data research investigated the current state of health and medical policies as policies in related fields.

The Laotian government came up with a plan to reform the health sector from 2013 to 2020.¹⁰⁴ The 2013 reform plan divided the period up to 2020 into three phases, aiming to achieve Millennium Development Goals (MDGs) in Phase 1 from 2013 to 2015, improve access to health care, and protect it financially in Phase 2 from 2016 to 2020, and ensure universal health coverage in Phase 3 from 2021 to 2025.

In 2016, the Laotian government established the Eighth Health Sector Development Plan, 2016-2020 (hereinafter 8th HSDP). The 8th HSDP was set to have 8 key areas, – 1) promoting hygiene and health, 2) expanding quarantine and disease control, 3) improving health services, 4) securing food, medicine, and medical equipment safety, 5) developing health human resources and strengthening health science research, 6) securing health finance, 7) expanding health planning and international cooperation based on health digitization, and 8) strengthening health administration and health monitoring – 42 project, and 94 tasks.¹⁰⁵

The Laotian government has set the following six directions for future medical system improvement. The Laotian government is planning to implement the following – 1) upgrading and standardizing hospitals, 2) modernizing diagnostic and therapeutic medicine, 3) providing technical support to local hospitals, 4) improving hospital operation management, 5) privatizing public hospitals, and 6) promoting Public-Private Partnership (PPP) projects. The PPP project can be only promoted in one area, such as the supply of equipment, facilities, and medical experts, and it seems to increase private sector

¹⁰⁴ MOH(2013)

¹⁰⁵ Pacific Rim Innovation and Management Exponents, Inc.(2018); Korea Institute of Procurement (2021)

participation rather than the public role in order to establish a medical system within a short period.¹⁰⁶

In the future, it is expected that the medical service infrastructure improvement will be actively carried in Lao PDR, with a focus on expanding human resources and facilities. In Lao PDR, hospitals located in Thailand have been used generally in the event of health problems or injuries, but awareness of the need to expand medical facilities in the country has increased after experiencing border shut-downs due to COVID-19. The Laotian government also includes investment plans, fostering talent in the related healthcare sectors, and fostering healthcare-related service industries in their 9th National Socio-Economic Development Plan (9th NSEDP). Currently, the Laotian government is actively promoting the improvement of health care quality through upgrading hospitals and expanding medical personnel. Lao PDR currently has 1,056 public health centers, 125 municipal hospitals, 17 provincial hospitals, and five general hospitals, with a total of nine public general hospitals in progress and three of them scheduled to open by 2022. In addition, there are 32 private hospitals and 1,050 private clinics in operation with three new hospitals expected to open by 2021.¹⁰⁷

¹⁰⁶ Pil-Seong Kim (2021b)

¹⁰⁷ Pil-Seong Kim (2021b)

<Table 9-27> Ongoing public and private hospital construction projects in Lao PDR

Type		Name of hospital	Number of beds	Location	Progress rate	Note
Public	General hospital	Mahosot	600	Vientiane Capital	95%	Chinese investment, Open in 2021
		Huaphan Provincial	200	Huaphan	95%	Vietnamese investment, Open in 2022
		Xiengkvang Provincial	200	Xiengkvang	98%	Vietnamese investment, Open in 2022
		Yot Ou District	N/A	Phongsaly	50%	
		Namor District	N/A	Oudomxay	90%	
		Nonghaed District	N/A	Xiengkvang	64%	
		Kuan District	N/A	Huaphan	100%	
		Khong District	N/A	Champasack	90%	
		Dakcheung District	N/A	Sekong	50%	
Private	General hospital	Kasemrad	254	Vientiane Capital	100%	Open in August 2021
	Orthopedics	Xieng Jiang Co., Ltd.	100	Champasack	90%	Open in 2021
	Kidney, hemodialysis	Savannakhet International Hospital	50	Savannakhet	85%	Open in 2021

Source: Reconfiguration by referring to Pil-Seong Kim (2021b)

3.

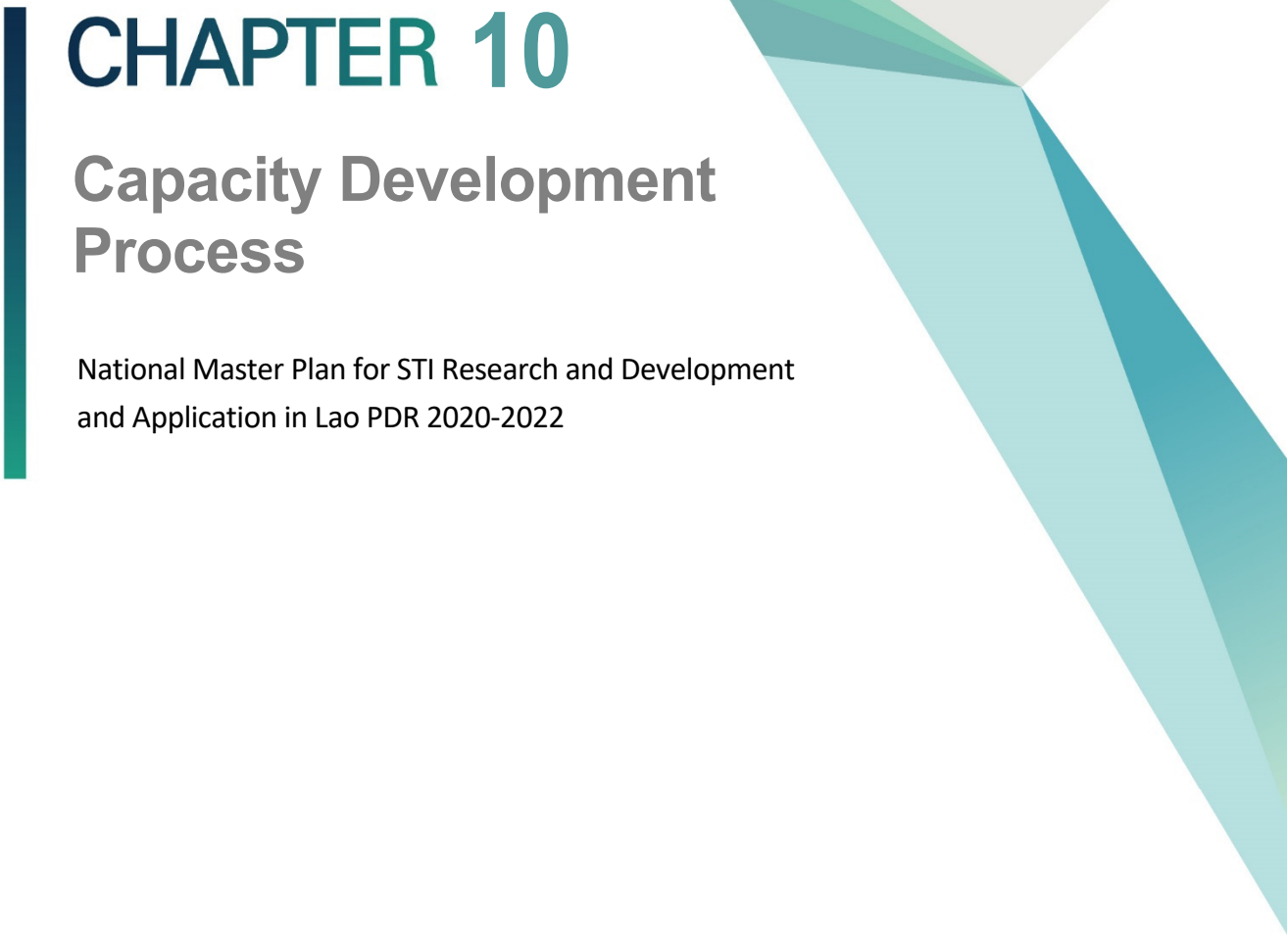
Implications of STI in Agriculture and Biotechnology Sector in Lao PDR

Although Lao PDR is an agricultural-oriented country with the proportion of agricultural workers of more than 60%, the agricultural productivity is very low and the proportion of agricultural GDP is less than 20%, while the non-agricultural sector also lacks growth. This is a different situation from the 1960s and 1970s when Korea's agriculture suffered a sharp drop in the labor force due to migration and slower growth compared to the non-agricultural sector. In addition, Lao PDR still lacks economic growth, so the government does not have enough financial resources to invest in agricultural development, and the market is also not significantly formed. This is different from the fact that the Korean government was able to implement relatively active agricultural and rural protection policies such as food production growth, rural development, price support policies, and agricultural structure transformation, and that the demand and market for agricultural products expanded rapidly.

However, there are also clues to solving the problem. First of all, since Lao PDR has a gifted natural agricultural environment and sufficient labor force, if new varieties development, cultivation technology development, and mechanization are carried out effectively, the effect of agricultural growth through improving agricultural productivity is expected to be very large. Moreover, the lack of processing and distribution infrastructure, as well as the nonsystematic structure of the distribution market, seems to play a role in the lack of commercialization and sales of agricultural products in Lao PDR. As in Korea in the past, if the distribution network reform is promoted to lower distribution margin and increase the bargaining power of producers, it is expected that farmers will be encouraged to produce agricultural products and participate in the market. It is necessary to establish a system in which agricultural producers can take the lead in responding to changes in domestic and foreign demand for agricultural products.

Although the Laotian government has been implementing a comprehensive agricultural strategy, it has recently been shown that the government is promoting key projects focused on urgent tasks, which is considered a desirable approach to implement effective policies. Universities and research institutes are the institutions that need to play a pivotal role in improvement of agricultural productivity and the development and dissemination of processing technologies and distribution/logistics technologies necessary for reforming the distribution market in Lao PDR, but at the moment, it appears that there is insufficient support for training technology development personnel and R&D expenses in each field, which seems to be the most urgent part for the Laotian government to support.

It seems difficult for the Laotian government to focus on the biotech sector as an advanced technology field, considering the current circumstances of Lao PDR. However, as biotechnology and biotech industry are widely applied not only to agriculture but also to health and medical fields such as medicines and medical devices, it is judged necessary to cultivate personnel in the basic and applied fields of the biotech sector such as molecular biology, biotechnology, and food processing technology, and support technology development for the industry, centering on public research institutes.

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CHAPTER 10

Capacity Development Process

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 10. Capacity Development Process

1. Preparations for the Second-Year Project

1.1 Preparatory Meetings

1.1.1 Setting the Outline of the Project through email for 2021 Project

Since it is the second year of the Official ODA Program with Lao PDR, the STEPI team contacted the research team in the Ministry of Technology and Communications (MTC), Lao PDR to discuss the expected yearly schedule, activities throughout the year based on the information negotiated in the PCPs. The project aims to establish a national science and technology master plan within three years project, but the first year's project was merely carried out through online meetings and emails due to the COVID-19. In order to achieve its objectives, active research and gathering baseline information and data of current status of STI in Lao PDR was needed. It was difficult to attain needed information through only secondary sources that could be attained through internet without fieldwork or interview for the first year.

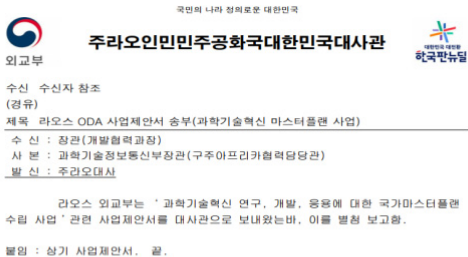
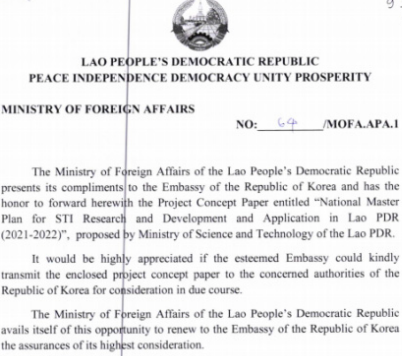
Therefore, to facilitate and activate the second year project in Lao PDR, the research team communicated through email to make contingency plan for the COVID-19, in case where travelling to Lao PDR or Korea is continuously restricted for the second year. While communicating through emails in February and March, STEPI and MOST Lao PDR planned to have working group meeting as soon as possible to discuss for the further plans.

1.1.2 Official Letter and PCPs for 2022

Starting at the end of 2020, the process of receiving official documents from the Ministry of Foreign Affairs of Lao PDR and PCP proposals for the third year project is in progress.

- Information on business plans, official documents, and official documents to the Ministry of Science and Technology of Lao PDR, the host department, and official documents to guide them to receive official documents through the Korean Embassy in Lao PDR and STEPI.

As of February 15, 2021, it received an official letter from the Ministry of Foreign Affairs of Lao PDR and an official letter of completion of receipt of the 2022 Project Proposal (PCP) from the Ministry of Science and Technology of the Republic of Korea.

1) PCPs	2) Official Letter by MOFA, Lao PDR
 국민의 나라 참외로 대한민국 외교부 주라오인민민주공화국대한민국대사관 수신 수신자 참조 (경유) 제목 라오스 ODA 사업제안서 송부(과학기술혁신 마스터플랜 사업) 수신 : 장관(개발협력과장) 사본 : 과학기술정보통신부장관(구주아프리카협력담당관) 발신 : 주라오대사 라오스 외교부는 '과학기술혁신 연구, 개발, 응용에 대한 국가마스터플랜 수립 사업' 관련 사업제안서를 대사관으로 보내왔는데, 이를 별첨 보고함. 붙임 : 상기 사업제안서, 2. 주 라 오 수신자 개발협력과장, 과학기술정보통신부장관(구주아프리카협력담당관) 2월11일 2021.2.11. 2월11일 이백희 공시참사관 서행삼 특명전권대사 임우홍 참조자 시행 주라오인민민주공화국대 (2021. 2. 11.) 협수 구주아프리카협력담당관- (2021. 2. 15.) 한민국대사관-875 384 우 01000 Lao P.D.R. 20 Samsenthai Road, Nongduang Nua Village, / http://lao.mofa.go.kr Sikhotsabong District, Vientiane Capital 전화번호 (856) 21 352 031 팩스번호 (856) 21 352 035 / hlee11@mofa.go.kr / 비공개(5)	 LAO PEOPLE'S DEMOCRATIC REPUBLIC PEACE INDEPENDENCE DEMOCRACY UNITY PROSPERITY MINISTRY OF FOREIGN AFFAIRS NO: 64/MOFA.APA.1 The Ministry of Foreign Affairs of the Lao People's Democratic Republic presents its compliments to the Embassy of the Republic of Korea and has the honor to forward herewith the Project Concept Paper entitled "National Master Plan for STI Research and Development and Application in Lao PDR (2021-2022)", proposed by Ministry of Science and Technology of the Lao PDR. It would be highly appreciated if the esteemed Embassy could kindly transmit the enclosed project concept paper to the concerned authorities of the Republic of Korea for consideration in due course. The Ministry of Foreign Affairs of the Lao People's Democratic Republic avails itself of this opportunity to renew to the Embassy of the Republic of Korea the assurances of its highest consideration. Vientiane, February 2021 Embassy of the Republic of Korea, Vientiane

2021 K-Innovation Partnership Program with Lao PDR

3) PCPs


Lao people's Democratic Republic
Peace Independence Democracy Unity Prosperity

Ministry of Science and Technology
Department of Technology and Innovation

No: **583** /MOS/DTI
Vientiane Capital, Date: **29 DEC 2020**

To: Inkyoung SUN
Head of Office of Development Cooperation Research
Science and Technology Policy Institute (STEP)
5-7th Fl, Building B, Sejong National Research Complex
170 Sicheong-daero, Sejong, 30147
Republic of Korea

Subject: Formal Application for the K-Innovation Partnership Program

The Department of Technology and Innovation, Ministry of Science and Technology, Lao PDR, we are pleased to submit this official letter as formal application for the K-Innovation Partnership Program, along with our Project Concept Paper titled "National Master Plan for STI Research and Development and Application in Lao PDR (2021-2022)".

Thank you for the opportunity to submit our application, we appreciate your kind consideration and look forward to a favorable response in the near future.

Sincerely,


Mr. Souman CHOUAMANY
Director General
Department of Technology and Innovation

Ministry of Science and Technology, Department of Technology and Innovation, Tel: +85621 732 461, Fax: +85621 233 472

4) PCP

 Let's make a change together for a better world!

Project Concept Paper
(K-Innovation Partnership Program)

National Master Plan for STI Research and Development and Application in Lao PDR (2021-2022)
Lao PDR

15, 12, 2020

Contact Information of the Applicant	
Name	Mr. Souman CHOUAMANY
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Affiliation 1	Mrs. Keophaivath DAUNGSAVANH Director General of Cabinet of Academy of Science and Technology Ministry of Science and Technology (MOST)
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Telephone	856-20-55510744 (Mr. Souman) 856-20-99800669 (Mrs. Keophaivath) 856-20-98899988 (Mr. Khamphe)
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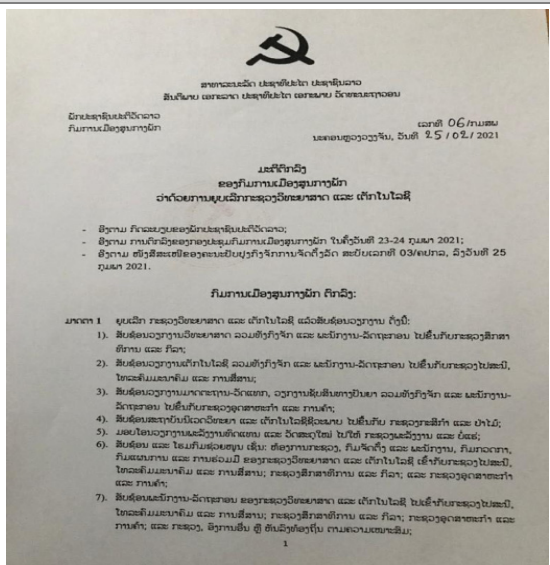
1.1.3 Dissolution of MOST and Conference Call for Discussion

Due to the government restructuring in Lao PDR, the main counterpart of the project, Ministry of Science and Technology (MOST) in Lao PDR was dissolved into five different ministries. STEP and MOST had an online meeting to consult on the further project and its counterparts (21, April 2021).

According to the change in major ministries in Lao PDR on February 25, 2021, the existing Ministry of Science and Technology (MOST) was dismantled and the existing Ministry of Science and Technology was distributed as below:

- Delegate science work 1) to the Ministry of Education and Sports.
- Delegation of technology-related tasks 2) former Ministry of Post and Telecommunications

- Delegate Standard-Measure and Intellectual Property tasks 3) to the Ministry of Trade and Industry (Ministry of Commerce and Industry).
- Moved the Institute of Biotechnology to the Ministry of Agriculture and Forestry (4).
- Delegate renewable energy and new materials tasks to 5) Ministry of Energy and Mining.
- All relevant support departments (ministerial cabinet, administrative and audit departments, planning departments of international relations, etc.) are moved to the above-mentioned departments. Officials and officials from all existing departments of the Ministry of Science and Technology are divided into the above-mentioned departments.

Letter for Informing the Dissolutions	English Translation
 ສາທາລະນະລັດ ປະຊາທິປະໄຕ ປະຊາຊົນລາວ ສັນຕິພາບ ເອກະລາດ ປະຊາທິປະໄຕ ເອກະພາບ ວັດທະນະຖາວອນ ມີຄຳສະຖານະລັດຖະມົນຕີ ກົມການເມືອງສູນກາງພັກ ເລກທີ 06/ກມສສ ນະຄອນຫຼວງວຽງຈັນ, ວັນທີ 06/04/2021 ມາດຕະການ ຂອງກົມການເມືອງສູນກາງພັກ ວ່າດ້ວຍການຍຸບລົງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ - ສ້າງຕັ້ງ ກົດຂັບຂີ່ຂອງກົມປະຊາທິປະໄຕ ແລະ ກົດຂັບຂີ່ຂອງກົມການເມືອງສູນກາງພັກ ໃນວັນທີ 23-24 ກຸມພາ 2021; - ສ້າງຕັ້ງ ສະໜັບສະໜູນຂອງກົມປະຊາທິປະໄຕ ແລະ ກົດຂັບຂີ່ຂອງກົມການເມືອງສູນກາງພັກ ໃນວັນທີ 03/04/2021, ວັນທີ 25 ກຸມພາ 2021. ກົມການເມືອງສູນກາງພັກ ຕົກລົງ: ມາດຕາ 1 ຍຸບລົງ ກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ ແລະ ສັນຍາຂອງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ 1) ສັນຍາຂອງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ ແລະ ກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ 2) ສັນຍາຂອງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ ແລະ ກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ 3) ສັນຍາຂອງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ ແລະ ກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ 4) ສັນຍາຂອງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ ແລະ ກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ 5) ສັນຍາຂອງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ ແລະ ກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ 6) ສັນຍາຂອງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ ແລະ ກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ 7) ສັນຍາຂອງກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ ແລະ ກະຊວງວິທະຍາສາດ ແລະ ທັກໂນໂລຊີ	The Politburo of the Central Party agreed : Article 1. Cancel Ministry of Science and Technology and allocate the works as below: 1. Move Science work to Ministry of Education and Sports. 2. Move Technology work to Ministry of Post and Telecommunications. 3. Move the works of Standard-Measures, Intelligent Properties to Ministry of Commerce and Industry. 4. Move the Institute of Bio-Technology to Ministry of Agriculture and Forestry. 5. Move the works of Renewable Energy and New Materials to Ministry of Energy and Mining. 6. Merge all supporting Departments (Ministry Cabinet, Administrative Dep. Inspection Dep. International Relations and Planning Dep.) to those of the above mentioned Ministries. 7. Allocate all staff and servants of Ministry of Science and Technology to the above mentioned Ministries.

1.1.4 Korea-Lao PDR Expert Group Meeting

STEPI research team conducted expert group meeting to discuss and share ideas and information on Lao PDR on 27 April 2021.

2021 K-Innovation Partnership Program with Lao PDR

- Invited Korean experts to promote understanding of Lao PDR' economy, science, and education as a whole and consulting advice and share experts' presentation and discussion.

Program: K-Innovation expert group meeting in Korea

Time	Contents	Name	Position/Institutes
13:30~13:40	Opening Introduction for K-Innovation Project	Inkyoung Sun	STEPI
13:40~13:50	2020 K-Innovation Project Outputs for 2020 program	Jiyong An	STEPI
13:50~14:20	cooperation experience with Lao PDR	Myunghwan Oh	Korea-Lao PDR Friendship Association
14:20~14:50	Sharing experience in National University in Lao PDR	Jejeong Lee	Former professor in University, Lao PDR
14:50-15:20	Sharing working experience in ICT, Lao PDR	Youngjoong Jang	Former advisor
15:20-15:30	Break		
15:30-16:00	Sharing working experience in MOST Lao PDR	Hyungjoon Lee	Former advisor
16:00-16:30	Education and STI in Lao PDR	Jisoon Chang	Sangmyung University
16:30-17:00	Discussion and Wrap up		





1.1.5 Visit to the Embassy of Lao PDR to the Republic of Korea

Visiting the Lao PDR Embassy in Korea by introducing K-Innovation and Lao PDR Science and Technology Innovation Policy Advisory Projects and proposing related meetings through e-mail contact with the Lao PDR Embassy in Korea.

- Request for cooperation in project-related projects and future projects through meetings with secretaries in charge of economic trade at the Lao PDR Embassy in Korea.
- Request for a list of international students or office workers living in Lao PDR living in Korea and for cooperation when inviting a launching meeting.
- When conducting a demand survey related to science and technology innovation in Lao PDR, request for cooperation so that it can be conducted for international students and experts living in Lao PDR in Korea.

The Lao PDR Embassy in Korea also understands the difficulties of reorganizing the government in the progress of the project, and decided to help the project proceed with

local ministries through smooth information sharing and cooperation.

Photos at the Embassy of Lao PDR



2. High-Level Forum and Kick-off Meeting

2.1 High-Level Forum and Kick-off Meeting

2.1.1 Overview

1) Objectives

The purpose of the kick-off meeting is to discuss and share ideas on the the project and for knowledge sharing. The high level forum and kick-off meeting involves project counterparts, MTC, and STEPI to discuss the project purpose, activities, and expected outputs. Various stakeholders including government officials from the relevant ministries and the directors of the research institutions were invited to the event to build consensus on the importance of developing the national master plan for STI Research and Development and Application in Lao PDR. This event discovered policy demands for the main institutions and ministries.

2) Kick-off Meeting's Main Activities and Agendas

The research team is responsible for holding the kick-off meeting. The research team conducted interviews with the stakeholders from MOST and other research institutions and decision makers of Lao PDR.

- Participants: STEPI researchers and Korean experts, researchers from Ministry of Technology and Communication (MTC) Lao PDR, local experts, the embassy of Lao PDR to the Republic of Korea, and government officials and international laotian students studying abroad in Korea, and STI related organizations.

Instead of travelling to both countries, STEPI and MTC decided to host online hybrid kick-off meeting where participants from each sides host the event online while inviting some participants to offline as well. The online hybrid high-level forum and kick-off meeting was

held on September 1, 2021 through online meeting system link. The embassy of Lao PDR to the Republic of Korea attended the meeting offline along with a number of Laotian students and government officials studying abroad in Korea. The online kick-off meeting is followed by introduction and welcome remarks and congratulatory remarks of the both sides. The presentations on the project introduction, current status of STI in Lao PDR and also key note presentations on the Korea's STI development and economic growth. About 100 participants attended the meeting both online and offline. The team exchanged Q & As regarding the presentations.

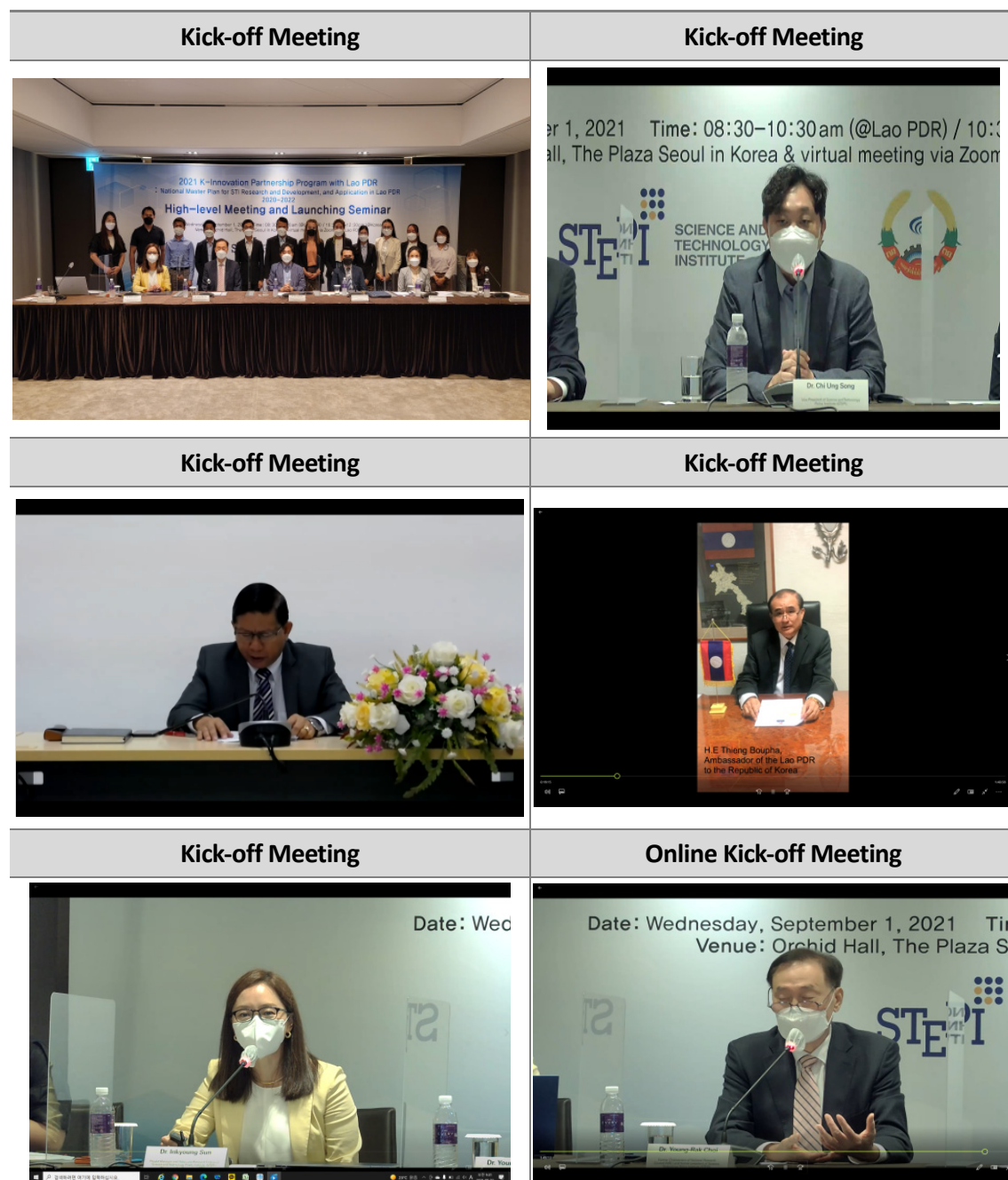
The following are the main objectives of the online kick-off meeting:

- Confirm the project purpose and expected outcome
- Collect the necessary information for assessing Lao PDR's national STI strategic planning and technology foresight capacities
- Share Korean experiences in national strategic planning and technology foresight
- Exchange ideas and opinions on the project schedule and activities

The main activities of the workshop will be as follows:

- Remarks and introduction of the project by the STEPI project investigator on the project overview including the purposes, scope, methodology, implementation, schedule, and results
- Remarks and Q&As by Lao PDR and sharing ideas on the project by the research team in Lao PDR
- Presentations by the Lao PDR experts on Lao PDR's current status of STI and STI policy and a few cases, to be followed by a lively discussion on key issues and the exchange of ideas and opinions
- Follow-up in-depth meetings with stakeholders to collect their views

[Figure 10-1] Kick-off Meeting



2021 K-Innovation Partnership Program with Lao PDR

Launching Seminar

1. Project Outline

2021 K-Innovation Partnership Program with Lao PDR	
Year	January – December 2021
Partner Ministry	Ministry of Technology and Communications (MOST), Lao PDR
Organizers	Science and Technology Policy Institute (STEPI)
Project Name	National Master Plan for STI Research and Development, and Application in Lao PDR 2020-2022
Sub-topics	(1) Current Status of National Innovation System (NIS) in Lao PDR
	(2) STI Governance and STI Roadmap
	(3) Capacity Building and Knowledge Exchange on STI Policymaking

2. Launching Seminar Overview

- Date: Mid-July 2021/ 08:30-10:30 am (@Lao PDR) / 10:30-12:30 pm (@Korea)
- Official Language: English
(A simultaneous Lao PDR-English translation will be provided by STEPI, if necessary)
- Venue: Virtual meeting via Zoom

3. Objectives and Activities

- Mark the official launch of the project by presenting project proposal
- Identify policy priorities of project stakeholders by presenting the overall research scope and direction
- Improve understanding regarding the project (Project outline, Project scope, Research Teams of Lao PDR's side and Republic of Korea's side, Selection of Local consultants, etc.) among project stakeholders
- Identify the project stakeholders and their role
- Identify specific needs, expectations and Diagnose current status and challenges of Lao PDR side
- Discuss on collection and sharing for data and background information
- Confirm the project theme among stakeholders
- (Deliverable) Final Research Topic and Scope, Launching Seminar PPT Slides including

preliminary analysis of Lao PDR, Final list of project participants including national consultants

4. Program Schedule

Testing Equipment and Setting (08:00-08:30)	
Opening Session Moderator: 000 (Institution)	
08:30-08:40	Opening and Welcome Remarks - Opening Remarks: Dr. Mi Ock Mun, President of STEPI (TBD) - Welcoming Remarks: Mr. Boviengkham Vongdara, Minister of Technology and Communication (TBD) Group Photo
Launching Seminar Moderator: 000 (Institution)	
08:40-08:50	Introduction of Participants - All Participants both from Korea and Lao PDR
08:50-09:10	Introduction of STEPI, K-Innovation Partnership Program and Program with Lao PDR - Presented by Dr. Inkyoung Sun (Associate Research Fellow, STEPI)
09:10-09:30	Introduction of Lao PDR's STI Governance (TBD) - Presented by Mr. Soumana Choulamany (Director, MTC)
09:30-09:45	Q&As and discussion
09:45-10:15	Special Lecture: STI Development in Korea - Presented by Dr. Sung-Chul Chung (Former President of STEPI)
10:15-10:25	Q&As and Discussion
Closing Moderator: 000 (Institution)	
10:25-10:30	Wrap up and Closing Remarks - Mr. Soumana Choulamany (Director, MTC) - Dr. Inkyoung Sun (Associate Research Fellow, STEPI)

3. Online K-Innovation Dissemination Workshop

An online expert workshop was held on 15th December 2021. During the workshop, STEPI and Korean experts made a presentation on methodology to develop national masterplan on STI strategies and share experiences in the identified topics, including national strategic planning for STI. At the same time, experts and government officials from Lao PDR will also share their current status of STI and STI policy and demands in Lao PDR in order to enhance in-depth understanding of Lao PDR's current status and experiences. Afterward, STEPI and MTC will make consensus on the expected results and outcome. STEPI will present the results and achievement of the cooperative project with Lao PDR and follow-up plan for 2022.



4. STI Demand Survey

4.1 Overview

(Target) The subjects of the survey were 46 people, including civil servants and experts in the field of science and technology in Lao PDR. Among the respondents, there were more males (26 persons, 56.5%) than females.

(Working Period) 89.2% of the respondents were those with less than 15 years of working period, of which 50% (23 persons) were from 5 to 10 years or less. Most of them worked in the STI field for less than 10 years, and more than half accounted for less than 5 years. Specifically, 54.3% (25 people) worked for less than 5 years and 89.9% if 10 years or less were included

(Academy) The composition of the respondents' degrees consisted of only two Ph.D., 20 masters, 24 bachelors, and so on.

(Major) At the field of major, there were 10 people in the STI field, but 36 people in other fields. However, there were actually many fields related to science such as biomedical engineering, environmental engineering, health science, forestry science, biology, chemical technology, road-bridge engineering, computer science, natural science, mathematics etc. And there are 2 Ph.D. such as regional development and medicine

(International experience) The result of international collaboration experiences is to 'yes (43%)', 'no (57%)'. The bilateral collaboration had Korea and Vietnam (5), Thailand (4), China (3), Singapore, Japan, Canada, Sweden, Luxembourg (1), and multilateral collaboration has ASEAN, ADB, WHO, FAO (1)

4.2 Results

(Key Challenges in Science and Technology) Lao PDR have NSEP(National Socio-Economic Plan, 2021-2025), and will achieve the objectives as ‘To fully focus on socio-economic development based on the existing potentials in order to help the country effectively achieve the LDC’s criteria through quality, inclusive and green growth and achieve the SDGs by 2030’. The key challenges of NSEP have 6 challenges and will overcome them for achievement of the objectives and outcome. The 6 key challenges are i) insufficient economic base or infrastructure, ii) disparity between urban and rural community, iii) insufficient human resources, iv) reflecting sustainability in the national development, v) inability to mobilize necessary financial resources on stable manner, vi) lack of Science & Technological capacity. The top 3 priority of key challenges in NSEP(2021-202) is to ‘lack of science & technological capacity(26.9)’, ‘insufficient human resources(23.5%)’, and ‘insufficient economic base or infra-structure(21%)’.

(Necessary capacity of science and technology) The most lack of capacity in science and technology is to ‘STI hardware (41.3%)’, ‘STI human resources (19.6%)’, ‘STI software and financial resources (15.2%)’, and ‘STI strategies (8.7%)’. For the capacity development of science and technology, it needs to invest to infrastructure, R&D and lab facilities, science and technology university as well as to develop the human resources. On base of hardware and software in science and technology, government will execute financial support and making the strategies.

(Insufficient human resources) The most significant resources for solution of insufficient human resources are to ‘policy makers (43.5%)’, ‘Ph.D. level researchers (19.6%)’ and low skilled technicians (19.6%)’, and ‘university professor (15.2%)’. The capacity buildings of the policy makers will be the priority program of cooperation programs, such as study visit, workshop and conference, consulting and training programs for policy makers in STI sector. And it needs to strengthen the leadership of governance and structure of government. Furthermore, it needs to make incentive for recruitment of qualified human resource to government body from other sectors or organizations. The incentive will include salary, working condition and motivation etc.

(Disparity between urban and rural) The disparity between urban and rural is also big challenge for national development. The response is to ‘mobilization of population (47.8%)’ and ‘sewage and waste management (30.4%)’. It is important to solve them on the role of STI, In particular, STI give rural and urban to solve the sewage and waste management. When it establishes the plant on sewage and waste management, it needs to technical and engineering and human resources. Furthermore, it put in the science and technology for the reduction of pollution, such as air, water, and hygiene etc.

(Priority Issues in NSEP) The priority for NSEP is to 6 issues, i)Quality, inclusive and sustainable growth, ii)Improve human resources, research and apply science- technology, to promote diversified and value- added production, iii)Improve the living conditions of the people, iv)Green and environmental friendly development, v)Enhance international cooperation, integration and connectivity, vi)Effective government administration and rule of law. The top 3 priority is to ‘Improve human resources, research and apply science-technology, to promote diversified and value- added production (30.9%)’, ‘Enhance international cooperation, integration and connectivity (17.9%)’, and ‘Improve the living conditions of the people (13.8%)’. It is very urgency issues that develop the human resource and apply science technology to provide diversities productions.

(Main Challenge for development of STI) The most suitable issues of main challenges in STI is to ‘limited system to support R&D commercialization & technology transfer (17.1%)’, ‘no comprehensive open data on R&D equipment and facilities (16.4%)’, and ‘limited Ph.D. researchers (13.6%)’.

(The urgent STI activities) The activities of STI consist of basic R & D- applied research and development-demonstration for technology scale-up and pilot projects-technology transfer. The major activities in STI lifecycle is to ‘applied research and development (32.3%)’, ‘basic R&D (24.0%)’, and ‘demonstration for technology scale-up and pilot projects (21.9%)’.

(S & T for economic development) The most necessity item of science and technology for economic development is to ‘appropriate technology (50.0%)’, ‘emerging technology (26.1%)’, and ‘technology transfer (23.9%)’. It shall need to develop the appropriate technology for economic development. Or it needs to consider the appropriated technology

for introduction of science and technology to Lao PDR from overseas. For this, it shall share with knowledge and technology through workshops, seminars, and forum, and put together capacity development, specially Ph.D. holder.

(Obstacles or Risk for STI) The obstacles or risks for STI hardware is to 'R&D equipment and lab facilities (16.6%)', 'infrastructure (12.6)', and 'science and technology university (11.2%)'. The obstacles and risks for STI software is to 'international experts network (12.1%)', 'job training & technology training (7.6%)', 'R&D facility maintenance (5.4%)'. The obstacles and risks for STI strategies is to 'National STI policy planning (10.8%)', 'law and institutional framework (4.9%)'. In conclusion, it shall need to develop the R & D equipment and lab facilities for overcoming of obstacles of the STI hardware. And for the alleviation of the risks of the STI software, it needs to establish the international expert networks. The international cooperation for overcoming the obstacles in science and technology is to mixture bilateral cooperation and multilateral cooperation, triangular cooperation, support for professors or researchers, more education facilities, grant and R&D cooperation on applied ST. There is a lot of multilateral cooperation in Mekong region, so it needs to cooperate with variety international agencies.

(STI Sector) The most useful strategic STI sector is to 'ICT (33.3%)', 'agriculture (22.7%)', and 'bio technology (14.4%)'. Another sector is to health & science, environment, energy, and heavy industry. If it needed to select some sector in science and technology, it may select 3 areas. Lao PDR has a lot of bio diversity in world, so it need to convergence with ICT and agriculture, ICT and bio technology.

(STI Project in near future) The most significant project for STI project is to 'to develop more STI human resources (23.5%)', 'to develop STI infrastructure (20.6%)', and 'to develop or promote public awareness of STI (16.9%)'. Another project is to 'to promote more intensified international cooperation in STI sector', 'to prepare adequate STI regulations or laws', 'to establish more efficient management or service for STI sector', etc.

(STI development experience in Korea) The top priority of Korean STI development experiences is to 'to develop the human resources to meet demand of the industry (22.2%)', 'to establish the science and technology university for development high qualified human

resources (20.7%)', 'To establish the national research institutes for research and development (14.1%)'. 'to set up the cooperation network between entrepreneurs-university-institute (13.3%)', 'to set up the decision-maker for development of science and technology (10.4%)', 'to set up and manage to national research and development system (9.6%)'.

- **(To develop the human resources to meet demand of the industry)** It shall be network between industry-university-government as science/techno park in Korea and make commercialization in order to develop the human resources to meet demand of the industry. Furthermore, it may provide learning opportunities for public officials to develop their competencies. For example, it will study visit to Korea as well as join research in Korea for cultivation the capacity.
- **(To establish the science and technology university)** According to Korean's experiences on STI development, it is to establish the science and technology university for development high qualified human resources. The first, it shall cooperate between Ministry of Technology and Communication, Ministry of Education and Sports in Lao PDR and STEPI Korea, and create STI's policies, and then establish science and technology university. The university may give the training program or study visit in short term and long term for knowledge, awareness and lessons learned in STI.
- **(To establish the national research institute)** It would recommend Science & Technology Policy Institute (STPI) as one of the most efficient institute that provide knowledge on science, technology and innovation in order to boost in terms of IT innovation. Also, it is to establish the national research institute, which is equipment and facilities. If possible, Korea should provide STI expert to dispatch to Lao PDR in order to help and support Lao PDR to develop necessary STI planning, working system and so on. It will provide the advice and research result for set up and implementation of STI's policies, to networking and coordinating the stakeholders in Lao PDR, to make synergy through international cooperation

4.3 Implications

In order to implement of STI in Lao PDR through sharing Korea's experience, it should consider in following implications.

(Human Resource Development) First, it shall need to make an effort to human resources development in science and technology innovation. It needs to make policies and international cooperation for balance demand and supply on science and technology among strategies for national human resources development plan. Also, it needs to supplement system for prevention of brain drain as well as cultivation of short term and long term HRD.

(Enhance Policy Capacity) Second, it shall develop the program for strengthening of policymakers and make it use of sustainability. Eventually, since the development of science and technology is achieved through the planning and implementation of policymakers, the development will be determined by the capacity of policymakers.

(Establishment Research Institute) Third, it shall establish the research institutes for science and technology by stages. This establishment is to make a system the think tank for national development, and mutual link and synergy. It has something in common of efficiency utilization of human resources.

(Improvement Higher Education) Forth, it shall improve the higher education for high qualified human resources in science and technology as well as establish the university of science and technology. In particular, it needs to improve the existing higher education institutes, and establish new concept university for new technology and trend. It needs to set up master plan for science and technology in Lao PDR as well as make feasibility study for the establishment of university.

(Comprehensive Science and Technology Strategy) Fifth, STI in Lao PDR shall build the comprehensive science and technology strategy as module of development stage, and it should consider on human resources development and sustainable development. After review on economic society development plan in Lao PDR, it shall make sustainable strategies for science and technology. And, it needs to introduce new access method and analysis for existing strategies. As the contents of capacity development, it needs to make an implementation plan through discussion and debate with policymakers and scientists.

5. Basic Statistics Information

5.1 Name list

	Name	Organization	Position	Degree
1	Mr Xayasith Phomchaleun	SVK Hospital	Chief of Department	Master
2	Mr. Phankham MONGPHACHAN	MoPWT	Technical officer	Master
3	Mr. Anousone Singkhamvongsa	MoFA	Official	Master
4	Ms. Talany KHOUNYOTHA	MoH	Dentist - Teacher	Bachelor
5	Ms. Soudchai Sengpathoumvanh	MoTC	Officer	Bachelor
6	Mr. PHOMMACHITH SONEXAI	MoTC	Technical	Master
7	Mr. Pisa Keomanilay	MoTC	Technical officer	Bachelor
8	Mr. Kongsamay XAIYAVONGSA	MoTC	Technical Officer	Bachelor
9	Ms. Houangsa Honglikith	MoTC	Officer	Master
10	Mr.bountham Sithixay	MoTC	Technical officer	Bachelor
11	Mr. Phongsavanh Khattiyavong	MoTC	Technical	Bachelor
12	Ms. Phennapha PHOMMACHANH	MoTC	Deputy director	Bachelor
13	Ms. Souphaphone SIRIVATH	MoTC	Technical	Bachelor
14	Mr. Chanthajohn MOUNMANIVONG	MHS Hospital	Engineer	Bachelor
15	Mr. Phoukhong SONEVONGXAY	LNCCI	Deputy Director	Master
16	Mr. Vongthong Xayyapheth	MoES	Director	Master
17	Ms. Anny PHOMMACHANH	MoES	Technical Staff	Master
18	Mr. Keophouxone PHANMALA	MoTC	Technical Officer	Bachelor
19	Mr. Viradeth Sixanonh	MoTC	Technical officer	Master
20	Mr. Souliya Chansomphou	Embassy	Economics counselor	Bachelor
21	Mr. Phoumpaseud SIVONGSAY	MoTC	Technical Officer	Master
22	Mr. Athinanh Manivong	MoTC	Deputy Director	Bachelor
23	Ms. Manilath KINPHOUNSINH	MoTC	Head of Division	Master
24	Mr. Sithong Sikhao	MoES	Deputy Director General	Master

	Name	Organization	Position	Degree
25	Ms. Phainidar Phetleuangxay	MoTC	Technical Officer	Bachelor
26	Mr. Chongnengxiong NENGXAY	MoTC	Officer	Master
27	Ms. phanthanome DIDARAVONG	MoES	Deputy Head	Master
28	Ms. Anousone PHENGSOBATH	MoH	Technical Staff	Master
29	Mr. Sonephet Sonemany	MoES	Technical official	Bachelor
30	Ms. Ketnaphone	MoAF	Technical officer	Bachelor
31	Ms. CHANTHALANGSY Panima	UHS	Lecturer, Doctor	Bachelor
32	Ms. Viengvilay THONGMANYLA	MoTC	Deputy Director	Bachelor
33	Ms. Kaisamone BOUNYARITH	MoH	Technical staff	Bachelor
34	Ms Khamphat	MoIC	Head of Division	Master
35	Ms. Keophayvanh Douangsavanh	MoES	Permanent Secretary	Master
36	Dr. Phonevilay Sinavong	NAFRI	Researcher	Ph.D.
37	Mr. khatiphone chanhphpaseuth	MoTC	Head of Division	Bachelor
38	Mr. Kanya Amkhavanh	MoTC	Deputy of Division	Bachelor
39	Mr. Taokham CHANTHAVONG	MoTC	Technical officer	Bachelor
40	Mr. Thanva SESOULATH	MoTC	Head of Division	Master
41	Ms. Niphalay THONGKHAM	MoH	Student	Bachelor
42	Mr. Semgchanh PHASAYASENG	MoTC	Deputy Director	Master
43	Ms. Phonesathien XAYYALASY	MoH	Technical official	Bachelor
44	Ms. Amphone Chanthavong	Embassy	Third Secretary	Bachelor
45	Dr. Khamphong NAMMAVONGMIXAY	NUOL	Vice President	Ph.D.
46	Mr. Souksamai THOUMBOULOM	PWTTI	Deputy director	Master

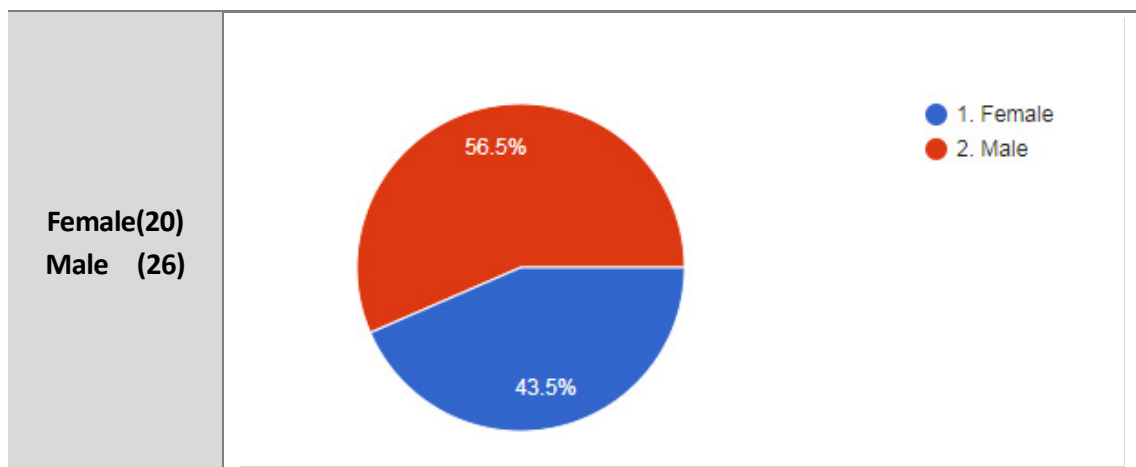
■ **Abbreviations:**

MoTC	Ministry of Technology and Communication
MoPWT	Ministry of Public Works and Transport
MoAF	Ministry of Agriculture and forestry
MoIC	Ministry of Industry and Commerce
LNCCI	Lao National Chamber of Commerce and Industry

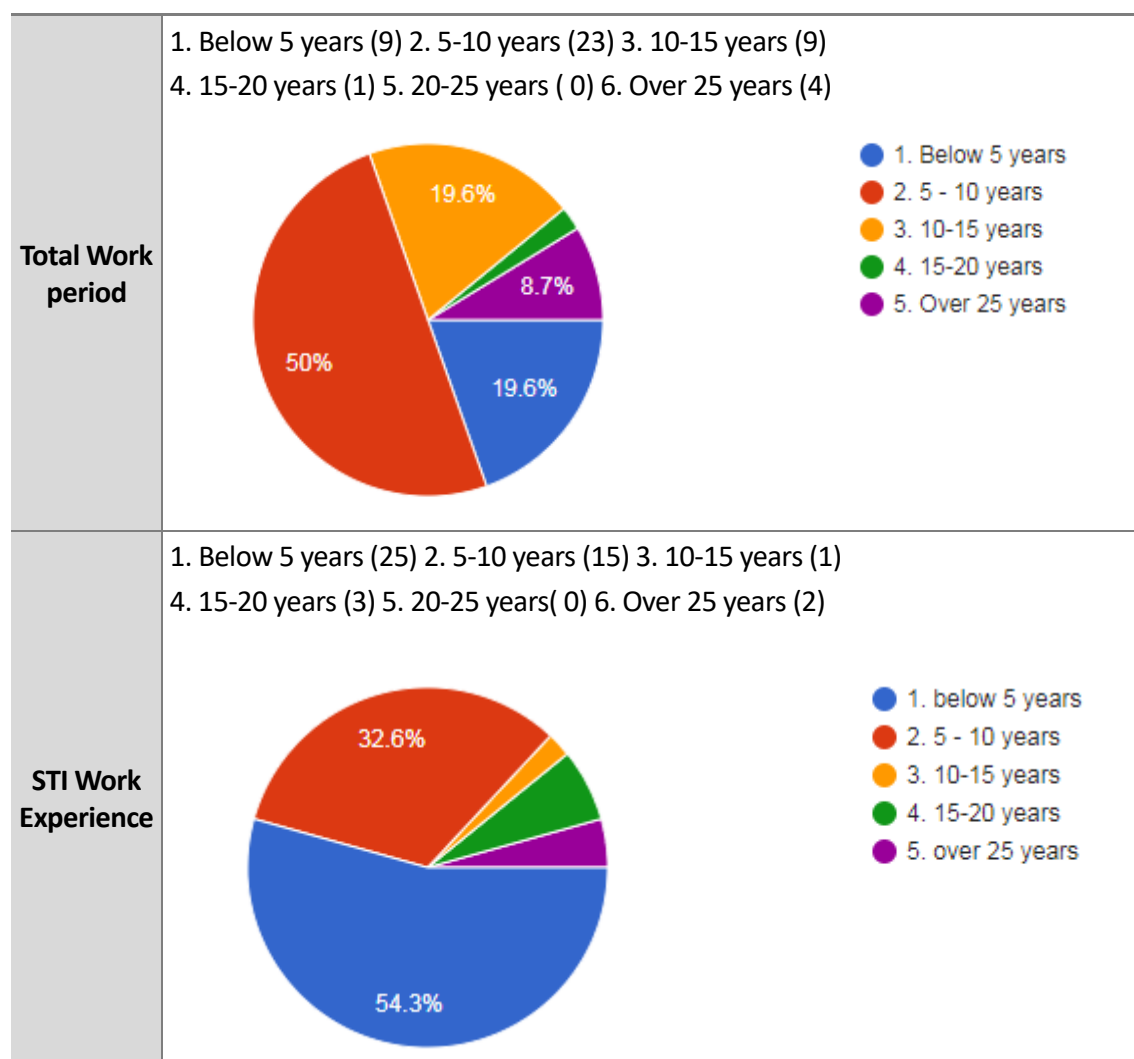
NAFRI	National Agriculture and Forestry Research Institute
PWTTI	Public Works and Transport Training Institute
NUOL	National University of Lao PDR (NUOL)
UHS	University of Health Science
SVK Hospital	Savanakhet Hospital
MHS Hospital	Mahosot Hospital

5.2 Basic Statistics

- **(Target)** The subjects of the survey were 46 people, including civil servants and experts in the field of science and technology in Lao PDR. Among the respondents, there were more males (26 persons, 56.5%) than females.



- **(Working Period)** 89.2% of the respondents were those with less than 15 years of working period, of which 50% (23 persons) were from 5 to 10 years or less. Most of them worked in the STI field for less than 10 years, and more than half accounted for less than 5 years. Specifically, 54.3% (25 people) worked for less than 5 years and 89.9% if 10 years or less were included.
- **(Academy)** The composition of the respondents' degrees consisted of only two Ph.D., 20 masters, 24 bachelors, and so on.



- **(Major)** At the field of major, there were 10 people in the STI field, but 36 people in other fields. However, there were actually many fields related to science such as biomedical engineering, environmental engineering, health science, forestry science, biology, chemical technology, road-bridge engineering, computer science, natural science, mathematics etc.
- **1 Ph.D.** : International and regional development, rural development, agricultural development, and policy research
- **1 Ph.D.** : Medicine. Assoc. Prof. of Pediatrics

Biomedical Engineering	2
Environmental	3
International Studies	3
Health Science	5
materials confluence and system engineering	
Forestry Science, Environmental Science, GIS.	
Biology	
Responsibility of management technology	
Responsibilities about services high technology	
Support MSME in Lao PDR on digital transformation.	
Business Management	2
Chemical Technology	
Road-Bridge Engineering	
Computer Science	
Technology and Innovation statistic system development	
Digital Library and Information Science	
Financial	
Innovation development for SMEs and others.	
Education Planning and Management	
Economic	3
Researching and making the legislation on Technology and Innovation	
Rural Community Development	2
Medical Doctor	2
political science	4
Natural Science	
Planning	
Mathematics	
Master of Law	
Logistic management	

6. Result

6.1 International Collaboration Experiences

The result of international collaboration experiences is to 'yes (43%)', 'no (57%)'.

Key challenges for economic development	Respondents	Percentage (%)
① Yes	20	43.5
② No	26	56.5
Total	46	100.0

The bilateral collaboration had Korea and Vietnam (5), Thailand (4), China (3), Singapore, Japan, Canada, Sweden, Luxembourg (1), and multilateral collaboration has ASEAN, ADB, WHO, FAO (1)

Main collaboration program is follows;

Korea	- Dr. Lee Jong-Wook Seoul project, Master degree program, KOFIH - Study on Materials confluence and system engineering, Changwon National University - Knowledge Sharing Programs,
Vietnam	- Strengthening Capacity Building for Science and Technology Research by Academy of Science and Technology
Thailand	- Science Smart Village Project by Ministry of Higher Education, Science, Research and Innovation - Project Research on Science and Technology Usage for Water Recourse Management by Hydro Informatics Institute (Ministry of Higher Education, Science, Research and Innovation),,

China	- Project Research on Science and Technology Usage for Water Recourse Management by Lancang-Mekong Cooperation Special Fund - China-ASEAN Technology Transfer center, Guangxi Science and Technology department, Nanning, Guangxi Zhuang Autonomous region, PR China on Talented Young Scientist Program (TYSP)
Japan	- SAKURA SCIENCE Exchange Program, Japan Science and Technology Agency
Luxembourg	- Organization Heart surgery project by Lao PDR-Luxembourg Heart Centre
Sweden	- Geo-coding ODA mapping projects for Lao PDR PDR support by Swedish government and others
Others	- ASEAN Health Cluster by Tobacco Industry Demoralization and National Nutrition Committee Secretariat - Economic Policy Support for Enhancing Productivity and Employment, Outcome 2: Innovation Policy Framework, Asian Development Bank (ADB)

6.2 Key challenges in National Socio-Economic Plan (2021-2025)

Lao PDR have NSEP(National Socio-Economic Plan, 2021-2025), and will achieve the objectives as ‘To fully focus on socio-economic development based on the existing potentials in order to help the country effectively achieve the LDC’s criteria through quality, inclusive and green growth and achieve the SDGs by 2030’. The outcome of NSEP(2021-2025) is to i) Quality, inclusive and sustainable growth, ii) Improve human resources, research and apply science technology and intellectual to promote diversified and value added production and service, iii) Improve the living conditions of the people to raise the wellbeing at individual and social level, iv) Green and environmental friendly development, v) Enhance strong and modern regional and international cooperation, integration and connectivity, vi) Effective Government administration and rule of law. The key challenges of NSEP have 6 challenges and will overcome them for achievement of the objectives and outcome. The 6 key challenges are i) insufficient economic base or infrastructure, ii) disparity between urban and rural community, iii) insufficient human resources, iv) reflecting sustainability in the national development, v) inability to mobilize necessary financial resources on stable manner, vi) lack of Science & Technological capacity.

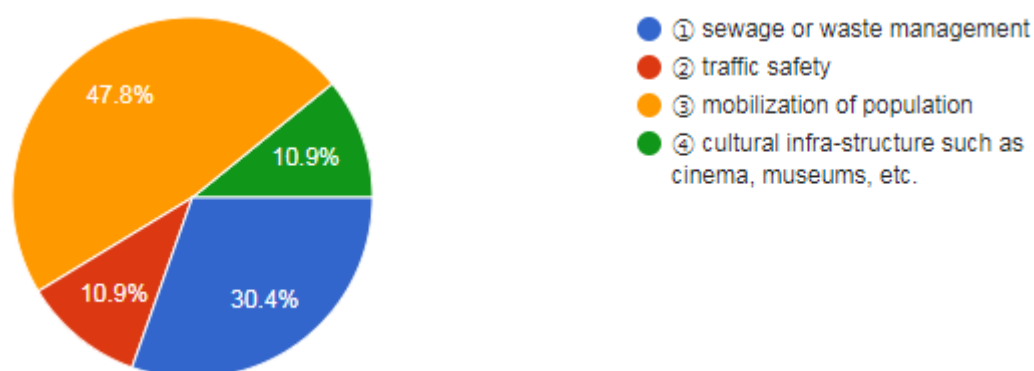
The top 3 priority of key challenges in NSEP(2021-202) is to 'lack of science & technological capacity(26.9)', 'insufficient human resources(23.5%)', and 'insufficient economic base or infra-structure(21%)'.

Key challenges for economic development	Respondents	Percentage (%)
① Insufficient economic base or infra-structure	25	21.0
② Disparity between urban and rural community	15	12.6
③ Insufficient human resources	28	23.5
④ Reflecting sustainability in the national development	8	6.7
⑤ Inability to mobilize necessary financial resources on stable manner	11	9.2
⑥ Lack of Science & Technological capacity	32	26.9
Total	119	100.0

6.2.1 Disparity between urban and rural community

The disparity between urban and rural is also big challenge for national development. The response is to 'mobilization of population (47.8%)' and 'sewage and waste management (30.4%)'. It is important to solve them on the role of STI, In particular, STI give rural and urban to solve the sewage and waste management. When it establishes the plant on sewage and waste management, it needs to technical and engineering and human resources. Furthermore, it put in the science and technology for the reduction of pollution, such as air, water, and hygiene etc.

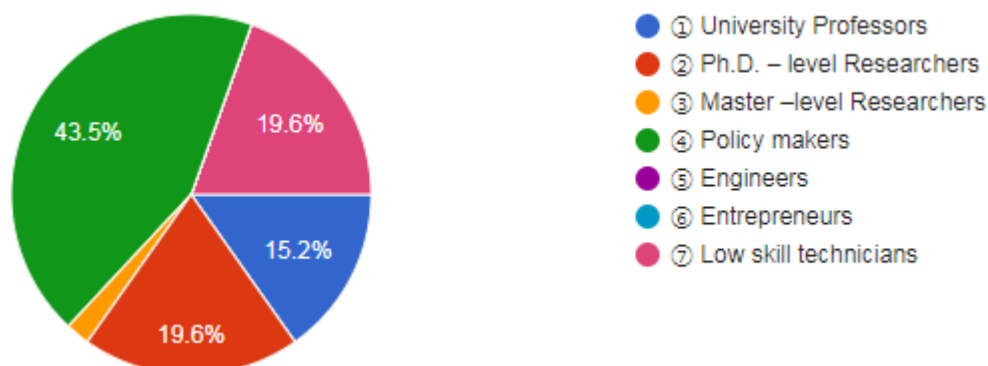
Disparity between urban and rural community	Respondents	Percentage (%)
① sewage or waste management, ,	14	30.4
② traffic safety	5	10.9
③ mobilization of population	22	47.8
④ cultural infra-structure such as cinema, museums, etc.	5	10.9
Total	46	100.0



6.2.2 Insufficient human resources

The most significant resources for solution of insufficient human resources are to 'policy makers (43.5%), 'Ph.D. level researchers (19.6%) and low skilled technicians (19.6%)', and 'university professor (15.2%'. The capacity buildings of the policy makers will be the priority program of cooperation programs, such as study visit, workshop and conference, consulting and training programs for policy makers in STI sector. And it needs to strengthen the leadership of governance and structure of government. Furthermore, it needs to make incentive for recruitment of qualified human resource to government body from other sectors or organizations. The incentive will include salary, working condition and motivation etc.

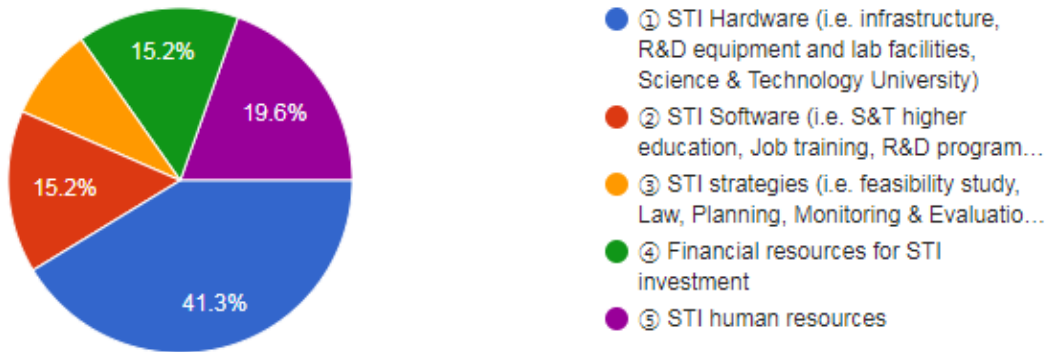
Insufficient human resources	Respondents	Percentage (%)
① University Professors	7	15.2
② Ph.D. – level Researchers	9	19.6
③ Master –level Researchers	1	2.2
④ Policy makers	20	43.5
⑤ Engineers	0	0.0
⑥ Entrepreneurs	0	0.0
⑦ Low skill technicians	9	19.6
Total	46	100.0



6.2.3 Lack of Science & Technological capacity

The most significant capacity for solution of insufficient human resources are to 'set up for STI hardware, 41.3%', 'development of human resources in STI sector(19.6%)', and others (STI program design, international network, and finance, investment etc.). The hardware of STI is to infrastructure, R&D equipment and lab facilities, science & technology university etc., it has foundation as well as condition for development of STI. The software of STI is to the contents such as S&T higher education, Job training, R&D program design & operation, international expert network etc., it has catalyst as well as driving force for development of STI.

Lack of Science & Technological capacity	Respondents	Percentage (%)
① STI Hardware (i.e. infrastructure, R&D equipment and lab facilities, Science & Technology University)	19	41.3
② STI Software (i.e. S&T higher education, Job training, R&D program design & operation, international experts network)	7	15.2
③ STI strategies (i.e. feasibility study, Law, Planning, Monitoring & Evaluation, Coordination)	4	8.7
④ Financial resources for STI investment	7	15.2
⑤ STI human resources R&D	9	19.6
⑥ Budget or financial resources	0	0.0
⑦ Access to market	0	0.0
Total	46	100.0



6.3 The country's priorities listed in National Socio-Economic Plan (2021-2025)

The priority for NSEP is to 6 issues, i)Quality, inclusive and sustainable growth, ii)Improve human resources, research and apply science- technology, to promote diversified and value- added production, iii)Improve the living conditions of the people, iv)Green and environmental friendly development, v)Enhance international cooperation, integration and connectivity, vi)Effective government administration and rule of law.

The top 3 priority is to 'Improve human resources, research and apply science- technology, to promote diversified and value- added production (30.9%)', 'Enhance international cooperation, integration and connectivity (17.9%)', and 'Improve the living conditions of the people (13.8%)'. It is very urgency issues that develop the human resource and apply science technology to provide diversities productions.

Country's priorities listed in National Socio-Economic Plan	Respondents	Percentage (%)
① Quality, inclusive and sustainable growth	16	13.0
② Improve human resources, research and apply science- technology, to promote diversified and value- added production	38	30.9
③ Improve the living conditions of the people	17	13.8
④ Green and environmental friendly development	14	11.4
⑤ Enhance international cooperation, integration and connectivity	22	17.9
⑥ Effective government administration and rule of law	16	13.0
Total	123	100.0

6.3.1 Recommendation for overcome the obstacles

The suggestions for overcome the obstacles for development of national plan are as followings;

1) Form Korea's experience

- The KOREA government has played significant role in the development of Korea's Science and Technology. As the main features of the Korean policy measure,
 - First, the Korean government focused on the expansion of R&D investment and human resource of S&T from the very early stage of development, which is considered a key factor that contributed to the rapid growth of Korea's S&T.
 - Second, the government put much effort to determine the policy direction and optimal growth path.
 - Third, it focused on fostering the R&D like: Government Research Institutes (GRI); private companies, and universities. Moreover, it actively promoted the national R&D program providing support measures as well as infrastructures at the same time.
 - Finally, the Korean government sought the productivity of input resource by implementing measures of selecting the strategic sector and concentrating investment on the available resources.

2) Effective government administration and rule of law

- The creating appropriate policies to support both government and private sector, ensuring the confidential environment on trade that allow creativities, competitiveness and cooperation.
 - List of key activities:
 - Creating appropriate policies and laws
 - Financial support from government
 - Human recourse development
 - Establish central learning for all (websites, mobile application, R&D institute)

- Support information and cooperation to entrepreneurs
- Expected results:
 - Having confidential action toward partners, economic plans and investor.
 - Allowing small business to growth
 - Having potential governance officer
 - Allowing people to improve their skill through the application of central learning
 - Good environment on trade

3) Obstacle: lack of employment in rural areas and limited knowledge on STI

- Potential action: move resources to rural areas and encourage local entrepreneurs to create more jobs for local people
- Activities: Organize workshop for local people related to STI
- Expected results: improve living conditions of people in rural areas, increase employment

4) Low quality of Human resource.

- Government needs to invest more skill development among young generation.
- Key activities including:
 - Improving the quality of educational and vocational institutions
 - Partnership with private sector
 - Collaborate with international educational institution in human resource development
- High skill labor force who can improve and strengthen the STI in the country.

5) Improve human resources, research and apply science- technology, to promote diversified and value- added production

- To improve human resources we also have obstacle is the Financial sources to support the implementation of STI and improve human resource on STI and activities are limited',

- For Potential action to overcome is we should have a finance support and joint working with many institutions and university in public and private since central until local ',
- After we have joint working we will discussion and list detail of key activities together for creating new things support to STI policy and contribution to social economic development plan as well as', '
- We expected our country will have many experts on STI and human resource on STI was developed more '.

6) Improve the living conditions of the people

- obstacle is knowledge , Technical skill and finance support are limited
- public and private should close up working more and take output knowledge and technical skill to local for awareness and training to them
- we will list detail of key activities for them which they will take these knowledge application to their life and creating their business and they will have a good life condition
- the people in hold country will get civilization'.

7) Enhance international cooperation, integration and connectivity

- obstacle is knowledge, understanding and fund
- internal and external cooperation have to understanding need both parties and relevant organizations , institutions public and private
- then we will have list detail of key activities in these cooperation ',
- we will have a good cooperation, good partner

H) Quality

- The shortage of human resource and limited of financial support

- allocated the number of staff working at the Central and provinces, and financial budget. Regularly monitoring and supporting from relevant sector, organize the lesson learn meeting to exchange the implemented experience with other countries.
- Enhancing the capacity of staff who has been working in this area, requesting the budget support more from the government and development partners.
- abdicated staff with the knowledge and skills working in this area.

8) Base on the LAO HEALTH SECTOR REFORM 2013-2025 improving the human resources is one of the priorities and urgent issues that need to be implemented .

- knowledge and skills of staff on administrative, managing and implementing plan is limited.
- The government needs to improving employment capacity both the short and long term, including training in the country and attending training oversea, organize the lesson learn workshop with the related sector and development partners.
- Capacity building of staff will be improved.

■ Others

The Government of Lao PDR has attached great important of the human resources development in the health and science sector as our priority to advance the national strategic development policy of the country, particularly in the areas of the rural development and poverty reduction of Lap Government to prepare Lao PDR for graduating from the Least Developed Country status in 2024. The development of human resources in Lao PDR in the past years from the central to local government has continuously enhanced and upgrade their knowledge in specialties which improves the quality of working. However, there is still insufficiency in demand of society due to it may face challenges of the process to progress the improvement in institutions

- # challenges

- The institutions are lacking of experiences of human resources development in the quantities and qualities
 - The Government is facing difficulties in financial support to allocate new workers in this sector especially after their gradations
 - The descent welfare policies towards are not in level that be able to influence, guarantee and make them to improve their performance as well as their responsibility in the workplace
 - Investing in human resources development in science-technology sectors are still limited.
- # solution
 - Improving and developing the comprehensive educational system in infrastructure, curriculum and resources such as lecture and expertise. Also, improving coordination mechanism between the educational institute and training place to make availabilities for students to get real practice experience.
 - Improving, reform the Human resource management that currently exists to a higher level by enhance specific knowledge both short and long-term courses.
 - Continuing cooperation with development partners, international organization to share knowledge and share experience in new technologies and attract more financial assistance to use in development sector for human resources in the Lao PDR.

Improve human resources, research and apply science- technology, to promote diversified and value- added production the reason choose this tittle because of indispensable to develop industrial human resources and to secure a transition from the educational system to the world of work in order to promote industry diversification and advancement, while it has pointed out the importance of industrial human resource development in its development policies. Expected results to consider industrial sectors with high growth potential, which are expected to boost the economy of Lao PDR.

According to National Socio Economic plan 2021 to 2025 and vision 2030 we focus on

Improve human resource, protect and reuse natural resource, Green environment and sustainable energy by add more faculty relative in this field on the University level, add more ministry of natural resource and environment in line of government, improving those policy and strategy about environment and sustainable energy but we still weak on laboratory equipment, limited number of professional on this way and have gaps on cooperation between government and private sector.

6.4 Main challenges for development of STI in Lao

The most suitable issues of main challenges in STI is to 'limited system to support R&D commercialization & technology transfer (17.1%)', 'no comprehensive open data on R&D equipment and facilities (16.4%)', and 'limited Ph.D. researchers (13.6%)'.

Main challenges for development of STI	Respondents	Percentage (%)
① No action plan to implement STI policy	15	10.7
② No legal framework to enforce STI policy	15	10.7
③ Gaps between national development plan and STI policy	10	7.1
④ No high-level body to coordinate, monitor, evaluation to R&D program	8	5.7
⑤ Inequality of policy-design or revision-activities	4	2.9
⑥ Limited Ph.D. researchers in research institutes	19	13.6
⑦ No comprehensive open data on R&D equipment & facilities, talents, entrepreneurs, companies	23	16.4
⑧ Limited system to support R&D commercialization & technology transfer	24	17.1
⑨ Few market to commercialize R&D results	8	5.7
⑩ Few collaboration between university-industry-research	14	10.0
Total	140	100.0

6.4.1 Major activities in the STI lifecycle

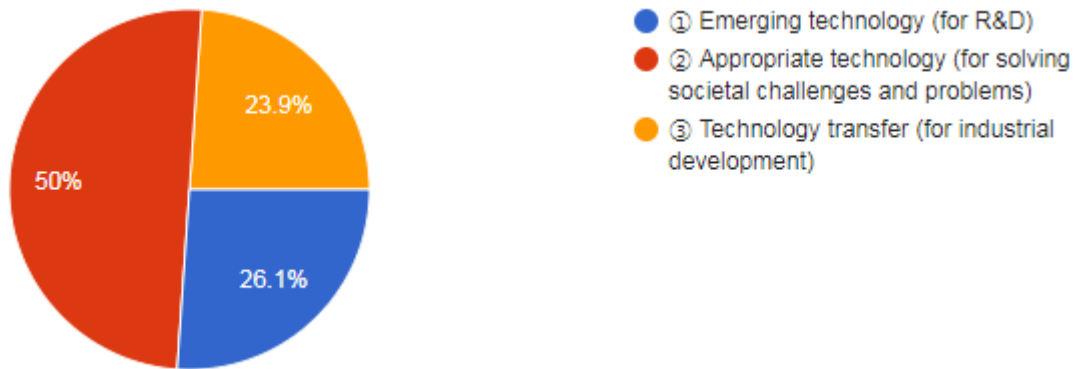
The major activities in STI lifecycle is to 'applied research and development (32.3%)', 'basic R&D (24.0%)', and 'demonstration for technology scale-up and pilot projects (21.9%)'.

Main issues for Science and Technology	Respondents	Percentage (%)
① Basic research and development	23	24.0
② Applied research and development	31	32.3
③ Demonstration (technology scale-up and pilot projects)	21	21.9
④ Technology commercialization	6	6.3
⑤ Technology deployment/diffusion	15	15.6
⑥ others ()	0	0.0
Total	96	100.0

6.4.2 The list of needed Science and Technology items for economic development in Lao PDR

The most necessity item of science and technology for economic development is to 'appropriate technology (50.0%)', 'emerging technology (26.1%)', and 'technology transfer (23.9%)'. It shall need to develop the appropriate technology for economic development. Or it needs to consider the appropriated technology for introduction of science and technology to Lao PDR from overseas. For this, it shall share with knowledge and technology through workshops, seminars, and forum, and put together capacity development, specially Ph.D. holder.

Needed Science and Technology items for economic development	Respondents	Percentage (%)
① Emerging technology (for R&D)	12	26.1
② Appropriate technology (for solving societal challenges and problems)	23	50.0
③ Technology transfer (for industrial development)	11	23.9
④ others ()	0	0.0
Total	46	100.0



6.4.3 Domestic obstacles or risks to develop Science and Technology

The obstacles or risks for STI hardware is to 'R&D equipment and lab facilities (16.6%)', 'infrastructure (12.6)', and 'science and technology university (11.2%)'. The obstacles and risks for STI software is to 'international experts network (12.1%)', 'job training & technology training (7.6%)', 'R&D facility maintenance (5.4%)'. The obstacles and risks for STI strategies is to 'National STI policy planning (10.8%)', 'law and institutional framework (4.9%)'. In conclusion, it shall need to develop the R & D equipment and lab facilities for overcoming of obstacles of the STI hardware. And for the alleviation of the risks of the STI software, it needs to establish the international expert networks.

Domestic obstacles or risks to develop Science and Technology		Respondents	Percentage (%)
STI hardware (select one)	① infrastructure	28	12.6
	② R&D equipment and lab facilities	37	16.6
	③ Science & Technology University	25	11.2
STI software (select two)	④ S & T higher education	11	4.9
	⑤ Job training & technology training	17	7.6
	⑥ R&D program design and operation	12	5.4
	⑦ R&D facility maintenance	4	1.8
	⑧ International experts networks (i.e. exchange programs)	27	12.1

Domestic obstacles or risks to develop Science and Technology		Respondents	Percentage (%)
STI strategies (select two)	⑨ Feasibility Study	2	0.9
	⑩ Law and institutional framework	11	4.9
	⑪ National STI policy planning	24	10.8
	⑫ Policy Monitoring and Evaluation	10	4.5
	⑬ Coordination among Ministries or different stakeholders	15	6.7
Total		223	100.0

6.4.4 Recommendation for overcome the obstacles.

Obstacle	Technology for Education
Potential action to overcome	To achieve these goals, the country will need to improve labour skills. T&I is very essential for economic and social development
List of key activities:	Young entrepreneurs in Lao PDR have innovative business model but the main obstacle for them are lack of financial support and appropriate policies including a stage to perform their ideas
Expected results	The developers and programmers have to creating innovative technology, Disruptive technology to address the problems and challenge that society is facing with.

Obstacle	Management technology
Potential action to overcome	Make Experience and development knowledge In Lao PDR, most of them are importing technology, so there is a problem of transfer and use of technology
List of key activities:	Very limited technology transfer especially in MSME sector
Expected results	create give more application and applying new technology for social service, give startup and research results awards

6.5 The types of international cooperation

The international cooperation for overcoming the obstacles in science and technology is to mixture bilateral cooperation and multilateral cooperation, triangular cooperation, support for professors or researchers, more education facilities, grant and R&D cooperation on applied ST. There is a lot of multilateral cooperation in Mekong region, so it needs to cooperate with variety international agencies.

1) Type: Bilateral cooperation; Multilateral cooperation; Mixture of Bi and Multilateral cooperation

Types of international cooperation that can be used to overcome the obstacles described in Q12	Respondents	Percentage (%)
① Bilateral cooperation	9(9)	19.6(17.6)
② Multilateral cooperation	13(16)	28.3(31.4)
③ Mixture of Bi and Multilateral cooperation	24(26)	52.2(51.0)
Total	46(51)	100.0(100.0)

Note: Numbers in the Parentheses reflect multiple answers.

2) Type: South-South cooperation; South-North cooperation; Triangular cooperation

Types of international cooperation that can be used to overcome the obstacles described in Q12	Respondents	Percentage (%)
① South-South cooperation	8(8)	17.4(16.0)
② South-North cooperation	15(17)	32.6(34.0)
③ Triangular cooperation	23(25)	50.0(50.0)
Total	46(50)	100.0(100.0)

Note: Numbers in the Parentheses reflect multiple answers.

3) Type: Intensified support for STI students; Intensified support for STI professors or researchers ; Intensified support for STI policy makers

Types of international cooperation that can be used to overcome the obstacles described in Q12	Respondents	Percentage (%)
① Intensified support for STI students	15(15)	32.6(27.8)
② Intensified support for STI professors or researchers	22(26)	47.8(48.1)
③ Intensified support for STI policy makers	9(13)	19.6(24.1)
Total	46(50)	100.0(100.0)

Note: Numbers in the Parentheses reflect multiple answers.

4) Type: More education facilities in Lao PDR; Training in advanced countries

Types of international cooperation that can be used to overcome the obstacles described in Q12	Respondents	Percentage (%)
① More education facilities in Lao PDR	34(34)	73.9(64.2)
② Training in advanced countries	10(15)	21.7(28.3)
③ Wrong answers	2(4)	4.3(7.5)
Total	46(53)	100.0(100.0)

Note: Numbers in the Parentheses reflect multiple answers or wrong answers.

5) Type: STI training at secondary education; STI training at tertiary education; STI training at post graduate

Types of international cooperation that can be used to overcome the obstacles described in Q12	Respondents	Percentage (%)
① STI training at secondary education	18(18)	39.1(32.7)
② STI training at tertiary education	18(19)	39.1(34.5)
③ STI training at post graduate	10(18)	21.7(32.7)
Total	46(50)	100.0(100.0)

Note: Numbers in the Parentheses reflect multiple answers.

6) Type: More grant; Mixture of grant and loans; More loans

Types of international cooperation that can be used to overcome the obstacles described in Q12	Respondents	Percentage (%)
① More grant	26(26)	56.5(52.0)
② Mixture of grant and loans	19(23)	41.3(46.0)
③ More loans	1(1)	2.2(2.0)
Total	46(50)	100.0(100.0)

Note: Numbers in the Parentheses reflect multiple answers.

7) Type: R&D cooperation on basic ST; R&D cooperation on applied ST

Types of international cooperation that can be used to overcome the obstacles described in Q12	Respondents	Percentage (%)
① R&D cooperation on basic ST	24(24)	52.2(45.6)
② R&D cooperation on applied ST	20(27)	43.5(47.4)
③ Wrong answers	2(4)	4.3(7.0)
Total	46(50)	100.0(100.0)

Note: Numbers in the Parentheses reflect multiple answers or wrong answers.

6.6 The strategic STI sectors to design a Blue print for STI in Lao PDR

The most useful strategic STI sector is to 'ICT (33.3%)', 'agriculture (22.7%)', and 'bio technology (14.4%)'. Another sector is to health & science, environment, energy, and heavy industry. If it needed to select some sector in science and technology, it may select 3 areas. Lao PDR has a lot of bio diversity in world, so it need to convergence with ICT and agriculture, ICT and bio technology.

Strategic STI sectors to design a Blue print for STI	Respondents	Percentage (%)
① ICT (Information, Communication and Technology)	44	33.3
② Agriculture	30	22.7
③ Bio-technology	19	14.4
④ Energy	7	5.3
⑤ Health & medicine	18	13.6
⑥ Environment	9	6.8
⑦ Heavy industry	3	2.3
⑧ Others ()	2	1.5
Total	132	100.0

6.6.1 Recommendation for ideas on STI sector (i.e. action name, a list of activities and expected results).

Strategic STI Sector	ICT
Name of potential action	innovative technology to adapted or solution that help increase the quality and quantity of product especially during Covid-19 pandemic
List of key activities:	-establish sustainable model for agriculture framework -STI adapted to smart farmer and marketing strategy and public outreach
Expected results	- having easy access to central platform to increase their market demand - linking information throughout central and local organization. - raise awareness on STI application

- Sector: ICT (Government and Private sectors related to STI)a. Name of potential action: ICT for development to contribute national social economic development plan.
- b. Key Activities: Create ICT policy and strategy for facilitating, Capacity building on ICT policy and development
- c. Expected results: Having Policy and Strategy on ICT, Having strengthening on ICT capacity building for sustainable Development

- Improvement of the organization, systematization of management mechanism on STI's activities
 - a. Apply R&D in priority area and sectors (IT, Biotech, Renewable and new material, Machinery and automation technology)
 - b. Provide ST services for society and people in remote areas, and
 - c. Cooperation with international
- sector: bio-technology
 - a. action name: R&D on bio-technology
 - b. activities: financial and collaboration support from international institutes
 - c. results: Lao PDR can produce bio-technology products so that Lao people can afford
- ICT
 - a. Develop existing STI policy in order to align with national and international master plan
 - b. cooperation among related stakeholders through at meeting to come up with a best policy solution which fits into need.
 - c. sustainability and easily to implement, decrease the confusion and overlapping with others
- Agriculture sector
 - a. Improve the quality of agricultural product
 - b. Activities:
 - Research in new product
 - Develop new agricultural technology
 - c. High quality product with low cost
- Sector: Bio-technology
 - a. Action name: R&D on bio-technology
 - b. Activities: financial and collaboration support from international institutes
 - c. Results: Lao PDR can produce bio-technology products so that Lao people can afford

- ICT: ICT is very useful for life being nowadays example: knowing covid 19 epidemic information on time by social media or applications which reminds us to be mindful etc., However, ICT in Lao is not vastly used in the industrial manufacture such: robotic, AI, due to the lacking of knowledge and funding capacity. Moreover, ICT is still be applied everywhere else such as ICT for the farmer who works on rubber tree operating, On this work running forecasting is essential because they need to avoid running. One thing I would suggest to apply ICT the most is monitoring the issuing governmental certificate permits (land certificate, penalty notice certificate, address certificate etc.,)
- Agriculture is a priority of the government currently because Lao PDR has riches on the water and is favorable for plants. This role is under the Ministry of Agriculture and Forest however I choose agriculture to be said here due to the plantations in Lao grant very low quantity which compare to other countries as such Vietnam and Thai land. Even we are in similar weather, As we know and see currently that Lao PDR still use conventional technologies in planting. Therefore accessing high-tech and applying in the industrial process is needed. This barrier needs the organizations in charge (MTC) to address and boost up the conventional technology to a high-tech level. There are many ways to overcome this obstacle, firstly the government should provide technology transfer and making the pilot project and then it will be widely diffused around the country
- Energy: Lao is planning to be the battery of ASEN due to the favourable landscape. However, we need to concerns the sustainable development and global climate change. Nowadays even if Lao has generated electricity but most of the Hydropower dam is foreign's stock and Lao people still pay high tax on the electric consumption so I have a suggestion that the Lao government should make a priority on renewable energy instead of expanding of the Hydropower dam. the purpose of ideas is to concern about the risk of flooding, drought, and dam erosion, I, therefore, see that Lao PDR has potential for renewable energy sources that are available all area in Lao.
- The Information and Communication Technology as an increasingly crucial tool for achieving socio-economic development, aims to bring the country into the information

age by increasing general access to ICT with the provision of modern telecommunications infrastructure and computer networks; by fostering enterprise and industry; promoting research and development in the fields of ICT; and by developing the necessary human resources and institutional capacities, expected to results ensure that the necessary institutional, human capacity, sectoral conditions, and legal frameworks are in place for leveraging and applying ICT to meet the challenges and needs of Lao PDR

- Agriculture;
 - a. apply science and technology to agricultural commercialize production in rural areas of Lao PDR;
 - b. R& D research:
 - c. Suitable technology for agricultural development in Lao PDR
- Improvement of the organization, systematization of management mechanism on STI's activities
 - a. Apply R&D in priority area and sectors (IT, Biotech, Renewable and new material, Machinery and automation technology)
 - b. Provide ST services for society and people in remote areas, and
 - c. Cooperation with international

6.6.2 Potential STI projects in Lao PDR

The most significant project for STI project is to 'to develop more STI human resources (23.5%)', 'to develop STI infrastructure (20.6%)', and 'to develop or promote public awareness of STI (16.9%)'. Another project is to 'to promote more intensified international cooperation in STI sector', 'to prepare adequate STI regulations or laws', 'to establish more efficient management or service for STI sector', etc.

Potential STI projects	Respondents	Percentage (%)
① To develop or promote public awareness of STI	23	16.9

② To prepare adequate STI regulations or laws	10	7.4
③ To develop more STI infrastructure	28	20.6
④ To develop more STI human resources	32	23.5
⑤ To establish more efficient management or services for STI sector	10	7.4
⑥ To promote more intensified international cooperation in STI sector	20	14.7
⑦ To establish closer link between central and local government	13	9.6
Total	136	100.0

6.6.3 The suggestion for STI sector (i.e. action name, a list of activities and expected results)

- To develop more STI human resources (Education sector)
 - a. STI training at tertiary education
 - b. added up STI lesson into school textbook and focus more on prototype action.
 - c. improving hard skills and critical thinking of human recourse
- To develop or promote public awareness of STI (Technology and Communications and Educations Sectors (Both Government and Private)).
 - a. STI for development.
 - b. Seminar, Workshop, Training.
 - c. Society have more understanding about STI, people enhancing knowledge on STI
- Sector: Promote awareness
 - a. action name: promote public awareness of STI
 - b. activities: organize seminars of STI to public
 - c. results: People of all levels have basic knowledge of STI
- to promote more intensified international cooperation in STI Sector
 - a. exchange and share experiences, now how, and way forward for STI

- b. meeting, training, short and long-term cooperation project,
 - c. expand cooperation network, grant aid, technologies, and so on.
- Science sector
 - a. Develop human resource
 - b. Activities:
 - Establish ST institution
 - Provide scholarship for talented student to take ST course
 - c. More innovations
 - Agriculture development;
 - a. R&D pilot project;
 - b. project implementation, evaluation;
 - c. knowledge sharing

Lao PDR has enjoyed strong economic growth over the past two decades, thanks largely to policies which have exploited its rich natural capital (forestry, hydropower and minerals) and strategic position in a fast-growing region. In order to maintain current growth rates and set the country on a more sustainable development path, Lao PDR is seeking to diversify its productive system by encouraging the production of higher-value goods and services.

Agricultural innovation has the potential to improve livelihoods of farmers and rural people, improve food and nutrition security, and allow for sustainable management of natural resources. An agricultural innovation system is described as a network of individuals, organizations and enterprises, with supporting institutions and policies in agricultural and related sectors that brings together existing or new products, processes and forms of organization to create additional socio-economic value. While farmers are the ultimate innovators and many innovations start locally, they require conducive policy, institutional support and facilitated partnership with different actors in order to scaled up and out sustainably. Farmers are willing and able to innovate through adoption of improved practices, building facilities and adapting technologies such as crops production, livestock breeding, seed varieties, and water management for crops and aquatic production if they are supported by the government and development partners.

Develop more STI human resource. In terms of sustainability, local human resources is needed for developing STI in the country instead of relying in the international TA. Therefore, the provision of STI in Higher education is required, the training and partnership with all stakeholders especially with the private sector is definitely required in order to strengthen capacity for Lao youth, Lao workers and Lao researchers.
To develop or promote public awareness of STI to long-term socio-economic development goals. In particular, SDGs and MDGs, green-sustainable. Because we need to promoting sustainable development and achieving the Millennium Development Goals (MDGs). Indeed, science, technology and innovation can play a critical role in each and every MDGs, including by: fostering access to knowledge, increasing productivity, industrialization, economic growth and the creation of decent jobs, develop more STI infrastructure and develop more STI human resources. In order to ensure a comprehensive, coherent report on STI and culture.
STI sector needed to disseminate the Science and Technology content to the public so the people would understand and know the importance of the Science and Technology work. Also, the Human resources need to be improved such by training and gaining a higher degree (PhD degree) from the developed countries. Furthermore, the international cooperation is also important for improvement of Science and technology sector for Lao PDR.
To promote more intensified international cooperation in STI sector. Lao government has a plan for cooperation with many countries and bilateral both party as department of technology and innovation, ministry of technology and communication cooperation with STEPI, Korea country for creating master plan STI for Lao and we also have been stilling cooperation with Thailand on science smart village project

6.6.4 Key subjects for STI strategic capacity development for government officials in Lao PDR

The priority orders of the most significant for STI strategic capacity development is to 'national STI policy planning', 'law and institutional framework', 'coordination among Ministries or different stakeholders', 'policy monitoring and evaluation', and 'feasibility study'.

Key subjects for STI strategic capacity development for government officials in Lao PDR	Respondents	Percentage (%)
① Feasibility Study	20	14.7
② Law and institutional framework	29	21.3
③ National STI policy planning	39	28.7
④ Policy Monitoring and Evaluation	21	15.4
⑤ Coordination among Ministries or different stakeholders	27	19.9
Total	136	100.0

6.7 Benchmarking of Korean STI development experiences

The top priority of Korean STI development experiences is to ‘to develop the human resources to meet demand of the industry (22.2%)’, ‘to establish the science and technology university for development high qualified human resources (20.7%)’, ‘To establish the national research institutes for research and development (14.1%)’.

Korea’s STI development experiences	Priority()	
	Respondents	Percentage (%)
① To develop the human resource to meet the demand of the industry	30	22.2
② To establish the science and technology university for development high qualified human resources	28	20.7
③ To establish the national research institutes for research and development	19	14.1
④ To develop the specific area to each regional universities	13	9.6
⑤ To set up the decision-maker for development of science and technology	14	10.4
⑥ To set up the cooperation network between entrepreneurs-university-institute	18	13.3
⑦ To set up and manage to national research and development system	13	9.6
Total	135	100.0

6.7.1 The necessity to Korea's STI experiences

The necessity on Korea's STI experiences as followings; 69.6% of 'to develop the human resource development to meet the demand of the industry' is very necessary. 47.8% of 'to establish the science and technology university for development high qualified human resources' is necessary. 47.8% of 'to establish the national research institutes' is very necessary. 73.9% of 'to develop the specific area to each regional universities' is necessary. 65.2% of 'to set up the decision-maker for development of S&T' is necessary. 60.9% of 'to set up the cooperation network between entrepreneurs-university-institute' is necessary. 60.9% of 'to set up and manage to national R&D system' is necessary.

Korea's STI development experiences	Necessary()			
	NO (%)	NN (%)	N (%)	VN (%)
① To develop the human resource to meet the demand of the industry	0 (0.0%)	0 (0.0%)	14 (30.4%)	32 (69.6%)
② To establish the science and technology university for development high qualified human resources	2 (4.3%)	1 (2.2%)	22 (47.8%)	21 (45.7%)
③ To establish the national research institutes for research and development	2 (4.3%)	1 (2.2%)	21 (45.7%)	22 (47.8%)
④ To develop the specific area to each regional universities	0 (0.0%)	4 (8.7%)	34 (73.9%)	8 (17.4%)
⑤ To set up the decision-maker for development of science and technology	2 (4.3%)	3 (6.5%)	30 (65.2%)	11 (23.9%)
⑥ To set up the cooperation network between entrepreneurs-university-institute	0 (0.0%)	3 (6.5%)	28 (60.9%)	15 (32.6%)
⑦ To set up and manage to national research and development system	0 (0.0%)	3 (6.5%)	28 (60.9%)	15 (32.6%)
Total				

*NO: No opinion, NN: Never Necessary, N: Not Necessary, N:Necessary, VN: Very necessary

6.7.2 The suggestions for the most urgent Korean experiences (i.e. action name, a list of activities and expected results).

- According to Korean's experiences on STI development to establish the science and technology university for development high qualified human resources
 - a. STI cooperation between Ministry of Technology and Communication, Ministry of Education and Sports and STEPI Korea
 - b. list of activities:
 - creating STI's policies,
 - establish science and technology university
 - c. expected results:
 - having qualified STI policies to support economic growth
 - preparing work forces for the developing demand
- To establish the science and technology university for development high qualified human resources.
 - a. STI knowledge for Students
 - b. Awareness STI knowledge, Lesson learn of STI, Training activities (Short term - Long term).
 - c. Students have more understanding and knowledge on STI as well as the participants who attend the training or seminar.
- To establish the national research institutes for research and development
 - a. Lao-Korea STI R&D
 - b. build up facilities, filled up with required equipment, Korean STI's Experts, local related government officials, put it a educational curriculum or higher.
 - c. skilled human resources, promote STI policy and its importance, effective implementation, and motivate a STI market both locally and internationally.
- To develop the human resource to meet the demand of the industry
 - a. Provide learning opportunity to government officials and private sectors
 - b. Activities: study trip in Korea, conduct research in Korea
 - c. Results: improve local capacity to develop what the country needs

- Develop the human resource
 - a. base of STI
 - b. create law and policy, set up the decision-maker and manage to national & development system
 - c. national STI development plan, policy makers, R&D equipment and lab facilities.
- To develop the human resource to meet the demand of the industry
 - a. To establish the science and technology university for development high qualified human resources
 - b. Develop ST courses, provide high quality ST lab and equipment
 - c. More qualified and quality researchers.
- To develop the human resource to meet the demand of the industry
 - a. Provide learning opportunity to government officials and private sector
 - b. Activities: study trip in Korea, conduct research in Korea
 - c. Results: improve local capacity to develop what the country needs
- To develop the human resource to meet the demand of industry;
 - a. Human Resource Capacity building for STI;
 - b. Training in related fields, ability competition;
 - c. able to develop and upgrade knowledge, more interest in such a sector and the number of human resources in this sector has increased

I don't have much idea about this area. But as far as I understand, the STI management strategy policy should be developed clearly and in line with the government's development plan.

I would recommend Science & Technology Policy Institute (STPI) as one of the most efficient institute that provide knowledge on science, technology and innovation in order to boost in terms of IT innovation.

I believe it is important to establish the cooperation network between industry-university-government to promote national economic growth. For example: Korea's Science Park/Techno Parks adapt this model, pushing innovative products that lead to commercialization.

In conclusion, I strongly believe that, once I obtained cooperation of Korea, I will try my best to hope experiences to help My country I have learnt from my study towards my career improvement back in my home country especially where I work for as much as possible.
In my opinion I think the Korean's experience about establish the science and technology university for development high qualified human resources is very important and useful for Lao PDR to develop the STI, nowadays Lao PDR don't have the specific university about STI so we can learn the experience from Korea how to process
To establish the cooperation network between industry-university-government can help promote local products or services into commercialization and therefore can improve our economic growth.
To establish the cooperation network between industry-university-government can help promote local products or services into commercialization and therefore can improve our economic growth.
Regarding the most urgent Korean experiences are all very necessary for us but depend on with many facilities as vision and direction of leader, fund,human resource , knowledge and others
I think firstly, Lao PDR needs the STI experts to lead the directions and then establish some specific institute that doing research on the frontier technology and innovation that meet contributes to the social needs or real solutions.
Public Private Partnership in one of the best practice to strengthen the STI system. therefore concerned regulations have to be formulated and PPP has to be promoted
Lao PDR has achieved constant economic growth for several decades and has been expected to contribute to the development of the ASEAN region as an attractive investment destination. However, it is also true that Lao PDR needs to address several urgent issues such as industrial diversification and advancement, and development of relevant systems and policies.
I believe that the sharing experience from Korea benefit us to achieve the goal especially establishing the master plan on STI. However, I think focusing on creating national policy and infrastructure of STI, it should be the first priority because of demanding the STI development and deployment in social-economy is increase day by day.
Developing the human resource is one of the key elements to achieving the settled goal, along with updated technology and technical assistance from development partner and other country. I strongly believe that we can go over the problem and led to sustainable growth.
To establish the science and technology university for development high qualified human resources, the Human resources need to be improved by training and seeking for the expertise. as a result, if the Human resource of Lao PDR have an expertise in Science and

Technology field, it would enhance the living condition and the economic of the country to be in the better way.
I believe that the sharing experience from Korea benefit us to achieve the goal especially establishing the master plan on STI. However it should be the first priority because of the demanding the STI development and deployment in the social - economy is increase day by day
I believe that the sharing experience from Korea benefit us to achieve the goal especially establishing the master plan on STI. However, I think focusing on creating national policy and infrastructure of STI, It should be the first priority because of the demanding the SIT development and deployment in social-economy is increase day by day.
Regarding Korean experiences to establish the science and technology for developing highly qualified human resources and support the private sector to develop productivity is could be apply for STI programme.
Regarding the experience, to establish the science and Technology University for Development High Qualified human resources are very necessary for Lao PDR. Due to the lack of STI experts in Lao PDR, which is very important for the current socio-economic development, the establishment of universities in the STI field is necessary to make everyone aware of and understand the importance of STI in order to be involved in STI competition. For implementation, there is the need for cooperation and exchange of lessons with countries that have advanced in STI, especially Republic of Korea by intergovernmental cooperation.

■ Survey Form



Questionnaire

This survey is a part of STEPI's study on "National Master Plan for STI R&D and Application in Lao PDR" through '2021 K-Innovation Partnership Program.

Science and Technology Policy Institute(STEPI) is a Korean government-funded research institute devoted to research and development of S&T policy and policy alternatives to promote

* STI: Science, Technology, Innovation; R&D: Research & Development

This survey aims to collect views of Lao PDR's STI experts on Korea's STI development experiences. Your answers and comments will be used to ensure the impact and sustainability as well as baseline data of the outcome of the project, as well as to identify future ODA projects on STI sector. We invite your sincere co-operation based on the best of your knowledge. Please note that all your answers will be kept confidential.

Should you have any enquiries regarding this survey, please contact Prof. CHANG, Ji-Soon at nowis21c@gmail.com or +82 (0)10 5625 1472.

Please return your reply by 10 September 2021 to Prof. CHANG at nowis21c@gmail.com.

Please check (✓) or fill out the followings on your basic information

Name	1. Female ()		
	2. Male ()		
Telephone Number		Email	
Total Work period	1. Below 5 years () 2. 5-10 years () 3. 10-15 years () 4. 15-20 years () 5. 20-25 years () 6. Over 25 years ()		
Organization Name			
Type of organization	1. Government () 2. Research Institute () 3. Academia () 4. Private company () 5. NGO () 6. Others ()		
Position at Organization		STI Work Experience	() years
Degree	1. Diploma () 2. Bachelor () 3. Masters () 4. Ph.D. ()		
Major in	1. STI () 2. Others ()		

1. Do you have international collaboration experiences with Foreign Government, University, NGO, Research Institute, or Private Company engaged in Science and Technology sector?

① Yes () ② No ()

1.

⑤ R&D professionals ^②	☐
⑥ Budget or financial resources ^②	☐
⑦ Access to market ^②	☐

7. Regarding country's priorities listed in National Socio-Economic Plan (2021-2025), please check the most urgent priority in the Table below. ^②

Country's priorities listed in National Socio-Economic Plan ^②	Check (✓) ^②
① Quality, inclusive and sustainable growth ^②	☐
② Improve human resources, research and apply science- technology, to promote diversified and value- added production ^②	☐
③ Improve the living conditions of the people ^②	☐
④ Green and environmental friendly development ^②	☐
⑤ Enhance international cooperation, integration and connectivity ^②	☐
⑥ Effective government administration and rule of law ^②	☐

8. Please find below a list of main challenges for development of STI in Lao. Please check 3 most suitable issues in the Table below. ^②

Main challenges for development of STI ^②	Check (✓) ^②
① No action plan to implement STI policy ^②	☐
② No legal framework to enforce STI policy ^②	☐
③ Gaps between national development plan and STI policy ^②	☐
④ No high-level body to coordinate, monitor, evaluation to R&D program ^②	☐
⑤ Inequality of policy-design or revision-activities ^②	☐
⑥ Limited Ph.D. researchers in research institutes ^②	☐
⑦ No comprehensive open data on R&D equipment & facilities, talents, entrepreneurs, companies ^②	☐
⑧ Limited system to support R&D commercialization & technology transfer ^②	☐
⑨ Few market to commercialize R&D results ^②	☐
⑩ Few collaboration between university-industry-research ^②	☐

9. Please find below a list of major activities in the STI lifecycle. Please check the most urgent issue in the Table below. ^②

Main issues for Science and Technology ^②	Check (✓) ^②
① Basic research and development ^②	☐
② Applied research and development ^②	☐
③ Demonstration (technology scale-up and pilot projects) ^②	☐
④ Technology commercialization ^②	☐
⑤ Technology deployment/diffusion ^②	☐
⑥ others () ^②	☐

10. Please find below a list of needed Science and Technology items for economic development

If 'Yes', please answer below; if 'No', go directly to Question 3

2. Please complete the Table below (in the order of the most important experience)

Supporting Foreign		Project Title	Period (m/y-m/y)
Country	Organization		

3. Please find below a list of key challenges in National Socio-Economic Plan (2021-2025).

Please check 3 most key challenges in the Table below.

Key challenges for economic development	Check (✓)
① Insufficient economic base or infra-structure	
② Disparity between urban and rural community	
③ Insufficient human resources	
④ Reflecting sustainability in the national development	
⑤ Inability to mobilize necessary financial resources on stable manner	
⑥ Lack of Science & Technological capacity	

4. Regarding your response on 'Disparity between urban and rural community' in Q3, please check (✓) the most significant in the Table below.

① sewage or waste management	
② traffic safety	
③ mobilization of population	
④ cultural infra-structure such as cinema, museums, etc.	

5. Regarding your response on 'Insufficient human resources' in Q3, please check (✓) the most significant in the Table below.

① University Professors	
② Ph.D. – level Researchers	
③ Master –level Researchers	
④ Policy makers	
⑤ Engineers	
⑥ Entrepreneurs	
⑦ Low skill technicians	

6. Regarding your response on 'Lack of Science & Technological capacity' in Q3, please check (✓) the most significant in the Table below.

① Infra-structure	
② Regulations or Institutional Frameworks (i.e. laws, policies) laws	
③ High skilled labor	
④ Low skilled labor	

in Lao PDR. Please check the most needed item in the Table below.

Needed Science and Technology items for economic development	Check (✓)
① Emerging technology (for R&D)	
② Appropriate technology (for solving societal challenges and problems)	
③ Technology transfer (for industrial development)	
④ others ()	

11. Please find below a list of domestic obstacles or risks to develop Science and Technology. Please check each category in the Table below.

Domestic obstacles or risks to develop Science and Technology	Check (✓)
STI hardware (select one)	① infrastructure
	② R&D equipment and lab facilities
	③ Science & Technology University
STI software (select two)	④ S & T higher education
	⑤ Job training & technology training
	⑥ R&D program design and operation
	⑦ R&D facility maintenance
	⑧ International experts networks (i.e. exchange programs)
STI strategies (select two)	⑨ Feasibility Study
	⑩ Law and institutional framework
	⑪ National STI policy planning
	⑫ Policy Monitoring and Evaluation
	⑬ Coordination among Ministries or different stakeholders

12. Regarding the obstacles you have selected in Q11, please suggest your ideas to overcome the obstacles.

Obstacle	
Potential action to overcome	
List of key activities	
Expected results	

13. Please find below 7 types of international cooperation that can be used to overcome the obstacles described in Q12. Please check (✓) the most relevant item per type.

- Type: Bilateral cooperation (); Multilateral cooperation (); Mixture of Bi and Multilateral cooperation ()
- Type: South-South cooperation (); South-North cooperation (); Triangular cooperation ()
- Type: Intensified support for STI students (); Intensified support for STI professors or researchers; Intensified support for STI policy makers ()
- Type: More education facilities in Lao PDR (); Training in advanced countries ()
- Type: STI training at secondary education (); STI training at tertiary education (); STI training at post graduate ()
- Type: More grant (); Mixture of grant and loans (); More loans ()

⑦ Type: R&D cooperation on basic ST (); R&D cooperation on applied ST ()

14. Please find below a list of strategic STI sectors to design a Blue print for STI in Lao PDR.

Please check 3 most useful strategic STI sectors in the Table below

Strategic STI sectors to design a Blue print for STI	Check (✓)
① ICT (Information, Communication and Technology)	
② Agriculture	
③ Bio-technology	
④ Energy	
⑤ Health & medicine	
⑥ Environment	
⑦ Heavy industry	
⑧ Others ()	

15. Regarding 3 strategic STI sectors you have selected in Q 14, please suggest your ideas for STI sector (i.e. action name, a list of activities and expected results).

Strategic STI Sector	
Name of potential action	
List of key activities:	
Expected results	

16. Please find below a list of potential STI projects that can be pursued for the next 5 years in Lao PDR. Please check 3 most significant projects in Table below.

Potential STI projects	Check (✓)
① To develop or promote public awareness of STI	
② To prepare adequate STI regulations or laws	
③ To develop more STI infrastructure	
④ To develop more STI human resources	
⑤ To establish more efficient management or services for STI sector	
⑥ To promote more intensified international cooperation in STI sector	
⑦ To establish closer link between central and local government	

17. Regarding 3 potential STI projects that you have selected in Q16, please suggest your ideas for STI sector (i.e. action name, a list of activities and expected results).

Strategic STI Sector	
Name of potential action	
List of key activities:	
Expected results	

18. Please find below a list of key subjects for STI strategic capacity development for government officials in Lao PDR. Please select priority orders from the most significant (5) to

5.

the least significant (1).⁴

① Feasibility Study ⁴	⁴	..
② Law and institutional framework ⁴	⁴	..
③ National STI policy planning ⁴	⁴	..
④ Policy Monitoring and Evaluation ⁴	⁴	..
⑤ Coordination among Ministries or different stakeholders ⁴	⁴	..

19. Please find below a list of Korean STI development experiences that can be benchmarked by Lao PDR. This question is intended to collect two types of information: priority and necessary. ⁴

1) Please check 3 most urgent experiences under the 'Priority' column, and⁴

2) Please check the most suitable box under the 'Necessary' column according to the following criteria:⁴

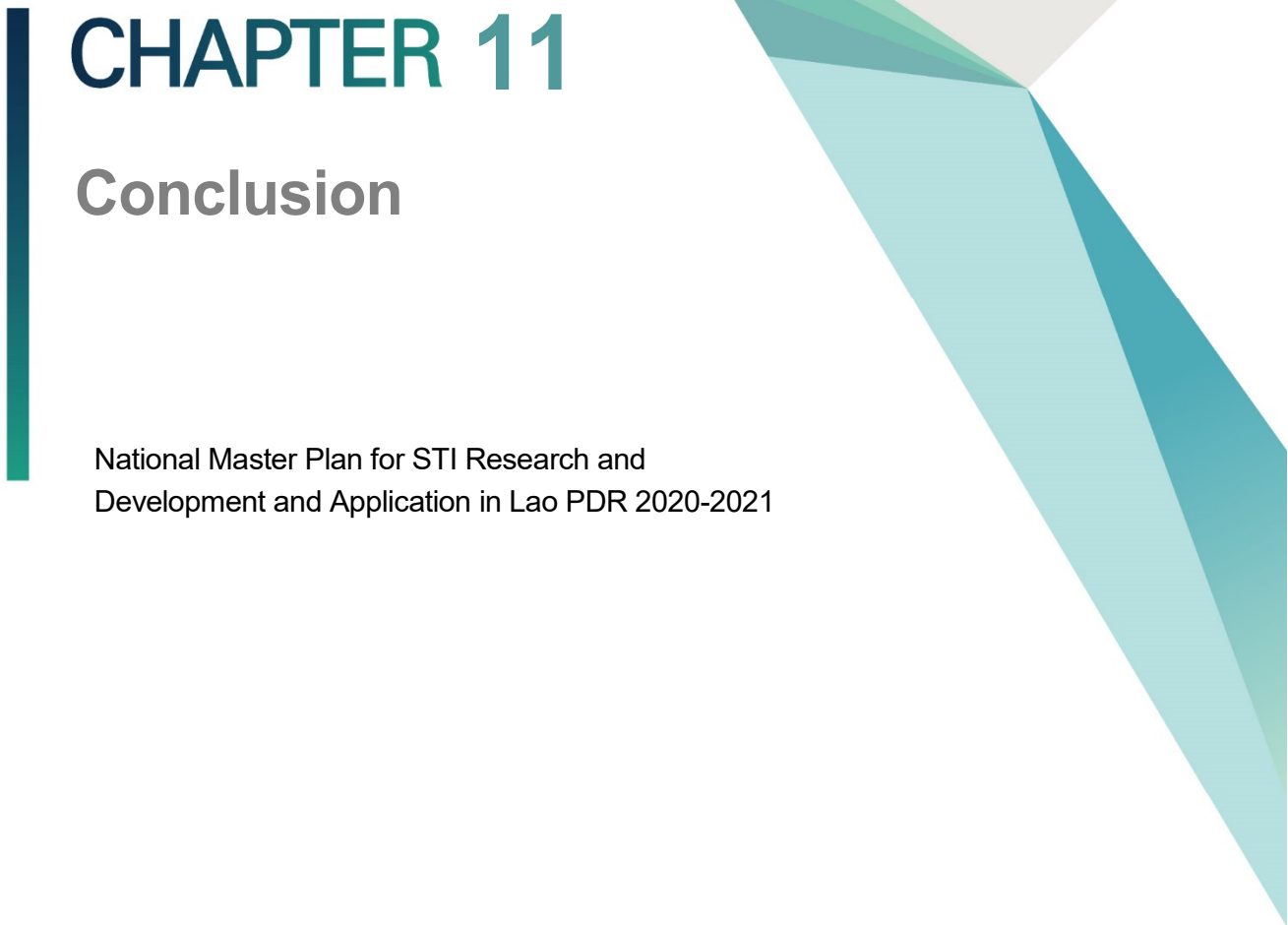
① No opinion ② Never Necessary ③ Not Necessary ④ Necessary ⑤ Very necessary⁴

Korea's STI development experiences ⁴	Priority ⁴ (✓) ⁴	Necessary(✓) ⁴					..
① To develop the human resource to meet the demand of the industry ⁴	⁴	⁴	⁴	⁴	⁴	⁴	..
② To establish the science and technology university for development high qualified human resources ⁴	⁴	⁴	⁴	⁴	⁴	⁴	..
③ To establish the national research institutes for research and development ⁴	⁴	⁴	⁴	⁴	⁴	⁴	..
④ To develop the specific area to each regional universities ⁴	⁴	⁴	⁴	⁴	⁴	⁴	..
⑤ To set up the decision-maker for development of science and technology ⁴	⁴	⁴	⁴	⁴	⁴	⁴	..
⑥ To set up the cooperation network between entrepreneurs-university-institute ⁴	⁴	⁴	⁴	⁴	⁴	⁴	..
⑦ To set up and manage to national research and development system ⁴	⁴	⁴	⁴	⁴	⁴	⁴	..

20. Regarding the most urgent Korean experiences you have selected in Q19, please suggest your ideas (i.e. action name, a list of activities and expected results).⁴

Benchmarking ⁴	⁴	..
Name of potential action ⁴	⁴	..
List of key activities: ⁴	⁴	..
Expected results ⁴	⁴	..

*Thank you so much for sparing your precious time on this questionnaire!*⁴

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a light grey triangle pointing left, and a larger teal shape composed of several overlapping triangles pointing right.

CHAPTER 11

Conclusion

National Master Plan for STI Research and
Development and Application in Lao PDR 2020-2021

Chapter 11. Conclusion

1. Conclusion

The objective of the K-Innovation Partnership Program with the Lao PDR (2020-2022) is to support the development of the first National Master Plan for STI Research & Development and Application in Lao PDR. Since the new administration of President Thongloun Sisulith restructured the government in 2021, the K-Innovation project with Lao PDR has sought to strengthen the policy capacity of key institutions of the Ministry of Technology and Communications (MTC) in the mid-and long-term STI strategic planning in Lao PDR, as well as the development of the master plan.

The Department of Technology and Innovation (DTI) is the key actor in STI policy planning with the functions of the secretariat for MTC on the development and macro-management of policies and strategies on technology and innovation. DTI has been considering Technology and Innovation Plan (2021-2025) under the 9th National Socio-Economic Development Plan, with a special focus on; increasing R&D on technology and innovation framework; developing standard of technology and innovation; developing technology projects; increasing awareness of technology and innovation; providing technology and innovation services; strengthening coordination and cooperation between different sectors in both international and domestic partners; and lastly, human resources development for higher quantity and quality of knowledge and know-how.

To strengthen the international technology and innovation network, DTI has been carrying out international cooperation projects with Asian countries such as China, Thailand, Vietnam as well as international organizations. In particular, the STEPI-MTC project focused on developing the master plan for research, development, and application, which would be

the foundation for STI-oriented national socio-economic development of the Lao PDR.

As MTC has just started implementing Digital Economic Development Plan (2021-2025) and Startup Master Development Plan, the issue in developing the master plan during the project is to effectively incorporate other strategies, plans, laws on high-tech into the newly developed master plan. The other challenges identified by DTI are the broad functions and duties of DTI with limited policy experience of its staff and insufficient financial resources.

Technology and innovation are, however, recognized as main factors in contributing to socio-economic development to enhance the competitiveness of the business sector and also contribute to GDP growth in Lao PDR. In the Fourth Industrial Revolution, there are widespread understanding and hope among people that technology will lead to a better life. In addition, there is a consensus among STI key stakeholders that human resources development, R&D infrastructure, and international networks are essential and urgent needs to enhance the national STI and socio-economic development.

The research team finds out a strong demand of MTC for developing effective STI policy to provide coherent and integrated insights to specific and unique circumstances of the country as well as capacity building of STI experts and international partnership. Therefore, it suggests that the development of highly qualified human resources can be one of the main focuses of the National STI Master Plan that the project has to develop in the final year. Given that DTI will work on the e-government system within five years, it also recommends that the master plan may include the issue of establishing an STI data system.

In order to produce an effective and feasible STI master plan that will eventually contribute to the sustainable development of Lao PDR, the research team composed of experts both from Korea and Lao PDR should interact rigorously next year to clarify what strategic areas and factors among many STI issues to be focused in the master plan. Considering this is the first STI Policy collaboration between Korea and Lao PDR, this project would be a good launch pad for a future bilateral partnership that benefits both countries for strengthening STI national competitiveness.

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